From Human Markets to Machine Markets

Certify, Don't Identify: Safety and Identification in Markets with Adaptive Trading Agents

Pratik Niroula

Department of Computer Information Science, Minnesota State University, Mankato

Thesis blueprint, October 2026

What this document is. The intended final form of a three-paper thesis. Papers 1 and 2 are finished and appear as Chapters 2 and 3; Paper 3 (Chapter 4) is new, as are the introduction and conclusion. Every result in the new chapters carries a tag: PROVED IN THIS THESIS (with its proof), PROVED IN CH. 2/3 (proved in a paper chapter), TARGET: STATEMENT FIXED, PROOF TO BE WRITTEN (the statement the thesis will prove, with a proof plan), CONJECTURE, or PENDING DATA (an empirical test specified but not yet run). No number in the new chapters is invented; Appendix A lists the status of every main result.

Abstract

Electronic markets are feedback systems in which many adaptive trading algorithms respond to prices that their own trades move. Regulators often respond by asking to know which algorithms are trading. This thesis asks whether that knowledge is needed for safety, and answers that it generally is not. In the closed-loop markets studied here, prices reveal the agents only through a merged measure of their memories and weights; separating two agents whose memory rates differ by δ costs trading capital of order δ^{-4}, separating K unknown agents costs a number of experiments of order $(\Delta\eta)^{-(4K-2)}$, and with general memory the accuracy of any method improves only as $1/\log N$. Safety, by contrast, depends on the agents only through one number, the summed price response, which is certified at the parametric rate $\sigma^2\epsilon^{-2}\log(1/\epsilon)$ whatever their number and spacing, provided memory is bounded, and cannot be certified at all without such a bound. A pre-trade guard built on the certificate is the largest safe admission rule and is minimax-optimal in lost capacity up to a logarithmic factor. A third paper organises these results under one framework in which every question an observer can ask is a functional of the agents' merged measure, and its cost is set by how the functional behaves as memory becomes long. It extends the safety results to sector-bounded nonlinear agents and to agents that keep learning, for which a certificate has a computable lifetime. In data, trader-class disclosures from Binance and the CFTC pin down memories that prices leave unrestricted, and in pre-registered tests the summed-response statistic forecasts adverse price excursions in thin crypto books and, more weakly, in thin US stocks; tests of a leverage or liquidation mechanism fail.

Statement of co-authorship

Chapter 2 reproduces the paper *Auditable Memory and Financial Identification Limits in Electronic Limit Order Books*, joint with Prashant Niroula. Chapter 3 reproduces the paper *From Human Markets to Machine Markets: Transient Amplification, an Exact Depth Condition, and a Maximal Pre-Trade Guard*, joint with Suboh Alkhushayni, Yuhao Chen, Mansi Bhavsar and Prashant Niroula. *[To be completed with the co-authors: the candidate's contribution to each chapter, and permission to include the papers.]* Chapter 4 (Paper 3), the introduction and the conclusion are the candidate's own work.

Contents

Abstract i

Statement of co-authorship ii

1 Introduction 1

1.1 The question . . . 1
1.2 The answer in one paragraph . . . 1
1.3 Three papers . . . 2
1.4 Status of the results . . . 3
1.5 Relation to existing work . . . 3
1.6 Reading guide . . . 4

2 Auditable memory and identification limits 5

2.1 Introduction . . . 6
2.2 Model and microfoundations . . . 11
2.3 The identification boundary . . . 14
2.4 Probing resources, memory resolution and money . . . 15
2.5 Account-level capital . . . 19
2.6 Unknown mixtures and passive data . . . 26
2.7 Robustness to nonlinear and adaptive trading rules . . . 35
2.8 What disclosure buys . . . 38
2.9 A spectral blind spot . . . 46
2.10 Empirical feedback and its measurement . . . 47
2.11 Implications for supervision and market design . . . 51
2.12 Conclusion and an open problem . . . 53
2.13 Appendix: Proofs . . . 56
2.14 Appendix: Notation . . . 63
2.15 Appendix: Further results . . . 63
2.16 Appendix: Reproducibility: source of every number . . . 66

3 Safety without identification: transient amplification and a maximal pre-trade guard **69**
3.1 Introduction . . . 70
3.2 The closed-loop market . . . 77
3.3 Spectral stability is not solvency . . . 80
3.4 When does a book amplify? An exact condition . . . 82
3.5 Identification and financial safety . . . 86
3.6 An identification-free pre-trade guard . . . 91
3.7 The value of disclosure for safe trading . . . 101
3.8 Who provides reversion? An equilibrium . . . 102
3.9 Evidence from Binance futures depth . . . 105
3.10 Evidence from spot prices . . . 109
3.11 Evidence from thinner books . . . 111
3.12 Tests of the information-safety results . . . 129
3.13 Policy implications . . . 144
3.14 Conclusion . . . 145
3.15 Appendix: Proofs . . . 149
3.16 Appendix: Notation . . . 154
3.17 Appendix: Further results . . . 155
3.18 Appendix: Reproducibility: source of every number . . . 156

4 Certify, don't identify: a theory of safety without identification **161**
4.1 Markets with adaptive agents . . . 162
4.2 Questions are functionals of the merged measure . . . 162
4.3 Observation contracts and the cost of a question . . . 163
4.4 The certification–identification dichotomy . . . 163
4.5 Disclosure changes the contract, not the question . . . 167
4.6 A robust safety bound: nonlinear and changing agents . . . 167
4.7 Certificates expire: learning agents . . . 169
4.8 Markets of trading algorithms that learn . . . 171
4.9 Agents who respond to the guard . . . 172
4.10 What the framework does not cover . . . 173

5 Conclusion and open problems **174**
5.1 Evidence that would settle the thesis . . . 174
5.2 Open problems . . . 177

A Status of results and notation across chapters **179**
A.1 Status register . . . 179

A.2 Notation across chapters 180

Bibliography for Chapters 1, 4 and 5 **182**

Chapter 1

Introduction

1.1 The question

Markets are moving from human traders to machines. Where prices were once set by people reading news and order books, they are now set mostly by algorithms that read prices, trade on them in milliseconds, and increasingly learn from the market while they trade in it. This transition changes what can be known about a market and what it takes to keep it safe, and it is the subject of this thesis.

Electronic markets are feedback systems. Trading algorithms watch prices, trade on what they see, and move the prices they watch. The algorithms are private, heterogeneous and changing, and a growing share of them learn from the market while they trade in it. Whoever is responsible for safety in such a market, a regulator, an exchange or a dealer, faces agents it cannot see.

The usual response is to ask to see inside. Under MiFID II, firms must flag algorithmic orders and identify the algorithm behind each decision; the US Consolidated Audit Trail records who stands behind every order. Behind these requirements sits an implicit claim: that a market cannot be kept safe without knowing who is trading in it.

This thesis tests that claim. Its central question is:

> *When can a market with unknown adaptive agents be certified safe without identifying the agents, and what does each kind of knowledge cost?*

1.2 The answer in one paragraph

The information that certifies safety is far smaller and far cheaper than the information that identifies the agents. Prices reveal the agents only through a merged measure of their memory rates and weights (Chapter 2). Recovering the agents from that measure is a moment problem whose cost explodes as agents become similar: the trading capital needed to separate two

memory rates δ apart grows as δ^{-4}, the number of experiments needed to separate K agents as $(\Delta\eta)^{-(4K-2)}$, and with general memory no method does better than accuracy $1/\log N$. Safety needs much less. The worst-case transient loss of a position depends on the agents only through one number, the summed price response, and that number is certified at the parametric rate whatever the number and spacing of the agents, provided their memory is bounded (Chapter 3). Chapter 4 explains why: every question an observer can ask is a functional of the merged measure, and its cost is decided by how that functional behaves as memory becomes long. "Is the market safe?" is a smooth question and cheap; "who is trading?" needs the fine structure of the measure and is expensive; questions that are singular at infinite memory cannot be answered from finite data at all. The safety half survives when agents are nonlinear and keep learning, at the price of certificates that expire.

1.3 Three papers

The thesis consists of three papers, framed by this introduction and a conclusion.

Paper 1: identification limits (Chapter 2). In a linear cohort model whose reduced form arises from Kalman filtering and regularized online learning, prices identify only the merged rate–weight measure. Separating a known pair costs executed quantity of order δ^{-2} and capital of order δ^{-4}, with explicit constants and a finite-horizon certificate; K unknown cohorts cost $(\Delta\eta)^{-(4K-2)}$ experiments, matched by Prony's method; general completely monotone memory gives the logarithmic rate; passive data cost the same polynomial order. Class-tagged disclosure removes the barrier and is worth a factor $(\Delta\eta)^{-(4K-4)}$ in samples. Trader-class positions from Binance and the CFTC pin down memories that prices leave unrestricted.

Paper 2: safety without identification (Chapter 3). In a limit order book with adaptive cohorts, cumulative transient risk has the closed form $S = \tilde{\kappa}/(m - q\tilde{\kappa})$, a book overshoots if and only if depletion outweighs impact-weighted reversion, and a pre-trade guard on the summed impulse response is the largest safe admission rule. S is certified with $O(\sigma^2\epsilon^{-2}\log(1/\epsilon))$ experiments; a finite window certifies it only under a memory bound; the certified guard is minimax-optimal in lost capacity up to a logarithmic factor; disclosure has a computable value for safe capacity; and reversion is under-supplied in equilibrium. In a pre-registered empirical program the statistic forecasts adverse excursions in thin books, while mechanism tests fail.

Paper 3: a theory of certification without identification (Chapter 4). Questions are treated as functionals of the merged measure and observation contracts as noise models on its moments. For linear questions the paper proves an upper bound on the cost of certification in terms of the question's tail at long memory and its coefficient energy, shows that truncation at that depth is necessary, and shows that questions singular at infinite memory cannot be certified

at all (Theorem 4.4.1(i)–(iii), PROVED IN THIS THESIS); the matching lower bound is the thesis's main open technical target. The rates of Papers 1 and 2 appear as cases of one picture. A robust safety bound then shows that the summed-gain guard stays safe for nonlinear agents whose rules, memories and weights change over time, given only a bound on their total sensitivity (Theorem 4.6.2, PROVED IN THIS THESIS), and a corollary gives the lifetime of a certificate when agents keep learning (Corollary 4.7.1, PROVED IN THIS THESIS). Whether the guard survives agents who adapt to it is stated as a conjecture.

1.4 Status of the results

Table 1.1: Main results of the thesis and their status

Result	Where	Status
Prices identify only the merged measure	Thm 2.3.1	proved
Capital of order δ^{-4}; $(\Delta\eta)^{-(4K-2)}$ experiments; $1/\log N$	Ch. 2, §2.4–2.6	proved
Safety depends only on S; guard is maximal	Lemma 3.6.1	proved
S certified at the parametric rate; impossible without a memory bound	Thms 3.5.3, 3.5.5	proved
Cost of certification for any linear question: upper bound, truncation, impossibility	Thm 4.4.1(i)–(iii)	proved
Matching lower bound	Thm 4.4.1(iv)	target; uniform refuted
Guard safe for nonlinear, time-varying agents	Thm 4.6.2	proved
Several assets (induced-norm and sharp bounds)	Props 4.6.4, 4.6.5	proved
Certificate lifetime under learning	Cor. 4.7.1	proved
Guard under strategic adaptation	Remark 4.9.2	existence refuted
Learning-agent simulation	§4.8	simulation, pre-registered
Ground-truth and prospective tests	§5.1	pre-registered design

1.5 Relation to existing work

The thesis connects four literatures that rarely meet. *System identification* (Ljung, 1999) asks how well a system's parameters can be recovered from input–output data; Paper 1 shows that in closed-loop markets the relevant parameters are zeros of the transfer function and are recoverable only at polynomial or logarithmic rates. *Mixture and moment problems* in statistics give the closest mathematical relatives: the optimal rate for estimating a k-atom mixing distribution is $n^{-1/(4k-2)}$ (Heinrich and Kahn, 2018; Wu and Yang, 2020), super-resolution has the same Vandermonde geometry (Candès and Fernandez-Granda, 2014; Moitra, 2015), and the estimation of smooth *functionals* of a distribution can be far easier than estimating the distribution itself (Lepski et al., 1999; Cai and Low, 2011). The theory of Paper 3 is the counterpart of that last observation for safety questions in feedback markets. *Robust and positive control* (Megretski

and Rantzer, 1997; Farina and Rinaldi, 2000; Rantzer, 2015) supplies the comparison arguments of Paper 3 and the ℓ_1 gain that bounds worst-case loss. *Partial identification* in econometrics (Manski, 2003) provides the stance: when the structure is not identified, ask which decisions are nonetheless determined. In finance, the closest work is on transient impact and resilience, liquidity spirals and automated-market crashes, reviewed in Chapters 2 and 3 (Brunnermeier and Pedersen, 2009; Kirilenko et al., 2017).

1.6 Reading guide

Chapters 2 and 3 are self-contained papers and keep their own notation, appendices and references. Chapter 4 can be read after either of them: it states the common framework, and each of its results points to the chapter where the special case is proved. Chapter 5 sets out the evidence that would settle the thesis's empirical claims and the open problems. Appendix A lists the status of every main result and maps the notation across chapters.

Chapter 2

Auditable memory and identification limits

This chapter, the first paper of the thesis, reproduces the paper *Auditable Memory and Financial Identification Limits in Electronic Limit Order Books*, joint with Prashant Niroula. It establishes the identification half of the thesis: what prices reveal about the agents of a closed-loop market, and what it costs to learn more. Its notation is its own; Appendix A maps it to the notation of Chapter 4.

Summary. When trading algorithms condition on recent prices, price formation is a closed-loop system. In a linear cohort model with exponential memory, prices identify only a merged rate–weight measure. Separating two memory rates differing by δ requires executed quantity $\Theta(\delta^{-2})$ and trading capital $\Theta(\delta^{-4})$, with matching attainability, adaptive bounds and a finite-horizon certificate ($[61.41, 64.48]$ par units at the BTCUSDT calibration). Separating K unknown cohorts requires $\Theta((\Delta\eta)^{-(4K-2)})$ experiments, while general completely monotone memory reduces minimax accuracy to $\Theta(1/\log N)$. Passive data incur the same polynomial cost; online learners trained in the closed loop obey the same δ^{-4} scaling when their feedback weights match (slopes -4.01 to -3.97), whereas hard thresholds conceal cohorts. Class-tagged disclosure reduces sample complexity by $\Theta((\Delta\eta)^{-(4K-4)})$. Empirically, trader-class positions tighten memory-rate intervals 27.6 to 124.0 times across 30 Binance perpetuals and 9.1 to 71.4 times across eight CFTC-regulated futures markets.

Reproducibility. Every number in the paper is produced by a script or notebook cell that writes a result file from public data or from a closed form; Appendix 2.16 maps each table, figure and number to its source. Every simulation number was re-run for this version. The specifications of the October 2026 follow-up tests (Sections 2.7 and 2.8.3) were committed to the repository before their test data were downloaded; because each specification reached the public repository in the same push as its results, that order is documented by the commit history rather than by an independent timestamp.

2.1 Introduction

Regulators increasingly require firms to say which algorithm sent each order. Under MiFID II, EU investment firms must flag algorithmic orders and identify the algorithm responsible for each investment and execution decision in their transaction reports (ESMA, 2016); the US Consolidated Audit Trail records who stands behind every order (SEC, 2012). These regimes are costly to build and to comply with, and they rest on an implicit claim: that a supervisor could not learn the same thing from public prices and order flow. This chapter tests that claim, and prices the answer.

The setting is a closed-loop market. Algorithms update on recent prices, their orders move prices, and prices feed back into the algorithms. We model K cohorts of algorithms that track the price with exponential memory, the reduced form of steady-state Kalman filtering and of regularized online gradient descent (Lemma 2.2.1), and ask what an outside observer can learn about them from market data. The answer is sharp. Prices reveal only a merged summary of the algorithms. Separating two algorithms whose memories differ by δ costs trading capital of order δ^{-4}, and separating K unknown algorithms costs a number of experiments of exact order $(\Delta\eta)^{-(4K-2)}$; with general memory, accuracy improves only as $1/\log N$, so halving the error squares the number of experiments. Class-tagged disclosure removes the barrier, and its value can be stated in samples: a factor of order $(\Delta\eta)^{-(4K-4)}$. In data, Binance's published long/short ratios by trader class, a coarse form of class tagging, pin down memories that prices and flow leave unrestricted in each of 30 futures markets, and the CFTC's mandated trader-class reports do the same in eight regulated U.S. futures markets.

1. An identification gap. Prices identify only the merged rate–weight measure $M = \sum_i q_i \eta_i \, \delta_{1-\eta_i}$ of the algorithms (Theorem 2.3.1). Cohorts that share a memory rate are observationally equivalent, whatever their identities or wealth. Nearby rates can be separated, but at a cost that rises quickly as they approach each other.

2. A resource gap. For a known symmetric pair separated by δ, price-only identification with deterministic probes requires executed quantity of order δ^{-2}, and a single pulse attains this order (Theorem 2.4.1). Because a probe moves the price against the prober, the quantity order becomes a capital order δ^{-4}, whether capital is supplied by a counterparty under a margin contract (Theorem 2.4.3) or by the probe trader itself (Lemma 2.4.5). We then compute what an account actually needs. A spectral argument gives a lower bound on the trading loss of any flat round trip (Theorem 2.5.2); Wald's identity extends it to adaptive, sequential probing (Theorem 2.5.3); and a Rayleigh–Hölder dual bounds the finite-horizon optimum (Theorem 2.5.4). At the calibrated parameters the bracket is $[61.41, 64.48]$ par units. For K unknown cohorts the number of independent experiments is exactly of order $(\Delta\eta)^{-(4K-2)}$: a two-point lower bound (Theorem 2.6.4) is matched by Prony's method for every K (Theorem 2.6.5). For general com-

pletely monotone memory the minimax accuracy is of exact order $1/\log N$: a lower bound built from moment-matched cohort pairs (Lemma 2.6.6) is matched by a moment estimator solved as a linear program (Theorem 2.6.7), so the exponential-memory model is the optimistic case.

3. A measurement gap. Passive data do not escape the barrier. The closed-loop transfer function is minimum phase, so the price spectrum identifies the merged measure (Theorem 2.6.8), and natural excitation can separate a pair with zero controlled quantity (Proposition 2.15.2). But the information rate falls at the same polynomial order as for active probing, $(\Delta\eta)^{4K-2}$ for every number of cohorts K (Theorem 2.6.9), and in-loop price noise leaves the capital order of active probing intact up to horizons of that same order (Theorem 2.5.2(iv)). Only richer disclosure removes the dependence on rate spacing: tagged flow (Proposition 2.8.1) or audited equity (Proposition 2.4.4). Tagging reduces the sample requirement by a factor of order $(\Delta\eta)^{-(4K-4)}$ for rates (Theorem 2.8.2). On Binance, published long/short ratios by trader class pin down class memories 28 to 124 times more precisely than prices and flow in each of 30 futures markets (Section 2.8.3), and calibrating the theorem to the BTCUSDT class memories makes them worth about 333 times as much data as prices alone.

4. What feedback indices measure. Standard reduced-form feedback indices are dominated by filtering mechanics and clock alignment; a placebo pipeline reproduces the mechanical persistence exactly; measured feedback rises with market depth across 30 perpetuals; and two fee events that multiply trading volume by 2.6 and divide it by 4.9 leave measured feedback unchanged (Section 2.10).

5. A spectral blind spot. Even without identification, could a supervisor monitor risk through eigenvalues or volatility? No. Markets with identical eigenvalues, one-step volatility and stationary price variance can have arbitrarily large worst-case transient loss (Theorem 2.9.1). This motivates the open question with which the paper ends: what can be measured and controlled without identifying the algorithms? Chapter 3 takes it up.

What we do not claim. The identification results hold for the stated linear model and observation contracts. They are not universal limits on surveillance with participant-level records, which can be far more informative (Propositions 2.4.4 and 2.8.1). The matching upper bound holds on a class with separated rescaled rates and weights bounded away from zero. Capital figures depend on the observation-noise calibration, which we estimate rather than assume. The empirical results are reduced-form diagnostics; they do not identify structural learning parameters. Our pre-registered real-data test of the identification boundary failed its criterion on one of its three parts: on 42 days of data the merged response was precisely estimated in only 5 of 30 markets (Section 2.8.3). Binance's class ratios are positions, not tagged order flow, so they are the closest public analogue to the disclosure of Proposition 2.8.1, not an instance of it.

2.1.1 Related literature

Price formation and impact. Our price equation is a discrete linear impact model in the tradition of Kyle (1985), Glosten and Milgrom (1985) and Hasbrouck (1991), in which order flow moves prices and prices revert toward value. Transient impact and the cost of trading follow Almgren and Chriss (2001), Obizhaeva and Wang (2013), Gatheral (2010) and Bouchaud et al. (2004); Alfonsi et al. (2016) show that absence of price manipulation requires positive-definite decay kernels, which is the property Lemma 2.4.2 delivers in our cohort economy. Bouchaud et al. (2018) document order splitting and long memory in order flow, which is the main confounder in Section 2.10. Relative to this literature we treat the impact kernel not as a primitive but as the closed-loop output of adaptive traders, and ask what it reveals about them.

Algorithmic trading and feedback. Algorithms that react to recent prices can amplify or dampen moves. Brunnermeier and Pedersen (2009) and Shleifer and Vishny (1992) study liquidity spirals and fire sales, and Kirilenko et al. (2017) document the behaviour of automated intermediaries in the 2010 Flash Crash. Sornette et al. (2023) show that non-normal interactions can generate transient bubbles. Our contribution is on the measurement side: we show what a closed-loop observer can and cannot learn about the feedback, and that standard reduced-form indices are dominated by filtering and timing.

Closest work. Three literatures are closest. Super-resolution and moment-based mixture estimation give rates for recovering clustered point sources and mixing measures from noisy moments (Heinrich and Kahn, 2018; Batenkov et al., 2021; Liu and Ammari, 2022; Wu and Yang, 2020; Kong and Valiant, 2017). Our K-cohort rate is, in that language, the square of the amplitude condition number $(\Delta\eta)^{-(2K-1)}$ for K clustered nodes (Heinrich and Kahn, 2018; Wu and Yang, 2020), and we do not claim the exponent as new. What changes when the mixture sits inside a closed-loop financial market is fourfold (Table 2.1): first, in mixture estimation and open-loop system identification the unknown rates are locations or poles of the observed law, whereas in a closed-loop market the cohorts feed back into the price, making the memory rates transmission *zeros* of the observable transfer function $H(z)$ that must first be extracted from the loop (Theorem 2.3.1); second, passive price observations incur the exact same $(\Delta\eta)^{-(4K-2)}$ sample-complexity barrier as active probing because $H(z)$ is minimum phase and the Whittle likelihood inherits the moment gap (Theorems 2.6.8 and 2.6.9); third, in a market statistical sample size translates into executed quantity $\Theta(\delta^{-2})$ and round-trip trading loss or margin capital $\Theta(\delta^{-4})$ via the self-impact Toeplitz quadratic form $\frac{1}{2}v^{\top}T_h v$ (Theorems 2.4.1–2.5.4); and fourth, class-tagged disclosure breaks the feedback loop cohort by cohort and cuts the sample requirement by $(\Delta\eta)^{-(4K-4)}$ (Proposition 2.8.1 and Theorem 2.8.2). Beyond the K-cohort case, we prove matching lower and upper bounds of exact logarithmic order $\Theta(1/\log N)$ for general completely monotone memory (Theorem 2.6.7) and price the resulting disclosure savings against

Table 2.1: Comparison with finite-mixture estimation and open-loop identification

Feature	Finite-mixture estimation (Heinrich and Kahn, 2018; Wu and Yang, 2020)	Open-loop exponential fitting (Roy and Kailath, 1989; Batenkov et al., 2021)	Closed-loop market identification (this chapter)
Where rates live	Component locations / moments of density	Poles of impulse response	Transmission zeros of closed-loop $H(z)$ (Thm 2.3.1)
Active vs. passive data	i.i.d. draws only	Designed input vs. output spectrum	Same $\Theta((\Delta\eta)^{-(4K-2)})$ order via minimum-phase Whittle gap (Thm 2.6.9)
Economic unit of cost	Sample size N	Signal-to-noise ratio	Quantity $\Theta(\delta^{-2})$, trading loss and capital $\Theta(\delta^{-4})$ via $\frac{1}{2}v^\top T_h v$ (Thms 2.4.1–2.5.4)
Effect of class tagging	Observed component labels ($\Theta(1)$)	Separate sensor per mode	Breaks closed loop cohort-by-cohort; saves $\Theta((\Delta\eta)^{-(4K-4)})$ (Thm 2.8.2)

audit-trail budgets.

Statistical identification. The identification results use two-point testing arguments (Le Cam, 1986; Tsybakov, 2009). The moment-matching construction of Section 2.6 is the dynamical analogue of the slow rates in finite-mixture estimation (Manole and Ho, 2022) and of super-resolution of near-colliding sources (Liu and Ammari, 2022; Batenkov et al., 2021); the general-memory result combines best-approximation duality (Bernstein, 1914; Rivlin, 1981) with the moment comparisons used by Kong and Valiant (2017) and Wu and Yang (2020), and its logarithmic rate parallels supersmooth deconvolution (Fan, 1991); recovery of exponentials connects to ESPRIT and matrix-pencil methods (Roy and Kailath, 1989; Hua and Sarkar, 1990). Minimum-phase spectral factorization and the Whittle likelihood are classical (Whittle, 1953). What is new here is the mapping from these statistical rates into executed quantity and money through a financially specified account, and the result that the relevant rates are zeros, not poles, of the observable transfer function.

Non-normal dynamics and supervision. Transient growth of stable non-normal systems is classical (Trefethen and Embree, 2005; Kreiss, 1962). We use it to show that supervision through eigenvalues and second moments of prices is blind to transient risk (Theorem 2.9.1), which motivates the identification-free control of Chapter 3. On the regulatory side, pre-trade risk controls for broker-dealers with market access are governed by SEC Rule 15c3-5 (SEC, 2010), and participant-level audit trails are the natural disclosure regime for Proposition 2.8.1. A large literature studies how pre- and post-trade transparency affects liquidity and trading costs (Pagano and Röell, 1996; Boehmer et al., 2005; Bessembinder et al., 2006); we ask a different question, what disclosure lets a supervisor learn about the algorithms that trade, and give the

answer in samples. Crypto venues, whose market structure is studied by Makarov and Schoar (2020), are a natural laboratory for this question because they publish class-level positioning every five minutes, while the CFTC publishes it weekly for regulated futures.

Partial fractions, Gaussian testing, convolution inequalities and positive-system gains are established tools. Our contribution is to connect them into a quantitative chain from memory resolution, through probing resources, to money, with matching orders, explicit constants and a supervisory interpretation, and then to measure the relevant quantities in data.

Table 2.2: Map of results

Question	Answer	Where
What do prices identify?	The merged rate–weight measure only	Thm 2.3.1
Quantity to separate a known pair	$\Theta(\delta^{-2})$, exact error formula	Thm 2.4.1
Capital to separate a known pair	$\Theta(\delta^{-4})$, counterparty or own funds	Thm 2.4.3, Lem. 2.4.5
Constant at the calibration	$[61.41, 64.48]$ par units; $\approx 10^5$ USDT	Thms 2.5.2–2.5.4, Table 2.7
Does adaptivity help?	At most by a factor ≈ 0.49 in expected loss	Thm 2.5.3
Unknown mixtures	$(\Delta\eta)^{-(4K-2)}$, tight for every K	Thms 2.6.4, 2.6.5
General memory	Accuracy $\asymp 1/\log N$, lower and upper bounds	Lem. 2.6.6, Thm 2.6.7
Passive data	Identifies M; same polynomial cost for every K	Thms 2.6.8, 2.6.9, Prop. 2.15.2
Nonlinear and adaptive rules?	Settled online learners obey δ^{-4}; fast adaptation or thresholds break it	Table 2.15
What removes the barrier?	Tagged flow, audited equity	Props 2.4.4, 2.8.1
What is tagging worth?	Factor $(\Delta\eta)^{-(4K-4)}$; $\approx 333\times$ for BT-CUSDT	Thm 2.8.2
Disclosure in dollars?	Annual probing loss exceeds CAT budget once $K \geq 3$ or $\Delta\eta \leq 0.04$	Table 2.17
Do class data pin down memories?	Yes: 28–124× tighter in 30 of 30 markets	Table 2.18
Can eigenvalues certify safety?	No	Thm 2.9.1
What do feedback indices measure?	Mostly filtering and timing; small real feedback	Tables 2.22, 2.24
Does feedback track activity?	No: fees move volume 2.6–4.9×, not $\hat{Q}$	Table 2.26

2.1.2 Outline

Section 2.2 sets out the model. Section 2.3 gives the identification boundary. Section 2.4 links memory resolution to quantity and money for a known pair. Section 2.5 computes account-level capital. Section 2.6 treats unknown mixtures and passive data. Section 2.7 checks the results against stylized nonlinear and adaptive trading rules. Section 2.8 studies disclosure and compares its cost in dollars with the U.S. Consolidated Audit Trail. Section 2.9 gives the spectral

blind spot. Section 2.10 reports the empirical evidence. Section 2.12 concludes. Proofs are in Appendix 2.13; Appendix 2.15 collects additional results, and Appendix 2.16 maps every number to its source.

2.2 Model and microfoundations

Time is discrete. Let p_t be the price, p^* the fundamental value and $x_t = p_t - p^*$ the pricing error. Competitive dealers absorb net order flow with price impact $\kappa > 0$, and arbitrage pulls the error toward zero with persistence $a \in [0, 1)$:

$$x_t = a x_{t-1} + \kappa \sum_{i=1}^{K} q_i y_{i,t-1} + \kappa v_t + w_t, \tag{2.1}$$

where v_t is an executed probe quantity and w_t a price innovation. Cohort i holds a state $y_{i,t}$ with exponential memory and trades $q_i y_{i,t}$:

$$y_{i,t} = (1 - \eta_i)\, y_{i,t-1} + \eta_i x_t, \qquad \eta_i \in (0, 1). \tag{2.2}$$

The ordering is *price first*: the price responds to the probe, then cohorts update. Write $r_i = 1 - \eta_i$ for the memory rate and $w_i = q_i \eta_i$ for the merged weight. The feedback kernel and its transform are

$$k_m = \sum_i w_i r_i^m, \qquad G(z) = \sum_i \frac{w_i z}{z - r_i}, \tag{2.3}$$

and the transfer function from probes to prices, $H(z) = \sum_{j \geq 0} h_j z^{-j}$, is

$$H(z) = \frac{\kappa z}{z - a - \kappa G(z)} = \frac{\kappa z D(z)}{(z - a) D(z) - \kappa N(z)}, \qquad G = \frac{N}{D}, \quad D(z) = \prod_i (z - r_i), \tag{2.4}$$

with impulse response $h_0 = H(\infty) = \kappa, h_1, h_2, \ldots$; in particular $H(1) = \kappa/(1 - S) = \sum_j h_j$. Stacking $z_t = (x_t, y_{1,t}, \ldots, y_{K,t})^\top$ gives $z_t = J z_{t-1} + b v_t$, where, because cohorts update on the same period's price, the probe enters every state on impact through

$$b = \kappa\, (1, \eta_1, \ldots, \eta_K)^\top, \tag{2.5}$$

and

$$J = \begin{pmatrix} a & \kappa q_1 & \cdots & \kappa q_K \\ \eta_1 a & 1 - \eta_1 + \eta_1 \kappa q_1 & \cdots & \eta_1 \kappa q_K \\ \vdots & & \ddots & \vdots \\ \eta_K a & \eta_K \kappa q_1 & \cdots & 1 - \eta_K + \eta_K \kappa q_K \end{pmatrix}, \qquad S \equiv a + \kappa \sum_i q_i. \tag{2.6}$$

In the *positive economy* ($q_i \geq 0$), J is entrywise nonnegative, its price row sums to S, and the economy is Schur stable iff $S < 1$.

Lemma 2.2.1 (Filtering and learning microfoundations)**.** *(i)* Kalman memory. *A desk that tracks a latent random-walk trend $v_{t+1} = v_t + \zeta_{t+1}$, $\zeta \sim N(0, \sigma_\zeta^2)$, from signals $r_t = v_t + \nu_t$, $\nu \sim N(0, \sigma_\nu^2)$, using the steady-state Kalman filter, updates as $\hat{v}_{t+1} = (1-\eta)\hat{v}_t + \eta r_t$ with*

$$\eta = \frac{-\sigma_\zeta^2 + \sqrt{\sigma_\zeta^4 + 4\sigma_\zeta^2\sigma_\nu^2}}{2\sigma_\nu^2} \in (0,1). \tag{2.7}$$

(ii) Gradient memory. *Online gradient descent with L_2 regularization $\gamma \in (0,1)$, step size $\alpha > 0$ and loss $\frac{1}{2}H(x_t - \theta_t)^2$ gives $\theta_{t+1} = (1-\eta)\theta_t + \eta q x_t$ with $\eta = \gamma + \alpha H \in (0,1)$ and $q = \alpha H/\eta$. Tabular Q-learning with learning rate η has the same form.*

Remark 2.2.2 (Why exponential memory). Lemma 2.2.1 shows that two different design philosophies, Bayesian filtering and regularized online learning, produce the same observable reduced-form belief state $y_{i,t}$. When desk i chooses order flow $u_{i,t}$ subject to $I_{i,t} = (1-\delta_{\text{inv}})I_{i,t-1} + u_{i,t}$ and an inventory cap $|I_{i,t}| \leq \bar{I}_i$ to maximize expected continuation return $\beta_i y_{i,t} u_{i,t}$ net of quadratic execution cost $\frac{1}{2}\lambda_{\text{tc},i}u_{i,t}^2$ and quadratic inventory holding cost $\frac{1}{2}\gamma_{\text{inv},i}I_{i,t}^2$, its interior first-order condition gives $u_{i,t}^* = q_i y_{i,t} - \psi_{\text{inv},i} I_{i,t-1}$ with market-response weight $q_i = \beta_i/(\lambda_{\text{tc},i} + \gamma_{\text{inv},i})$ and inventory mean-reversion $\psi_{\text{inv},i} = \gamma_{\text{inv},i}(1-\delta_{\text{inv}})/(\lambda_{\text{tc},i} + \gamma_{\text{inv},i})$, which shrinks toward zero as trading costs or inventory penalties tighten and destabilizes the market when the closed-loop spectral radius reaches one (`microfoundation_and_baselines.py`). This filtering-and-control recursion specifies an individual agent's decision rule, not a complete economic equilibrium (dealer–trader equilibrium is analysed in Chapter 3). The memory rate encodes the desk's signal-to-noise ratio or regularization, which is the private information a supervisor would like to recover; Section 2.3 shows why prices reveal it only in merged form.

Lemma 2.2.3 (Positive price-response geometry)**.** *In the positive, stable economy the impulse response is a nonnegative mixture of decaying exponentials,*

$$h_j = \sum_\ell \omega_\ell \lambda_\ell^j, \qquad \omega_\ell \geq 0, \quad \sum_\ell \omega_\ell = \kappa, \quad 0 \leq \lambda_\ell < 1.$$

It is completely monotone and nonincreasing; its peak is the direct impact κ, even though J is non-normal. In price/memory coordinates the price-only gain is $\sup_t \|e_x^\top J^t\|_2 = 1$.

Numerically, $(-1)^n \Delta^n h_j \geq 0$ for $n \leq 4$ held in all 892 stable random positive economies tested. Lemma 2.2.3 matters twice: it makes probing costs positive (Section 2.4), and it shows that in this economy transient amplification of the price response cannot arise from cohort feedback alone. Chapter 3 locates amplification in order-book depth instead.

2.2.1 Discussion of the assumptions

Linearity. The model is linear in prices, flows and memory states. Linearity is what makes the merged-measure equivalence exact and the testing problems Gaussian. Nonlinear strategies (for example threshold rules or position limits) can break the equivalence class, but they do so in regions of the state space that a small probe does not visit; for small probes the linearization around the steady state is the relevant object.

Price-first timing. We order the price update before the cohort update. Under the alternative lagged timing the transition matrix becomes $\left(\begin{smallmatrix} a & 1 \\ \eta Q & 1-\eta \end{smallmatrix}\right)$ and its eigenvalues differ, so stability statements must be computed under the timing actually used. The identification results depend on the transfer function and survive either convention.

Exponential memory. Lemma 2.2.1 justifies exponential memory for filtering and regularized learning. Neural and reinforcement-learning policies are not exponential filters. Rules whose orders are smooth functions of an exponential-memory state are exponential filters to first order, so the results hold for small probes; Section 2.7 checks this in stylized simulations and finds where it fails, and how closely production algorithms follow such rules remains an empirical question we do not settle. Mixtures of exponentials can approximate any completely monotone kernel, so a power-law memory corresponds to a continuum of cohorts, for which Theorem 2.6.7 shows that accuracy improves only logarithmically in the data.

Algorithms that react to being probed. The probing results assume that the cohorts keep their memories while they are probed. An algorithm that detects a probe could pause, shorten its memory or trade against the probe. Four observations limit what this does to our conclusions. First, a reaction that all cohorts share does not separate them, and an algorithm that hides by mimicking another rate makes the hypotheses closer, so neither lowers the costs we derive; a reaction that differs across cohorts is itself a signature, but one the observer must model, which is a new identification problem rather than a free one. Second, the passive route needs no probe: Theorem 2.6.9 shows that natural price variation identifies the same separation at the same polynomial order in sample length, so an observer who fears detection pays in time, not in rate. Third, detection is itself a testing problem against the same noise: the generalized-eigenvector probe of Section 2.5 trades at most 1.25 par units in any period, and a probe sized within normal flow variation is hard to tell from noise, so the threat bites mainly for large single pulses, which our ledger shows are not optimal anyway. Fourth, what we cannot rule out is an algorithm designed to defeat surveillance, which turns identification into a game between observer and observed; our results are the benchmark against which such a game would be measured, and we leave it to future work.

Known primitives. The resource theorems take κ, a and the noise law as known. Unknown primitives can only make separation harder, so the lower bounds remain valid, while the upper bounds require estimating them, which is cheap relative to the memory rates because they enter at first order.

Observation contracts. Each result names what the observer sees. A price-only observer, an observer with executed quantities, one with noisy statements, one with audited equity and one with tagged flow face different experiments. The paper's central message is that these differences, not the observer's statistical skill, determine what can be learned.

2.3 The identification boundary

A central question in algorithmic market surveillance is whether an observer of executions and prices can identify the individual strategies active in the market.

Theorem 2.3.1 (Merged rate–weight equivalence)**.** *Let $\kappa > 0$ and $a \in [0, 1)$ be known. Use zero initial states and a common probe contract, and observe the modeled price state x_t.*

(i) Equivalence class. *Two economies $\mathcal{E}^A = \{(q_i^A, \eta_i^A)\}$ and $\mathcal{E}^B = \{(q_j^B, \eta_j^B)\}$ produce identical price responses for every probe sequence if and only if their merged measures coincide:*

$$\sum_{i:\eta_i^A=\eta} q_i^A \eta \;=\; \sum_{j:\eta_j^B=\eta} q_j^B \eta \qquad \forall \eta \in (0,1). \tag{2.8}$$

For $a > 0$ the minimal realization has dimension $1 + |\mathrm{supp}(M)|$, counting distinct rates with nonzero merged weight after signed cancellation; for $a = 0$ the price carries no state of its own and the dimension is $|\mathrm{supp}(M)|$. A cohort with $\eta = 1$ merges into the anchor a without adding a pole.

(ii) Separation complexity under average power. *Consider testing a single cohort with rate $\bar{\eta}$ against two cohorts with rates $\bar{\eta} \pm \delta$ and the same total weight, on a compact family that is uniformly Schur stable with rates bounded away from 0 and 1. Observe $\tilde{x}_t = x_t + \nu_t$ with $\nu_t \sim N(0, \sigma_\nu^2)$ and use deterministic probes with $\frac{1}{T}\sum_{t<T} v_t^2 \le P$. Any test with worst-case error at most $\alpha \in (0, 1/4)$ needs*

$$T \;\ge\; \frac{4\sigma_\nu^2(1-2\alpha)^2}{K^2 P\, \delta^4} = \Omega(\delta^{-4}), \tag{2.9}$$

where K depends only on the family.

The memory rates are poles of G and *zeros* of H, not closed-loop poles. This has a practical consequence we verify in Section 2.6: pole-fitting methods applied directly to the price impulse

response recover closed-loop eigenvalues (0.4256 and 0.8224 for $K = 1$, $\eta = 0.3$, $a = 0.5$, $\kappa = 0.8$, $q = 0.2$) and miss the memory rate 0.70.

Remark 2.3.2 (Scope). The result concerns the stated model and observation contract. Cohorts sharing an exact rate enter prices through their merged weight; this does not determine identities or wealth allocations. Nearby rates can be distinguishable with enough information. The δ^{-4} statement in (ii) is a lower bound on a declared family, with family-dependent constants. Individual algorithms may choose their rates strategically, but any unilateral change maps into the same observer-facing merged measure, which reinforces rather than circumvents the boundary.

2.4 Probing resources, memory resolution and money

Observation length alone cannot buy identification when the lifetime quantity of controlled executions is fixed. This section connects the merged-memory experiment to a financially specified input/output contract. Throughout, Φ is the standard normal distribution function.

2.4.1 Quantity

Theorem 2.4.1 (Sharp probing-resource orders for a known symmetric pair)**.** *Fix $0 \le a < 1$, $\kappa, q > 0$, $S = a + \kappa q < 1$ and $\eta \in (0, 1)$. Compare a single cohort of weight q and rate η with two cohorts of weight $q/2$ and rates $\eta \pm \delta$, both in $[\epsilon, 1 - \epsilon]$. All parameters are known. The price receives executed quantity through κv_t, starts from zero, has no process noise, and is observed with independent $N(0, \sigma^2)$ noise. Probes are deterministic and common to both alternatives. Let $d_T = (h^\delta_j - h^0_j)_{j<T}$ and $K_* = 2\kappa^2 q/[\epsilon^3(1 - S)^2]$.*

(i) For every probe with $\sum_t |v_t| \le V$,

$$D_{\mathrm{KL}}(P^v_0 \| P^v_\delta) \le \frac{K_*^2 V^2 \delta^4}{2\sigma^2}, \qquad e_{\mathrm{worst}} \ge \frac{1}{2}\Big(1 - \frac{K_* V \delta^2}{2\sigma}\Big)_+. \tag{2.10}$$

(ii) The optimal worst-case error over these probes and all classifiers is exactly

$$e^*_T(V) = \Phi\left(-\frac{V\|d_T\|_2}{2\sigma}\right), \tag{2.11}$$

attained by a pulse at the first input and the likelihood-ratio test. Waiting adds information but $\|d_T\|_2 \uparrow \|d_\infty\|_2 < \infty$, so it cannot make the error arbitrarily small at fixed V. Error at most $\alpha < 1/2$ requires $V \ge 2\sigma(1 - 2\alpha)/(K_*\delta^2)$ independently of T.

(iii) The first two taps coincide and $d_2 = -\kappa^2 q \delta^2$. A single pulse of size

$$A = \frac{2\sigma\Phi^{-1}(1 - \alpha)}{\kappa^2 q \delta^2} \tag{2.12}$$

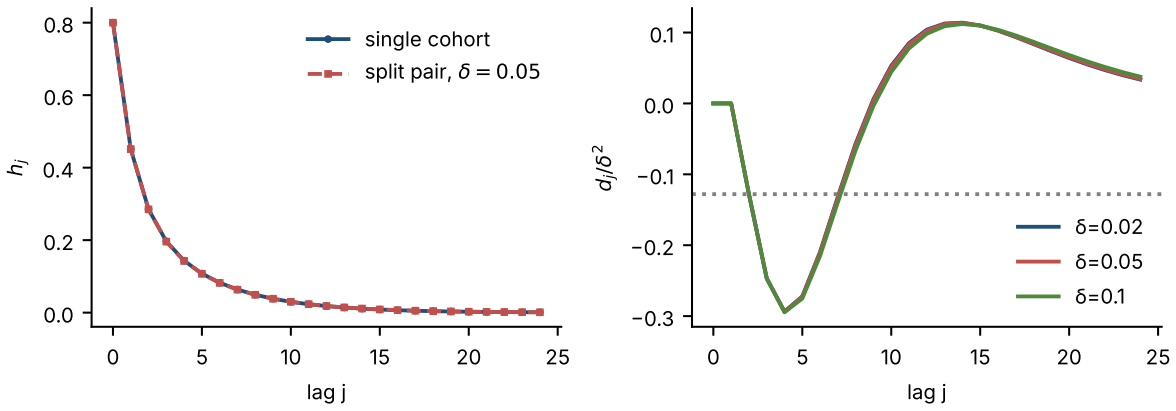


Figure 2.1: Left: impulse responses of the single cohort and the split pair ($\delta = 0.05$) are nearly indistinguishable. Right: the difference scaled by δ^2 collapses onto one curve, with first nonzero value $-\kappa^2 q$ at lag 2 (dotted). Calibration $a = 0.5$, $\kappa = 0.8$, $q = 0.2$, $\eta = 0.4$.

Table 2.3: Exact known-pair error: formula versus simulation

Pulse V	Horizon T	$\Phi(-V\|d_T\|_2/2\sigma)$	Simulated error, H_0	Simulated error, H_δ
20	3	0.3745	0.3734	0.3762
40	3	0.2611	0.2599	0.2603
40	10	0.0029	0.0030	0.0027
80	20	1.8×10^{-10}	0	0

200,000 paths per hypothesis; $a = 0.5$, $\kappa = 0.8$, $q = 0.2$, $\eta = 0.4$.

and the third observation achieve error exactly α. The necessary and sufficient quantity order is δ^{-2}.

(iv) Under average power $\sum_{t<T} v_t^2 \le PT$, the necessary horizon order is δ^{-4}, attained by the pulse whenever $T \ge \max\{3, \lceil A^2/P \rceil\}$.

(v) For a fixed monetary price exposure $\lambda \neq 0$, additive reserve L and cost envelope $c\sum|v_t|$, the sharp robust quantity allowance under budget $B > L$ is $V_B = (B - L)/(|\lambda|\kappa + c)$.

The identity in (iii) is verified to machine precision for $\delta \in \{0.02, 0.05, 0.10\}$: the difference is -1.2×10^{-4}, -7.5×10^{-4} and -3.0×10^{-3} at $\kappa = 1$, $q = 0.3$, and -3.2×10^{-4} at the calibration $\kappa = 0.8$, $q = 0.2$, $\delta = 0.05$. The theorem does not solve estimation of an unknown mixture (Section 2.6).

A Monte Carlo check of the exact error. Formula (2.11) is exact for the likelihood-ratio test. We simulated 200,000 noisy paths per hypothesis at the calibration ($\sigma = 0.01$, $\delta = 0.05$) for several pulse sizes and horizons (Table 2.3); simulated errors match the formula to Monte Carlo precision under both hypotheses.

Figure 2.1 shows why the problem is hard. The two economies have the same direct impact

and the same first-lag response; they first differ at lag two, by an amount proportional to δ^2, and the whole difference sequence scales with δ^2. A test therefore needs the signal-to-noise ratio $V\|d\|/\sigma$ to grow like δ^{-2}, which is the quantity order, and because the adverse mark of a probe grows with the square of its size, capital grows like δ^{-4}.

2.4.2 Execution cost

A probe changes inventory as well as price. We therefore specify cash and execution timing. Price has units of currency per asset, and κ currency per squared quantity.

Lemma 2.4.2 (Fair execution)**.** *For a quote schedule refreshed after the uncontrolled state update, execution at the average price of a linear book costs, relative to the constant fundamental mark,*

$$C(v) = \tfrac{1}{2} v^\top T_h v + c\|v\|_1, \qquad (T_h)_{ij} = h_{|i-j|}. \tag{2.13}$$

In the positive stable economy T_h is positive definite, so no nonzero round trip produces a deterministic gain.

We verify the ledger identity (2.13) against an explicit cash account to 5.7×10^{-13} on 1,000 random probes, and the smallest eigenvalue of T_h over 1,000 random stable positive economies is 0.141. A direct-only proxy $\sum \frac{1}{2}\kappa v_t^2$ omits cross-time impact: for $v = (+1, -1)$ it gives 0.80 against a true Toeplitz loss of $h_0 - h_1 = 0.3488$.

2.4.3 Capital supplied by a counterparty

Theorem 2.4.3 (Sharp robust quantity cap from cash and margin)**.** *Use the positive, stable, reset economy with constant fundamental price $P_* > 0$ and no process noise. A designated dealer fills only nonnegative probes $v_t \geq 0$ at the average price $P^0_{t+1} + \kappa v_t/2$ of a linear schedule, holds inventory $I_{t+1} = I_t - v_t$ and cash $K_{t+1} = K_t + v_t(P^0_{t+1} + \kappa v_t/2) - cv_t$, starts with equity $B > 0$, and must satisfy $\mu|I_t|P_t \leq E_t$ at every time. For the robust contract admitting every probe with $\sum_t v_t \leq V$, the largest allowance guaranteeing margin is*

$$V_B = \frac{2B}{\beta + \sqrt{\beta^2 + 4(\mu + \frac{1}{2})\kappa B}}, \qquad \beta = c + \mu P_*, \tag{2.14}$$

equivalently $(\mu+\frac{1}{2})\kappa V^2 + (c+\mu P_*)V \leq B$. The cap holds across all positive cohort representations with $S < 1$, without resolving individual rates. An observer that receives only noisy prices and executed quantities has best error $\Phi[-V_B\|d_T\|_2/(2\sigma)]$ within this allowance; fixed error $\alpha < 1/2$ requires capital of order at least δ^{-4}, and a single positive pulse attains that order.

Even with zero initial inventory, the probe creates a quadratic adverse mark that a fixed initial price exposure would miss. Figure 2.2 illustrates the experiment.

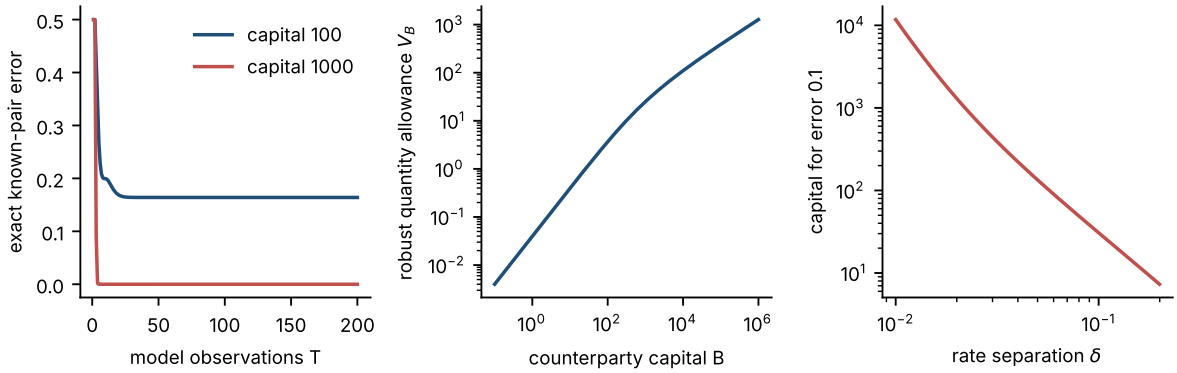


Figure 2.2: Known-pair experiment funded by the counterparty cash/margin contract. (a) Exact Gaussian error at fixed capital 100 and 1,000, separation $\delta = 0.05$. (b) The sharp robust quantity allowance (2.14). (c) Capital for error 0.1 with 200 observations. Parameters: $a = 0.5$, $\kappa = 0.8$, $q = 0.2$, $\eta = 0.4$, observation s.d. 0.003, reference price 100, margin 0.25, cost 0.02. No process noise.

Proposition 2.4.4 (Exact financial disclosure changes the experiment)**.** *In this noiseless-state account, exact disclosure of the third equity mark after a single positive pulse $0 < A \le V_B$ distinguishes the known pair with zero error for every $\delta > 0$.*

Exact marked equity is extra information, not a free observation assumed by the price-only theorem. Real account marks have their own precision: with reports rounded to 0.01, a guaranteed change requires a larger trade (Section 2.8).

2.4.4 Capital paid by the probe trader

Lemma 2.4.5 (Time-domain Toeplitz coercivity: capital order for funded probes)**.** *Use the same economy and execution law. The statistical observer receives only noisy prices and executed quantities. The probe trader starts with cash $B > 0$ and zero inventory; a finite signed sequence v changes inventory by $+v_t$ and cash by $K_{t+1} = K_t - v_t(P^0_{t+1} + \kappa v_t/2) - c|v_t|$, after which trading stops and the trader must remain solvent as the price returns to P_*. Put*

$$r = 1 - \epsilon(1 - S) < 1, \qquad m = \kappa \frac{1 - r}{1 + r} > 0, \qquad K_* = \frac{2\kappa^2 q}{\epsilon^3 (1 - S)^2}.$$

Every solvent strategy satisfies $D_{\mathrm{KL}}(P^v_0 \| P^v_\delta) \le K_*^2 B \delta^4/(\sigma^2 m)$, so error at most $\alpha < 1/2$ requires

$$B \;\ge\; \frac{2\sigma^2 m (1 - 2\alpha)^2}{K_*^2 \delta^4}, \tag{2.15}$$

without a quantity cap and at any horizon. Conversely, for any margin fraction $0 < \mu \le 1$, the pulse A in (2.12) achieves error α and remains margin-solvent at every later mark with capital $B_A = \frac{\kappa}{2} A^2 + (c + \mu P_*) A$. Necessary and sufficient capital orders are δ^{-4}.

Lemma 2.4.5 is an order statement: an elementary time-domain coercivity argument for Toeplitz forms that establishes the δ^{-4} rate before the exact frequency-domain optimization of Section 2.5. Its explicit lower constant is correspondingly loose: at the calibrated parameters it evaluates to 4.9×10^{-4} par units at $\delta = 0.05$ with $\epsilon = 0.35$, five orders of magnitude below any design we can execute, and because $m/K_*^2 \propto \epsilon^7(1-S)^5$ it collapses near the boundary of the family. We therefore state it as a lemma; Section 2.5 computes the constant that matters, within 5% of a feasible design at every separation.

2.4.5 A worked example: two Kalman desks

To make the orders concrete, consider two desks that track the same trend with Kalman filters (Lemma 2.2.1(i)) and differ only in their signal-to-noise ratios: $\sigma_\zeta^2/\sigma_\nu^2 = 0.09$ and 0.16. Their gains are $\eta_1 = 0.2584$ and $\eta_2 = 0.3279$, so $\bar{\eta} = 0.2931$ and $\delta = 0.0348$. With equal weights, the economy is observationally the split pair of Theorem 2.4.1 around $\bar{\eta}$, and the response difference first appears at lag two, $d_2 = -\kappa^2 q \delta^2 = -1.55 \times 10^{-4}$ (verified).

A supervisor who wants to tell this market from a single desk with gain $\bar{\eta}$ at 5% error using a single pulse and the third observation needs $A = 2\sigma\Phi^{-1}(0.95)/(\kappa^2 q\delta^2) = 212$ units of executed quantity, and, under a 0.5% margin and 4 basis points of fees, single-pulse capital of 18,166 par units. The pulse is very inefficient: by the δ^{-4} scaling of Theorem 2.5.2, the optimal flat probe needs about $61.41 \times (0.05/0.0348)^4 \approx 262$ par units. Converting with the measured Roll noise (factor $(0.40/0.01)^2$), the optimal probe costs roughly 4×10^5 USDT and the naive pulse about 3×10^7 USDT. Two desks whose signal-to-noise ratios differ by almost a factor of two are therefore expensive, but not impossible, to tell apart by probing; three or more desks are much harder (Theorem 2.6.4).

2.4.6 Natural excitation and the cost of ambiguity

Natural price innovations can also separate the pair with zero controlled quantity: an exact finite-sample test on independent three-step blocks needs order δ^{-4} blocks (Internet Appendix, Proposition 2.15.2), the same order as the passive Whittle bound of Section 2.6. Non-identification also has a price on the quote side: when the relevant set of economies includes costlier ones, a robust dealer widens its quote, which Chapter 3 analyses; cohort non-identification alone does not establish that the worst relevant cost exceeds the informed one.

2.5 Account-level capital

Lemma 2.4.5 gives the order but not the constant. This section computes what a trader actually needs under a calibrated linear-impact environment, then converts it to Bitcoin dollars.

2.5.1 Observation design and the tri-region partition

An observer distinguishes two candidate mechanisms with rates θ_0 and $\theta_1 = \theta_0 + \delta$ over N periods, submitting probes $V = (V_1, \ldots, V_N)^\top$ that generate inventory $I_t = \sum_{s \le t} V_s$. Under contract I_0 (active probing, Gaussian observation noise) $p_t = (h_i * V)_t + \epsilon_t$, $\epsilon_t \sim N(0, \sigma^2)$, so

$$D_{\mathrm{KL}}(P_0^{\otimes N} \| P_1^{\otimes N}) = \frac{1}{2\sigma^2} \sum_{t=1}^{N} \big((h_1 * V)_t - (h_0 * V)_t\big)^2. \tag{2.16}$$

The observer minimizes capital $C(N, V) = \sum_t (\lambda V_t^2 + \phi |V_t|) + \max_t \mu p_0 |I_t|$ subject to (i) separation $D_{\mathrm{KL}} \ge \frac{1}{2}(z_\alpha + z_\beta)^2$, (ii) caps $|V_t| \le V_{\mathrm{eff}}$ and $|I_t| \le I_{\max}$, and (iii) a solvency budget $\max_t \mathrm{Loss}_t(V) \le B$. Here $V_{\mathrm{eff}} = \min(V_{\max}, D_{\max}/\kappa)$ combines resting depth and a price-displacement limit.

Proposition 2.5.1 (Tri-region partition and finite-session resolution floor)**.** *Under observation horizon T per pulse, (δ, B) space splits into three regions.*

(i) Feasible single pulse. *For $\delta \ge \delta_{\min}(T)$ and $B \ge C^*(\delta)$ one pulse within capacity suffices. For a simple rate shift ($\eta_1 = \eta_0 + \delta$) the response difference is $\Theta(\delta)$ and $C^* = \Theta(\delta^{-2})$; for symmetric splitting the first-order variation cancels, the difference is $\Theta(\delta^2)$ and $C^* = \Theta(\delta^{-4})$.*

(ii) Sequential bounded probing. *For $\delta < \delta_{\min}(T)$ the required quantity exceeds V_{eff}. Capped pulses each supply divergence $V_{\mathrm{eff}}^2 \|d_T\|^2/(2\sigma^2)$, so*

$$N_{\mathrm{pulses}}(T, \delta) = \Big\lceil \frac{4\sigma^2 z_\alpha^2}{V_{\mathrm{eff}}^2 \|d_T\|_2^2} \Big\rceil. \tag{2.17}$$

(iii) Infeasible with isolated pulses. *If the budget cannot finance sequential costs, peak inventory breaches $I_{\max}$, or a session deadline binds, the finite-session floor $\delta_{\min}(T_{\mathrm{session}})$ is the root of $V_{\mathrm{eff}} \|d_{T_{\mathrm{cycle}}}(\delta)\|_2 = 2\sigma z_\alpha / \sqrt{\lfloor T_{\mathrm{session}}/T_{\mathrm{cycle}} \rfloor}$, with small-$\delta$ approximation $\delta_{\min} = \Theta(T_{\mathrm{session}}^{-1/4})$.*

At the calibration $\kappa = 0.8$, $q = 0.2$, $\sigma = 0.01$, $V_{\max} = 50$, $D_{\max} = 5$ ($V_{\mathrm{eff}} = 6.25$), the single-tap floor is $\delta_{\min}(3) = \sqrt{2\sigma z_\alpha/(V_{\mathrm{eff}} \kappa^2 q)} = 0.2028$; extended observation lowers it to 0.1376 ($T = 4$), 0.0973 ($T = 10$), 0.0914 ($T = 20$) and 0.0905 ($T = 100$). At $\delta = 0.05$ the simplified one-leg benchmark (cost $\frac{1}{2}\kappa V_{\mathrm{eff}}^2 + \phi P_* V_{\mathrm{eff}} = 15.875$ per pulse plus initial margin 3.125) needs 271 pulses and 4,305.25 par units with $T = 3$, 58 pulses and 923.88 with $T = 4$, but only 12 pulses and 193.63 with $T = 20$ and 11 pulses and 177.75 with $T = 100$; optimizing the pulse size (6.06 rather than 6.25) lowers the $T = 20$ figure to 182.03. Most of the saving comes from observing the transient response rather than a single tap (Figure 2.3).

These are costs within an isolated-pulse design class, not minimum capital; the next subsection replaces the formula by a ledger.

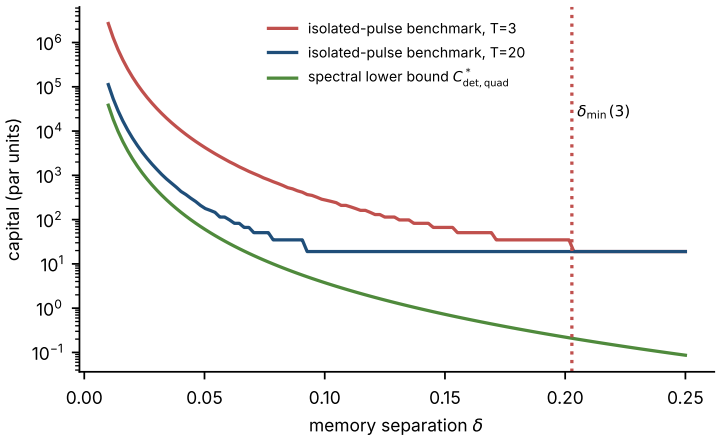


Figure 2.3: Capital against separation: isolated-pulse benchmarks with three and twenty observations per pulse, and the spectral lower bound of Theorem 2.5.2. The dotted line is the single-tap resolution floor $\delta_{\min}(3) = 0.2028$. Recomputed in this revision.

Table 2.4: Account-level initial capital at $\delta = 0.05$ (par units)

Design	C_0	Error	End inv.	Peak margin	Max $\lvert v_t \rvert$	Trades
12 buys, $T = 20$ apart, never unwound	229.13	0.0578	75.00	39.38	6.25	12
6 buy/sell round trips	189.45	0.0308	0.00	3.28	6.25	12
5 buy/sell round trips	157.89	0.0442	0.00	3.28	6.25	10
12 buys, each unwound after 20 steps	378.82	0.0040	0.00	3.28	6.25	24
Generalized-eigenvector probe ($L = 120$)	65.06	0.0500	0.00	2.74	1.24	120
Generalized-eigenvector probe ($L = 160$)	64.86	0.0500	0.00	2.17	1.09	160
Fee-refined probe ($L = 240$)	64.48	0.0500	0.00	—	1.30	240

The pulse designs place a pulse every 40 steps and close it 20 steps later; the fee-refined probe is re-optimized by sequential quadratic programming over $L = 240$. The certified lower bound of Theorem 2.5.2 is 61.41; the explicit constant of Lemma 2.4.5 is 4.9×10^{-4}.

2.5.2 Account-level capital and the gap to the lower bound

We run each candidate design through an explicit synthetic ledger at reference price $P_* = 100$: fills at the average linear-book price, taker fees of 4 basis points, marks at the prevailing price, and 0.5% maintenance margin. Initial capital is $C_0 = \max_t(\text{margin}_t - \text{equity}_t)$, maximized over both hypotheses, and the test uses the full price path including unwind legs.

Table 2.4 gives three conclusions. First, twelve buys that are never unwound fail the 5% target once the test uses the joint price path (divergence $4.95 < 5.41$) and end with 75 units of inventory. Second, if every unwind leg is itself informative, five or six round trips meet the target with zero terminal inventory. Third, pulses are not optimal: a generalized-eigenvector probe, which maximizes divergence per unit of fee-free round-trip cost, needs about 65 par units with maximum trade 1.25 and maximum displacement 2.23, well inside limits.

For flat fee-paying policies $v \in \mathbb{R}^L$ with $\sum_t v_t = 0$, the capital problem is

$$\min_{v} \max_{i\in\{0,1\}} L_i(v) + \sum_t \phi P_* |v_t| + \max_t \mu P_* |I_t| \quad \text{s.t.} \quad \|(H_1 - H_0)v\|_2 \geq 2\sigma z_{1-\alpha}, \tag{2.18}$$

where $L_i(v) = \frac{1}{2} v^\top T_{h_i} v$ is the exact Toeplitz round-trip loss. Zero-padding embeds horizon L into $L+1$ at identical cost and information, so the minimum is non-increasing in L.

Theorem 2.5.2 (Spectral dissipation lower bound)**.** *Let $\hat{H}_i(z) = \sum_{j\geq 0} h_{i,j} z^{-j}$, $\Delta\hat{H} = \hat{H}_1 - \hat{H}_0$, and define the dissipation kernel*

$$g_i(\omega) = \operatorname{Re} \hat{H}_i(e^{i\omega}) - \kappa/2 > 0 \qquad \forall \omega \in [-\pi, \pi]. \tag{2.19}$$

(i) Toeplitz energy identity. *For any flat round trip, $L_i(v) = \frac{1}{2\pi} \int_{-\pi}^{\pi} g_i(\omega) |V(e^{i\omega})|^2 d\omega$.*

(ii) Lower bound. $\|(H_1 - H_0)v\|_2^2 \leq \sup_\omega \frac{|\Delta\hat{H}(e^{i\omega})|^2}{g_i(\omega)} L_i(v)$, *so every deterministic round trip with error at most α has*

$$L_i(v) \;\geq\; 4\sigma^2 z_{1-\alpha}^2 \inf_\omega \frac{g_i(\omega)}{|\Delta\hat{H}(e^{i\omega})|^2} \;\equiv\; C^*_{\mathrm{det,quad}}(\delta). \tag{2.20}$$

(iii) Bracket. *At $\delta = 0.05$, $C^*_{\mathrm{det,quad}} = 61.14$ at $\omega^* = 0.2526$; adding the unavoidable fee $\phi P_* \cdot 2\sigma z / \sup_\omega |\Delta\hat{H}|$ gives the certified bound $C_{\mathrm{det}} \geq 61.41$. A feasible 240-step sequence attains 64.48, so the fee-paying optimum lies in* $[61.41, 64.48]$.

(iv) Process noise. *Let the price also receive the innovations $w_t \sim N(0, \sigma_w^2)$ of (2.1), independent of observation noise $\nu_t \sim N(0, \sigma_\nu^2)$, so that from the reset state the observer sees $\tilde{x}_t = (h_i * v)_t + \kappa^{-1}(h_i * w)_t + \nu_t$, $t < T$. For the known pair of Theorem 2.4.1, put $\sigma^2_{\mathrm{eff}} = \sigma_\nu^2 + \sigma_w^2/4$, $\bar{M} = 1/(1-S)$ and $C_w = 2\sigma_w^4 \bar{M}^2 K_*^2 (\sigma_\nu^2 + \sigma_w^2 \bar{M}^2)/(\kappa^2 \sigma_\nu^2 \sigma^4_{\mathrm{eff}})$. Every deterministic round trip whose test has errors α and β under the two hypotheses satisfies*

$$L_i(v) \;\geq\; 2\sigma^2_{\mathrm{eff}} \big[\mathrm{kl}(\alpha, 1-\beta) - C_w T \delta^4\big]_+ \inf_\omega \frac{g_i(\omega)}{|\Delta\hat{H}(e^{i\omega})|^2}, \tag{2.21}$$

with kl *as in Theorem 2.5.3. In-loop noise therefore leaves the δ^{-4} capital order intact for every horizon $T \leq \mathrm{kl}(\alpha, 1-\beta)/(2C_w\delta^4)$; at horizons of order δ^{-4} and longer, the noise itself separates the pair without a probe (Theorem 2.6.9).*

Positivity of g_i follows from Lemma 2.2.3: $g_i(\omega) = \frac{1}{2}\sum_\ell \omega_\ell \frac{1-\lambda_\ell^2}{1 - 2\lambda_\ell \cos\omega + \lambda_\ell^2} > 0$. We compute $C^*_{\mathrm{det,quad}}$ across separations (Table 2.5): 61.14 at $\delta = 0.05$, with $\delta^4 C^*_{\mathrm{det,quad}}$ between 3.67×10^{-4} and 3.84×10^{-4}, confirming the δ^{-4} law. The fee term, the feasible upper bounds and the adaptive bound of Theorem 2.5.3 are reported in the same table.

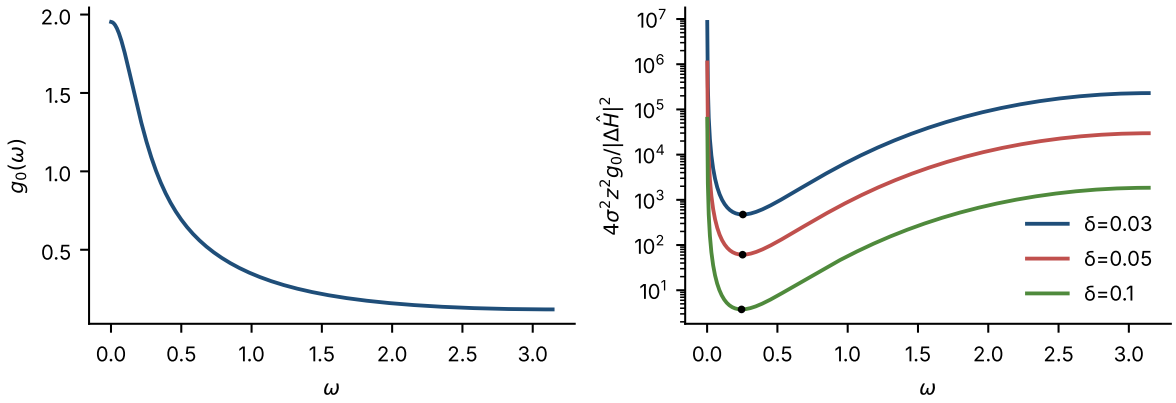


Figure 2.4: Left: the dissipation kernel $g_0(\omega)$ is strictly positive. Right: the capital ratio in (2.20); dots mark its minimum, near $\omega^* \approx 0.25$ for every separation. Recomputed in this revision.

Theorem 2.5.3 (Adaptive and sequential probing)**.** *Let $(v_t)_{t\le\tau}$ be any non-anticipating policy adapted to the price filtration, with an almost surely finite stopping time and $\sum_{t\le\tau} v_t = 0$. Then*

$$\mathbb{E}_0[L_0(v)] \;\ge\; 2\sigma^2\,\mathrm{kl}(\alpha, 1-\beta)\inf_{\omega}\frac{g_0(\omega)}{|\Delta\hat{H}(e^{i\omega})|^2}, \tag{2.22}$$

where $\mathrm{kl}(\alpha,1-\beta) = \alpha\log\frac{\alpha}{1-\beta} + (1-\alpha)\log\frac{1-\alpha}{\beta}$. *Hence adaptation can lower expected fee-free loss by at most the factor* $\mathrm{kl}(\alpha,1-\beta)/(2z_{1-\alpha}^2) \approx 0.4897$ *at* $\alpha=\beta=0.05$, *and cannot alter the* δ^{-4} *order.*

At $\delta = 0.05$ the adaptive bound is 29.94, or 30.22 with the minimal fee. Pairing it with the feasible deterministic 64.48 gives a certified ratio of at least 0.464. In 20,000 simulated sequential probability-ratio tests, which split the optimal probe into ten copies of equal information, mean loss was 39.5 under H_0, between the two bounds, but losses vary with the stopping time: the 95th and 99th percentiles of required account capital were 91.7 and 128.4, and funding a trader with only the mean loss gives a 37.5% insolvency rate before the test ends. Expected-loss savings are not solvency savings.

Theorem 2.5.4 (Finite-horizon Rayleigh–Hölder duality)**.** *For a horizon of $L \ge 4$ trading periods, let $E \in \mathbb{R}^{L\times(L-1)}$ be the first-difference basis of flat probes ($\sum_t v_t = 0$), $Q_i = \frac{1}{2}T_{h_i}$ (so that $L_i(v) = v^\top Q_i v$), $A_{\mathrm{diff}} = (H_1 - H_0)^\top(H_1 - H_0)$, and let $G_i = E^\top Q_i E \succ 0$ and $M = E^\top A_{\mathrm{diff}} E \succeq 0$ be the reduced execution-loss and information matrices, with G_i-orthonormal generalized eigenvectors e_j and eigenvalues $0 \le \lambda_1 \le \cdots \le \lambda_{L-3} < \lambda_{L-2} \le \lambda_{L-1}$; put $c_{\mathrm{req}} = 4\sigma^2 z_{1-\alpha}^2$ and $c_{\mathrm{fee}} = \phi P_*$. For $\theta \in [0,\pi)$ let $d(\theta) = \cos\theta\, e_{L-2} + \sin\theta\, e_{L-1}$, $\lambda(\theta) = \lambda_{L-2}\cos^2\theta +$*

Table 2.5: Capital lower bounds and feasible upper bounds across separations (par units)

δ	ω^*	$C^*_{\mathrm{det,quad}}$	$\delta^4 C^*_{\mathrm{det,quad}}$	Certified C_{det}	Feasible UB	Adaptive LB
0.02	0.2546	2397.85	3.84×10^{-4}	2399.58	2428.25	1176.04
0.03	0.2541	473.19	3.83×10^{-4}	473.97	484.50	232.51
0.05	0.2526	61.14	3.82×10^{-4}	61.41	64.48	30.22
0.08	0.2488	9.26	3.79×10^{-4}	9.37	10.38	4.64
0.10	0.2453	3.77	3.77×10^{-4}	3.83	4.44	1.91
0.15	0.2331	0.73	3.67×10^{-4}	0.76	1.01	0.38

Calibration $a = 0.5$, $\kappa = 0.8$, $q = 0.2$, $\eta = 0.4$, $\sigma = 0.01$, $\alpha = 0.05$. From `recompute_theory.py` and `recompute_p1_capital.py`; the certified column adds the fee lower bound to the fee-free infimum; the feasible column is the best flat probe found over $L \le 240$; the adaptive column adds the fee lower bound to the adaptive bound.

$\lambda_{L-1} \sin^2\theta$, $s_\theta = \mathrm{sign}(Ed(\theta))$, *and*

$$F_{\mathrm{dom}}(\theta) = c_{\mathrm{fee}} \sqrt{\frac{c_{\mathrm{req}}}{\lambda(\theta)}}\, \|Ed(\theta)\|_1, \qquad W_\perp(\theta) = c_{\mathrm{fee}}^2 \sum_{j \le L-3} \frac{(s_\theta^\top E e_j)^2}{1 - \lambda_j/\lambda(\theta)},$$
$$\gamma(\theta) = \frac{\lambda(\theta)\,\lambda_{L-3}}{(\lambda(\theta) - \lambda_{L-3})\, c_{\mathrm{req}}}. \tag{2.23}$$

Then every flat probe meeting the separation constraint of (2.18) *satisfies*

$$L_i(v) + c_{\mathrm{fee}}\|v\|_1 \;\ge\; \min_\theta \Big\{ \frac{c_{\mathrm{req}}}{\lambda(\theta)} + \Psi(\theta) \Big\} \;\equiv\; C_{\mathrm{RH}}(L) \;\ge\; \frac{c_{\mathrm{req}}}{\lambda_{L-1}}, \tag{2.24}$$

$$\Psi(\theta) = \min_{x \ge 0} \Big\{ x^2 + \Big[F_{\mathrm{dom}}(\theta)\big(1 - \gamma(\theta)x^2\big)_+ - \sqrt{W_\perp(\theta)}\, x \Big]_+ \Big\} \;\ge 0,$$

a piecewise-quadratic minimization in closed form; when $\gamma F_{\mathrm{dom}} < 1$ *and the minimizer has* $\gamma x^2 \le 1$, $\Psi = \max\big(0, F_{\mathrm{dom}} - W_\perp/(4(1 - \gamma F_{\mathrm{dom}}))\big)$.

The bound is computable for any horizon. Table 2.6 reports its quadratic part $c_{\mathrm{req}}/\lambda_{L-1}$, which decays toward the infinite-horizon value 61.14 at rate $O(L^{-1})$, the full bound $C_{\mathrm{RH}}(L)$, and the best feasible fee-paying probe we find at each horizon. At $L = 80$ the full bound is 64.72 against a best probe of 64.76 on the same loss-plus-fee objective, so the dual is essentially tight at short horizons; at longer horizons the fee term of the dual weakens and the bracket widens to about 5%. The fee of unpadded full-support eigenvectors grows as $\Theta(\sqrt{L})$, which is why zero-padded or L_1-localized probes are needed at long horizons.

2.5.3 Robustness: heteroskedastic impact and probing protocols

Corollary 2.5.5 (Invariance under impact-induced heteroskedasticity)**.** *Suppose probing inflates innovation variance as* $\sigma^2_{i,t}(v) = \sigma^2(1 + \xi[(h_i * v)_t]^2)$ *and slippage scales as* $\kappa_{\mathrm{eff}} = \kappa(1 + \xi_0|v_t|)$. *For capacity-bounded probes,* $D_{\mathrm{KL}}(P_0^v \| P_1^v) \le \frac{1+O(\xi^2 V_{\mathrm{eff}}^2)}{2\sigma^2} \|(H_1 - H_0)v\|_2^2 = O(\|v\|_2^2 \delta^4)$, *and convex impact only raises dissipation. Required capital remains* $\Theta(\delta^{-4})$.

Table 2.6: Finite-horizon capital at $\delta = 0.05$ (par units)

Horizon L	Quadratic part $c_{\mathrm{req}}/\lambda_{L-1}$	Full bound $C_{\mathrm{RH}}(L)$	Best feasible probe	Gap to $C_{\mathrm{det}} = 61.41$
80	62.55	64.72	65.02	5.9%
120	61.84	63.39	64.59	5.2%
160	61.54	62.46	64.50	5.0%
240	61.32	61.76	64.48	5.0%
320	61.25	—	≤ 64.48	$\leq 5.0\%$
480	61.19	—	≤ 64.48	$\leq 5.0\%$

Quadratic bound $c_{\mathrm{req}}/\lambda_L$ from the generalized eigenproblem of Theorem 2.5.4 on flat probes (equivalently $c_{\mathrm{req}}/(2\lambda'_L)$ with λ' computed against T_{h_0} rather than $\frac{1}{2}T_{h_0}$); it converges to the infinite-horizon value 61.14. The full bound minimizes (2.24) over 3,600 values of θ (`hardening/rh_bound.py`); at $L = 80$ and 120 a direct numerical minimization of loss plus fee under the separation constraint gives 64.76 and 64.43, above the bound. Feasible probes: top generalized eigenvector refined by sequential quadratic programming on the full ledger; at $L = 320$ and 480 the $L = 240$ probe padded with zeros is feasible at the same cost, so the best feasible value can only be lower.

Table 2.7: Capital lower bound under alternative observation-noise calibrations (USDT)

Noise calibration	$\delta = 0.02$	$\delta = 0.05$	$\delta = 0.10$
Roll estimator, median (0.40 USDT)	3,874,696	99,193	6,200
One-second price-change s.d., median (1.98 USDT)	93,688,369	2,398,422	149,902
Earlier assumption (4 USDT, upper scenario)	383,812,500	9,825,600	614,100

$C(\sigma, \delta) = 61.41 \times (\sigma/0.01)^2 \times (0.05/\delta)^4$, with σ the unrounded median. Noise estimated on 11 days of checksum-verified Binance BTCUSDT one-second klines.

Four probing protocols (unconstrained single pulse, repeated capped pulses, a sequential test with dynamic throttling, and a predetermined generalized-eigenvector trajectory) all show capital slopes near -4 in $\log \delta$: -3.90, -3.84, -3.47 and -3.81 over $\delta \in [0.03, 0.12]$, where the sequential slope is flattened by the 1.7 pulses needed at $\delta = 0.12$ (Appendix Table 2.28). At equal observation horizon, sequential stopping saves 14–49% of capital relative to repeated capped pulses, but most of the apparent saving from the single-tap baseline comes from observing the transient response, not from adaptivity.

2.5.4 From par units to Bitcoin dollars

Capital scales as $\sigma^2\delta^{-4}$, so the dollar magnitude hinges on the observation noise. Treating the 4 USDT ten-second price volatility as observation noise would give 9.8 million USDT at $\delta = 0.05$, but that conflates price volatility with measurement error. We now estimate observation noise from Binance BTCUSDT one-second closes on January 10–20, 2023, with the Roll covariance estimator $\hat{\sigma} = \sqrt{-\mathrm{cov}(\Delta p_t, \Delta p_{t-1})}$ (Roll, 1984). Daily estimates range from 0.09 to 0.77 USDT, median 0.40 USDT.

At realistic noise, separating rates that differ by 0.05 costs about 9.9×10^4 USDT, which is not prohibitive for a well-capitalized desk (Table 2.7). The barrier becomes large only for finer separation (3.9 million USDT at $\delta = 0.02$) and, through Theorem 2.6.4, when several cohorts

Table 2.8: Daily observation-noise estimates, Binance BTCUSDT one-second closes

Date	Roll $\hat{\sigma}$ (USDT)	One-second change s.d. (USDT)	Mean price (USDT)
2023-01-10	0.152	0.679	17,288
2023-01-11	0.092	0.891	17,456
2023-01-12	0.232	2.292	18,343
2023-01-13	0.348	1.762	19,075
2023-01-14	0.773	3.511	20,818
2023-01-15	0.482	1.790	20,778
2023-01-16	0.629	2.357	21,036
2023-01-17	0.457	1.976	21,194
2023-01-18	0.695	2.541	21,122
2023-01-19	0.402	1.316	20,850
2023-01-20	0.316	2.038	21,306
Median	0.402	1.976	

The Roll estimator measures the negative first-order autocovariance of price changes, which reflects bid-ask bounce and recording noise rather than price volatility.

are unknown. Two factors we do not model would raise these figures: walking several levels of a real book costs more than the linear top-of-book impact assumed here, and impact-induced heteroskedasticity raises the noise a probe must overcome. The capital figures are therefore lower bounds for the stated model.

2.6 Unknown mixtures and passive data

Theorem 2.4.1 concerns a *known* pair. A supervisor faces an unknown number of cohorts with unknown rates and weights. This section asks how fast the cost of recovering them grows.

2.6.1 Why naive pole fitting fails

Because the memory rates are zeros of H (Theorem 2.3.1), extracting poles directly from the price impulse response by matrix-pencil or Prony methods targets closed-loop eigenvalues. A four-step inversion repairs this: (1) decouple $g_t = h_{t+1} - ah_t$; (2) form a Hankel matrix of g and compute a minimal realization by Kung's eigensystem realization; (3) compute its transmission zeros, which are the roots of the denominator of G; (4) project reconstructed flow onto the recovered memory filters to obtain the weights (Appendix 2.15.3). Without noise this recovers $K = 0$, $K = 1$ and well-separated two-cohort systems exactly (Appendix 2.15.3); with noise, per-cohort weights become unreliable once rates are close. The reason is information-theoretic, as the rest of this section shows.

2.6.2 Local information: the Cramér–Rao benchmark

Proposition 2.6.1 (Cramér–Rao benchmark)**.** *Let θ collect the weights $w_1, \dots, w_K$ and rates $\eta_1, \dots, \eta_K$. Suppose the impulse response $h_t(\theta)$, $t \le T$, is observed with i.i.d. Gaussian error from*

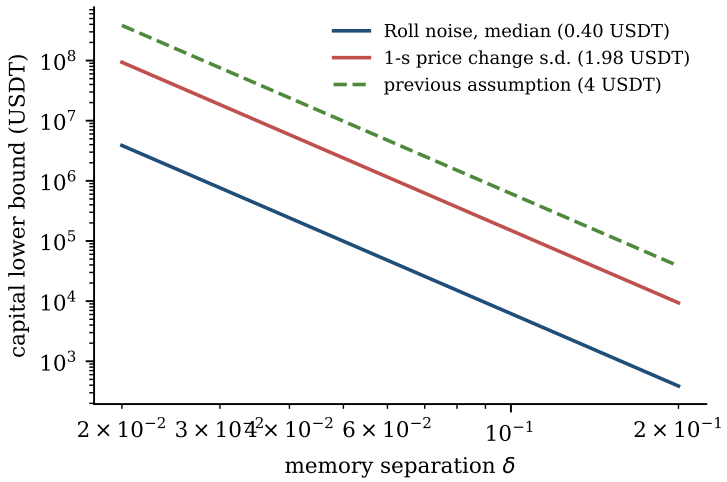


Figure 2.5: Capital lower bound against memory separation for three observation-noise calibrations.

Table 2.9: Cramér–Rao sample sizes for multi-cohort weight recovery ($T = 120$ taps)

Scenario	K	N^* probes	σ_{crit}	ML success ($f = 1$)	RMSE/CRB
$\eta = 0.30$	1	0.1	6.05×10^{-3}	94%	1.02
$\Delta\eta = 0.30$	2	47.8	2.31×10^{-4}	94%	1.01
$\Delta\eta = 0.20$	2	597	6.55×10^{-5}	96%	0.98
$\Delta\eta = 0.10$ ($\delta = 0.05$ pair)	2	40,374	7.96×10^{-6}	93%	1.03
$\Delta\eta = 0.05$	2	2,620,320	9.88×10^{-7}	95%	1.03
spacing 0.10	3	17,349,791	3.84×10^{-7}	98%	1.22
spacing 0.20	3	27,038	9.73×10^{-6}	98%	0.88

Tolerance ± 0.01 at 95%. σ_{crit} is the single-probe noise at which one probe suffices; operational single-probe noise is 1.6×10^{-3}. ML columns: least squares over all weights and rates from six generic starts (never the true values), 200 replications per row at σ_{crit}; success means every weight within ± 0.01. Fitted log–log slope of N^* in $\Delta\eta$: -6.04.

N reset probes of size V_{probe} against innovation noise σ_p, so the error has standard deviation $\sigma_h = \sigma_p/(V_{\text{probe}}\sqrt{N})$. With Jacobian $J_{tj} = \partial h_t/\partial\theta_j$ and $F_{\text{unit}} = J^\top J$, any regular unbiased estimator has $\mathrm{Var}(\hat{w}_k) \geq \sigma_h^2 [F_{\text{unit}}^{-1}]_{kk}$*, and a 95% interval of half-width* tol *requires*

$$N^* \geq \left(\frac{1.96}{\text{tol}}\right)^2 \left(\frac{\sigma_p}{V_{\text{probe}}}\right)^2 [F_{\text{unit}}^{-1}]_{kk}. \tag{2.25}$$

For $K = 2$ with spacing $\Delta\eta$, $[F_{\text{unit}}^{-1}]_{kk} = \Theta((\Delta\eta)^{-6})$, so $N^* = \Theta((\Delta\eta)^{-6})$.

At the critical noise level maximum likelihood attains the bound: its RMSE for the hardest weight is between 0.88 and 1.22 times the Cramér–Rao value and every weight is within tolerance in 93% to 98% of replications (Table 2.9), so the critical noise levels, not estimator failure, are what make close cohorts expensive. Under ordinary surveillance budgets ($N \leq 50$) unbiased estimators cannot resolve close weights.

2.6.3 A global two-point bound for any estimator

Proposition 2.6.1 applies to regular unbiased estimators. Could biased, ridge, shrinkage or Bayesian estimators escape the $(\Delta\eta)^{-6}$ scaling? No.

Theorem 2.6.2 (Global two-point bound and matching upper bound, $K = 2$)**.** *Fix total weight $q = 0.20$, mean rate $\bar{\eta} = 0.40$, target weight separation $\tau = 0.01$, and $\Delta\eta \in (0, 8/11)$. Let $p_{\mp} = \frac{1}{2} \mp \tau/q$ and*

$$\theta_0 = (qp_-, qp_+, \bar{\eta} - p_+\Delta\eta, \bar{\eta} + p_-\Delta\eta), \qquad \theta_1 = (qp_+, qp_-, \bar{\eta} - p_-\Delta\eta, \bar{\eta} + p_+\Delta\eta). \tag{2.26}$$

(i) Label cohorts in increasing order of rate, so that weight vectors are compared rate by rate (as unordered sets the two weight vectors coincide). Both economies have spacing $\Delta\eta$, weight separation $\|w(\theta_1) - w(\theta_0)\|_\infty = 2\tau$ in this labelling, and equal rate moments $M_m = \sum_k w_k \eta_k^m$ for $m \in \{0, 1, 2\}$, while

$$M_3(\theta_1) - M_3(\theta_0) = 4\tau p_- p_+ (\Delta\eta)^3. \tag{2.27}$$

Hence $\|h(\theta_1) - h(\theta_0)\|_2 = \Theta(\tau(\Delta\eta)^3)$.

(ii) For any measurable estimator (including biased, ridge, James–Stein or MAP) that reports weights in the same rate order, the supremum being over the class of two-cohort economies with total weight q, mean rate $\bar{\eta}$ and spacing $\Delta\eta$,

$$\sup_\theta P_\theta(\|\hat{w} - w(\theta)\|_\infty \geq \tau) \;\geq\; 1 - \Phi\Big(\frac{\sqrt{N}\|h(\theta_1) - h(\theta_0)\|_2}{2\sigma_h}\Big), \tag{2.28}$$

so failure probability at most 0.05 requires $N \geq 4\sigma_h^2 z_{0.95}^2 / \|h(\theta_1) - h(\theta_0)\|_2^2 = \Omega((\Delta\eta)^{-6})$. The same bound holds without any labelling convention for the permutation-invariant loss

$$d_\infty(\hat{\theta}, \theta) = \min_{\pi \in S_2} \max_k \big(|\hat{w}_{\pi(k)} - w_k| + |\hat{\eta}_{\pi(k)} - \eta_k|\big)$$

in place of $\|\hat{w} - w(\theta)\|_\infty$, whenever $\Delta\eta \geq 2\tau q/(q + 2\tau)$, because then $d_\infty(\theta_0, \theta_1) = \min\{2\tau(1 + \Delta\eta/q), (1 + 2\tau/q)\Delta\eta\} \geq 2\tau$.

(iii) Conversely, on the known-midpoint family a linear estimator that projects the averaged impulse response on the normalized third-moment direction and inverts (2.27) has error at most $|\hat{M}_3 - M_3|/(2p_-p_+(\Delta\eta)^3)$, giving $N = O(\tau^{-2}(\Delta\eta)^{-6}\log(1/\delta_{\text{fail}}))$. On the separated class with weights and rates bounded away from the boundary, Toeplitz deconvolution followed by a Prony–Hankel root step has Hankel determinant $g_1 g_3 - g_2^2 = w_1 w_2 \eta_1 \eta_2 r_1 r_2 (\Delta\eta)^2 > 0$ for the deconvolved kernel $g_m = \sum_i w_i \eta_i r_i^m$, and perturbation analysis gives the same order.

Table 2.10: Two-point sample complexity for two-cohort weight recovery

$\Delta\eta$	ΔM_3	$\|\Delta h\|_2$	N^*_{LeCam} (95%)	N^*_{CRB}
0.30	2.67×10^{-4}	1.98×10^{-3}	7.1	47.8
0.20	7.92×10^{-5}	5.60×10^{-4}	88.2	597.0
0.14	2.72×10^{-5}	1.89×10^{-4}	775.3	5,267.9
0.10	9.90×10^{-6}	6.84×10^{-5}	5,927.6	40,374.2
0.08	5.07×10^{-6}	3.49×10^{-5}	22,740.5	155,048.9
0.06	2.14×10^{-6}	1.47×10^{-5}	128,333.0	875,732.3
0.05	1.24×10^{-6}	8.50×10^{-6}	383,859.1	2,620,321.2
0.04	6.34×10^{-7}	4.35×10^{-6}	1,466,361.5	10,012,662.4

Worst-case failure at $\Delta\eta = 0.10$*,* $N = 50$*, 1,000 trials per hypothesis*: bound 44.1%; ML 94.5%, ridge 47.5%, positive-part shrinkage 100%, heavy prior 73.9%.

$\tau = 0.01$, $\sigma_h = 1.6 \times 10^{-3}$, $a = 0.5$, $\kappa = 0.8$, 121 taps. Recomputed columns from `recompute_theory.py`. Estimators re-implemented in `recompute_p1_estimators.py`: ML is least squares from four starts; ridge and heavy prior are posterior modes with prior standard deviations $(0.05, 0.10)$ and $(0.005, 0.02)$ for (weights, rates) around the symmetric centre of the two hypotheses; positive-part shrinkage pulls the ML weights toward their mean. Failure means some weight is off by at least τ.

We recomputed the two-point quantities exactly (Table 2.10): the moment gap equals $4\tau p_- p_+ (\Delta\eta)^3$ to machine precision, and the Le Cam requirement at $\Delta\eta = 0.10$ is 5,927.6 resets. The worst-case failure rates of four re-implemented estimators all lie above the lower bound (Table 2.10).

Remark 2.6.3 (The hard directions carry no risk). The two economies of Theorem 2.6.2 have the same total weight $\sum_i q_i = q$, so their summed price response $\sum_t h_t = \kappa/(1 - a - \kappa \sum_i q_i)$ is identical, 2.353 at the calibration of Table 2.10: the directions in which prices are least informative about the cohorts are directions in which cumulative price risk does not move. The summed response is a linear functional of the response and is estimated at the parametric rate whatever the spacing of the memories; Chapter 3 turns this into a certification theorem, a minimax-optimal pre-trade guard and a decision value of disclosure (Chapter 3). Identification and safety ask different questions of the same data.

2.6.4 Many unknown cohorts

Theorem 2.6.4 (K-cohort two-point lower bound)**.** *Fix* $K \geq 2$*, a center* $r_c \in (0, 1)$ *and total merged weight* $W > 0$*. There exist two* K*-cohort economies with rates in* $r_c + \Delta\eta[-1, 1]$*, positive weights, and merged measures whose moments of order* $0, \ldots, 2K-2$ *coincide while the moment of order* $2K-1$ *differs by a constant times* $(\Delta\eta)^{2K-1}$*. Their impulse responses differ by* $\|h^A - h^B\|_2 \asymp (\Delta\eta)^{2K-1}$*, so recovering the weights from* N *independent noisy reset experiments with worst-case error probability below* $1/4$ *requires*

$$N \geq c\,\sigma^2 (\Delta\eta)^{-(4K-2)}.$$

Table 2.11: Gauss and Gauss–Radau rules on $[-1, 1]$ used in Theorem 2.6.4

K	Gauss nodes / Gauss–Radau nodes	Max gap, orders $< 2K-1$	Gap at order $2K-1$
2	± 0.5774 / $-1,\ 0.3333$	6×10^{-17}	0.2222
3	$0,\ \pm 0.7746$ / $-1,\ -0.2899,\ 0.6899$	2×10^{-16}	0.0533
4	$\pm 0.3400,\ \pm 0.8611$ / $-1,\ -0.5753,\ 0.1811,\ 0.8228$	4×10^{-16}	0.0131

All Gauss–Radau weights are positive.

Table 2.12: Numerical check of Theorem 2.6.4

	$K=2$	$K=3$	$K=4$
Theoretical exponent of N^*	-6	-10	-14
Fitted log–log slope (small $\Delta\eta$)	-5.96	-9.96	-14.01

Le Cam bound $N^* \propto \sigma^2/\|h^A - h^B\|_2^2$ with $\sigma = 1$, $a = 0.5$, $\kappa = 1$, total merged weight 0.10, rates centered at 0.70, responses over 400 lags.

The same requirement holds for the permutation-invariant loss $W_1(\hat{M}, M)$ between estimated and true merged measures at accuracy $c_K W \Delta\eta$, where $2c_K > 0$ is the Wasserstein distance between the two normalized quadrature rules, so no relabelling of atoms escapes it.

The construction uses the K-point Gauss and Gauss–Radau rules of the same weight function: both are positive, have K atoms and integrate polynomials of degree $2K-2$ exactly. For $K = 2$ it reproduces the order of Theorem 2.6.2. Table 2.11 lists the two rules: their low moments agree to machine precision and their moment of order $2K-1$ differs by 0.222, 0.0533 and 0.0131 for $K = 2, 3, 4$. The shrinking gap explains why constants deteriorate with K on top of the exponent.

Table 2.12 and Figure 2.6 confirm the order numerically.

Theorem 2.6.5 (Matching upper bound for every K)**.** *Fix $K \geq 2$, a compact set $\mathcal{S} \subset \{s \in [-1,1]^K : s_1 < \cdots < s_K\}$ of ordered rescaled rates with $\min_{i \neq j} |s_i - s_j| \geq \gamma > 0$, and weights in $[\omega_{\min}, \omega_{\max}]$ with $\omega_{\min} > 0$. Consider the class of K-cohort economies with rates $r_c + \Delta\eta\, s$, $s \in \mathcal{S}$, and N independent reset experiments whose responses carry Gaussian noise with variance σ^2. There are constants $c_0, C > 0$ depending only on $(K, \mathcal{S}, \gamma, \omega_{\min}, \omega_{\max}, a, \kappa, r_c)$ such that, whenever $\epsilon_N \equiv \sigma N^{-1/2} (\Delta\eta)^{-(2K-1)} \leq c_0/\sqrt{\log(2/\beta)}$, Prony's method applied to the first $2K$ estimated kernel values, with its output atoms listed in increasing order of rate, recovers the weights with*

$$\max_i |\hat{w}_i - w_i| \ \leq\ C\sqrt{\log(2K/\beta)}\ \epsilon_N$$

with probability at least $1 - \beta$. Hence $N = C'\sigma^2 (\Delta\eta)^{-(4K-2)}$ experiments suffice for any fixed accuracy and confidence, and together with Theorem 2.6.4 the minimax sample requirement is $N^* \asymp \sigma^2 (\Delta\eta)^{-(4K-2)}$.

Proof. The kernel values $k_m = \sum_i w_i r_i^m$ are the power moments of the merged measure.

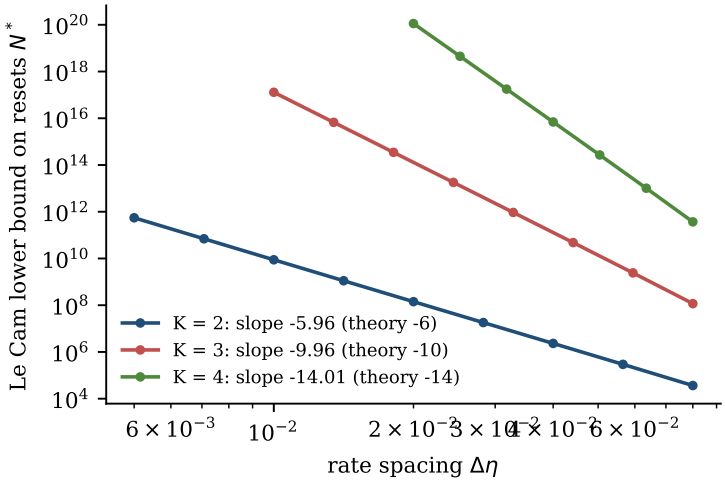


Figure 2.6: Two-point lower bound on the number of reset experiments needed to separate K-cohort economies, as a function of the rate spacing.

The closed-loop recursion $h_t = ah_{t-1} + \kappa \sum_m k_m h_{t-1-m}$ (plus κ at $t = 0$) can be solved for $k_0, \dots, k_{2K-1}$ from $h_0, \dots, h_{2K}$ with $h_0 = \kappa \neq 0$; the map is smooth with Lipschitz constant independent of $\Delta\eta$, so the estimated kernel values have error $O_P(\sigma N^{-1/2})$. Centring at r_c is a fixed triangular linear map, $\mu_m = \sum_{j \le m} \binom{m}{j} (-r_c)^{m-j} k_j$, and $\mu_m = (\Delta\eta)^m t_m$ with $t_m = \sum_i w_i s_i^m$. The map $F : (s, \omega) \mapsto (t_0, \dots, t_{2K-1})$ does not depend on $\Delta\eta$; its Jacobian is a confluent Vandermonde matrix with determinant $\prod_i \omega_i \prod_{i<j} (s_i - s_j)^4$, bounded away from zero on the compact class. By the quantitative inverse function theorem there are $\rho_0 > 0$ and $L < \infty$, uniform over the class, such that F^{-1} exists and is L-Lipschitz on the ball of radius ρ_0 around every point of $F(\text{class})$. An error e in μ_m is an error $e(\Delta\eta)^{-m}$ in t_m, largest at $m = 2K - 1$. The kernel errors are Gaussian with standard deviation of order $\sigma N^{-1/2}$, so a union bound over the $2K$ coordinates gives $\|\hat{t} - t\|_\infty \le C_1 \sqrt{\log(2K/\beta)}\, \epsilon_N$ with probability at least $1 - \beta$. When this is below ρ_0, $\hat{t}$ lies in the ball where F^{-1} is L-Lipschitz. The moment map is invariant to relabelling the atoms, so it is injective only on ordered nodes; on the class of ordered, γ-separated nodes it is injective, and the inverse function theorem applies there. Prony's method computes the algebraic inverse of the moment map up to the order of the atoms, so, listing its output in increasing order of rate (which the error bound below keeps unambiguous once $C\sqrt{\log(2K/\beta)}\, \epsilon_N < \gamma/2$), its output is $F^{-1}(\hat{t})$ and the stated bound follows with $C = LC_1$ and $c_0 = \rho_0 / C_1$. The matching statement requires the class to contain the Gauss and Gauss–Radau pair of Theorem 2.6.4 (ordered nodes with separation at least the smaller of their two minimum gaps, and weights within $[\omega_{\min}, \omega_{\max}]$ after scaling to total weight W); for any such class the orders match, with constants that depend on the class. □

Table 2.13 confirms the order numerically. For numerical stability the estimator works in centred and rescaled coordinates (the estimator is the same; only its floating-point evaluation

Table 2.13: Sample requirement of Prony's method against the lower-bound order

	$K=2$	$K=3$	$K=4$	$K=5$
Lower-bound order $-(4K-2)$	-6	-10	-14	-18
Fitted slope of N^* in $\Delta\eta$ (Prony)	-5.94	-10.00	-13.93	-18.34

Rates $0.7+\Delta\eta\, s$ with s the K-point Gauss nodes, total weight 0.1, $\Delta\eta \in [0.03, 0.2]$; N^* is the number of experiments for a weight RMSE of 10% of the smallest weight, from 400 simulated estimates per point.

changes). The argument parallels the stability analysis of super-resolution for clustered complex nodes (Batenkov et al., 2021), here for real nodes, and it reproduces the ESPRIT slope of -6.15 reported for $K=2$ in the earlier version.

2.6.5 General memory: the minimax rate is logarithmic

Exponential memory is a modelling choice. Suppose instead that the feedback kernel is completely monotone, $k_m = \int_{[0,\rho]} r^m \, d\mu(r)$ with $\rho < 1$, where μ is any nonnegative measure of total mass W: a finite or infinite mixture of exponential memories with merged weight W. Write $\mathcal{M}(\rho, W)$ for this class, assume the closed loop is uniformly stable, $s \equiv a + \kappa W/(1-\rho) < 1$, and measure accuracy by

$$d(\mu, \mu') = \sup\Big\{ \int f \, d(\mu - \mu') : \ f \text{ 1-Lipschitz on } [0,\rho], \ f(0) = 0 \Big\},$$

which equals the Wasserstein-1 distance between measures of equal mass and also compares estimates whose mass is not exactly W. Let $E_L(|x|)$ denote the error of the best uniform approximation of $|x|$ on $[-1,1]$ by polynomials of degree L, and let $c_* = \inf_{L \geq 1} L\, E_L(|x|)$. Bernstein (1914) showed that $L\, E_L(|x|)$ converges to a positive constant (0.2802; Varga and Carpenter, 1985), so $c_* > 0$; numerically $c_* = 1/4$, attained at $L = 2$ (Table 2.14).

Lemma 2.6.6 (Moment-matched cohort pairs)**.** *For every $n \geq 2$ there are two probability measures on $[0,\rho]$, each with exactly n atoms, whose moments of orders $0, \dots, 2n-2$ coincide and whose Wasserstein-1 distance is at least $\rho\, E_{2n-2}(|x|) \geq c_* \rho/(2n-2)$.*

The atoms are the points where the best approximation of $|r - \rho/2|$ by polynomials of degree $2n-2$ attains its maximal error with alternating signs, and the weights are those of the divided difference on these points, which annihilates every polynomial of degree $2n-2$ (Appendix 2.13). Each measure is an n-cohort economy.

Theorem 2.6.7 (Minimax rate for general memory)**.** *Let N independent reset experiments observe the impulse response with Gaussian noise of variance σ^2 per tap, and let $B(\rho) = (1 + \sqrt{2})(1 + 2/\rho)$ and $C_h = \kappa^2/(1-s)^2$.*

(i) Lower bound. *Let* $\Lambda_N = \log(N/\sigma^2) + \log\big(2C_h^2 W^2/((1-\rho)\rho)^2\big)$. *Whenever* $\Lambda_N \geq 4\log(1/\rho)$, *every estimator* $\hat{\mu}$ *satisfies*

$$\sup_{\mu \in \mathcal{M}(\rho, W)} P_\mu\Big(d(\hat{\mu}, \mu) \geq \frac{c_* \, \rho W \log(1/\rho)}{\Lambda_N}\Big) \;\geq\; \frac{1}{4e},$$

and the bound already holds over n*-cohort economies, for a finite* n *that grows with* N.

(ii) Upper bound. *With* a *and* κ *known, invert the averaged response into kernel values* $\hat{k}_0, \ldots, \hat{k}_L$, *with* $L + 1 = \lfloor \log(N/\sigma^2)/(4 \log B(\rho)) \rfloor$, *and let* $\hat{\mu}$ *be any measure on* $[0, \rho]$ *minimizing* $\max_{j \leq L} \big| \int r^j d\hat{\mu} - \hat{k}_j \big|$ *(a linear program). For every* $\beta \in (0, 1)$ *there is* N_0 *depending only on* $(\rho, W, a, \kappa, \beta)$ *such that for* $N/\sigma^2 \geq N_0$,

$$\sup_{\mu \in \mathcal{M}(\rho, W)} P_\mu\Big(d(\hat{\mu}, \mu) > \frac{3\pi\rho W \log B(\rho)}{\log(N/\sigma^2)}\Big) \;\leq\; \beta.$$

Hence the minimax accuracy is of exact order $1/\log(N/\sigma^2)$, *and reaching accuracy* w *requires and suffices* $N/\sigma^2 = \exp(\Theta(1/w))$.

Proof sketch. (i) Put the measures of Lemma 2.6.6, scaled to mass W, into the model. Their difference annihilates polynomials of degree $2n-2$, so the generating function of the kernel difference is $\int (1 - rz)^{-1} d(\mu_A - \mu_B) = z^{2n-1} \int r^{2n-1} (1 - rz)^{-1} d(\mu_A - \mu_B)$; on $|z| = 1$ it is at most $2W\rho^{2n-1}/(1-\rho)$, and Parseval bounds $\|k^A - k^B\|_2$ by the same quantity. The response difference is $H_A H_B$ times the shifted kernel difference, so $\|h^A - h^B\|_2 \leq C_h \|k^A - k^B\|_2$. Choosing the smallest n for which the Kullback–Leibler divergence of the N-sample experiments is at most one, the Bretagnolle–Huber form of Le Cam's lemma gives the bound with half the Wasserstein separation of the lemma. (ii) For a 1-Lipschitz f with $f(0) = 0$, Jackson's theorem gives a polynomial q of degree L with $\sup |f - q| \leq \pi\rho/(4(L+1))$, and expanding q in Chebyshev polynomials of $2r/\rho - 1$ bounds the sum of its monomial coefficients by $\rho B(\rho)^{L+1}$. Hence $d(\hat{\mu}, \mu) \leq \pi\rho(W + \delta)/(2(L+1)) + 2\delta\rho B(\rho)^{L+1}$, where δ is the largest kernel error. The kernel is recovered from the response by a series reciprocal, $K(z) = (1 - az - \kappa/H(z))/(\kappa z)$, whose error is at most $2((1+s)/\kappa)^2(L+2)$ times the largest response error, so $\delta = O_P(\sigma N^{-1/2} L\sqrt{\log L})$. The choice of L makes $\delta B(\rho)^{L+1} = O_P((\sigma^2/N)^{1/4} L\sqrt{\log L})$, negligible against the first term. Full proofs are in Appendix 2.13. □

Theorem 2.6.7 replaces a polynomial barrier with a logarithmic one. With exponential memory and K known cohorts, halving the error multiplies the required number of experiments by a constant; with general memory, halving the error squares it. The lower bound holds already for finite mixtures, so knowing that the memory is a finite mixture of exponentials does not help unless the number of cohorts and their spacing are also known. The constants of the two bounds differ, by a factor that grows as $\rho \to 1$, so the dependence on how long memory lasts is not pinned down. The source of the gap is visible in the proof: as $\rho \to 1$ the moment-matched

Table 2.14: General memory: moment-matched cohort pairs and the moment estimator

A. Best approximation of $\lvert x\rvert$						
L	2	4	6	10	20	30
$L\,E_L(\lvert x\rvert)$	0.2500	0.2705	0.2756	0.2785	0.2797	0.2800
B. Pairs of Lemma 2.6.6						
Atoms per measure n	2	4	6	8	10	12
$W_1(\mu_A,\mu_B)$, $\rho=0.9$	0.2250	0.1037	0.0669	0.0494	0.0391	0.0324
$n\,W_1$, $\rho=0.9$	0.450	0.415	0.402	0.395	0.391	0.388
$\lVert k^A-k^B\rVert_2$, $\rho=0.9$	4.41×10^{-2}	1.03×10^{-3}	4.08×10^{-5}	1.96×10^{-6}	1.05×10^{-7}	5.95×10^{-9}
$\lVert k^A-k^B\rVert_2$, $\rho=0.99$	7.41×10^{-2}	5.84×10^{-3}	1.07×10^{-3}	2.76×10^{-4}	8.33×10^{-5}	2.76×10^{-5}
C. Linear-programming estimator, continuous memory distribution, $\rho=0.9$						
$\log_{10} N$	4	8	12	16	20	24
Moments used L	2	3	5	6	7	8
Error $d(\hat\mu,\mu)$	0.00878	0.00759	0.00432	0.00445	0.00382	0.00399
Error $\times\log(N/\sigma^2)$	0.202	0.245	0.179	0.226	0.229	0.276

A: Remez algorithm; the limit is Bernstein's constant 0.2802. B: moments of orders $0,\dots,2n-2$ agree to 10^{-61} (60-digit arithmetic); every W_1 exceeds $\rho E_{2n-2}(\lvert x\rvert)$. C: true merged measure Beta(2,3) on $[0,0.9]$ with mass 0.1; kernel values observed with noise $\sigma N^{-1/2}$, $\sigma=10^{-3}$; L from the rule in Theorem 2.6.7(ii); measure fitted on a 401-point grid; 20 replications. Kind: recomputed (`general_minimax.py`).

pairs separate their kernels more slowly (ρ^{2n-1} decays slowly), which weakens the lower bound through $\log(1/\rho)$, while the upper bound pays $\log B(\rho)$, which stays bounded away from zero, for the conditioning of polynomial approximation on $[0,\rho]$. Logarithmic rates of this kind are familiar from deconvolution with supersmooth errors (Fan, 1991) and from moment methods for mixing distributions (Kong and Valiant, 2017; Wu and Yang, 2020); here they arise from the closed loop of a market.

Table 2.14 checks the construction and the estimator. The pairs of Lemma 2.6.6 have exactly n atoms, agree in all moments below order $2n-1$ to 10^{-61}, and remain at Wasserstein distance about $0.4/n$, while the distance between their kernels shrinks by a factor of about 0.2 per atom at $\rho=0.9$. The linear-programming estimator applied to a continuous memory distribution has error times $\log(N/\sigma^2)$ between 0.18 and 0.28 for N from 10^4 to 10^{24}: a 10^{20}-fold increase in data halves the error. The exponential-memory model of the previous subsections, where the cost is polynomial in $1/\Delta\eta$, is therefore the optimistic case: relaxing the parametric form makes identification from prices much harder, not easier.

2.6.6 Passive data

Theorem 2.6.8 (Passive identification: minimum phase)**.** *In any stable economy with $\lvert r_i\rvert<1$ and nonzero merged weights, the zeros of H are exactly the memory rates r_i, and H is minimum phase. With a single price innovation, the price spectrum identifies H up to scale, hence the memory rates; with κ and a known, as in Theorem 2.3.1, it identifies the merged measure.*

The result holds for signed weights (mean-reverting cohorts): in 1,733 stable random economies with weights of both signs, the computed zeros equalled the memory rates and lay

inside the unit disk. With a second independent shock in the cohorts' memories, the Jacobian of the price spectrum with respect to six model parameters has full rank, with singular values from 34.51 down to 0.0698. Passive noise helps identification rather than hindering it.

Theorem 2.6.9 (Passive cost)**.** *Suppose the price is observed passively as the stationary output $x_t = H_\theta(L)\xi_t$ of a single innovation with variance $\sigma^2 > 0$, for θ in a compact, uniformly Schur-stable family with rates in $[\eta_{\min}, \eta_{\max}] \subset (0,1)$, and let $D(\theta_1\|\theta_0)$ be the Kullback–Leibler divergence rate per observation.*

(i) For a single cohort at $\bar\eta$ against the symmetric pair at $\bar\eta \pm \delta$ with the same total weight, $D = \Theta(\delta^4)$.

(ii) For the moment-matched two-cohort economies of Theorem 2.6.2, $D = \Theta(\tau^2(\Delta\eta)^6)$.

(iii) For every $K \geq 2$, for the K-cohort Gauss and Gauss–Radau economies of Theorem 2.6.4, $D = \Theta((\Delta\eta)^{4K-2})$.

Hence any test based on T passive observations with worst-case error at most $\alpha < 1/4$ needs $T = \Omega(\delta^{-4})$, $T = \Omega(\tau^{-2}(\Delta\eta)^{-6})$ and $T = \Omega((\Delta\eta)^{-(4K-2)})$ respectively, and the likelihood-ratio test attains these orders as $T \to \infty$: passive data cost the same polynomial order as active probing for every number of cohorts.

The proof (Appendix 2.13) combines the Whittle form of the divergence rate with the resolvent identity $H_1 - H_0 = H_1 H_0 (G_1 - G_0)$: the two spectra differ by the leading unmatched moment, of order δ^2 for the pair and $\tau(\Delta\eta)^3$ for the moment-matched economies, and the divergence rate is the square of that difference. Numerically, the fitted slopes of the required sample length are -4.21 and -5.93, against the theoretical -4 and -6; for part (iii) the fitted slopes of the Whittle divergence rate in $\Delta\eta$ are 5.91, 9.98 and 14.05 for $K = 2, 3, 4$, against 6, 10 and 14 (`hardening/p1_checks.py`).

Passive surveillance is possible in principle but costs the same polynomial order as active probing. Proposition 2.15.2 is the exact finite-sample version for the known pair.

2.7 Robustness to nonlinear and adaptive trading rules

The theorems of Sections 2.3–2.6 use the linear exponential-memory recursion (2.2). Real desks cap positions, trade only beyond thresholds, adapt their gains and learn online. A small probe of size A moves the state by $O(A)$, so any rule whose order is a twice-differentiable function of an exponential-memory state is linear to first order and differs from it by $O(A^2)$; the theorems should then hold for small probes and fail for large ones. We check this, and locate where it fails, in simulations of five stylized rules at $a = 0.5$, $\kappa = 0.8$, total weight $q = 0.2$ and mean rate $\bar\eta = 0.4$, over 240 periods after a pulse probe.

The rules are: (1) *momentum with a position limit*, the cohort's order a smooth saturating (tanh) function of its memory state $y_{i,t}$, normalized to unit slope at the operating point; (2) a *threshold rule* that trades only when $|y_{i,t}| > \tau = 0.05$, in two versions, an exact dead zone and one averaged over background state variation of standard deviation 0.04; (3) an *adaptive-gain rule* in the form of recursive least squares with forgetting factor $1-\eta_i$, whose gain falls when the regressor is large; (4) an *online neural readout*, a linear readout of a fixed two-layer tanh feature map with eight hidden units, trained by regularized stochastic gradient steps of size η_i; and (5) a *tabular value rule* that updates a 21-bin table at rate η_i, in two versions, kernel-smoothed across bins and a single-bin lookup. These are stylized: each is built so that its small-signal dynamics are an exponential filter with rate η_i, and they are not calibrated to any production system.

Two checks follow the theory. *Test A (merged measure):* one cohort with weight q and position limit 1 against two cohorts with weights $q/2$ at the same rate and limits 0.55 and 1.65, so the two economies have the same merged measure; the raw response gap $\|X^{(1)}(A)-X^{(2)}(A)\|_2$ should be $O(A^2)$ (slope 2 in $\log A$) and the probe-normalised gap $\|X^{(1)}(A)-X^{(2)}(A)\|_2/A$ should be $O(A)$ (slope 1 in $\log A$) over $A \in [0.002, 0.064]$. *Test B (δ^{-4} law):* one cohort at $\bar{\eta}$ against the symmetric pair at $\bar{\eta} \pm \delta$, $\delta \in [0.015, 0.12]$, at probe size $A = 0.005$; the normalized response difference $\|\Delta h\|_2$ should scale as δ^2 and the certified capital of Theorem 2.5.2 as δ^{-4}. The saturation threshold A_{sat} is the smallest probe at which $\|\Delta h\|_2$ at $\delta = 0.05$ departs by more than 10% from its small-probe value.

To check that these conclusions do not depend on hand-built rules, we pre-registered a second experiment (`P1_trained_learners.json`, SHA-256 `3e8bf30cbfb6...`) in which online learners are trained from scratch inside the closed-loop market for $T_{\text{burn}} = 5{,}000$ periods of flow and noise innovations; their learned parameters are saved and reloaded into the probing simulator without rescaling. Four learning rules are used: (i) recursive least squares with forgetting factor $\lambda = 0.995$, learning its return-forecast slope; (ii) a two-layer tanh network with eight hidden units whose readout is trained by regularized online stochastic gradient descent on trading profit; (iii) a scalar policy-gradient rule trained on trading profit; and (iv) a tabular Q-learner with 21 states and 21 actions, TD(0) Bellman backups ($\gamma = 0.90$) and Boltzmann action choice. Each learner acts on an exponential-memory state $y_{i,t}$ with rate η_i: training sets its response gain and curvature, not its memory rate, so the experiment asks whether learned policies preserve the merged-measure and δ^{-4} predictions, not whether learning produces exponential memory. The specification fixed four predictions: (a) a single exponential fits each trained learner's small-probe impulse response with $R^2 \geq 0.90$; (b) two trained populations with the same merged measure differ by at most 1% in relative L_2 response at $A = 0.005$; (c) the certified capital to separate trained populations whose memory rates differ by 2δ scales with log–log slope in $[-4.15, -3.85]$; and (d) we report where equivalence and the δ^{-4} law fail when learning continues during the probe. A first implementation of this experiment did not pass the trained parameters to the probing simulator; the results below come from the corrected implementation (`validate_`

Table 2.15: Merged-measure equivalence and the δ^{-4} law under stylized nonlinear rules and closed-loop trained online learners

Rule	Test A raw ($O(A^2)$)	Test A norm. ($O(A)$)	Test B slope ($\|\Delta h\|_2$)	Test B slope (capital)	Capital at $\delta = 0.05$	A_{sat} / R^2_{exp}
Panel A. Stylized nonlinear and adaptive rules (fixed operating point)						
Linear benchmark	—	—	2.007	−4.001	61.42	—
1. Momentum (tanh limit)	2.015	1.015	2.007	−4.001	61.47	0.834
2. Threshold rule, averaged	2.043	1.043	2.007	−4.001	61.42	0.049
3. Adaptive gain (RLS)	2.064	1.064	2.007	−4.001	61.46	0.751
4. Online neural readout	2.011	1.011	2.007	−4.001	61.48	0.751
5. Tabular value, smoothed	2.039	1.039	2.007	−4.001	61.44	1.271
Panel B. Learners trained in the closed loop, saved and reloaded (frozen during the probe; shared training stream)						
6. Recursive least squares	3.000	2.000	2.008	−4.014	54.58	$R^2 = 0.993$
7. Two-layer network (SGD)	1.878	0.878	2.007	−4.012	65.19	$R^2 = 0.994$
8. Policy gradient on profit	3.000	2.000	2.004	−4.006	58.05	$R^2 = 0.995$
9. Tabular Q-learner (TD(0))	2.108	1.108	1.994	−3.973	186,821	$R^2 = 1.000$
Panel C. Where the trained-learner results change (range over the four learners)						
Independent training streams	rel. gap $\leq 10^{-5}$		0.13 to 0.92	−1.75 to −0.38	—	(c) fails
Learning during the probe	rel. gap $\leq 1.1 \times 10^{-3}$		1.99 to 2.01	−4.02 to −3.98	—	(c) holds

Exact dead zone ($\tau = 0.05$): $\|\Delta h\|_2 = 0$ at $A = 0.005$; the probe moves the state past τ only for $A > 0.139$.
Single-bin value lookup: $\|\Delta h\|_2 = 0$ at $A = 0.005$; the probe leaves the central bin only for $A > 0.149$.

Panel A: deterministic simulations around the operating point; Test A reports the raw response-gap slope in $\log A$ ($O(A^2)$) and the probe-normalised gap slope $\|X^{(1)}(A) - X^{(2)}(A)\|_2/A$ ($O(A)$, plotted in Figure 2.7). Panel B: learners trained for 5,000 periods, parameters saved and reloaded without rescaling and frozen during the probe; the two compared populations are trained on the same innovation stream; relative equivalence gaps at $A = 0.005$ are at most 10^{-5}. A raw Test A slope above 2 means the gap is of higher order than A^2. Panel C: the same learners trained on independent innovation streams (first row) or updated at learning rate 0.35 during the probe (second row). Capital in par units, certified bound of Theorem 2.5.2 (61.42 for the linear model); the Q-learner settles on a small response gain (0.022), so separating it costs far more capital. Kind: recomputed (Panel A: `recompute_p1_learners.py` → `R_P1_learners.json`; Panels B–C: `validate_trained_learners_hardened.py` → `R_P1_trained_learners_hardened.json`).

`trained_learners_hardened.py`), which reloads the saved parameters.

Table 2.15 and Figure 2.7 show both where the linear reduction holds and where it breaks down. In Panel A, for small probes every smooth or dithered rule behaves like the linear model: economies with the same merged measure have raw response gap $O(A^2)$ (slopes 2.011 to 2.064) and probe-normalised gap $O(A)$ (slopes 1.011 to 1.064), while the response difference and certified capital scale as δ^2 (slope 2.007) and δ^{-4} (slope −4.001), with capital at $\delta = 0.05$ within 0.1% of the linear 61.42. In Panel B, predictions (a)–(c) pass for all four trained learners: a single exponential explains $R^2 \geq 0.993$ of each small-probe impulse response, populations with the same merged measure differ by at most 10^{-5} in relative response at $A = 0.005$, and certified capital scales with slopes −4.014 to −3.973, inside the registered band. The level of capital differs across learners because training sets the response gain.

Panel C shows where this changes. When the two populations are trained on independent innovation streams, prediction (c) fails: the capital slopes are −1.75, −0.97, −1.62 and −0.38.

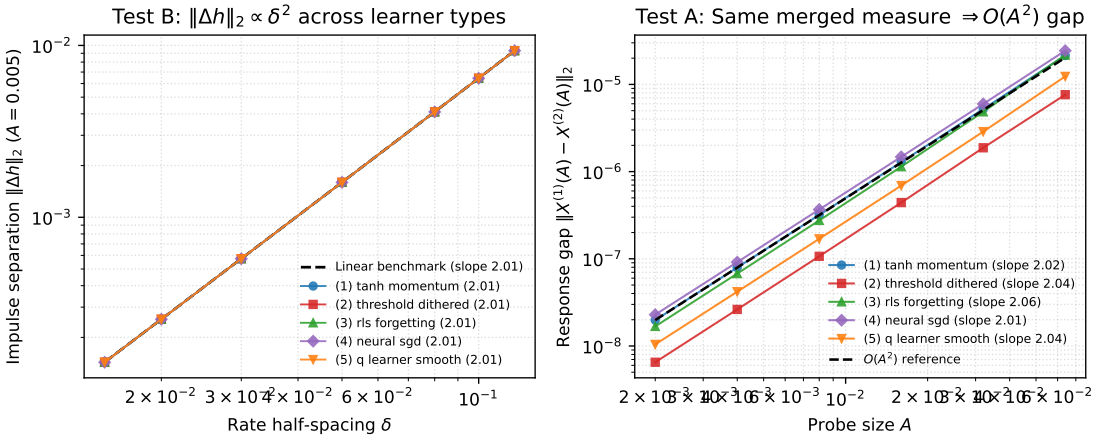


Figure 2.7: Left: Test B, probe-normalised response difference $\|\Delta h\|_2$ against the rate half-spacing δ at probe size $A = 0.005$, five stylized rules and the linear benchmark. Right: Test A, probe-normalised response gap $\|X^{(1)}(A) - X^{(2)}(A)\|_2/A$ between two economies with the same merged measure against the probe size A, with an $O(A)$ reference line (corresponding to an $O(A^2)$ raw gap).

Independently trained populations end with learned gains that differ by sampling error of order $T_{\text{burn}}^{-1/2}$, so they differ in merged weight as well as in memory rate, and a difference in weight is cheaper to detect than an $O(\delta^2)$ difference in rates. The δ^{-4} law therefore prices the separation of memory rates between populations whose feedback weights match; it is not a claim about telling apart any two trained populations. When learning continues during the probe at rate 0.35, the law survives (slopes -4.02 to -3.98) and the merged-measure gap rises only to 1.1×10^{-3}, so the breakdown we anticipated under prediction (d) did not occur in this design.

The linear theory fails in two further places. First, saturation, adaptive gains and the neural readout leave the small-signal regime once probes exceed $A_{\text{sat}} \in [0.751, 1.271]$ par units, and the dithered threshold rule leaves it already at $A_{\text{sat}} = 0.049$, the threshold itself. Second, rules with hard discontinuities are different in kind: an exact dead zone ($\tau = 0.05$) or a single-bin lookup does not respond at all to probes smaller than 0.139 and 0.149 par units ($\|\Delta h\|_2 = 0$), concealing those cohorts from small probes altogether. Neither smooth nonlinearity nor hard thresholds offers a cheaper route to separating memory rates than the linear benchmark; populations whose learned weights differ are told apart by their weights, not their rates.

2.8 What disclosure buys

Because active probing is costly and passive data are slow, the natural policy question is what a supervisor gains from disclosure. We compare six observation contracts at $\delta = 0.05$.

- I_0: public prices only.

Table 2.16: Disclosure contracts at $\delta = 0.05$ (pulse-design capital, par units)

Contract	Channel	Pulses	Benchmark	Saving vs. $T = 20$
I_0: prices, $T = 3$	public prices	271	4,305.25	−2,123.5%
I_0^*: prices, $T = 20$	full price path	12	193.62	0.0%
I_1: prices and signed flow	public flow	12	193.62	0.0%
I_2: noisy equity ($\sigma_E = 0.5$)	statement	12	193.62	0.0%
I_3: exact equity, 5-unit lot	audited equity	1	12.70	93.4%
I_3: exact equity, 0.01 rounding	rounded report	1	15.52	92.0%
I_4: timestamps only	no likelihood	12	193.62	0.0%
I_5: audit registry	internal states	0	0.00	100.0%

Closed-form values. Noisy equity adds information linearly in the number of resets, $D_{\mathrm{KL}} = N\Delta_E^2/(2\sigma_E^2)$, and saves only 1.5% against the $T = 3$ baseline. The equity difference is $\Delta E = \kappa^2 q \delta^2 V^2$; with reports rounded to 0.01, a guaranteed change needs $V \geq 5.59$.

- I_1: prices and aggregated executed flow.
- I_2: prices and one independent noisy report of dealer marked equity per reset ($\sigma_E = 0.5$).
- I_3: prices and exact audited equity (Proposition 2.4.4), also with reports rounded to 0.01.
- I_4: algorithm update timestamps without the update values.
- I_5: a supervisory audit registry with access to internal algorithm states.

Table 2.16 shows that disclosure is not one thing. Timestamps without update values carry no information about δ in this model. Noisy periodic statements barely help. Exact audited equity collapses the requirement to a single trade, limited only by lot sizes and report precision, and an audit registry removes the trading requirement altogether.

The cleanest form of useful disclosure is flow tagged by participant class.

Proposition 2.8.1 (Tagged flow removes the barrier)**.** *Let the price be driven by i.i.d. innovations of variance σ_ξ^2 in a positive, stable economy, with stationary autocovariances $\gamma_x(s) = \sum_\ell c_\ell \lambda_\ell^{|s|}$ ($c_\ell \geq 0$, $0 \leq \lambda_\ell < 1$ the closed-loop modes). Suppose the observer receives prices and each cohort's flow $\tilde{u}_{i,t} = u_{i,t} + \varepsilon_{i,t}$, $u_{i,t} = q_i y_{i,t}$, with i.i.d. measurement noise of variance σ_f^2 independent of prices. Then $\tilde{u}_{i,t} = r_i \tilde{u}_{i,t-1} + w_i x_t + e_{i,t}$ with $e_{i,t} = \varepsilon_{i,t} - r_i \varepsilon_{i,t-1}$, and the just-identified IV estimator of $\beta_i = (r_i, w_i)$ with instruments $Z_t = (x_{t-1}, x_t)^\top$ satisfies $\sqrt{T}(\hat{\beta}_i - \beta_i) \to N(0, V_i)$,*

$$V_i = \Sigma_i^{-1} \Omega_i \Sigma_i^{-\top}, \qquad \Sigma_i = \begin{pmatrix} w_i A_0 & \gamma_x(1) \\ w_i A_1 & \gamma_x(0) \end{pmatrix}, \qquad \Omega_i = \sigma_f^2 \big[(1 + r_i^2)\Gamma_0 - r_i(\Gamma_1 + \Gamma_1^\top) \big], \quad (2.29)$$

where $A_m = \sum_{s \geq 0} r_i^s \gamma_x(s+m)$, $\Gamma_0 = \left(\begin{smallmatrix} \gamma_x(0) & \gamma_x(1) \\ \gamma_x(1) & \gamma_x(0) \end{smallmatrix}\right)$ and $\Gamma_1 = \mathbb{E}[Z_t Z_{t-1}^\top]$. The Jacobian determinant is

$$\det \Sigma_i = w_i \sum_{\ell,m} c_\ell c_m \frac{1 - \lambda_\ell \lambda_m}{1 - r_i \lambda_m} \; > \; 0. \quad (2.30)$$

V_i depends only on cohort i's own (w_i, r_i), σ_f^2 and the price autocovariances, which are bounded and nondegenerate uniformly on a uniformly stable family whatever the rate spacing; so the

merged weights are estimated at rate $T^{-1/2}$ with a constant that does not depend on $|r_i - r_j|$.

Each tagged flow follows $u_{i,t} = r_i u_{i,t-1} + w_i x_t$, a regression whose information about w_i does not involve the other cohorts' rates; the closed form (2.30) matches the simulated Jacobian (0.1330 against 0.1322 in a two-cohort economy with 4×10^5 periods; `hardening/p1_checks.py`). In simulation ($T = 5{,}000$, 60 replications) the root-mean-square error of the recovered weights is flat in $\Delta\eta$ (fitted slope 0.011, error between 5.8×10^{-4} and 6.6×10^{-4}). This is the formal content of a disclosure regime such as a consolidated audit trail: tagging flow by participant class turns a polynomial barrier into an ordinary estimation problem.

2.8.1 How much disclosure is worth

Proposition 2.8.1 says that tagging removes the barrier; the next result says by how much.

Theorem 2.8.2 (Value of class-tagged disclosure)**.** *In the class of Theorem 2.6.5, let N_p be the number of experiments needed with prices alone and N_t the number needed with each class's flow observed with noise σ_f. To recover the weights to fixed accuracy, $N_p/N_t \asymp (\sigma/\sigma_f)^2(\Delta\eta)^{-(4K-2)}$; to resolve the rates to accuracy proportional to their spacing, $N_p/N_t \asymp (\sigma/\sigma_f)^2(\Delta\eta)^{-(4K-4)}$.*

Proof. $N_p \asymp \sigma^2(\Delta\eta)^{-(4K-2)}$ by Theorems 2.6.4 and 2.6.5. With tagged flow each class is a single-exponential problem whose Fisher information for (w_i, r_i) does not depend on the other classes, so $N_t \asymp \sigma_f^2$ for the weights and $N_t \asymp \sigma_f^2(\Delta\eta)^{-2}$ for rates resolved to a fraction of $\Delta\eta$. □

With the same noise and lags for both observers, the Cramér–Rao requirements give fitted slopes of -6.08 for prices and -4.34 for the ratio at $K = 2$ (theory -6 and -4). Calibrated to the class memories estimated below, prices alone would need about 333 times as many days as the published class ratios to separate the top-trader and all-account memories in BTCUSDT (rates 0.9889 and 0.9986 per five minutes; about 6,000 days against 18), and about 19 times as many in ETHUSDT (rates 0.9793 and 0.9972).

2.8.2 Disclosure in dollars

Theorem 2.8.2 states the value of class-tagged disclosure in samples. Theorems 2.5.2 and 2.6.4 let us translate that value into a flow of round-trip trading losses and set it beside the official operating budgets of the U.S. Consolidated Audit Trail (CAT). Two adjustments make the comparison exact and like-for-like:

(i) *Exact K-cohort moment-gap constant.* Rather than extrapolating from Cramér–Rao ratios, we evaluate the spectral round-trip trading loss bound of Theorem 2.5.2,

$$L^*_{\mathrm{rt}}(K, \Delta\eta) \;=\; \frac{2\sigma^2 z^2_{1-\alpha}}{\sup_\omega\big(|\Delta\hat{H}_{K,\Delta\eta}(e^{i\omega})|^2/g_0(\omega)\big)},$$

Table 2.17: Price-only round-trip probing loss per market and annualised weekly-renewal cost against official CAT budgets

	Loss / probe (USDT)		Annual (52×, USDT)		Mkts ($228.3M / $1.7B)	
Economy ($\Delta\eta$)	Roll	1-sec	Roll	1-sec	Roll	1-sec
$K = 2$, known (0.20)	6,200	149,901	322,400	7,794,852	708.1 / 5,273.0	29.29 / 218.09
$K = 2$, known (0.10)	99,192	2,398,422	5,157,984	124,717,944	44.3 / 329.6	1.83 / 13.63
$K = 2$, known (0.04)	3,874,696	93,688,369	201,484,192	4,871,795,188	1.1 / 8.4	0.05 / 0.35
$K = 2$, Radau (0.20)	4,292	103,788	223,184	5,396,976	1,022.9 / 7,617.0	42.30 / 314.99
$K = 2$, Radau (0.10)	283,488	6,854,612	14,741,376	356,439,824	15.5 / 115.3	0.64 / 4.77
$K = 3$, Radau (0.20)	804,403	19,450,107	41,828,956	1,011,405,564	5.5 / 40.6	0.23 / 1.68
$K = 3$, Radau (0.10)	1,137,325,880	27,500,017,407	59,140,945,760	1,430,000,905,164	0.0 / 0.0	0.00 / 0.00
$K = 4$, Radau (0.20)	156,632,086	3,787,291,895	8,144,868,472	196,939,178,540	0.0 / 0.2	0.00 / 0.01

Separation variable $\Delta\eta$ throughout ($\Delta\eta = 2\delta$ for the symmetric known pair). Gauss–Radau rows use the exact K-cohort quadrature pair of Theorem 2.6.4 and Table 2.11 (moment gaps $\Delta_2 = 0.222222$, $\Delta_3 = 0.053333$, $\Delta_4 = 0.013061$ at $W = 0.08$, $r_c = 0.60$). Loss per probe is the spectral round-trip trading loss bound of Theorem 2.5.2 at the median BTCUSDT Roll noise (0.40 USDT) and median one-second price-change standard deviation (1.98 USDT); annual loss multiplies by 52 weekly renewals. The final columns divide the revised CAT LLC 2025 operating budget ($228.3M; Nov. 2024 filing $248.8M, SEC Release No. 34-101901, p. 4) and the SEC 2016 annual industry estimate ($1.7B, Release No. 34-79318, p. 363) by the annual probing loss per market. Probing cost is a lower bound on the cost of learning from prices, not a social cost. Kind: recomputed (`recompute_p1_cat_costs.py` → `R_P1_cat_disclosure_dollars.json`).

directly for the K-point Gauss versus Gauss–Radau economies of Theorem 2.6.4 and Table 2.11, whose moments below order $2K-1$ coincide and whose $(2K-1)$-th moment gap has the exact constant $\Delta_2 = 0.222222$, $\Delta_3 = 0.053333$, and $\Delta_4 = 0.013061$ at spacing $\Delta\eta$ (with $\Delta\eta = 2\delta$ for the symmetric known pair).

(ii) *Flow against flow.* A single probe loss L^*_{rt} is a one-time certification cost, whereas an audit trail is an ongoing annual expenditure. Because algorithm populations turn over and regulatory surveillance is periodic, we compare annual CAT operating costs against the annualised round-trip trading loss $52\times L^*_{\mathrm{rt}}$ under weekly certification renewal (matching the CFTC weekly reporting cycle of Section 2.8.4). We benchmark against three primary official figures: the CAT LLC 2025 operating budget of $248.8 million in the November 2024 filing (SEC Release No. 34-101901, Dec. 12, 2024, p. 4), revised to $228.3 million in May 2025; the SEC's 2016 CAT NMS Plan Approval Order estimate of $2.4 billion in initial industry build costs and $1.7 billion per year in ongoing industry costs (SEC, 2016, Release No. 34-79318, Nov. 15, 2016, 81 FR 84696 at 84801, p. 363); and the SEC's Rule 613 proposing and adopting releases estimating $4.0 billion initial and $2.1 billion annual ongoing costs (SEC, 2012, Release No. 34-62174, May 26, 2010, 75 FR 32556 at 32598, 32600, pp. 147, 153; Release No. 34-67457, July 18, 2012, 77 FR 45722 at 45790–45795, pp. 220–238).

Probing cost is a lower bound on the cost of learning from prices, not a social cost. Within that interpretation, Table 2.17 shows how rapidly the polynomial exponent $(\Delta\eta)^{-(4K-2)}$ and the shrinking Gauss–Radau moment-gap constant Δ_K turn the comparison in favour of disclosure.

For a known pair at coarse separation $\Delta\eta = 0.20$ ($\delta = 0.10$), weekly price-only renewal costs \$0.32 million per market per year at Roll noise and \$7.79 million at one-second noise, so matching the \$228.3 million CAT LLC operating budget requires 708.1 or 29.29 markets. At $\Delta\eta = 0.10$ ($\delta = 0.05$) the unknown $K = 2$ Gauss–Radau pair costs \$14.74 million (Roll) or \$356.44 million (one-second) per market per year, matching the CAT LLC budget in 15.5 or 0.64 markets and the SEC's \$1.7 billion industry figure in 115.3 or 4.77 markets. Once a supervisor must separate $K = 3$ unknown cohorts at $\Delta\eta = 0.20$, annual probing loss rises to \$41.83 million (Roll) and \$1.01 billion (one-second) per market (5.5 and 0.23 markets at the CAT LLC budget; 40.6 and 1.68 markets at \$1.7 billion), and at $K = 3, \Delta\eta = 0.10$ or $K = 4, \Delta\eta = 0.20$ a single market's annual probing loss exceeds both the CAT LLC operating budget and the SEC's total annual industry estimate. The renewal frequency is an assumption: break-even market counts scale inversely with it, so monthly renewal multiplies every count in the last two columns by 4.3 and renewal on each of 252 trading days divides them by 4.8.

2.8.3 Disclosure in real data

Binance publishes, every five minutes, long/short ratios for its top-trader group, by position and by account, and for all accounts. These are class-level positions, the closest public analogue to the tagged data of Proposition 2.8.1. We treat a class's reported net position as the observable counterpart of its cohort state: it obeys the same one-lag recursion class by class, so the block-diagonal information structure of Proposition 2.8.1 carries over. The mapping is an assumption, because the reports are positions rather than flows and returns stand in for the unobserved deviation x_t. For each class c we estimate $n_{c,t} = r_c n_{c,t-1} + w_c \,\mathrm{ret}_t$, with $n = (R-1)/(R+1)$ from the ratio R. The price-only alternative regresses signed five-minute taker flow on twelve lagged five-minute returns and fits two exponentials to the kernel; it estimates the memory rates without class information. Intervals come from a day bootstrap with 200 draws.

Table 2.18 gives three results. First, tagged class data pin down memory rates that prices and flow leave essentially unrestricted: the price-only intervals are 28 to 124 times wider in every one of 30 symbols on fresh days, and cover most of the admissible range. Part of this ratio is mechanical, because a series of positions identifies its own persistence directly while the price-only route must fit two exponentials to a weakly identified kernel. Restricting attention to the five symbols in which the price-only fit does identify the merged response (ACH, API3, OGN, ZEN and DASH) does not change the conclusion: the price-only rate intervals are still 0.64 to 0.92 wide and the ratios run from 27.6 to 62.5 (median 47.5). We therefore read the ratios as a measure of how little prices reveal about class memory, not as a horse race between two estimators of equal standing. Second, the classes differ where theory says the information should be: in BTCUSDT and ETHUSDT the top-trader group's positioning has a half-life of about three to five hours against about one to two days for all accounts, while in thin symbols the classes are often indistinguishable. Third, the pre-registered test fails its 80% criterion, on the

Table 2.18: Memory rates with and without class-tagged data

Symbol	Days	Top position $\hat{r}$ [95% CI]	All accounts $\hat{r}$ [95% CI]	Width ratio	Classes differ
BTCUSDT	48	0.9889 [0.9842, 0.9935]	0.9986 [0.9955, 1.0005]	121.7	yes
ETHUSDT	48	0.9793 [0.9616, 0.9907]	0.9972 [0.9934, 0.9997]	55.6	yes
CELRUSDT	48	0.9884 [0.9830, 0.9922]	0.9882 [0.9804, 0.9914]	81.6	no
CVXUSDT	33	0.9822 [0.9732, 0.9908]	0.9829 [0.9694, 0.9900]	45.9	no

Pre-registered test on fresh days, 30 USD-M perpetuals, 2025–2026

(i) width ratio > 10	30 of 30 (range 27.6–124.0)
(ii) price-only rate interval wider than 0.25	30 of 30 (range 0.65–0.99)
(iii) merged response $\sum_m k_m$ identified	5 of 30
Classes differ	11 of 30

Top panel: training days 2023–2024; width ratio is the mean price-only rate-interval width over the mean tagged width. The price-only intervals for both rates cover most of $[0, 0.99]$. Bottom panel: specification registered before the fresh days were downloaded (SHA-256 `65acda361861...`); a symbol passes if all three of the first rows hold, and the test required 80% of symbols to pass.

part of the prediction that concerns what prices *do* reveal: the merged response is significantly different from zero in only 5 of 30 symbols. Theorem 2.6.8 says the merged measure is identified; on 42 days of thin-market data it is not estimated precisely. Prices reveal even less than the identification theory guarantees, which strengthens rather than weakens the case for class-tagged disclosure.

To test directly whether the memory rates identified by class-tagged positions improve out-of-sample return predictions after large order-flow shocks, we pre-registered a dedicated out-of-sample test (`P1_disclosure_oos.json`, SHA-256 `f0182ce50978...`) before downloading the 22nd-of-month data for January 2025–September 2026 across 19 thin perpetuals (6,137 top-decile five-minute flow shocks on 2024 training days; 4,579 shocks on untouched 2025–2026 test days). On the training sample we estimate each symbol's two memory rates $\hat{r}_1, \hat{r}_2$ either from the class-tagged positions or from the price-only two-exponential fit, construct the filtered memory states $y_{k,t} = \hat{r}_k y_{k,t-1} + (1 - \hat{r}_k)\text{ret}_t$, and forecast the 30-minute post-shock cumulative return $R_{t+1\to t+6}$ out of sample relative to an AR(1) return-and-flow baseline. The primary hypothesis fails: on the untouched test shocks, the price-only memory states yield $\Delta R^2 = -0.199$ percentage points relative to the baseline, the class-tagged memory states yield $\Delta R^2 = -1.288$ points, and the tagged-minus-price-only difference is $\Delta R^2 = -1.089$ points (95% date-block bootstrap CI $[-2.605, +0.376]$, Clark–West $t = 0.96$, $p = 0.1684$). Likewise, adding the level and one-hour change of the top-trader and all-account positions to the hourly tail-risk controls of Chapter 3 leaves out-of-sample R^2 unchanged (-0.19 points, CI $[-0.99, 0.83]$).

After this test failed, we registered a follow-up (`P1_disclosure_state_oos.json`, SHA-256 `5136cc6b846d...`) on untouched 2nd-and-9th-of-month data for January 2025–September 2026 across the same 19 thin perpetuals (12,046 top-decile five-minute flow shocks on 24 training dates in 2024; 9,875 shocks on 42 test dates; test opened once). It asks the narrower question that

Theorem 2.3.1 and Proposition 2.8.1 answer directly: can prices and flow track the class states themselves? A two-exponential price-and-flow filter ($\alpha_{\text{fast}} = 0.50$, $\alpha_{\text{slow}} = 0.10$ on five-minute returns and signed flow, plus shock return, shock flow and symbol fixed effects) is compared with a filter that adds the current class states and their one-bar changes. Both registered primaries pass by a wide margin (Table 2.19): for the 30-minute-ahead divergence between top-trader and all-account positions, out-of-sample R^2 is -4.8% with prices and flow and 46.0% with class data; for the top-trader state it is 4.4% and 54.8%.

Table 2.19: Pre-registered out-of-sample cohort-state trajectory test on untouched 2nd-and-9th-of-month days (`P1_disclosure_state_oos`), 2025–2026

30-min post-shock target (19 thin)	R^2_{pf}	R^2_{tag}	ΔR^2 (pts) [95% CI]	CW t (p)	Res.
Prim. 1: Divergence d_{t+6} (9,875 shocks)	−4.766	45.962	+50.728 [+45.291, +55.918]	21.27 ($< 10^{-16}$)	pass
Prim. 2: Top state $n_{\text{top},t+6}$ (9,875 shocks)	4.441	54.845	+50.404 [+47.226, +53.724]	21.85 ($< 10^{-16}$)	pass
Sec.: Divergence d_{t+6}, all (210,144 bars)	−0.586	37.053	+37.639 [+36.232, +39.107]	72.75 ($< 10^{-16}$)	—
Sec.: Top state $n_{\text{top},t+6}$, all (210,144 bars)	0.186	46.779	+46.593 [+45.555, +47.643]	80.44 ($< 10^{-16}$)	—

Out-of-sample R^2 in percent. Trained on 24 dates in 2024 (12,046 shocks, 120,384 bars); evaluated once on 42 untouched test dates (2nd and 9th of each month, Jan 2025–Sep 2026); 2,000-draw date-block bootstrap 95% CIs; Holm $p < 10^{-16}$ across both primaries. Spec SHA-256 `5136cc6b846d...` (`R_P1_disclosure_state_oos.json`).

The gain is almost entirely persistence. In an exploratory decomposition on the same test shocks (`hardened_empirical_audits.py` → `R_hardened_empirical_audits.json`), an AR(1) on the current level alone accounts for 100.5% and 99.1% of the improvement, and when the target is the 30-minute *change* in positions the class data add −0.26 and +1.41 points. Because the class filter observes the series it forecasts, this is a weak test: it confirms that prices and flow cannot reconstruct where the classes are positioned, which Theorem 2.3.1 implies, but it does not show that class data reveal how their positions will move, and neither registered test finds that class data improve return forecasts.

2.8.4 A second disclosure regime: the CFTC's trader-class reports

Binance publishes its class ratios voluntarily, for one venue. The U.S. Commodity Futures Trading Commission requires large traders in regulated futures markets to report their positions and publishes them weekly by class: leveraged funds, asset managers and dealers in financial futures, and managed money, producer/merchants and swap dealers in commodities. This is class-tagged disclosure mandated by law. We registered a test (SHA-256 `6f1d6f35ae88...`) before downloading any data, in eight markets that span asset classes: CME Bitcoin and Ether, the E-mini S&P 500, 10-year Treasury notes, the euro, the yen, WTI crude oil and Henry Hub natural gas, using every weekly report through October 2026 (282 to 1,039 weeks per market). For each class, $n_{c,t} = (\text{long} - \text{short})/\text{open interest}$ follows $n_{c,t} = \alpha_c + r_c n_{c,t-1} + w_c \,\text{ret}_t$, the weekly version of the cohort recursion. The price-only alternative fits a two-exponential kernel to twelve weekly return lags. Intervals come from a 13-week moving-block bootstrap. The

Table 2.20: Trader-class memory from the CFTC Commitments of Traders, eight regulated futures markets

Market	Weeks	Fast $\hat{r}$ [95% CI]	Slow $\hat{r}$	$\hat{r}_{\text{fast}} - \hat{r}_{\text{slow}}$ [95% CI]	$\hat{w}_{\text{fast}}$	Ratio
Bitcoin (CME)	434	0.951 [0.928, 0.969]	0.986	−0.035 [−0.062, −0.010]	−	23.7
Ether (CME)	282	0.890 [0.826, 0.932]	0.960	−0.070 [−0.143, −0.011]	−	9.1
E-mini S&P 500	1,039	0.914 [0.884, 0.937]	0.954	−0.040 [−0.071, −0.012]	−	18.3
10-year T-note	1,039	0.992 [0.985, 0.998]	0.993	−0.000 [−0.006, +0.007]	(0)	71.4
Euro	1,035	0.961 [0.947, 0.972]	0.996	−0.034 [−0.051, −0.022]	+	37.3
Yen	1,035	0.955 [0.939, 0.968]	0.984	−0.029 [−0.047, −0.013]	+	34.0
WTI crude oil	1,039	0.961 [0.946, 0.974]	0.999	−0.037 [−0.053, −0.023]	+	35.2
Natural gas	1,039	0.981 [0.969, 0.990]	0.980	+0.001 [−0.014, +0.017]	+	44.0

Pre-registered hypotheses: H1 (primary) 8 of 8, pass; H2 4 of 8, fail; H3 6 of 8, pass.

Fast class: leveraged funds (financial futures) or managed money (commodities); slow class: asset managers or producer/merchants. $\hat{w}_{\text{fast}}$: sign of the response to returns, + or − when the 95% interval excludes zero, (0) otherwise. Ratio: mean width of the two price-only rate intervals over the width of the fast-class interval; the price-only intervals span 0.95–0.96 of the admissible range $[0, 0.99]$ in every market. Weekly returns from Binance spot (Bitcoin, Ether) and Yahoo Finance (others; the Treasury return is minus the change in the 10-year yield). Specification SHA-256 `6f1d6f35ae88...`; test opened once. Kind: real data (Part 11).

registered hypotheses were: (H1, primary) the fast class's memory rate is pinned to an interval narrower than 0.25 while both price-only intervals are wider, in at least 75% of markets; (H2) the fast class trend-follows, $w_c > 0$, in at least 75%; (H3) the fast class has the shorter memory in at least half.

The primary hypothesis holds in all eight markets (Table 2.20). The class reports pin each fast class's memory rate to an interval between 0.013 and 0.106 wide, while weekly prices leave both memory rates anywhere in the admissible range; the width ratios run from 9 to 71. The ratio partly reflects how little weekly returns reveal, which is the point of the theory rather than an artefact of it: with prices alone the memories are not located at all. The classes differ as the cohort model allows. Leveraged funds and managed money have the shorter memory in six of eight markets, with half-lives of 6 to 18 weeks against months to years for asset managers and producers; in Treasuries and natural gas the two memories are indistinguishable.

The trend-following hypothesis fails. The fast class buys after rises in the euro, the yen, crude oil and natural gas, but sells after rises in Bitcoin, Ether and the S&P 500 futures. The first half matches the finding of Moskowitz et al. (2012) that speculators' positions tend to go with past returns. We did not predict this. One candidate explanation is that leveraged-fund positions in financial futures are dominated by the short futures leg of cash–futures basis trades, which a single memory per class cannot represent. A post-hoc check is consistent with it but not decisive (Part 13, `R6`): weekly position changes of leveraged funds and asset managers are strongly negatively correlated in Bitcoin, Ether, the S&P 500 and Treasuries (−0.37 to −0.67) but not in the euro or yen (−0.04 and −0.05), and holding the asset managers' change fixed removes the negative response in Bitcoin and halves it in Ether and the S&P 500. Because class

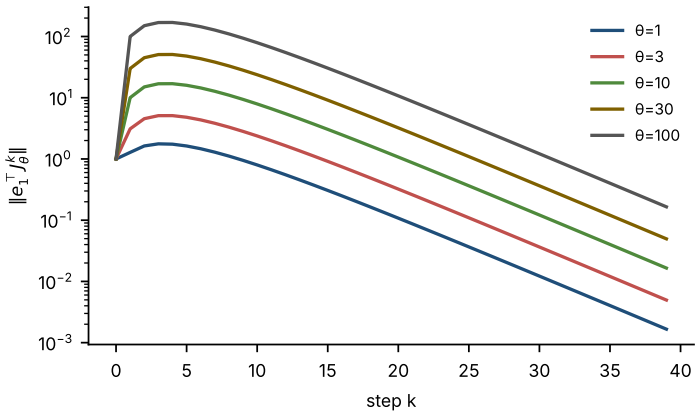


Figure 2.8: Price-row transient gain of the spectral-blindness family J_θ: eigenvalues $(0.8, 0.7)$, one-step price variance 1 and stationary price variance 5 are the same for every θ.

positions sum to zero across all reporting categories, some negative correlation is mechanical, so this is a signature, not a test. The failure sharpens rather than weakens the case for class tagging: class labels are not strategies, and a supervisor who wants to know how a class responds to prices needs its positions, which the price process does not reveal.

Together with Table 2.18, the evidence now covers two disclosure regimes, one voluntary and one mandated by law, in crypto and in regulated equity-index, rate, currency and commodity futures. In both, class-tagged positions locate memories that prices leave unrestricted.

2.9 A spectral blind spot

Even without identification, could a supervisor monitor risk through eigenvalues or volatility?

Theorem 2.9.1 (Spectral blindness)**.** *For any eigenvalues $0 < \lambda_2 < \lambda_1 < 1$, one-step price variance $s_1 > 0$ and stationary price variance V large enough, there is a family of two-state markets $J_\theta = \left(\begin{smallmatrix} \lambda_1 & \theta \\ 0 & \lambda_2 \end{smallmatrix}\right)$ with shock covariance $\mathrm{diag}(s_1, s_2(\theta))$ such that all members share the eigenvalues, the one-step price variance and the stationary price variance, while the worst-case transient gain $\sup_k \|e_1^\top J_\theta^k\|$ grows without bound in θ.*

At $\lambda = (0.8, 0.7)$, $s_1 = 1$ and $V = 5$, the gain is 1.77, 5.10, 16.95, 50.85 and 169.50 for $\theta = 1, 3, 10, 30, 100$, with stationary variance 5.000 in every case; the cumulative gain rises from 18.3 to 1,667.7. No monitor that depends only on eigenvalues and second moments of the price can be uniformly safe (Figure 2.8).

Two clarifications prevent over-reading. First, the cohort economy of Section 2.2 is itself non-normal, yet its price output does not amplify (Lemma 2.2.3). Large weighted gains can be manufactured by output weights alone: with $C_w = (1, 160)$, a price-and-memory functional peaks at 160.003 at $t = 0$, which is the static weight, not dynamic growth (Appendix 2.15.5). Second,

in a network of M identical cohort economies coupled by symmetric nonnegative cross-impact, the coupled operator stays symmetric in the coordinates of Lemma 2.2.3; with cross-impact 0.05, the spectral radius rises from 0.822 ($M = 1$) to 0.973 ($M = 10$) while the Kreiss constant stays at one, as it must for a stable symmetric operator (`recompute_oct20.py`). Non-normality is necessary for transient amplification of the price, not sufficient; where amplification comes from is the subject of Chapter 3.

2.10 Empirical feedback and its measurement

The theory says what prices can reveal. The empirical work has two roles, and neither is a test of the identification bounds themselves, which concern controlled experiments no outside researcher can run. First, Section 2.8.3 tests the theory's policy prediction directly: that class-tagged data pin down memories that prices leave unrestricted. Second, this section asks what the reduced-form feedback indices a supervisor would compute from public data actually measure, and how much of them is feedback rather than filtering, timing or trading activity.

2.10.1 Estimand and filtering mechanics

In theory $x_t = p_t - p_t^*$ is the deviation from an unobserved fundamental. Empirical work replaces p_t^* by a trailing filter. Let p_b be a log price proxy on uniform bars of length Δ and $m_b = (1-\alpha)m_{b-1} + \alpha p_b$ an exponential moving average with $\alpha = 2/(W/\Delta + 1)$. The empirical state is

$$x_b^{\mathrm{EMA}} \equiv p_b - m_b = (1-\alpha)(p_b - m_{b-1}) = \frac{1-\alpha}{1-(1-\alpha)L}\Delta p_b. \tag{2.31}$$

Because $x_{b+1}^{\mathrm{EMA}} = (1-\alpha)(p_{b+1} - m_b)$, a martingale price gives

$$\mathbb{E}[x_{b+1}^{\mathrm{EMA}} \mid \mathcal{F}_b] = (1-\alpha)x_b^{\mathrm{EMA}} : \tag{2.32}$$

the filtered state is an AR(1) with mechanical persistence $a_0 = 1 - \alpha$, which equals $29/31 \approx 0.9355$ for $W = 300$ seconds and $\Delta = 10$ seconds.

Let f_t be signed flow. The empirical VAR(1)

$$\begin{pmatrix} x_{t+1}^{\mathrm{EMA}} \\ f_{t+1} \end{pmatrix} = \begin{pmatrix} J_{11} & J_{12} \\ J_{21} & J_{22} \end{pmatrix} \begin{pmatrix} x_t^{\mathrm{EMA}} \\ f_t \end{pmatrix} + e_{t+1} \tag{2.33}$$

has reduced-form coefficients, not authenticated impact or learner parameters. When denominators are nonzero we use the scale-invariant indices

$$\hat{Q} = \frac{\hat{J}_{12}\hat{J}_{21}}{1-\hat{J}_{22}}, \qquad \hat{S} = \hat{J}_{11} + \hat{Q}, \qquad \hat{L} = \frac{\hat{Q}}{1-\hat{J}_{11}}. \tag{2.34}$$

Table 2.21: Samples

Sample	Coverage	Size	Source
Binance spot panel, 8 pairs	96 dates, 2021–2024	66.3M one-second bars	Binance archive
Binance spot fee-event windows, 8 pairs	±20 days around 2022-07-08 and 2023-03-22	640 pair-days	Binance archive
Binance USD-M perpetuals, 30 symbols	90 dates, 2023–2026	2,565 symbol-days	Binance archive
Binance BTCUSDT one-second panel	72 dates, 2021–2023	6.2M one-second bars	Binance archive
Binance BTCUSDT noise-calibration days	January 10–20, 2023	11 days	Binance archive

Binance files are verified against the published SHA-256 checksums (every file used). Sampled dates are the 10th and 24th of each month.

Table 2.22: Multi-year Binance spot panel: reduced-form indices under three timing constructions

Construction	$\hat{Q}$	Date interval	$\hat{S}$	Date interval	$\hat{Q} > 0$
Forward-paired flow (look-ahead)	0.0081424	[0.0072965, 0.0089882]	0.92074	[0.91665, 0.92482]	87.4%
Observed-flow EMA	0.0003733	[0.0002351, 0.0005116]	0.92072	[0.91640, 0.92504]	62.4%
Observed, 30 flow lags	0.0002580	[0.0001415, 0.0003745]	0.91738	[0.91167, 0.92308]	59.1%
Observed raw return	−0.0001555	[−0.0003732, 0.0000622]	−0.04051	[−0.05078, −0.03025]	49.2%
Placebo random walk	0.0000016	[−0.0000020, 0.0000052]	0.93510	[0.93485, 0.93534]	51.0%

768 study units (96 dates × 8 pairs), 6,634,752 ten-second transitions; equal-weight asset means within date, then date means with Student intervals. Mechanical persistence $29/31 = 0.93548$.

Equation (2.32) implies that $\hat{J}_{11}$ must be compared with $1 - \alpha$, not with zero: persistence that a naive reading attributes to feedback can be pure filtering.

2.10.2 Data

2.10.3 Clock alignment dominates reduced-form feedback

We estimate (2.33) on ten-second bars built from one-second Binance klines across a 96-date, eight-pair spot panel (the 10th and 24th of each month, 2021–2024) under three constructions (Table 2.22), and run the identical pipeline on a placebo random walk with each pair-day's volatility. Pairing the state with *next-bar* flow (a look-ahead) gives $\hat{Q} = 0.0081$; aligning flow with the bar in which it is observed lowers this by more than an order of magnitude, to 0.00037; and replacing the EMA state by raw returns gives an interval that includes zero. Mean $\hat{J}_{11}$ under observed flow is 0.9203, below the mechanical 0.9355, and the placebo reproduces the mechanical value (0.9351) with $\hat{Q}$ indistinguishable from zero.

Table 2.23: Asset means under forward-paired and observed flow

Pair	Forward-paired $\hat{Q}$	Observed $\hat{Q}$	$\hat{Q}_{30}$	Observed $\hat{S}$
BTC/USDT	0.0141870	0.0006073	0.0004891	0.93464
ETH/USDT	0.0130885	0.0005675	0.0002549	0.93511
SOL/USDT	0.0087763	0.0002129	0.0001100	0.91966
ADA/USDT	0.0067465	0.0003928	0.0003546	0.91452
DOGE/USDT	0.0058956	0.0002464	0.0001432	0.92496
BNB/USDT	0.0057420	0.0004479	0.0002762	0.89533
XRP/USDT	0.0057034	0.0002249	0.0001508	0.92362
LTC/USDT	0.0049997	0.0002870	0.0002853	0.91792

96 dates per pair. $\hat{Q}_{30}$ controls 30 ten-second bars of lagged flow. Placebo $|\hat{Q}| \leq 1.1 \times 10^{-5}$ for every pair.

Table 2.24: Feedback beyond filtering mechanics (Binance BTCUSDT, 72 days, 2021–2023)

	Real prices	Placebo (independent random walk)
Mean loop gain $\hat{Q}$	0.0045	0.0000009
95% CI (day bootstrap)	[0.0034, 0.0056]	[−0.0000006, 0.0000023]
Mean price persistence $\hat{J}_{11}$	0.9364	0.9357
Mechanical value $1 - 2/31$		0.9355

Placebo 95% band across days: $[-0.0000095, 0.0000156]$. This construction pairs the state with next-second flow; it is not comparable in level with the observed-flow ten-second index of Table 2.22.

2.10.4 Feedback beyond filtering mechanics

We re-estimate feedback on a separate sample and add a placebo. We use Binance BTCUSDT one-second klines on 72 sampled days in 2021–2023. The state x_t is the log price deviation from its EMA (span 30 one-second bins), f_t is next-second signed taker flow, and a VAR(1) gives $\hat{Q}$. Because the filter alone produces persistence, we run the identical pipeline on a placebo: a random walk with the same one-second volatility, independent of the real flow.

Table 2.24 gives two findings. Feedback, in the sense of a loop gain the filter cannot produce, is detectable: the real $\hat{Q}$ lies far outside the placebo band. It is also small, and the price persistence that a naive reading would attribute to feedback is almost entirely mechanical (0.9364 against 0.9355). Because this construction pairs the state with next-second flow, part of $\hat{Q}$ reflects the timing effect of Table 2.22; the placebo shows only that the filter does not produce it.

2.10.5 Order splitting and the estimation of feedback

Persistent order flow makes lagged-flow regressions inconsistent. Suppose flow follows $q_t = \gamma y_t + u_t$ with memory $y_t = (1-\eta)y_{t-1} + \eta \Delta p_{t-1}$, order splitting $u_t = \rho u_{t-1} + \nu_t$, and $\Delta p_t = \kappa q_t + \lambda z_t + e_t$ with an observed exogenous price shock z_t. Lagged returns are correlated with u_{t-1}, so OLS of flow on returns is biased even when $\gamma = 0$. Quasi-differencing ($q_t - \rho q_{t-1} = \gamma(y_t - \rho y_{t-1}) + \nu_t$)

Table 2.25: Feedback estimators under order splitting (simulation, 500 replications)

Estimator	Coverage at $\gamma = 0.2$	Rejection at $\gamma = 0$	Bias	RMSE
OLS	0.000	1.000	0.892	0.893
Cochrane–Orcutt, true ρ	0.944	0.054	0.001	0.039
Cochrane–Orcutt, estimated ρ	0.916	0.080	0.002	0.043
2SLS	0.938	0.044	−0.002	0.116
LIML	0.938	0.044	−0.003	0.117

Newey–West standard errors (10 lags), $T = 3{,}000$. Median first-stage $F = 224$; a placebo instrument (future z) has $F = 0.62$. Estimators outside 93–97% coverage give invalid inference.

or instruments that move y through prices but are independent of u ($z_{t-1}, z_{t-2}, z_{t-3}$) restore consistency. We compare estimators over 500 replications with $\eta = 0.3$ and $\rho = 0.8$ (Table 2.25).

OLS is badly biased. Cochrane–Orcutt with the true ρ, 2SLS and LIML have correct size and coverage; Cochrane–Orcutt with an estimated ρ under-covers (Cochrane and Orcutt, 1949; Staiger and Stock, 1997). These are properties of estimators in a simulated market. On real data, a 2024 sub-panel of 192 spot units shows why the correction matters little at our bar length: with one VAR lag, the flow-equation residuals have essentially no first-order serial correlation (median $\hat{\rho} = -0.0002$), and Cochrane–Orcutt quasi-differencing moves the mean loop gain from 3.15×10^{-4} to 3.21×10^{-4} (Part 13, `R7`, post hoc). Order splitting shows up instead at longer lags, which the 30-lag control of Table 2.23 addresses. A real-data instrumental-variables estimate needs an instrument whose exclusion can be defended, which we do not claim to have; funding-settlement times, for example, may affect order flow directly.

Controlling for 30 ten-second bars of lagged flow lowers $\hat{Q}$ further (Table 2.23, column $\hat{Q}_{30}$), consistent with persistent order splitting.

2.10.6 Feedback across 30 perpetual markets

Chapter 3 estimates the same loop gain on one-minute bars of 30 Binance USD-M perpetuals spread across the liquidity ranking (2023–2024 training days). Measured feedback rises with depth: its Spearman correlation with log depth within 1% of the mid is +0.70 [0.41, 0.87], from about 0.012 in BTCUSDT and ETHUSDT to 0.0016 on average in the thinnest third, and the placebo pipeline gives $\hat{Q} \approx 0$ for every symbol. Feedback in this reduced-form sense is a feature of deep, algorithmically traded books.

2.10.7 Two fee events

Binance set spot trading fees on BTC pairs to zero on July 8, 2022 and reinstated them on March 22, 2023. Each event changes the cost of trading BTCUSDT while leaving the seven other pairs unchanged, and the second reverses the first. For every pair and every day within 20 days of each event we compute daily quote volume and the observed-flow $\hat{Q}$, and estimate $Y_{it} = \alpha_i + \gamma_t + \beta\,\mathrm{ZeroFee}_{it} + e_{it}$ with pair and date fixed effects and date-clustered errors,

Table 2.26: Zero fees on BTCUSDT: activity and measured feedback

	Zero fees introduced (2022-07-08)	Zero fees removed (2023-03-22)	Pooled
Log quote volume, $\hat{\beta}$	0.945 (0.074)	1.599 (0.080)	1.272 (0.084)
Observed-flow $\hat{Q}$, $\hat{\beta}$	0.00058 (0.00032)	−0.00091 (0.00060)	−0.00016 (0.00036)
p-value	0.078	0.140	0.650
Pair-days	320	320	640

$\hat{\beta}$ is the effect of zero fees on BTCUSDT relative to seven control pairs; date-clustered standard errors in parentheses. For the second event ZeroFee equals one before March 22, 2023.

separately for each event and pooled (Table 2.26).

The fee change moved activity enormously: zero fees multiplied BTCUSDT volume relative to the controls by $e^{0.945} \approx 2.6$ when they began, and their removal divided it by $e^{1.599} \approx 4.9$. Measured feedback did not follow. The two point estimates have opposite signs, neither is significant, and the pooled effect is -0.00016 with an interval of $[-0.00087, 0.00055]$, against a sample mean of 0.00037. A strong first stage with no second-stage response says that $\hat{Q}$ is not a by-product of trading activity, and the reversal design removes the main weakness of a single treated event. With one treated asset the interval remains wide, so this is evidence that $\hat{Q}$ does not scale with activity, not a proof of invariance.

2.11 Implications for supervision and market design

What a supervisor can learn from public data. Prices and anonymous flow reveal the merged measure M, the closed-loop impulse response and the reduced-form operator, and nothing finer. A supervisor who wants to know *which* algorithms drive a feedback loop cannot get there from public data at reasonable cost when their memories are close. This is not a failure of econometric technique: Theorem 2.6.2 holds for every estimator, biased or not.

What probing costs. At realistic observation noise, an exchange or a large desk could separate two well-separated memory rates for about 10^5 USDT of capital. Probing is therefore feasible for coarse questions and infeasible for fine ones. Because capital grows as δ^{-4} and experiments as $(\Delta\eta)^{-(4K-2)}$, halving the separation multiplies the cost by 16 for a known pair and by 2^{4K-2} for K unknown cohorts.

What disclosure buys. Disclosure regimes differ sharply in value (Table 2.16). Timestamps without update values are worthless for this purpose. Noisy periodic statements are nearly worthless. Audited equity and, above all, flow tagged by participant class remove the barrier entirely (Proposition 2.8.1). This gives a quantitative argument for audit-trail regimes that record the participant class of each order, and a quantitative argument against relying on periodic

statements for the same purpose. Calibrated to Binance's published class ratios, separating the memories of the top-trader group and of all accounts in BTCUSDT would take prices alone about 333 times as long (Theorem 2.8.2). The class reports that the CFTC already requires have the same property in regulated markets (Table 2.20), so the argument concerns the granularity and frequency of existing disclosure, not a new kind of reporting.

When audit trails already exist. The US Consolidated Audit Trail (SEC Rule 613) gives the SEC and the self-regulatory organizations a record of every order in NMS stocks and listed options from origination to execution or cancellation, linked to an anonymized customer identifier and to the broker-dealer that handled it; since 2025 it no longer collects the names, addresses and years of birth of natural persons (SEC, 2012, 2016, 2024, 2025). Our results bear on such regimes in three ways. First, they price what the record is worth for the question of who drives a feedback loop: participant-level records remove a barrier that otherwise costs round-trip trading loss of order δ^{-4} and samples of order $(\Delta\eta)^{-(4K-4)}$ (Proposition 2.8.1, Theorem 2.8.2); set against the CAT LLC 2025 operating budget ($248.8 million in November 2024, revised to $228.3 million in May 2025) and the SEC's 2016 annual industry estimate ($1.7 billion), the annualised weekly-renewal probing loss that disclosure makes unnecessary is small for two well-separated classes and exceeds the entire CAT budget once $K \geq 3$ at $\Delta\eta \leq 0.10$ or $K = 4$ at $\Delta\eta = 0.20$ (Section 2.8.2, Table 2.17). For this purpose a stable account identifier is what matters and names are not needed, which is a quantitative reason to keep the account linkage while removing personal data. Second, an audit trail records accounts, not decision rules. One account can run several algorithms and one algorithm can trade through several accounts, so an observer with account-level records who wants rule-level memories faces our problem within each account's flow, with fewer cohorts but the same exponent; MiFID II's algorithm identifier (ESMA, 2016) is the closer analogue of the tagged flow in Proposition 2.8.1. Third, the audit trail does not reach the markets we study. Crypto perpetual futures are outside it, futures regulated by the CFTC have separate reporting, and its data are not available to exchanges, other participants or researchers. For these observers the price-only and coarse-disclosure contracts of this chapter are the relevant ones.

What to monitor instead. Because transient loss can be arbitrarily large at fixed eigenvalues and volatility (Theorem 2.9.1), supervision that wants to control risk rather than identify participants should monitor quantities that bound transient loss directly: the summed absolute impulse response and operator confidence sets, both of which are identified from public data. These are cheap where identification is expensive (Remark 2.6.3); Chapter 3 shows that the summed response is certified at the parametric rate from controlled experiments, and only under a declared bound on memory length from a finite window (Chapter 3).

How to measure feedback. Empirical feedback indices should (i) compare persistence with the mechanical value of the filter, (ii) align flow with the bar in which it is observed, (iii) control for order-splitting persistence or use instruments with a defended exclusion, and (iv) report placebo pipelines. Each of these changes the sign or the order of magnitude of published-style estimates in our data.

2.12 Conclusion and an open problem

Prices reveal a merged summary of algorithmic feedback. Recovering the algorithms behind it costs executed quantity of order δ^{-2} and trading capital of order δ^{-4} for a known pair, with a certified constant of about 61 par units at the calibration, and a number of experiments of exact order $(\Delta\eta)^{-(4K-2)}$ for K unknown cohorts, passively as well as actively; in simulations of stylized nonlinear rules and of online learners trained in the closed loop the same costs hold when the competing populations' feedback weights match, populations whose learned weights differ are told apart by those weights instead, and hard thresholds hide cohorts from small probes altogether. Without the exponential form, accuracy improves only as $1/\log N$, with matching lower and upper bounds. Class-tagged disclosure removes the barrier, and its value can be stated in samples and in dollars: a sample factor of order $(\Delta\eta)^{-(4K-4)}$, about 333 for the trader classes Binance already publishes in BTCUSDT, and an annualised weekly-renewal probing loss that matches the $228.3 million CAT LLC operating budget across 15.5 (Roll) or 0.64 (one-second) markets for an unknown two-cohort Gauss–Radau pair at $\Delta\eta = 0.10$ and exceeds it in a single market once $K \geq 3$ at $\Delta\eta = 0.10$; the CFTC's class reports locate memories that prices leave unrestricted in all eight regulated markets we examined, while a pre-registered out-of-sample test finds that class-tagged memory states do not improve 30-minute post-shock return forecasts ($\Delta R^2 = -1.09$ points). At realistic noise the capital barrier for a moderate separation is about 10^5 USDT. Measured feedback in crypto markets is real but small, rises with depth, does not move when fees change trading volume severalfold, and is easily confused with filtering and timing artefacts. Finally, eigenvalues and volatility cannot certify safety, because transient loss can be arbitrarily large at fixed spectrum and variance.

This leaves an open question for supervision: *what can be measured and controlled without identifying the algorithms?* The impulse response $g_m = c^\top J^m b$, the operator J and finite-sample confidence sets for it are identified from public data even when cohorts are not. Chapter 3 uses exactly this interface to build an identification-free pre-trade control.

Data and code availability. All data are public: Binance archives (`https://data.binance.vision`), verified against published checksums. The pre-registered specification of the disclosure test, its SHA-256 hash and its test-set unlock counter are in the replication package. Every number in this chapter is produced by a notebook cell or script that writes a result file (Appendix 2.16).

Use of AI tools. The authors used Claude (Anthropic), an AI assistant, to write and debug analysis code, run checks of derivations and numerical results, and draft and edit text. Every number in the paper is produced by code the authors executed and archived in the replication package. The authors reviewed all content and take full responsibility for it.

Disclosures. The authors declare no relevant financial interests.

References

Alfonsi, A., A. Schied, and F. Klöck (2016). Multivariate transient price impact and matrix-valued positive definite functions. *Mathematics of Operations Research* 41, 914–934.

Almgren, R., and N. Chriss (2001). Optimal execution of portfolio transactions. *Journal of Risk* 3, 5–39.

Batenkov, D., G. Goldman, and Y. Yomdin (2021). Super-resolution of near-colliding point sources. *Information and Inference* 10, 515–572.

Bernstein, S. (1914). Sur la meilleure approximation de $|x|$ par des polynômes de degrés donnés. *Acta Mathematica* 37, 1–57.

Bessembinder, H., W. Maxwell, and K. Venkataraman (2006). Market transparency, liquidity externalities, and institutional trading costs in corporate bonds. *Journal of Financial Economics* 82, 251–288.

Boehmer, E., G. Saar, and L. Yu (2005). Lifting the veil: An analysis of pre-trade transparency at the NYSE. *Journal of Finance* 60, 783–815.

Bouchaud, J.-P., Y. Gefen, M. Potters, and M. Wyart (2004). Fluctuations and response in financial markets: The subtle nature of "random" price changes. *Quantitative Finance* 4, 176–190.

Bouchaud, J.-P., J. Bonart, J. Donier, and M. Gould (2018). *Trades, Quotes and Prices: Financial Markets Under the Microscope*. Cambridge University Press.

Brunnermeier, M. K., and L. H. Pedersen (2009). Market liquidity and funding liquidity. *Review of Financial Studies* 22, 2201–2238.

Cochrane, D., and G. H. Orcutt (1949). Application of least squares regression to relationships containing auto-correlated error terms. *Journal of the American Statistical Association* 44, 32–61.

European Securities and Markets Authority (2016). Guidelines on transaction reporting, order record keeping and clock synchronisation under MiFID II. ESMA/2016/1452.

Fan, J. (1991). On the optimal rates of convergence for nonparametric deconvolution problems. *Annals of Statistics* 19, 1257–1272.

Gatheral, J. (2010). No-dynamic-arbitrage and market impact. *Quantitative Finance* 10, 749–759.

Glosten, L. R., and P. R. Milgrom (1985). Bid, ask and transaction prices in a specialist market with heterogeneously informed traders. *Journal of Financial Economics* 14, 71–100.

Hasbrouck, J. (1991). Measuring the information content of stock trades. *Journal of Finance* 46, 179–207.

Heinrich, P., and J. Kahn (2018). Strong identifiability and optimal minimax rates for finite mixture estimation. *Annals of Statistics* 46, 2844–2870.

Hua, Y., and T. K. Sarkar (1990). Matrix pencil method for estimating parameters of exponentially

damped/undamped sinusoids in noise. *IEEE Transactions on Acoustics, Speech, and Signal Processing* 38, 814–824.

Kirilenko, A., A. S. Kyle, M. Samadi, and T. Tuzun (2017). The Flash Crash: High-frequency trading in an electronic market. *Journal of Finance* 72, 967–998.

Kong, W., and G. Valiant (2017). Spectrum estimation from samples. *Annals of Statistics* 45, 2218–2247.

Kreiss, H.-O. (1962). Über die Stabilitätsdefinition für Differenzengleichungen die partielle Differentialgleichungen approximieren. *BIT Numerical Mathematics* 2, 153–181.

Kyle, A. S. (1985). Continuous auctions and insider trading. *Econometrica* 53, 1315–1335.

Le Cam, L. (1986). *Asymptotic Methods in Statistical Decision Theory*. Springer.

Liu, P., and H. Ammari (2022). Super-resolution of positive near-colliding point sources. arXiv preprint arXiv:2212.00536.

Makarov, I., and A. Schoar (2020). Trading and arbitrage in cryptocurrency markets. *Journal of Financial Economics* 135, 293–319.

Manole, T., and N. Ho (2022). Refined convergence rates for maximum likelihood estimation under finite mixture models. *Proceedings of the 39th International Conference on Machine Learning*, PMLR 162.

Niroula, P., S. Alkhushayni, Y. Chen, M. Bhavsar, and P. Niroula (2026). From human markets to machine markets: Transient amplification, an exact depth condition, and a maximal pre-trade guard. Working paper, Minnesota State University, Mankato.

Moskowitz, T. J., Y. H. Ooi, and L. H. Pedersen (2012). Time series momentum. *Journal of Financial Economics* 104, 228–250.

Obizhaeva, A. A., and J. Wang (2013). Optimal trading strategy and supply/demand dynamics. *Journal of Financial Markets* 16, 1–32.

Pagano, M., and A. Röell (1996). Transparency and liquidity: A comparison of auction and dealer markets with informed trading. *Journal of Finance* 51, 579–611.

Rivlin, T. J. (1981). *An Introduction to the Approximation of Functions*. Dover.

Roll, R. (1984). A simple implicit measure of the effective bid-ask spread in an efficient market. *Journal of Finance* 39, 1127–1139.

Roy, R., and T. Kailath (1989). ESPRIT—estimation of signal parameters via rotational invariance techniques. *IEEE Transactions on Acoustics, Speech, and Signal Processing* 37, 984–995.

U.S. Securities and Exchange Commission (2010). Risk management controls for brokers or dealers with market access. Release No. 34-63241.

U.S. Securities and Exchange Commission (2012). Consolidated Audit Trail; Final Rule (Release No. 34-67457, July 18, 2012, 77 FR 45722 at 45790–45795, pp. 220–238; proposing release No. 34-62174, May 26, 2010, 75 FR 32556 at 32598, 32600, pp. 147, 153).

U.S. Securities and Exchange Commission (2016). Joint Industry Plan; Order Approving the National Market System Plan Governing the Consolidated Audit Trail. Release No. 34-79318, November 15, 2016, 81 FR 84696 at 84801, p. 363.

U.S. Securities and Exchange Commission (2024). Notice of Filing of Proposed Rule Change Relating to the 2025 CAT Budget. Release No. 34-101901, December 12, 2024, p. 4 (revised May 2025).

U.S. Securities and Exchange Commission (2025). Order granting exemptive relief from certain requirements of the National Market System Plan governing the Consolidated Audit Trail. Release No. 34-

102386, February 10, 2025.

Shleifer, A., and R. W. Vishny (1992). Liquidation values and debt capacity: A market equilibrium approach. *Journal of Finance* 47, 1343–1366.

Sornette, D., S. Lera, J. Lin, and K. Wu (2023). Non-normal interactions create socio-economic bubbles. *Communications Physics* 6, 261.

Staiger, D., and J. H. Stock (1997). Instrumental variables regression with weak instruments. *Econometrica* 65, 557–586.

Trefethen, L. N., and M. Embree (2005). *Spectra and Pseudospectra: The Behavior of Nonnormal Matrices and Operators*. Princeton University Press.

Tsybakov, A. B. (2009). *Introduction to Nonparametric Estimation*. Springer.

Varga, R. S., and A. J. Carpenter (1985). On the Bernstein conjecture in approximation theory. *Constructive Approximation* 1, 333–348.

Whittle, P. (1953). Estimation and information in stationary time series. *Arkiv för Matematik* 2, 423–434.

Wu, Y., and P. Yang (2020). Optimal estimation of Gaussian mixtures via denoised method of moments. *Annals of Statistics* 48, 1981–2007.

2.13 Appendix: Proofs

Lemma 2.2.1. (i) For $v_{t+1} = v_t + \zeta_{t+1}$ and $r_t = v_t + \nu_t$, the steady-state prior variance P solves $P = P\sigma_\nu^2/(P+\sigma_\nu^2) + \sigma_\zeta^2$, i.e. $P^2 - \sigma_\zeta^2 P - \sigma_\zeta^2\sigma_\nu^2 = 0$, whose positive root is $P = \big(\sigma_\zeta^2 + \sqrt{\sigma_\zeta^4 + 4\sigma_\zeta^2\sigma_\nu^2}\big)/2$. The gain is $\eta = P/(P+\sigma_\nu^2) \in (0,1)$, which simplifies to (2.7), and the update $\hat v_{t+1} = \hat v_t + \eta(r_t - \hat v_t)$ has the stated form. (ii) $\nabla_\theta \tfrac12 H(x_t-\theta)^2 = -H(x_t-\theta)$, so $\theta_{t+1} = (1-\gamma)\theta_t + \alpha H(x_t - \theta_t) = (1-\eta)\theta_t + \eta q x_t$ with $\eta = \gamma + \alpha H$ and $q = \alpha H/\eta$.

Lemma 2.2.3. Omit zero-weight cohorts. For $a > 0$ put $b_i = \kappa q_i$ and

$$T = \operatorname{diag}\big(\sqrt{a}, \sqrt{b_1/\eta_1}, \ldots\big), \qquad u = \big(\sqrt{a}, \sqrt{b_1\eta_1}, \ldots\big)^\top.$$

The price-first transition becomes the symmetric matrix $J' = TJT^{-1} = \operatorname{diag}(0, r_1, \ldots) + uu^\top$, the input transforms to κu and $J'e_x = \sqrt{a}\,u$, so $h_j = (\kappa/a)e_x^\top (J')^{j+1} e_x$. The spectral decomposition of J' gives nonnegative weights summing to $h_0 = \kappa$, and Schur stability puts the eigenvalues in $[0,1)$. The case $a = 0$ follows by continuity. A nonnegative mixture of decaying geometric sequences is completely monotone. Since J is nonnegative with row sums at most one, every price row of J^t has ℓ_1, hence ℓ_2, norm at most one, attained at $t = 0$.

Theorem 2.3.1. With zero initial states, $y_{i,t} = \sum_{s\le t} \eta_i r_i^{t-s} x_s$, so $\sum_i q_i y_{i,t} = \sum_{s\le t} k_{t-s} x_s$ and every response is determined by (a, κ, k), hence by M. Conversely, equal responses for all probes imply equal H, hence equal G, and partial fractions of G identify the distinct poles with nonzero residues w_i, i.e. M. After cancellation, $H = \kappa z D(z)/[(z-a)D(z) - \kappa N(z)]$ with

$D = \prod(z - r_i)$ and $N(z) = \sum_i w_i z \prod_{j \neq i}(z - r_j)$, as in (2.4); at a retained r_i the denominator equals $-\kappa N(r_i) = -\kappa w_i r_i \prod_{j \neq i}(r_i - r_j) \neq 0$ (memory rates are positive because $\eta_i < 1$), so no root of D cancels. The remaining numerator factor z cancels if and only if the denominator vanishes at 0; since $N(0) = 0$ the denominator there equals $-aD(0)$, which is nonzero for $a > 0$ and zero for $a = 0$. Hence the McMillan degree, which is the dimension of a minimal realization, is $1 + |\mathrm{supp}(M)|$ for $a > 0$ and $|\mathrm{supp}(M)|$ for $a = 0$ (numerically, the rank of the Hankel matrix of $(h_j)_{j \geq 1}$ is $K + 1$ and K respectively for $K = 1, 2, 3$; `hardening/p1_checks.py`). For (ii), direct algebra gives

$$G_\delta(z) - G_0(z) = -\frac{qz(z-1)\delta^2}{(z - 1 + \bar{\eta})\big((z - 1 + \bar{\eta})^2 - \delta^2\big)}.$$

On $|z| = 1$ each factor has modulus at least ϵ. Uniform stability gives $M_H = \sup |H_\theta| < \infty$, and the identity $H_\delta - H_0 = H_\delta H_0 (G_\delta - G_0)$ gives an ℓ_2 gain at most $K\delta^2$. The mean difference of a power-constrained probe has squared norm at most $K^2\delta^4 PT$; the equal-noise KL is at most $K^2\delta^4 PT/(2\sigma_\nu^2)$, and Pinsker's inequality with total variation at least $1 - 2\alpha$ gives (2.9).

Theorem 2.4.1. In either positive economy $J \geq 0$ and $\|J\|_\infty \leq 1 - \epsilon(1 - S) < 1$, so the response is nonnegative with $\max h = \kappa$ and $\sum h = \kappa/(1 - S)$. From the display above, $|G_\delta - G_0| \leq 2q\delta^2/\epsilon^3$ on the circle; combined with the resolvent identity and positivity this gives $\|H_\delta - H_0\|_\infty \leq K_* \delta^2$, and Parseval bounds $\|h^\delta - h^0\|_2$ by the same constant. Young's inequality gives $\|\mu_\delta - \mu_0\|_2 \leq K_* \delta^2 \|v\|_1$, and Gaussian KL with Pinsker gives (2.10). For (ii), the mean difference is a lower-triangular convolution matrix times v; each column is a shifted copy of a prefix of d_T, so $\|\mu_\delta - \mu_0\|_2 \leq V \|d_T\|_2$ with equality for a pulse at $t = 0$. For two known equal-covariance Gaussian laws the minimax error is $\Phi(-\|\mu_\delta - \mu_0\|_2/(2\sigma))$, attained by the midpoint likelihood-ratio test. For (iii), the merged measures have equal mass and their first moments differ by $-q\delta^2$, which enters h_2 with factor κ^2 while h_0, h_1 coincide; a pulse A separates the third-observation means by $A\kappa^2 q\delta^2$. Part (iv) follows from the ℓ_2 gain bound, and (v) from superposition and $\max_j h_j = \kappa$, which give $\sup_t |\lambda x_t| \leq |\lambda| \kappa V$, attained by a single pulse.

Lemma 2.4.2. After the uncontrolled update, the pre-probe displacement at step j is $\sum_{i<j} h_{j-i} v_i$; average linear-book execution adds $\kappa v_j/2$. Summing signed execution costs gives $\frac{1}{2} v^\top T_h v$. Each kernel $\lambda^{|i-j|}$ is a positive-definite AR(1) covariance for $0 \leq \lambda < 1$, and a nonnegative mixture with positive total weight is positive definite. For a round trip the fundamental cash term cancels.

Theorem 2.4.3. Let $U_t = \sum_{j<t} v_j$. Positive responses give $0 \leq x_t \leq \kappa U_t$ and $I_t = -U_t$. Summing cash and marking inventory, $B - E_t = U_t x_t - \sum_{j<t} v_j x^0_{j+1} - \frac{\kappa}{2} \sum_{j<t} v_j^2 + cU_t$. Its quadratic part $v^\top A_t v$ has diagonal $h_{t-1-i} - \kappa/2$ and off-diagonal $\frac{1}{2}(h_{t-1-i} + h_{t-1-j} - h_{j-i})$ for $i < j$; monotonicity of h bounds every entry by $\kappa/2$. Hence $B - E_t \leq \kappa U_t^2/2 + cU_t \leq \kappa V^2/2 + cV$,

while position value is at most $V(P_* + \kappa V)$; the stated inequality guarantees margin. A single pulse attains both adverse bounds at its first response, so no larger allowance is robust. Solving the quadratic gives (2.14). The quantity requirement $V_B \gtrsim \delta^{-2}$ then forces $B \gtrsim \delta^{-4}$, and the pulse A funded with $(\mu + \frac{1}{2})\kappa A^2 + (c + \mu P_*)A$ attains it.

Proposition 2.4.4. The initial cash receipt is common to both alternatives and the dealer then holds $-A$. At time three $E_{3,r} = B + \kappa A^2/2 - cA - A^2 h_{r,2}$, so $E_{3,\delta} - E_{3,0} = A^2\kappa^2 q\delta^2 > 0$.

Lemma 2.4.5. Marking residual inventory at P_* gives $E_\infty = B - \frac{1}{2}v^\top T_h v - c\sum|v_t| \geq 0$. For each mode $\lambda \leq r$ the Toeplitz kernel has spectral density $(1-\lambda^2)/(1-2\lambda\cos\omega+\lambda^2) \geq (1-r)/(1+r)$, so $v^\top T_h v \geq m\|v\|_2^2$ and $\|v\|_2^2 \leq 2B/m$. The transfer bound gives mean-difference squared norm at most $K_*^2\delta^4\|v\|_2^2$, hence the KL bound; Pinsker gives (2.15). For one pulse, cash becomes $B_A - A(P_* + \kappa A/2) - cA$ and inventory A; at later marks $E_t = \mu A P_* + A^2 h_{t-1}$ and $\mu|I_t|P_t = \mu A P_* + \mu A^2 h_{t-1}$, so margin holds for $\mu \leq 1$.

Proposition 2.5.1. Parts (i) and (ii) follow from (2.16) with $\Delta h = \Theta(\delta)$ for a rate shift and $\Theta(\delta^2)$ for symmetric splitting, and from additivity of KL over independent pulse cycles. Part (iii) solves the session constraint for δ; substituting $d_T(\delta) \approx \delta^2 \tilde{d}_T$ gives the fourth-root approximation.

Theorems 2.5.2 and 2.5.3. For a flat round trip, telescoping the cash account gives $L_i(v) = \frac{1}{2}v^\top T_{h_i} v = \frac{1}{2\pi}\int g_i(\omega)|V(e^{i\omega})|^2 d\omega$, with $g_i > 0$ by Lemma 2.2.3. Parseval gives $\|(H_1 - H_0)v\|_2^2 = \frac{1}{2\pi}\int |\Delta\hat{H}|^2|V|^2 \leq \sup_\omega(|\Delta\hat{H}|^2/g_i)L_i(v)$, and the Neyman–Pearson requirement $\|(H_1 - H_0)v\|_2 \geq 2\sigma z_{1-\alpha}$ gives (2.20). For adaptive policies, Doob's optional stopping theorem applied to the log-likelihood-ratio martingale and the data-processing inequality give $\mathbb{E}_0[\Lambda_\tau] \geq \mathrm{kl}(\alpha, 1-\beta)$, while $\mathbb{E}_0[\Lambda_\tau] = \mathbb{E}_0\sum_t(\Delta\mu_t)^2/(2\sigma^2) \leq \sup(|\Delta\hat{H}|^2/g_0)\mathbb{E}_0[L_0]/(2\sigma^2)$, which gives (2.22).

Part (iv). Let G_θ be the $T \times T$ lower-triangular Toeplitz section of H_θ/κ, so the observation covariance is $\Sigma_\theta = \sigma_w^2 G_\theta G_\theta^\top + \sigma_\nu^2 I$. A finite section of a bounded analytic symbol has norm at most its supremum on the circle, so $\|G_\theta\| \leq \sup_\omega |H_\theta|/\kappa \leq \bar{M}$ because $|H_\theta(e^{i\omega})| \leq \sum_j h_j = \kappa/(1-S)$. The inverse of a lower-triangular Toeplitz section is the section of the reciprocal series; κ/H_θ is analytic on and outside the unit circle because H_θ is minimum phase (Theorem 2.6.8), and Lemma 2.2.3 gives $\mathrm{Re}\, H_\theta(e^{i\omega}) = \sum_\ell \omega_\ell(1-\lambda_\ell\cos\omega)/(1-2\lambda_\ell\cos\omega+\lambda_\ell^2) \geq \sum_\ell \omega_\ell/(1+\lambda_\ell) \geq \kappa/2$, so $\|G_\theta^{-1}\| \leq 2$ and $\Sigma_\theta \succeq \sigma_{\mathrm{eff}}^2 I$ (the smallest singular value of G_θ is at least 0.522 in 762 random positive economies and sections; `hardening/p1_checks.py`). The Gaussian divergence splits into a mean and a covariance part. The mean part is $\frac{1}{2}\Delta\mu^\top\Sigma_1^{-1}\Delta\mu \leq \|(H_1 - H_0)v\|_2^2/(2\sigma_{\mathrm{eff}}^2)$. For the covariance part, the eigenvalues λ_k of $A = \Sigma_1^{-1/2}\Sigma_0\Sigma_1^{-1/2}$ are at least $c = \sigma_\nu^2/(\sigma_\nu^2 + \sigma_w^2\bar{M}^2)$, and $\lambda - 1 - \log\lambda \leq (\lambda-1)^2/\lambda$, so it is at most $\|A - I\|_F^2/(2c) \leq \|\Sigma_0 - \Sigma_1\|_F^2/(2c\,\sigma_{\mathrm{eff}}^4)$. Writing $\Sigma_0 - \Sigma_1 = \sigma_w^2(\Delta G\, G_0^\top + G_1\Delta G^\top)$ and $\|\Delta G\|_F^2 \leq T\|h^\delta - h^0\|_2^2/\kappa^2 \leq TK_*^2\delta^4/\kappa^2$ (proof of Theorem 2.4.1) gives a covariance part of at most $C_w T\delta^4$.

The data-processing inequality turns errors (α, β) into $D_{\mathrm{KL}} \geq \mathrm{kl}(\alpha, 1-\beta)$, and part (ii) bounds $\|(H_1 - H_0)v\|_2^2$ by $\sup_\omega(|\Delta\hat{H}|^2/g_i)L_i(v)$, which gives (2.21).

Theorem 2.5.4. Write a feasible flat probe as $v = Eu$ and expand $u = \sum_j c_j e_j$ in the G_i-orthonormal eigenbasis, so $L_i(v) = \sum_j c_j^2$ and the separation constraint reads $\sum_j \lambda_j c_j^2 \geq c_{\mathrm{req}}$. Write the top-two component as $(c_{L-2}, c_{L-1}) = r(\cos\theta, \sin\theta)$ with $r \geq 0$; replacing θ by $\theta + \pi$ flips d and s_θ and leaves (2.23) unchanged, so $\theta \in [0, \pi)$ suffices. Put $L_0(\theta) = c_{\mathrm{req}}/\lambda(\theta)$ and $D = L_i(v) - L_0(\theta)$. The constraint gives $r^2\lambda(\theta) \geq c_{\mathrm{req}} - \sum_{j \leq L-3} \lambda_j c_j^2$, hence

$$D = r^2 + \sum_{j \leq L-3} c_j^2 - L_0(\theta) \;\geq\; \sum_{j \leq L-3} \Big(1 - \frac{\lambda_j}{\lambda(\theta)}\Big) c_j^2 \;\geq\; \Big(1 - \frac{\lambda_{L-3}}{\lambda(\theta)}\Big) \sum_{j \leq L-3} c_j^2 \;\geq 0,$$

and $r^2 \geq L_0(\theta) - \frac{\lambda_{L-3}}{\lambda(\theta)} \sum_{j \leq L-3} c_j^2 \geq L_0(\theta)\big(1 - \gamma(\theta)D\big)$. By Hölder's inequality with the sign vector s_θ, $c_{\mathrm{fee}}\|Eu\|_1 \geq c_{\mathrm{fee}} r \|Ed(\theta)\|_1 + c_{\mathrm{fee}} \sum_{j \leq L-3} c_j\, s_\theta^\top E e_j$, and Cauchy–Schwarz with weights $1 - \lambda_j/\lambda(\theta)$ bounds the second term below by $-\sqrt{D\, W_\perp(\theta)}$. Using $r \geq \sqrt{L_0(1-\gamma D)_+} \geq \sqrt{L_0}\,(1-\gamma D)_+$ and the fact that the fee is nonnegative, with $x = \sqrt{D}$,

$$L_i(v) + c_{\mathrm{fee}}\|v\|_1 \;\geq\; L_0(\theta) + x^2 + \Big[F_{\mathrm{dom}}(\theta)(1 - \gamma(\theta)x^2)_+ - \sqrt{W_\perp(\theta)}\, x\Big]_+ \;\geq\; L_0(\theta) + \Psi(\theta).$$

Minimizing over the unknown θ gives (2.24), and $\Psi \geq 0$, $L_0(\theta) \geq c_{\mathrm{req}}/\lambda_{L-1}$ give the last inequality. On the piece $\gamma x^2 \leq 1$ with $\gamma F_{\mathrm{dom}} < 1$ the bracket is a convex quadratic in x with minimum $F_{\mathrm{dom}} - W_\perp/(4(1 - \gamma F_{\mathrm{dom}}))$, which gives the closed form.

Corollary 2.5.5. Since $\Delta h_t = O(\delta^2)$, the variance difference $\sigma_{1,t}^2 - \sigma_{0,t}^2 = \sigma^2 \xi[(h_1 + h_0) * v]_t[\Delta h * v]_t = O(\delta^2)$, and the variance term of the Gaussian KL, $\frac{1}{2}\sum(x - 1 - \log x)$ with $x = \sigma_{0,t}^2/\sigma_{1,t}^2$, is quadratic in that difference. Convex impact only increases dissipation.

Proposition 2.6.1. Standard Cramér–Rao for a Gaussian mean model with Jacobian J. For $K = 2$ the columns of J for the two rates become collinear as $\Delta\eta \to 0$; expanding to third order, the smallest singular value of J is of order $(\Delta\eta)^3$, so $[F^{-1}]_{kk} = \Theta((\Delta\eta)^{-6})$.

Theorem 2.6.2. Expanding $M_m(\theta_1) - M_m(\theta_0)$ under (2.26) gives zero for $m \leq 2$ and $4\tau p_- p_+ (\Delta\eta)^3$ for $m = 3$. The map from moments to impulse responses is analytic with nonsingular derivative at the common limit, so $\|\Delta h\|_2 = \Theta(\tau(\Delta\eta)^3)$. Le Cam's two-point lemma with N independent Gaussian experiments gives (2.28). For the permutation-invariant loss, the identity matching of θ_0 and θ_1 costs $2\tau + (p_+ - p_-)\Delta\eta = 2\tau(1 + \Delta\eta/q)$ per atom and the swapped matching costs $\max(2p_+, 2p_-)\Delta\eta = (1 + 2\tau/q)\Delta\eta$; when both are at least 2τ, any estimate within τ of one economy in d_∞ is farther than τ from the other by the triangle inequality, and the two-point argument is unchanged. For (iii), the projection estimator of M_3

is unbiased with variance $\sigma_h^2/(N\|\nabla M_3 h\|^2)$, and sub-Gaussian concentration gives the sample size; the Prony–Hankel determinant identity follows from $g_m = w_1\eta_1 r_1^m + w_2\eta_2 r_2^m$.

Theorem 2.6.4. Let (s^A, ω^A) and (s^B, ω^B) be the K-point Gauss and Gauss–Radau rules for the uniform weight on $[-1, 1]$, normalized to unit mass. Both are positive and exact for polynomials of degree $2K-2$, and their moments of degree $2K-1$ differ by $g_K \neq 0$. With $r = r_c + \Delta\eta\, s$ and weights $W\omega$, the raw moments $\sum w_i r_i^m$ are polynomials in $\Delta\eta$ that agree below degree $2K-1$, so the kernels differ by $O((\Delta\eta)^{2K-1})$ with nonzero leading coefficient; analyticity of the kernel-to-response map gives $\|h^A - h^B\|_2 \asymp (\Delta\eta)^{2K-1}$, and Le Cam's lemma gives the bound. For the W_1 version, the two merged measures are the images of the fixed normalized rules under $s \mapsto r_c + \Delta\eta\, s$ scaled to mass W, so their distance is $W\Delta\eta$ times the distance $2c_K > 0$ between the rules, and the same two-point argument applies with the metric loss.

Lemma 2.6.6. Let $L = 2n-2$ and let p^* be the best uniform approximation of $|x|$ on $[-1, 1]$ by polynomials of degree L, with error $E = E_L(|x|)$. By Chebyshev's equioscillation theorem (Rivlin, 1981) there are points $t_0 < \cdots < t_{L+1}$ at which $|x| - p^*(x)$ equals $\pm E$ with alternating signs. The divided-difference weights $w_i = 1/\prod_{j\neq i}(t_i - t_j)$ alternate in sign and satisfy $\sum_i w_i p(t_i) = 0$ for every polynomial p of degree at most L; multiplying by -1 if necessary, w_i has the sign of $|t_i| - p^*(t_i)$ for every i. Exactly n weights are positive and n negative, and the positive and negative parts have equal mass because constants are annihilated. Normalizing each part to unit mass and mapping $x \mapsto \rho(x+1)/2$ gives n-atom probability measures π_A (positive part) and π_B (negative part) on $[0, \rho]$ with equal moments of orders $0, \dots, L$. For the 1-Lipschitz function $f(r) = |r - \rho/2| = (\rho/2)|x|$,

$$\int f\, d(\pi_A - \pi_B) = \frac{\rho}{2}\int \big(|x| - p^*(x)\big)\, d(\pi_A - \pi_B) = \frac{\rho}{2}(E + E) = \rho E,$$

because p^* integrates to the same value under both measures, $|x| - p^* = E$ on the support of π_A and $-E$ on that of π_B. By Kantorovich–Rubinstein duality $W_1(\pi_A, \pi_B) \geq \rho E \geq c_*\rho/L$.

Theorem 2.6.7. Write $K(z) = \sum_m k_m z^m$, $D = 1 - az - \kappa z K$ and $H = \kappa/D$, which is the generating function of the recursion $h_t = a h_{t-1} + \kappa \sum_m k_m h_{t-1-m}$ with $h_0 = \kappa$. Since $k_m \geq 0$ and $\sum_m k_m = \int (1-r)^{-1} d\mu \leq W/(1-\rho)$, the ℓ_1 norm of $az + \kappa z K$ is at most $s < 1$, so $\|H\|_1 \leq \kappa/(1-s)$ and $\|1/H\|_1 = \|D\|_1/\kappa \leq (1+s)/\kappa$.

(i) Let $\mu_A = W\pi_A$ and $\mu_B = W\pi_B$ with π_A, π_B from Lemma 2.6.6; both lie in $\mathcal{M}(\rho, W)$. With $\nu = \mu_A - \mu_B$, $\sum_m (k_m^A - k_m^B) z^m = \int (1 - rz)^{-1} d\nu(r) = z^{2n-1} \int r^{2n-1}(1 - rz)^{-1} d\nu(r)$, because the first $2n-1$ terms of the geometric series integrate to zero. On $|z| = 1$ the modulus is at most $2W\rho^{2n-1}/(1-\rho)$, so Parseval gives $\|k^A - k^B\|_2 \leq 2W\rho^{2n-1}/(1-\rho)$. Next, $H_A - H_B = z(K_A - K_B) H_A H_B$, so by Young's inequality $\|h^A - h^B\|_2 \leq \|H_A\|_1 \|H_B\|_1 \|k^A -$

$k^B\|_2 \le C_h\|k^A - k^B\|_2$. The Kullback–Leibler divergence between the N-sample Gaussian experiments is $N\|h^A - h^B\|_2^2/(2\sigma^2) \le (N/\sigma^2)\,2C_h^2W^2\rho^{2(2n-1)}/(1-\rho)^2$, which is at most one as soon as $2(2n-1)\log(1/\rho) \ge \Lambda_N - 2\log(1/\rho)$. Take n the smallest integer with this property and $n \ge 2$; then $2n - 2 < \Lambda_N/(2\log(1/\rho))$ when $\Lambda_N \ge 4\log(1/\rho)$. Since d is a norm on signed measures and $d(\mu_A, \mu_B) = W\,W_1(\pi_A, \pi_B) \ge Wc_*\rho/(2n-2) > 2c_*\rho W\log(1/\rho)/\Lambda_N$, the Bretagnolle–Huber form of Le Cam's lemma (Tsybakov, 2009, Lemma 2.6 and Theorem 2.2) gives $\max_{j\in\{A,B\}} P_j(d(\hat\mu, \mu_j) \ge c_*\rho W\log(1/\rho)/\Lambda_N) \ge \frac14 e^{-1}$.

(ii) *Approximation.* Let f be 1-Lipschitz on $[0,\rho]$ with $f(0) = 0$, so $|f| \le \rho$. The function $g(x) = f(\rho(x+1)/2)$ is $(\rho/2)$-Lipschitz on $[-1,1]$, and Jackson's theorem (Rivlin, 1981) gives a polynomial p of degree L with $\sup|g - p| \le (\pi/2)(\rho/2)/(L+1) \le \rho$, hence $\sup|p| \le 2\rho$. Its Chebyshev coefficients satisfy $|b_j| \le 2\sup|p| \le 4\rho$. The polynomial $q(r) = p(2r/\rho - 1)$ approximates f with the same error, and its monomial coefficients satisfy $\sum_j |q_j| \le \sum_{j\le L}|b_j|\,\|T_j(2r/\rho - 1)\|_{\mathrm{coef}} \le 4\rho\sum_{j\le L}B(\rho)^j \le \rho B(\rho)^{L+1}$, where $\|\cdot\|_{\mathrm{coef}}$ is the sum of absolute monomial coefficients, $\|T_j\|_{\mathrm{coef}} \le (1+\sqrt2)^j$, substituting $y = \alpha r - 1$ multiplies the coefficient sum of y^i by at most $(1+\alpha)^i$, and $B(\rho) - 1 \ge 4$. *Moment comparison.* If $\max_{j\le L}|\int r^j d\hat\mu - k_j| \le 2\delta$, then $\hat\mu$ has mass at most $W + 2\delta$ and

$$\int f\,d(\hat\mu - \mu) = \int (f-q)\,d(\hat\mu - \mu) + \sum_{j\le L} q_j\Big(\int r^j d\hat\mu - k_j\Big) \le \frac{\pi\rho(W+\delta)}{2(L+1)} + 2\delta\rho B(\rho)^{L+1}.$$

Kernel recovery. From $\kappa z K = 1 - az - \kappa/H$, the kernel values $k_0, \dots, k_L$ are the coefficients of $(1 - az - \kappa/H)/(\kappa z)$, which depend on $h_0, \dots, h_{L+1}$ only. Work in the algebra of power series modulo z^{L+2} with the ℓ_1 norm, which is submultiplicative. If the averaged response is $\hat H = H + E$ and $\|1/H\|_1\|E\|_1 \le \frac12$, then $1/\hat H = (1/H)(1 + E/H)^{-1}$ and $\|1/\hat H - 1/H\|_1 \le 2\|1/H\|_1^2\|E\|_1$, so $\max_j|\hat k_j - k_j| \le \delta \equiv 2((1+s)/\kappa)^2(L+2)\max_{t\le L+1}|E_t|$. Each E_t is Gaussian with variance σ^2/N, so with probability at least $1-\beta$, $\max_t|E_t| \le \sigma\sqrt{2\log(2(L+2)/\beta)/N}$. The true μ is feasible for the linear program with value at most δ, so the minimizer satisfies the moment condition above. *Choice of L.* With $L+1 = \lfloor\log(N/\sigma^2)/(4\log B(\rho))\rfloor$, $B(\rho)^{L+1} \le (N/\sigma^2)^{1/4}$, so $\delta B(\rho)^{L+1} = O((\sigma^2/N)^{1/4}L\sqrt{\log(L/\beta)})$ and $\delta/(L+1)$ are $o(1/\log(N/\sigma^2))$, while $\pi\rho W/(2(L+1)) \le 2\pi\rho W\log B(\rho)/(\log(N/\sigma^2) - 4\log B(\rho))$. For N/σ^2 larger than a threshold depending on $(\rho, W, a, \kappa, \beta)$ the side condition $\|1/H\|_1\|E\|_1 \le \frac12$ holds on the event and the total is below $3\pi\rho W\log B(\rho)/\log(N/\sigma^2)$, uniformly over $\mathcal{M}(\rho, W)$. The final statement combines (i) and (ii).

Theorem 2.6.9. For stationary Gaussian processes with spectral densities $f_j \propto |H_j(e^{i\omega})|^2$, the divergence rate is

$$D = \frac{1}{4\pi}\int_{-\pi}^{\pi}\big(\varrho - 1 - \log\varrho\big)\,d\omega, \qquad \varrho = f_1/f_0 = |H_1/H_0|^2.$$

By the resolvent identity, $H_1/H_0 - 1 = H_1 \Delta G$ with $\Delta G = G_1 - G_0$, and uniform stability bounds $|H_1|$ on the circle, so $u \equiv \varrho - 1 = 2\,\mathrm{Re}(H_1 \Delta G) + O(|\Delta G|^2)$. Write $G(z) = \sum_k w_k \phi(z, \eta_k)$, with w_k the cohort weights (any factor of η_k is absorbed in ϕ), where ϕ is analytic in η on a neighbourhood of $[\eta_{\min}, \eta_{\max}]$ uniformly for $|z| = 1$ (its pole $z = 1 - \eta$ stays inside the disk). Expanding ϕ around $\bar\eta$, $\Delta G = \sum_{j \geq 0} \frac{\partial_\eta^j \phi(z, \bar\eta)}{j!} \Delta \tilde M_j$, where $\tilde M_j = \sum_k w_k (\eta_k - \bar\eta)^j$. For the pair, $\Delta \tilde M_0 = \Delta \tilde M_1 = 0$ and $\Delta \tilde M_2 = q\delta^2$; for the moment-matched economies, $\Delta \tilde M_j = 0$ for $j \leq 2$ and $\Delta \tilde M_3 = 4\tau p_- p_+ (\Delta\eta)^3$ by (2.27); for the K-cohort Gauss and Gauss–Radau economies with rates $\bar\eta + \Delta\eta\, s$ and weights $W\omega$, both rules integrate polynomials of degree $2K-2$ exactly, so $\Delta \tilde M_j = W(\Delta\eta)^j \sum_k (\omega_k^A (s_k^A)^j - \omega_k^B (s_k^B)^j)$ vanishes for $j \leq 2K-2$ and equals $\pm W g_K (\Delta\eta)^{2K-1} \neq 0$ at $j = 2K-1$. Higher terms are smaller by a factor of the spacing: $|\Delta \tilde M_j| \leq 2W(\Delta\eta)^j$, and analyticity of ϕ in η on a disk of radius $\varrho > 0$ around $\bar\eta$, uniformly on the circle, gives $|\partial_\eta^j \phi| / j! \leq C \varrho^{-j}$, so for $\Delta\eta < \varrho/2$ the tail beyond the first nonzero term is $O((\Delta\eta)^{j_0+1})$. Hence $\sup_\omega |u| = O(\delta^2)$, $O(\tau(\Delta\eta)^3)$, respectively $O((\Delta\eta)^{2K-1})$, and $u = c(\omega)\epsilon + o(\epsilon)$ with $\epsilon = \delta^2$, $\tau(\Delta\eta)^3$ or $(\Delta\eta)^{2K-1}$ and $c(\omega)$ proportional to $2\,\mathrm{Re}\big(H_0\, \partial_\eta^{j_0} \phi(\cdot, \bar\eta)\big)/j_0!$, $j_0 = 2$, 3 or $2K-1$. The function c does not vanish identically: $H_0 \partial_\eta^j \phi$ is analytic outside the disk and non-constant, and an analytic function whose real part vanishes on the circle is an imaginary constant. For $|u| \leq \frac{1}{2}$, $u^2/4 \leq u - \log(1+u) \leq u^2$, so $D \asymp \int c(\omega)^2 d\omega\, \epsilon^2$, which gives (i)–(iii). The sample requirements follow from $D(P_1^T \| P_0^T) = TD + O(1)$ for stationary Gaussian processes with smooth spectra and the Bretagnolle–Huber inequality (Tsybakov, 2009, Lemma 2.6); attainment follows from Stein's lemma for the likelihood-ratio test.

Theorem 2.6.8. In the delay variable $\lambda = z^{-1}$, $H = \kappa \prod_i (1 - r_i\lambda) / \big[(1 - a\lambda) \prod_i (1 - r_i\lambda) - \kappa\lambda \sum_i w_i \prod_{j \neq i} (1 - r_j \lambda)\big]$, which is (2.4) divided through by z^{K+1}; its zeros are the memory rates, which do not cancel by the argument for Theorem 2.3.1, and stability puts the poles inside the disk. A rational spectral density $\sigma^2 |H(e^{i\omega})|^2$ has a unique minimum-phase factor up to a unimodular constant, which is H; Theorem 2.3.1 then gives the merged measure.

Proposition 2.8.1. With tagged flow, $u_{i,t} = r_i u_{i,t-1} + w_i x_t$ holds exactly and involves only cohort i, so $\tilde u_{i,t} = r_i \tilde u_{i,t-1} + w_i x_t + e_{i,t}$ with the MA(1) error $e_{i,t}$. Prices are independent of the measurement noise, so $\mathbb{E}[Z_t e_{i,t}] = 0$ and the IV estimator is consistent when $\Sigma_i = \mathbb{E}[Z_t (\tilde u_{i,t-1}, x_t)]$ is nonsingular. Since $u_{i,t-1} = w_i \sum_{s \geq 0} r_i^s x_{t-1-s}$, $\mathbb{E}[x_{t-1} u_{i,t-1}] = w_i A_0$ and $\mathbb{E}[x_t u_{i,t-1}] = w_i A_1$, which gives Σ_i. The price is a nonnegative mixture of geometric responses to its innovation (Lemma 2.2.3), so $\gamma_x(s) = \sum_\ell c_\ell \lambda_\ell^s$ with $c_\ell = \sigma_\xi^2 \kappa^{-2} \omega_\ell \sum_m \omega_m / (1 - \lambda_\ell \lambda_m) \geq 0$; substituting, $\det \Sigma_i = w_i (\gamma_x(0) A_0 - \gamma_x(1) A_1) = w_i \sum_{\ell,m} c_\ell c_m (1 - \lambda_\ell \lambda_m)/(1 - r_i \lambda_m)$, each term positive. The scores $Z_t e_{i,t}$ are one-dependent with $\mathbb{E}[Z_t Z_t^\top e_{i,t}^2] = \sigma_f^2 (1 + r_i^2) \Gamma_0$ and $\mathbb{E}[Z_t Z_{t-1}^\top e_{i,t} e_{i,t-1}] = -r_i \sigma_f^2 \Gamma_1$, so their long-run variance is Ω_i, and the central limit theorem for m-dependent sequences gives (2.29). On a uniformly stable family the modes and weights lie in a compact set with $\lambda_\ell \leq \bar\lambda < 1$, so $\det \Sigma_i \geq w_{\min} (1 - \bar\lambda^2) \gamma_x(0)^2 > 0$ and $\|\Omega_i\|$ is bounded, uniformly in the spacing.

Theorem 2.9.1. With J_θ upper triangular the eigenvalues are fixed. The stationary covariance solves $P = J_\theta P J_\theta^\top + \mathrm{diag}(s_1, s_2)$, and P_{11} is affine in s_2 with positive slope, so for each θ a unique $s_2(\theta) \geq 0$ gives $P_{11} = V$ when V exceeds the value at $s_2 = 0$. The one-step price variance is s_1 throughout, and $e_1^\top J_\theta e_2 = \theta$ gives $\sup_k \|e_1^\top J_\theta^k\| \geq |\theta|$.

2.14 Appendix: Notation

Symbol	Meaning
x_t	pricing error (price minus fundamental)
a	price persistence (anchor), $0 \leq a < 1$
κ	price impact per unit of executed quantity
$q_i,\ \eta_i,\ r_i$	cohort weight, adaptation rate, memory rate $r_i = 1 - \eta_i$
$w_i = q_i \eta_i$	merged weight
$M = \sum_i w_i \delta_{r_i}$	merged rate–weight measure
$k_m,\ G(z)$	feedback kernel and its transform
$H(z),\ h_j$	closed-loop transfer function and impulse response, $h_0 = \kappa$
$J,\ S$	state transition matrix and $S = a + \kappa \sum q_i$
$v_t,\ V$	executed probe quantity and total quantity $\sum \lvert v_t \rvert$
d_T	impulse-response difference between hypotheses, first T taps
σ	observation-noise standard deviation
$\delta,\ \Delta\eta$	half-separation of a known pair; spacing of unknown cohorts
K_*	transfer-difference constant $2\kappa^2 q / [\epsilon^3 (1 - S)^2]$
T_h	Toeplitz matrix $(h_{\lvert i-j \rvert})$ of execution cost
$g_i(\omega)$	dissipation kernel $\mathrm{Re}\, \hat{H}_i(e^{i\omega}) - \kappa/2$
$B,\ \mu,\ c,\ \phi$	capital, maintenance-margin fraction, proportional cost, taker fee
V_{eff}	per-trade execution capacity $\min(V_{\max}, D_{\max}/\kappa)$
$\hat{Q},\ \hat{S}$	reduced-form loop gain and persistence indices

2.15 Appendix: Further results

2.15.1 A finite-window price-surveillance bound

Theorem 2.15.1 (Finite-window abstention)**.** *Observe $x_t = s x_{t-1} + w_t$, $w_t \sim N(0, \sigma^2)$, $x_0 = 0$, $t \leq T$, with $s \in (0, 2)$. Let a possibly randomized rule Ψ output Stable, Unstable or Abstain, with $\sup_{s<1} P_s(\Psi = \text{Unstable}) \leq \alpha$ and $\sup_{s \geq 1} P_s(\Psi = \text{Stable}) \leq \alpha$, $\alpha < 1/2$. Choose $0 < c_0 < 1$*

with $c_0 e^{c_0}/(2\sqrt{2}) < 1 - 2\alpha$. *On* $B_T = (1 - c_0/T, 1 + c_0/T)$,

$$\inf_{s \in B_T} P_s(\Psi = \text{Abstain}) \;\geq\; 1 - 2\alpha - \frac{c_0 e^{c_0}}{2\sqrt{2}} > 0,$$

and for every finite T *the minimax worst-case error of binary rules over* $s \in (0, 2)$ *is at least* $1/2$.

Proof. Pair $s_0 \in (1 - c_0/T, 1)$ with $s_1 = s_0 + c_0/T$. Then $D_{\mathrm{KL}}(P_{s_0} \| P_{s_1}) = \frac{(s_1 - s_0)^2}{2\sigma^2} \sum_t \mathbb{E} x_{t-1}^2 \leq c_0^2 e^{2c_0}/4$, so Pinsker bounds total variation by $c_0 e^{c_0}/(2\sqrt{2})$. Error control gives $P_{s_0}(\text{Stable}) \leq \alpha + \text{TV}$ and $P_{s_0}(\text{Unstable}) \leq \alpha$, which yields the abstention bound. For binary rules, continuity in total variation and Le Cam's two-point bound give worst-case error tending to $1/2$. □

The interval B_T is an abstention layer that any error-controlled monitor must respect in the scalar Gaussian case; richer disclosure can remove it.

2.15.2 Passive distinction of a known pair in finite samples

Proposition 2.15.2 (Passive distinction with zero controlled quantity)**.** *Keep the known pair, but add independent uniform innovations* $w_t \in [-b, b]$ *of variance* $s_w^2 = b^2/3$ *at each price update, observe* $\tilde{x}_t = x_t + \nu_t$ *with* $\nu_t \sim N(0, \sigma^2)$*, use independent three-step reset blocks and no probes, with* $P_* > 3b$. *Let* $b_1 = a + \kappa q \eta$, $b_{2,0} = b_1^2 + \kappa q \eta (1 - \eta)$ *and* $b_{2,\delta} = b_{2,0} - \kappa q \delta^2$. *In each block the covariance of the first and third observations differs by* $\Delta = \kappa q s_w^2 \delta^2$. *With*

$$R = \max_{r \in \{0, \delta\}} \Big\{ (s_w^2 + \sigma^2) \big[s_w^2 (1 + b_1^2 + b_{2,r}^2) + \sigma^2 \big] + s_w^4 b_{2,r}^2 \Big\},$$

the midpoint test of the average product has worst-case error at most α *whenever* $n \geq 4R/(\alpha \Delta^2)$ *blocks are observed: zero controlled quantity and order* δ^{-4} *blocks.*

The Chebyshev bound is conservative but the order is real. With $b = 0.03$, observation s.d. 0.01 and $\delta = 0.10$, the bound asks for 3.9×10^7 blocks at $\alpha = 0.1$; in simulation the midpoint test still errs 32–38% of the time with 2×10^5 blocks. Independent resets and known innovation laws are extra conditions. The proposition does not show that anonymous public records identify participants; it shows that passive excitation can accumulate memory information while the probe account stays untouched.

Proof. In each block $x_1 = w_1$, $x_2 = b_1 w_1 + w_2$ and $x_3 = b_{2,r} w_1 + b_1 w_2 + w_3$, so $|x_t| \leq 3b$. Observation noise changes variances but not the covariance $s_w^2 b_{2,r}$. The product variance is the product of marginal variances plus the squared covariance plus a fourth-cumulant term $b_{2,r}^2 (\mathbb{E} w^4 - 3 s_w^4) \leq 0$ for uniform innovations, so it is at most R. Chebyshev bounds each midpoint error by $4R/(n\Delta^2)$. □

2.15.3 Blind inversion of the closed loop

1. *Impulse decoupling:* $g_t = h_{t+1} - ah_t$.
2. *Minimal realization:* form the Hankel matrix $H_{ij} = g_{i+j-1}$ and run Kung's eigensystem realization with a truncated SVD.
3. *Transmission zeros:* compute the zeros of the realization, which under known (κ, a) isolate the roots of the denominator of G.
4. *Weights:* with recovered rates, project $u_t = (x_t - ax_{t-1})/\kappa$ on the reconstructed filters $y_{k,t-1}$.

Without noise each step is exact: the realization has the order of the true system, its transmission zeros are the memory rates (Theorem 2.3.1), and the projection returns the weights, so the procedure removes the pole-shift bias of fitting poles directly to the price response. With noise, per-cohort recovery degrades as rates move together at the rate given by Theorems 2.6.4 and 2.6.5, which is why we do not report it as an estimator of individual cohorts.

2.15.4 Probing protocols

Table 2.28: Probing protocols at observation horizon $T = 4$ per cycle

Protocol	δ	Executed vol.	Max trade	Error (SE)	Capital
A: single pulse (unconstrained)	0.03	131.5	131.5	0.047 (0.003)	6,985.91
	0.05	47.3	47.3	0.054 (0.004)	921.73
	0.08	18.5	18.5	0.053 (0.004)	146.73
	0.12	8.2	8.2	0.044 (0.003)	31.45
B: repeated capped pulses	0.03	2,768.8	6.2	0.050 (0.003)	7,035.75
	0.05	362.5	6.2	0.054 (0.004)	923.88
	0.08	56.2	6.2	0.047 (0.003)	146.00
	0.12	12.5	6.2	0.038 (0.003)	34.88
C: sequential test, capped	0.03	1,410.2	6.2	0.045 (0.003)	3,585.14
	0.05	195.6	6.2	0.038 (0.003)	500.00
	0.08	35.9	6.2	0.032 (0.003)	94.30
	0.12	10.5	6.2	0.011 (0.002)	29.85
D: generalized eigenvector	0.03	2,359.7	3.7	0.050 (0.003)	2,632.77
	0.05	314.8	3.7	0.048 (0.003)	353.67
	0.08	49.2	3.7	0.046 (0.003)	57.53
	0.12	9.9	3.8	0.045 (0.003)	13.56

2,000 replications per hypothesis and cell. Log–log capital slopes: −3.90, −3.84, −3.47, −3.81. Protocols A and B reproduce the earlier version exactly. C and D are re-implemented because the earlier code was not retained: C is a Wald test with capped pulses; D repeats the top generalized eigenvector of a flat $(T + 1)$-step probe, scaled to a maximum trade of 3.75, and is cheaper than the earlier D. Kind: [RECOMPUTED] (`recompute_p1_protocols.py`).

2.15.5 Coordinate-weighted output

For $J = \left(\begin{smallmatrix} a & \kappa q \\ \eta a & 1-\eta+\eta\kappa q \end{smallmatrix}\right)$ with $a = 0.5$, $\kappa = 0.8$, $q = 0.2$, $\eta = 0.3$ ($\rho(J) = 0.8224$, non-normal), the price output has $\sup_t \|e_x^\top J^t\| = 1$. The weighted output $C_w = (1, \lambda/\mathrm{MMR}) = (1, 160)$ with $\lambda = 0.8$ and $\mathrm{MMR} = 0.005$ has $\sup_t \|C_w J^t\| = \|C_w\| = 160.003$, attained at $t = 0$, then 122.32, 98.32, 80.79 [RECOMPUTED IN CLOSED FORM]. The factor 160 is a static coordinate weight; $C_w z$ is not a margin quantity because z contains no inventory. Margin must be computed on a cash-and-inventory ledger.

2.16 Appendix: Reproducibility: source of every number

The replication package contains numbered Colab notebooks and stand-alone scripts; its README lists them in run order. "Part N" below and in table notes refers to notebook N (Part 1: theorem checks and spot tests; Part 2: BTCUSDT futures depth; Parts 4–13: the cross-section, replications, fresh-day tests, second venues, the CFTC test, the certified guard and post-hoc checks). Result identifiers such as `X6` or `U6` name the JSON file that the notebook writes. "Kind" distinguishes closed-form or simulation results (recomputed), results from public market data (real data), and audits.

Result	Source	Kind
Complete monotonicity, 892 economies	`L26_complete_monotonicity`	recomputed
$d_2 = -\kappa^2 q \delta^2$	`P1.7-T1_third_tap`	recomputed
Ledger identity; T_h positive definite	`R2-T1`, `R2-T3`	recomputed
Figure 2.2	`make_figs2.py`	recomputed
$\delta_{\min}(3)$, Table 2.5 recomputed columns, 61.14, 29.94	`recompute_theory.py` → `R_P1_theory_constants.json`	recomputed
Pulse benchmarks (271, 58, 12, 11 pulses), Figures 2.1–2.3	`make_figs3.py` → `R_P1_pulse_benchmarks.json`	recomputed
Table 2.3, Table 2.11, worked example, passive simulation	`recompute_extra.py` → `R_P1_extra.json`	recomputed
Daily noise (Table 2.8)	`P1.3-T1_sigma_obs`	real data
Table 2.10 recomputed columns	same	recomputed
K-cohort slopes (Table 2.12)	`T1_K_cohort_rate`	recomputed
ESPRIT slope −6.15	`P1.2-T1_esprit_rate`	recomputed
Minimum phase, 1,733 economies; rank 6 of 6	`T2a`, `T2b`	recomputed
Passive slopes −4.21, −5.93	`T3_passive_rate`	recomputed
Tagged-flow slope 0.011	`T4_disclosure`	recomputed

Result	Source	Kind
Spectral blindness gains	`T5_spectral_blindness`	recomputed
Noise and capital grid (Table 2.7)	`P1.3-T1_sigma_obs`	real data
Loop gain and placebo (Table 2.24)	`P1.6-T1_ema_placebo`	real data
Estimator table (Table 2.25)	`P1.5-T1`, `T2`, `T3`	recomputed
Data integrity (111 files)	`R2-T9_data_integrity`	audit
Tables 2.4, 2.5 (certified, feasible, adaptive columns)	`recompute_p1_capital.py` → `R_P1_capital_ledger.json`	recomputed
Table 2.16; resolution floors; Table 2.28	`recompute_p1_protocols.py` → `R_P1_protocols.json`	recomputed
SPRT capital figures	`recompute_p1_sprt.py` → `R_P1_sprt.json`	recomputed
Table 2.9 (N^*, σ_{crit})	`recompute_crb.py` → `R_P1_crb.json`	recomputed
Table 2.9 (ML columns), Table 2.10 (estimators)	`recompute_p1_estimators.py`	recomputed
Symmetric multi-asset network (Section 2.9)	`recompute_oct20.py`	recomputed
Proof audit: K_* bound and d_2 identity (Theorem 2.4.1), moment gaps (Theorem 2.6.2), spectral-blindness example (Theorem 2.9.1)	`recompute_audit.py` → `R_audit_oct2026.json`	recomputed
Theorem 2.6.5, Table 2.13	`kcohort_upper.py` → `R_P1_kcohort_upper.json`	recomputed
Lem. 2.6.6, Thm 2.6.7, Tab. 2.14	`general_minimax.py`	recomputed
Theorem 2.8.2 numbers	`disclosure_value.py` → `R_P1_disclosure_value.json`	recomputed
Tables 2.22, 2.23, 2.26	Part 8: `V2_spot_panel`, `V3_fee_events`	real data
Table 2.18, top panel	Part 4: `X9_disclosure_metrics`	real data
Table 2.18, bottom panel; class-position forecasts	Part 6: `Z6_C_identification`, `Z6_secondary`; `P6_fresh_days.json` (65acda36…)	real data
Table 2.20	Part 11: `C3_cot_data`, `C4_cot_disclosure`; `P11_cot.json` (6f1d6f35…)	real data

Result	Source	Kind
Basis-trade signature; real-data Cochrane–Orcutt	Part 13: `R6`, `R7`	real data (post hoc)
Feedback across 30 perpetuals	Part 4: `X3_T6_cross_section`	real data
Theorem 2.5.4 full bound (Table 2.6), direct minimization	`hardening/rh_bound.py`, `rh_check.py` → `R_P1_rh_explicit.json`	recomputed
Minimal dimension at $a = 0$; Toeplitz sections; passive slopes for $K \leq 4$; IV determinant	`hardening/p1_checks.py` → `R_P1_hardening_checks.json`	recomputed
Stylized nonlinear and trained online rules (Table 2.15, Figure 2.7)	`recompute_p1_learners.py` → `R_P1_learners.json` (Panel A); `validate_trained_learners_hardened.py` → `R_P1_trained_learners_hardened.json` (Panels B–C); specification `P1_trained_learners.json` (3e8bf30c…)	recomputed
Disclosure in dollars (Table 2.17)	`recompute_p1_cat_costs.py` → `R_P1_cat_disclosure_dollars.json`	recomputed
Out-of-sample post-shock return test (Section 2.8.3)	`recompute_prereg_oct21_22nd.py` → `R_P1_disclosure_oos.json`; `P1_disclosure_oos.json` (f0182ce5…)	real data
Out-of-sample cohort-state trajectory test (Table 2.19)	`recompute_prereg_oct21_followup.py` → `R_P1_disclosure_state_oos.json`; `P1_disclosure_state_oos.json` (5136cc6b…)	real data

Chapter 3

Safety without identification: transient amplification and a maximal pre-trade guard

This chapter, the second paper of the thesis, reproduces the paper *From Human Markets to Machine Markets: Transient Amplification, an Exact Depth Condition, and a Maximal Pre-Trade Guard*, joint with Suboh Alkhushayni, Yuhao Chen, Mansi Bhavsar and Prashant Niroula. It establishes the safety half of the thesis: what certifies a safe trading decision when the agents cannot be identified, and what that certificate costs.

Summary. Eigenvalues and volatility do not bound the losses a market produces before it settles. In a limit order book with adaptive traders, cumulative transient risk equals $S = \tilde{\kappa}/(m - q\tilde{\kappa})$, a book overshoots if and only if $\varphi\varepsilon > (1-a)\kappa$, and a pre-trade guard on the summed impulse response is the largest safe rule and does not depend on how traders split into cohorts. S is certified with $O(\sigma^2\epsilon^{-2}\log(1/\epsilon))$ experiments whatever the cohort spacing, and the certified guard is minimax-optimal in lost capacity up to a logarithmic factor. In 25 pre-registered tests, 15 pass: the statistic forecasts next-hour adverse excursions in thin Binance perpetuals beyond volatility and VPIN (+2.79 points of R^2, CI $[+1.43, +4.94]$), more weakly in thin US stocks, and not on Coinbase; tests of a liquidation mechanism fail.

Reproducibility. Every number in the paper is produced by a script or notebook cell that writes a result file from public data or from a closed form; Appendix 3.18 maps each table, figure and number to its source. Pre-registered specifications, their SHA-256 hashes and test-set unlock counters are in the replication package. Every simulation number was re-run for this version. The specifications of Parts 22–25 were committed to the repository before their test data were downloaded; because each reached the public repository in the same push as its results, that order is documented by the commit history rather than by an independent timestamp.

3.1 Introduction

On October 10, 2025, crypto derivatives markets liquidated an estimated $19 billion of leveraged positions in a single day, the largest such event on record (21Shares, 2025); on May 6, 2010, US equity prices fell and recovered by several percent within minutes (Kirilenko et al., 2017). In both episodes the losses that mattered were transient: positions were closed out on the path, before prices settled. This chapter is about that path risk in markets where algorithms trade with each other.

Algorithmic markets are closed-loop systems: trading moves prices, prices move algorithms, and algorithms trade. A common shortcut equates the stability of such a system with its eigenvalues lying inside the unit circle. That is the wrong object for a leveraged intermediary, whose survival depends on the path of losses before the system settles. A market can be spectrally stable and still push a dealer through its maintenance margin within seconds.

This chapter asks three questions. When does a limit order book amplify a flow shock, so that prices keep moving after the shock has passed? How can a dealer or a venue control transient loss without identifying who is trading, which Chapter 2 shows is expensive or impossible from public data? And when the traders cannot be identified, what information certifies a safe trading decision, what does that information cost, and how much safe trading must be given up when it is missing? We give exact answers in a linear book model and then confront them with data, reporting the results as they came out.

1. Spectral stability does not bound transient loss. The exact safe shock radius of a stable linear market is the loss budget divided by its peak transient gain, and that gain can be arbitrarily large at any fixed spectral radius (Theorem 3.3.1). Markets with identical eigenvalues, one-step volatility and stationary price variance can have arbitrarily large worst-case transient gain, as Chapter 2 shows. Supervision through eigenvalues or volatility alone is therefore insufficient in principle.

2. When a book amplifies. We distinguish peak overshoot, a response that rises above its direct impact, from cumulative amplification, a summed response far above it. When order flow depletes one side of the book and the book refills at rate r, the price response overshoots its direct impact if and only if $\varphi\varepsilon > (1-a)\kappa$ (Theorem 3.4.2(i)). With momentum traders the one-period margin $C_3 = \varphi\varepsilon + \kappa^2\eta q - (1-a)\kappa$ is sufficient (Theorem 3.4.2(iii)) but no one-period condition is exact (Remark 3.4.3); momentum alone cannot make a stable book overshoot (Theorem 3.4.2(iv)), and an explicit condition rules overshoot out (Theorem 3.4.2(v)). Cumulative risk has a closed form, $S = \tilde{\kappa}/(m - q\tilde{\kappa})$: long-run impact divided by reversion net of momentum (Theorem 3.4.4). It needs no depletion, and it is the form our evidence supports: crypto books show no peak overshoot, but thin books revert slowly, so their summed response is large.

3. The maximal safe guard. For bounded shocks, worst-case loss over a horizon is inventory times the shock bound times the summed absolute impulse response, the classical ℓ_1 gain of control theory. Under a declared risk contract (fixed inventory, per-period shock bound, mark-to-market loss), a pre-trade guard on this summed gain is necessary and sufficient for staying within the loss budget, so it admits the largest safe set of actions (Lemma 3.6.1). A guard on the peak response, the quantity closest to the eigenvalue intuition, admits unsafe actions. A Grönwall bound carries the guard to an estimated operator (Theorem 3.6.3), a contraction-block certificate makes it computable on a finite horizon (Lemma 3.6.2), and its inputs are identified from prices without identifying the algorithms (Proposition 3.2.2 and Theorem 3.5.5). We embed the guard in a dealer's balance sheet with margin, hedging and fill accounting (Theorem 3.6.8) and show that strategic re-optimization by filtering algorithms reinforces it (Proposition 3.6.11).

4. Information and safety. The information that certifies a safe decision is not the information that identifies the traders. Two markets can agree exactly on their early price responses and differ threefold in risk (Example 3.5.1), and any supervisor who must rarely admit an unsafe level then gives up a computable amount of safe capacity (Proposition 3.5.2). A finite window of responses certifies the summed response to an interval computed exactly from the Markov–Krein extremal measures, which shrinks geometrically with the window under a bound on memory length and is unbounded without one (Theorem 3.5.3). With noisy reset experiments the summed response is certified at the parametric rate, $N = O(\sigma^2\epsilon^{-2}\log(1/\epsilon))$, whatever the number and spacing of the traders' memories, while identifying them costs $(\Delta\eta)^{-(4K-2)}$ or more (Theorem 3.5.5); Chapter 2's hardest pairs have identical summed responses. A guard built on the certificate has a closed form, admits an interval of levels, flags infeasible states, and is minimax-optimal in lost capacity up to a logarithm (Theorem 3.6.5). When admission also changes the book, the safe levels need not form an interval, so certifying the largest one no longer certifies those below it; we certify the whole safe set by interval branch and bound (Proposition 3.6.6, Theorem 3.6.7). The window bounds are proved rather than computed (Lemma 3.5.4), and certificates that allow correlated noise keep their level where the simple one fails (Proposition 3.5.7).

5. The value of disclosure. Disclosure that rules out the extremal compatible economy raises safe capacity, and its value grows as memory lengthens, approaching the full oracle capacity; disclosure of a noisy proxy can be worth less than nothing (Theorem 3.7.1, Table 3.5).

6. Who controls the risk. Reversion is a public good. In equilibrium dealers under-provide it, momentum traders over-provide feedback, fewer dealers mean more transient risk, and the rebate that restores the first best is $(1-\theta/N)\Lambda S^2/\tilde{\kappa}$, a function of the guard's own statistic (Proposition 3.8.1). When dealers also choose depth, fewer dealers raise risk through larger impact while the unit-free multiple $S/\tilde{\kappa}$ stays fixed, the pattern we see across Binance perpetuals.

7. Evidence. We test the theory's empirical content in 25 pre-registered primary tests, each with a hashed specification and a test set opened once; 15 pass (Table 3.27). Where the theory predicts nothing, we find nothing: BTCUSDT, the deepest book, sits on the stable side of the amplification condition on 93.5% of days, shows no overshoot and fails a pre-registered forecasting test (Section 3.9); on a fixed price grid flow depletes the side it hits, and the opposite finding in Binance's depth bands is an artefact of their re-centring (Section 3.11.4). In thin books, where the amplifying regime is more frequent, six-hour estimates of the amplification margin forecast the size of the next hour's adverse excursion out of sample in every pre-registered forecasting test on thin Binance perpetuals, on new symbols and on untouched days (+0.85 to +4.2 points of R^2), beyond a full volatility-forecasting baseline and VPIN, and mostly through impact-weighted reversion, as Theorem 3.4.4 implies. Within a volatility quintile, hours in the top quintile of the statistic have excursions about 26 basis points larger than hours in the bottom quintile, enough to roughly double the hourly liquidation probability of a 20× position. The effect is about a fifth as large in 23 thin US stocks (Section 3.11.12), negligible on Bybit and absent on Coinbase spot (Section 3.12.4).

We have not identified why. The effect does not rise with open-interest leverage (Part 19, specified in advance but not hash-registered) and does not run through open-interest liquidations (Part 22, both primaries fail). A follow-up registered after Part 22 (Part 25) finds that the statistic forecasts price persistence over the next 15 to 60 minutes, which accounts for nearly all of its forecasting gain, but its placebo criterion fails, much of that mediation is mechanical, and the statistic is itself built from slow reversion. We therefore read the evidence as showing that thin books are risky because their prices revert slowly relative to impact, which is what the statistic measures, not as a test of a deeper mechanism.

As a risk control, the certified guard lowers tail loss against a static limit by 8% on untouched days and 23% on untouched symbols (Sections 3.12.2 and 3.12.3), but it does not beat a volatility cap at matched exposure. Combined with a volatility cap it lowers the 18-hour forced-liquidation rate of a 20× position by 7.4 percentage points (Part 24, registered after a capacity test failed), mostly because it admits 11% less exposure; scaled to the same mean exposure the two rules are indistinguishable (Section 3.12.12). Simulations with a known truth confirm the coverage, attainment and optimality claims and show where the certificate fails: when memory is longer than declared, and when feedback strengthens after certification (Section 3.12.1).

What we do not claim. The information-safety theorems concern declared observation and risk contracts (controlled reset experiments, exact windows, tagged or audited data); they are not finite-sample guarantees for observational order-book records, for which we report a bootstrap diagnostic and its out-of-sample performance. Our cohorts are any traders whose demand follows a linear filter of past prices, human or automated; nothing in the evidence attributes an effect to algorithms rather than to people. We do not claim that crypto books amplify shocks through depth: flow depletes the side it hits on a fixed price grid, but we observe no peak overshoot,

and the depth term carries no forecasting content. The thin-book forecasting result is reduced-form, and its size depends on the venue: largest in thin Binance books, about a fifth as large in thin US stocks, negligible on Bybit and absent on Coinbase; it improves tail classification only against the full volatility baseline. We do not claim that the guard replaces volatility in practical position sizing: its theorems are statements about worst-case loss under a certified operator, an estimated six-hour response adds nothing to volatility, and a three-day certified estimate improves a volatility limit only modestly (2.4%, a narrow pass) while the guard alone does worse. The October 10, 2025 episode is reported descriptively and supports no inference (Section 3.11.15). One test cell of the first thin-book test was executed twice with unchanged code and specification (Section 3.11.5). The equilibrium is a stylized model; thinner books do carry more cumulative risk per dollar of flow, as it predicts, but through larger impact rather than through the weaker reversion it names.

Table 3.1: Map of results

Question	Answer	Where
Does $\rho(J) < 1$ bound transient loss?	No; safe radius B/M_C can vanish at fixed ρ	Thm 3.3.1, Companion paper
Can volatility monitors help?	No; fixed variance, unbounded gain	Companion paper
When does a book amplify?	Iff $\varphi\varepsilon > (1-a)\kappa$; with momentum $C_3 > 0$ suffices, no exact one-period rule	Thm 3.4.2(i), Thm 3.4.2(iii), Rem 3.4.3
Can momentum alone amplify?	Not in a stable book	Thm 3.4.2(iv)
How large is cumulative risk?	$S = \tilde{\kappa}/(m - q\tilde{\kappa})$	Thm 3.4.4
Which guard is safe and largest?	Summed absolute gain; peak gain is unsafe	Lem 3.6.1
Robust to estimation error?	Yes, $M/(1-\epsilon S)$	Thm 3.6.3
Needs identification of algorithms?	No; inputs are invariant to cohort splits	Prop 3.2.2, Thm 3.5.5
Do learners undo the guard?	No; gains fall when admission falls, and every finite-horizon summed response rises with the gains in the cohort economy	Prop 3.6.11
Is reversion efficiently supplied?	No; under-provided, and the fix is priced by S	Prop 3.8.1
Is BTCUSDT on the amplifying side?	No: $\hat{C}_3 < 0$ on 93.5% of days	Table 3.7
Does depth carry an overshoot?	No; all overshoots negative	Figure 3.4
Does $\hat{C}_3$ forecast tail risk in BTCUSDT?	No (pre-registered)	Table 3.9
Is spot amplification above one?	No; unit-free gain ≈ 1.0000	Section 3.10
Does flow deplete thin books?	Yes on a fixed price grid; the apparent replenishment is a band-recentring artefact	Tables 3.15, 3.16
Does $\hat{C}_3$ forecast tail size in thin books?	Yes, in four pre-registered tests on Binance	Tables 3.17, 3.19, 3.21, 3.22
Is it just volatility?	No: +2.8 pts beyond HAR, EWMA and hour effects; S itself +2.1	Table 3.22

Is it just VPIN?	No: +2.0 pts beyond the baseline plus VPIN (post hoc); +0.56 on new symbols (pre-registered)	Section 3.11.10, Table 3.30
Is it tick size, impact or scale?	No; survives controls, unit-free versions work	Tables 3.18, 3.20, 3.21, 3.22
Does it hold on a second venue?	Bybit: absent with trades; with depth, significant but negligible	Section 3.11.11, Table 3.23
In a regulated market?	Thin US stocks: yes, +0.50 pts (pre-registered); mid-size stocks too	Table 3.24
Does the estimated response size positions better?	Six-hour: no; certified three-day: −2.4% tail loss, narrow pass; alone: worse	Tables 3.25, 3.26
How large is the effect?	top vs bottom quintile ≈ 26 bp at fixed volatility	Section 3.11.17
Can equal early data hide different risk?	Yes, threefold; a computable capacity loss follows	Ex. 3.5.1, Prop. 3.5.2
What does a finite window certify?	An exact Markov–Krein interval; nothing without a memory bound	Thm 3.5.3
Certify risk vs identify traders?	$\sigma^2\epsilon^{-2}\log(1/\epsilon)$ vs $(\Delta\eta)^{-(4K-2)}$	Thm 3.5.5, Table 3.3
Optimal guard under uncertainty?	Closed form, interval, minimax capacity loss	Thm 3.6.5
If admission changes the book?	Safe set can be two intervals; certified by branch and bound	Prop 3.6.6, Thm 3.6.7
If tap noise is correlated?	Simple certificate undercovers (75%); t and Hotelling certificates exact	Prop 3.5.7
What is disclosure worth?	Most when memory is long; noisy proxies can hurt	Thm 3.7.1, Table 3.5
On untouched days?	Yes: frozen model +4.2 pts; guard −8% vs static, not vs volatility (pre-registered)	Table 3.29
On untouched symbols?	Yes, on five: +0.73 pts; +0.56 beyond VPIN; guard −23% vs static (pre-registered)	Table 3.30
On a new venue?	No: Coinbase, −0.09 pts, fails (pre-registered)	Table 3.31
Is it order-flow persistence?	No: a placebo with the same flow persistence adds nothing on Binance (post hoc)	Table 3.32
Against peak, participation and margin rules?	Better than certified peak, margin and participation rules on Part 15; not than six-hour peak (post hoc)	Table 3.32
Is leverage or OI liquidation the mechanism?	No: OI drops mediate −6.8%; follow-up points to slow reversion, placebo criterion fails	Tables 3.33, 3.36, 3.39
Do zero fees change reversion or impact?	Primary reversion/multiple null; spot impact −18% (exploratory, 1 treated pair)	Table 3.34
What is the guard worth in dollars and capacity?	Beats a static limit; vs. volatility cap −7.41 pp liq. rate (Part 24), mostly from lower exposure, none at matched exposure	Tables 3.35, 3.37, 3.38

3.1.1 Related literature

Closest work. Three strands are closest. Models of transient impact and resilience (Obizhaeva and Wang, 2013; Bouchaud et al., 2004, 2018) take the decay of impact as given; we derive when depletion, refill and momentum make a book overshoot, give cumulative impact in closed form, and ask who supplies the reversion that bounds it. Models of liquidity spirals (Brunnermeier and Pedersen, 2009; Shleifer and Vishny, 1992) explain how funding constraints amplify shocks; our amplification is a property of book dynamics and does not need funding constraints, and our guard is the pre-trade control that bounds the resulting loss. Sornette et al. (2023) show that non-normal interactions create transient bubbles; we give a necessary and sufficient pre-trade control for such transients and test the conditions for them in order-book data.

Market microstructure and resilience. Transient impact and resilience follow Obizhaeva and Wang (2013), Bouchaud et al. (2004) and Bouchaud et al. (2018); optimal execution with impact is in Almgren and Chriss (2001) and Cartea et al. (2015). Depth imbalance as a short-horizon price predictor is documented by Cont et al. (2014). Biais et al. (2005) survey the field; price impact as a measure of information and illiquidity goes back to Kyle (1985) and Amihud (2002). Inventory-based dealer models go back to Ho and Stoll (1981), and Avellaneda and Stoikov (2008) give the standard high-frequency market-making model. Empirically, resiliency, the speed at which a book refills after a liquidity shock, is measured by Biais et al. (1995), Degryse et al. (2005) and Large (2007), who estimates it with a self-exciting point process for order events; Foucault et al. (2005) show that it is set by the mix of patient and impatient traders and the tick size. Our reversion term m is the reduced-form counterpart of resiliency, and $S = \tilde{\kappa}/(m - q\tilde{\kappa})$ shows how it combines with impact and momentum: a book that is resilient in this sense can still carry a large cumulative response when momentum feedback absorbs most of its reversion. Our book model is a minimal two-state version of resilience dynamics, chosen so that the amplification condition is exact.

Feedback, fire sales and automated markets. Liquidity spirals and fire sales are in Shleifer and Vishny (1992) and Brunnermeier and Pedersen (2009); automated-market crashes in Kirilenko et al. (2017); and the market-design case for discrete auctions in Budish et al. (2015). Menkveld (2013) documents high-frequency market makers as the new intermediaries, and algorithmic trading has been found to improve liquidity and price discovery in normal times (Hendershott et al., 2011; Brogaard et al., 2014); our question is what happens on the path after a shock. Sornette et al. (2023) show that non-normal interactions can generate transient bubbles in unleveraged systems. We add an exact condition for when a book amplifies, a control that does not need the identity of the traders, and evidence on whether the condition holds in a large market.

Early warning and liquidity crashes. The best-known early-warning metric for flow-driven crashes is VPIN (Easley et al., 2012), whose forecasting content beyond volume and volatility is disputed (Andersen and Bondarenko, 2014). Equilibrium models explain crashes through endogenous liquidity withdrawal (Huang and Wang, 2009) and illiquidity contagion across assets (Cespa and Foucault, 2014). Our statistic differs in kind: it is a structural quantity, the summed response of the book to a unit of flow, with a closed form and a control-theoretic meaning, and we evaluate it out of sample against a full volatility-forecasting baseline. In a post-hoc comparison on the same test hours, $\hat{C}_3$ keeps about 73% of its gain once VPIN is added to the baseline (Section 3.11.10). Crypto markets are studied as a laboratory for microstructure and arbitrage by Makarov and Schoar (2020).

Non-normal dynamics and robust control. Transient growth of stable non-normal systems and the Kreiss matrix theorem are classical (Trefethen and Embree, 2005; Kreiss, 1962); the sharp en constant is due to Spijker (1991). Standard H_∞ control bounds frequency-domain gain under known coordinates. The time-domain worst case under bounded disturbances, the ℓ_1 norm of the impulse response, is the object of ℓ_1-optimal control (Vidyasagar, 1986; Dahleh and Pearson, 1987; Dahleh and Diaz-Bobillo, 1995), and our guard's worst-case identity is that classical fact. What we add is its use as an admission rule on a dealer's balance sheet, its invariance to the unobservable cohort coordinates and identification from prices, and its link to who supplies reversion in equilibrium. Cross-impact under no dynamic arbitrage is studied by Schneider and Lillo (2019).

Regulation. SEC Rule 15c3-5 requires broker-dealers with market access to enforce pre-trade controls on credit and capital (SEC, 2010). Our guard is a candidate pre-trade control with a provable maximality property.

Forecast evaluation. Our predictive tests use Clark and West (2007) and Diebold and Mariano (1995) with Newey and West (1987) standard errors and block resampling in the spirit of Künsch (1989), the volatility baseline follows Corsi (2009) and J.P. Morgan/Reuters (1996), local projections follow Jordà (2005), and we correct for multiple testing as in Holm (1979). Following the case for pre-registration in science (Nosek et al., 2018) and the concerns about multiple testing in financial economics (Harvey et al., 2016; Harvey, 2017), we pre-register specifications, count every evaluation of a held-out period, and report the failures with the successes.

3.1.2 Outline

Section 3.2 sets out the closed-loop market. Section 3.3 shows why spectral stability is not solvency. Section 3.4 derives the amplification condition. Section 3.6 constructs and analyzes the guard, and Section 3.8 asks who supplies reversion in equilibrium. Sections 3.9 and 3.10

report the BTCUSDT evidence, and Section 3.11 the thin-book program, its replications and its failures. Section 3.13 discusses policy, and Section 3.14 concludes.

3.2 The closed-loop market

3.2.1 Price formation

Time is discrete with bars of length Δ. Let p_t be the price, $p_t^* = p_{t-1}^* + \xi_t$ the unobserved fundamental and $x_t = p_t - p_t^*$ the pricing error. Competitive dealers post aggregate depth $D > 0$, so price impact is $\kappa = 1/D$, and fundamental arbitrage removes a fraction $\kappa\varphi_f$ of the error each period. With algorithmic demand u_t and liquidity noise, the pricing error follows

$$x_{t+1} = ax_t + \kappa u_t + w_{t+1}, \qquad a = 1 - \kappa\varphi_f \in [0, 1). \tag{3.1}$$

Algorithmic cohorts trade $u_t = \sum_i q_i y_{i,t}$ with exponential memory $y_{i,t+1} = (1-\eta_i)y_{i,t} + \eta_i x_{t+1}$, a reduced form that arises when an adaptive desk updates its belief by steady-state Kalman filtering or regularized online gradient descent and maximizes expected continuation return $\beta_i y_{i,t} u_{i,t}$ minus quadratic execution cost $\frac{1}{2}\lambda_{\mathrm{tc},i} u_{i,t}^2$ and inventory penalty $\frac{1}{2}\gamma_{\mathrm{inv},i} I_{i,t}^2$ ($I_{i,t} = (1 - \delta_{\mathrm{inv}})I_{i,t-1} + u_{i,t}$, $|I_{i,t}| \le \bar{I}_i$), yielding $u_{i,t}^* = q_i y_{i,t} - \psi_{\mathrm{inv},i} I_{i,t-1}$ with $q_i = \beta_i/(\lambda_{\mathrm{tc},i} + \gamma_{\mathrm{inv},i})$ and $\psi_{\mathrm{inv},i} = \gamma_{\mathrm{inv},i}(1 - \delta_{\mathrm{inv}})/(\lambda_{\mathrm{tc},i} + \gamma_{\mathrm{inv},i})$ (Chapter 2; this single-agent filtering-and-control rule is distinct from the joint dealer–trader equilibrium of Section 3.8). The baseline equations set $\psi_{\mathrm{inv},i} = 0$, the limit in which inventory holding penalties are small relative to execution costs ($\gamma_{\mathrm{inv},i} \ll \lambda_{\mathrm{tc},i}$) or inventory is reset at the bar horizon; Proposition 3.2.3 below relaxes this and gives the closed-form summed response when cohorts carry inventory across bars with $\psi_{\mathrm{inv},i} > 0$. Stacking $z_t = (x_t, y_{1,t}, \ldots, y_{K,t})^\top$ gives $z_{t+1} = Jz_t + bw_{t+1}$, where $b = (1, \eta_1, \ldots, \eta_K)^\top$ because cohorts update on the same period's price (a unit of flow enters through κb), and

$$J = \begin{pmatrix} a & \kappa q_1 & \cdots & \kappa q_K \\ \eta_1 a & 1 - \eta_1 + \eta_1 \kappa q_1 & \cdots & \eta_1 \kappa q_K \\ \vdots & & \ddots & \vdots \\ \eta_K a & \eta_K \kappa q_1 & \cdots & 1 - \eta_K + \eta_K \kappa q_K \end{pmatrix}. \tag{3.2}$$

3.2.2 Stability and the role of depth

Proposition 3.2.1 (Stability with one cohort)**.** *With one cohort (q, η) and $a = 1 - \kappa\varphi_f$, the market is Schur stable if and only if*

$$q < \varphi_f, \qquad \kappa < \frac{4 - 2\eta}{(2-\eta)\varphi_f - \eta q} \ \text{ when } (2-\eta)\varphi_f > \eta q, \qquad |1 - \kappa\varphi_f|(1-\eta) < 1. \tag{3.3}$$

Higher impact κ (thinner depth) lowers price persistence a; feedback is destabilizing only when cohort aggressiveness q reaches the arbitrage strength φ_f.

The condition $q < \varphi_f$ is the familiar statement that momentum demand must be weaker than value arbitrage. The upper bound on κ rules out oscillatory instability when the book is very thin.

3.2.3 What an observer can use

Chapter 2 shows that prices identify only the merged rate–weight measure $M = \sum_i q_i \eta_i \delta_{1-\eta_i}$ of the cohorts. A pre-trade control must therefore depend on the market only through objects that are the same for every cohort representation of M.

Proposition 3.2.2 (Input–output invariance)**.** *Two populations with the same merged measure generate the same transfer function $H(z) = \sum_k g_k z^{-k} = \kappa z/[z - a - \kappa G(z)]$, $G(z) = \sum_i q_i \eta_i z/(z - 1 + \eta_i)$, and the same impulse coefficients $g_k = \kappa\, c_{\text{price}}^\top J^k b$ for every $k \geq 0$. Under any invertible change of state coordinates $z \mapsto Tz$, the Euclidean gain $\|CT^{-1}(TJT^{-1})^k\|_2$ depends on T, whereas the price path*

$$\Delta p_{t+k} = \sum_{j \geq 1} g_{k-1+j} u_{t-j} + \sum_{m=0}^{k-1} g_{k-1-m} u_{t+m}$$

and its worst case over $\|u\|_\infty \leq \bar{u}$ depend only on $\{g_k\}$ and are invariant to both T and cohort splitting.

Numerically, splitting a cohort of 1,000 random economies into two with the same memory rate leaves every impulse coefficient unchanged up to rounding ($\max_k |g_k(\mathcal{E}^A) - g_k(\mathcal{E}^B)| \leq 2.5 \times 10^{-12}$ over 200 lags; `recompute_oct20.py`). Proposition 3.2.2 is the reason we build the guard on impulse coefficients and avoid state norms in latent coordinates. When a state norm is needed, we fix units by observable calm-market innovation standard deviations, which is what makes the spot statistic of Section 3.10 unit-free.

Proposition 3.2.3 (Summed response with carried cohort inventory)**.** *Suppose each cohort i carries inventory $I_{i,t} = (1 - \delta_{\text{inv}}) I_{i,t-1} + u_{i,t}$ with $\delta_{\text{inv}} \in (0, 1]$ and submits order flow $u_{i,t} = q_i y_{i,t} - \psi_i I_{i,t-1}$ with $q_i \geq 0$ and $\psi_i \geq 0$. Then:*

(i) The closed-loop transfer function from a flow impulse to price is $H_{\text{inv}}(z) = \kappa z/\big[z - a - \kappa G_{\text{inv}}(z)\big]$ with

$$G_{\text{inv}}(z) = \sum_{i=1}^{K} \frac{q_i \eta_i z(z - 1 + \delta_{\text{inv}})}{(z - 1 + \eta_i)(z - 1 + \delta_{\text{inv}} + \psi_i)},$$

which is invariant to splitting any cohort that shares (η_i, ψ_i). Whenever the augmented

state matrix is Schur stable, the signed cumulative price response is

$$S(\psi) = \sum_{t=0}^{\infty} h_t(\psi) = H_{\text{inv}}(1) = \frac{\kappa}{1 - a - \kappa\Phi_{\text{eff}}(\psi)}, \qquad \Phi_{\text{eff}}(\psi) \equiv \sum_{i=1}^{K} \frac{\delta_{\text{inv}}}{\delta_{\text{inv}} + \psi_i} q_i \le \Phi. \tag{3.4}$$

(ii) If $0 \le \psi_i \le 1 - \delta_{\text{inv}}$ *and* $\delta_{\text{inv}} \ge \eta_i$ *for each* i *and* $a + \kappa\Phi_{\text{eff}}(\psi) < 1$*, every open-loop feedback tap* $k_m(\psi)$ *and hence every closed-loop impulse coefficient* $h_t(\psi)$ *is nonnegative, so the worst-case* ℓ_1 *gain equals the signed sum,* $\|h(\psi)\|_1 = \sum_{t=0}^{\infty} |h_t(\psi)| = S(\psi)$*. When* ψ_i *is large enough that inventory payback reverses the sign of late taps,* $\|h(\psi)\|_1 > S(\psi)$*.*

Inventory mean-reversion $\psi_i > 0$ acts as an automatic stabilizer that attenuates each cohort's long-run feedback weight from q_i to $\frac{\delta_{\text{inv}}}{\delta_{\text{inv}}+\psi_i} q_i$ (across 500 random stable economies (3.4) matches the simulated state-space impulse sum to 1.78×10^{-15}, and in the monotone regime $\|h(\psi)\|_1 = S(\psi)$ to 1.78×10^{-15}, reducing S from 2.7778 at $\psi = 0$ to 2.5000 at the calibrated worked example; in the sign-reversal regime $\|h(\psi)\|_1 = 2.0960 > S(\psi) = 1.7241$; `recompute_p2_step2_theory.py`).

Proposition 3.2.4 (Quote cost of ambiguity)**.** *For a one-unit ticket with adverse mark* $Z_\theta \sim N(\mu_\theta, \sigma_\theta^2)$ *and a CARA dealer with risk aversion* γ*, let* $c_\theta = \mu_\theta + \gamma\sigma_\theta^2/2$*. Under competition with zero robust certainty equivalent, the informed quote is* c_{θ_0} *and the ambiguous quote* $\sup_\theta c_\theta$*; with inverse demand* $(V_0 - V)/b$ *and marginal cost* c*, raising an interior quote from* s_0 *by* Δ *costs* $b(s_0 - c)\Delta + \frac{b}{2}\Delta^2$ *in surplus.*

Ambiguity about the algorithms is costly only when it changes the worst relevant cost. A guard that bounds worst-case transient exposure can therefore narrow the ambiguity premium; whether it does in a real market is an empirical question we do not answer here.

3.2.4 Discussion of the assumptions

Linearity and small shocks. All dynamics are linear. This is the right approximation for the transient response to flow shocks of ordinary size, which is what a pre-trade control must bound one step ahead; it is not a model of regime changes such as exchange outages, where nonlinear liquidity withdrawal dominates.

One book, one asset. Sections 3.4–3.6 treat a single instrument; Proposition 3.3.2 shows how directed cross-impact enlarges transient gains in a network. Cross-margined portfolios need the multi-asset operator, which we leave to future work.

Bounded shocks. The guard bounds losses for shocks in a declared envelope. Choosing the envelope is a risk-management decision; a larger envelope trades admitted flow for safety. The theorems hold for any envelope, and the empirical work does not depend on it.

Table 3.2: Safe radius and Kreiss bounds at a fixed spectral radius

θ	$\rho(J_\theta)$	M_C (price row)	$\sup_t \|J_\theta^t\|$	$\mathcal{K}(J_\theta)$	r^*_{exact} $(B=1)$
0	0.8	1.00	1.00	1.00	1.000
1	0.8	1.36	1.37	1.06	0.735
5	0.8	6.03	6.04	3.43	0.166
20	0.8	24.01	24.01	13.42	0.042
100	0.8	120.00	120.00	66.99	0.008

$J_\theta = \left(\begin{smallmatrix} 0.8 & \theta \\ 0 & 0.4 \end{smallmatrix}\right)$; Kreiss constant by grid search over $|z| > 1$.

Observables. Every input to the guard (impulse coefficients, an operator estimate and its confidence radius, inventory and margin) is observable to a dealer or venue. Nothing requires the identities or memory parameters of the algorithms.

3.3 Spectral stability is not solvency

Theorem 3.3.1 (Safe radius, Kreiss bounds and the failure of the spectral radius)**.** *Let $z_t = J^t z_0$ with $\rho(J) < 1$, loss budget $B > 0$ and loss output C.*

(i) The largest initial-state radius with $\sup_t \|CJ^t z_0\|_2 \le B$ for every $\|z_0\|_2 \le r$ is

$$r^*_{\text{exact}} = \frac{B}{M_C}, \qquad M_C \equiv \sup_{t\ge 0} \|CJ^t\|_2. \tag{3.5}$$

(ii) With $C = I$ and the Kreiss constant $\mathcal{K}(J) = \sup_{|z|>1}(|z|-1)\|(zI-J)^{-1}\|_2$, the Kreiss matrix theorem gives $\mathcal{K}(J) \le \sup_t \|J^t\|_2 \le e\,n\,\mathcal{K}(J)$ for $J \in \mathbb{R}^{n\times n}$, hence

$$\frac{B}{e\,n\,\mathcal{K}(J)} \;\le\; r^*_{\text{exact}} \;\le\; \frac{B}{\mathcal{K}(J)}. \tag{3.6}$$

(iii) For any $\rho \in (0,1)$ and any $B > 0$ there is a family J_θ with $\rho(J_\theta) = \rho$ for all θ and an output $C = (1,0)$ such that $r^*_{\text{exact}}(J_\theta) \to 0$ as $\theta \to \infty$.

Table 3.2 illustrates the theorem for $J_\theta = \left(\begin{smallmatrix} 0.8 & \theta \\ 0 & 0.4 \end{smallmatrix}\right)$: the spectral radius is 0.8 throughout, while the price-row gain rises from 1 to 120 and the safe radius falls by the same factor. The Kreiss constant tracks the gain within the factor en.

The theorem concerns a specified initial-state ball and output norm. Changing state coordinates changes the disturbance set, so latent-state units cannot be read as monetary losses; this is why Section 3.6 works with input–output gains instead.

Spectral blindness. Chapter 2 shows more: for any eigenvalues $0 < \lambda_2 < \lambda_1 < 1$, one-step price variance and sufficiently large stationary price variance, there are two-state markets

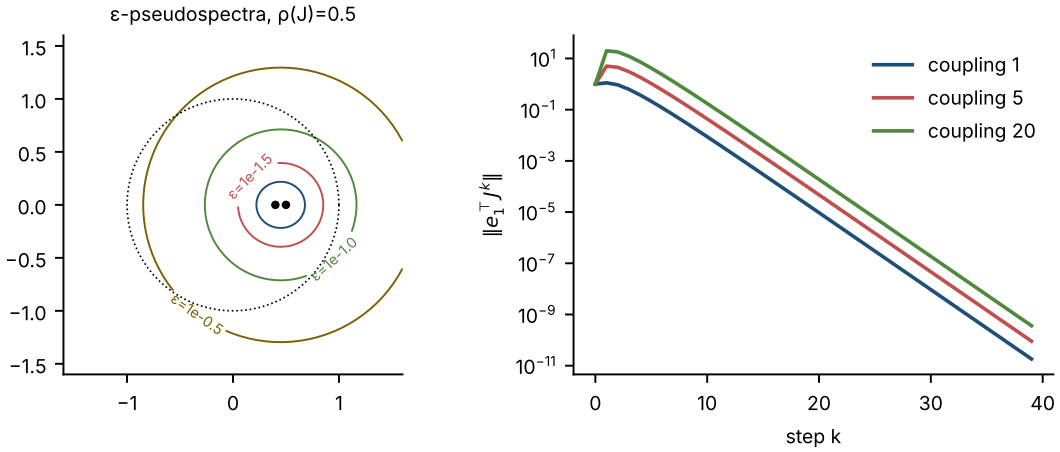


Figure 3.1: Left: ε-pseudospectra of $J = \left(\begin{smallmatrix} 0.5 & 5 \\ 0 & 0.4 \end{smallmatrix}\right)$; the eigenvalues (dots) are well inside the unit circle (dotted), but the $\varepsilon = 10^{-1}$ contour reaches beyond it. Right: price-row gain $\|e_1^\top J^k\|$ for couplings 1, 5 and 20.

sharing all three quantities whose worst-case transient gain is arbitrarily large (Chapter efch:p1, spectral-blindness theorem), so neither eigenvalue nor volatility monitors bound transient loss.

At $\lambda = (0.8, 0.7)$ with one-step variance 1 and stationary variance 5, the gain is 1.77, 5.10, 16.95, 50.85 and 169.50 as the off-diagonal coupling rises from 1 to 100. Figure 3.1 shows the geometry for a constructed matrix with spectral radius 0.5: its pseudospectra cross the unit circle, and the price-row gain grows with the coupling before decaying.

No amplification without depth. In the cohort economy (3.2) with nonnegative weights and $\rho(J) < 1$, the price impulse response is a nonnegative mixture of decaying exponentials with total weight κ; it is completely monotone and peaks at the direct impact (Chapter efch:p1, positive-geometry lemma).

This is a negative result with a positive use. The cohort matrix (3.2) is non-normal, yet its price output never amplifies. Non-normality is necessary for price amplification but not sufficient, and adaptive momentum feedback alone does not supply it. Section 3.4 shows that depth depletion does.

3.3.1 Multi-asset coupling

Market makers trade many correlated instruments. Let $x_t \in \mathbb{R}^d$ be pricing errors and $u_t \in \mathbb{R}^d$ flows, with $x_{t+1} = Ax_t + Ku_t + w_{t+1}$, $A = \mathrm{diag}(a_1, \ldots, a_d)$, a cross-impact matrix K, and momentum flow $y_{t+1} = (I - H)y_t + Hx_{t+1}$, $u_t = Qy_t$. The stacked Jacobian is

$$J_{\text{multi}} = \begin{pmatrix} A & KQ \\ HA & I - H + HKQ \end{pmatrix} \in \mathbb{R}^{2d \times 2d}. \tag{3.7}$$

Proposition 3.3.2 (Directed cross-impact shrinks the safe radius)**.** *Suppose a leader–satellite structure ($K_{12} = 0$, $K_{21} \geq 0$) with $A, Q, K \geq 0$ entrywise, $H \in (0,1)^d$, stable diagonal blocks and a fixed output $C \geq 0$. Let J_{diag} be the decoupled Jacobian with $K_{21} = 0$. Then every power J^t_{multi} is entrywise non-decreasing in K_{21}, so $M_C(J_{\text{multi}}) \geq M_C(J_{\text{diag}})$ and $r^*_{\text{exact}}(J_{\text{multi}}) \leq r^*_{\text{exact}}(J_{\text{diag}})$; likewise $\mathcal{K}(J_{\text{multi}}) \geq \mathcal{K}(J_{\text{diag}})$.*

The proposition concerns nonnegative downstream coupling under a fixed output and shock set. Hedged or signed monetary outputs need separate treatment, and the size of the effect requires an empirical calibration we do not provide. Persistence and transient amplification are nevertheless different things: in a network of identical cohort economies coupled by symmetric nonnegative cross-impact, which stays symmetric in suitable coordinates, cross-impact of 0.05 raises the spectral radius from 0.822 ($d = 1$) to 0.973 ($d = 10$) while the Kreiss constant stays at one (`recompute_oct20.py`). Directed coupling, as in Proposition 3.3.2, is what turns persistence into amplification.

3.4 When does a book amplify? An exact condition

Transient path risk takes two forms, and the distinction organizes both the theory and the evidence. The first is *peak overshoot*, a response that rises above its direct impact; Theorem 3.4.2 shows that it requires depth depletion to outweigh impact-weighted reversion. The second is *cumulative amplification*, a summed response S much larger than direct impact; Theorem 3.4.4 shows that it arises without any depletion whenever reversion is slow relative to impact and momentum. The guard of Section 3.6 bounds the second. The data of Sections 3.9–3.11 show no peak overshoot in crypto books, even though flow depletes the side it hits on a fixed price grid, and they support the second form in thin books.

3.4.1 A book with depletion and refill

Let x_t be the price displacement, d_t the signed depth imbalance (bid minus ask depth near the mid, relative to total) and u_t signed order flow. Flow moves the price directly and depletes one side of the book, which refills at rate r; depth imbalance moves the price:

$$x_t = a\, x_{t-1} + \varphi\, d_{t-1} + \kappa\, u_t, \tag{3.8}$$

$$d_t = (1 - r)\, d_{t-1} + \varepsilon\, u_t, \tag{3.9}$$

with $\kappa > 0$, $a \in [0, 1)$ and $r \in (0, 1]$. Write $b = 1 - r$. With the sign convention $d = (B - A)/(B + A)$, a buy that consumes ask depth raises d, so $\varepsilon > 0$ means *depletion* of the side that is hit, and $\varphi > 0$ means that a bid-heavy book pushes the price up. The impulse response

to a unit flow shock at $t = 0$ is $h_0 = \kappa$ and

$$h_k = a\, h_{k-1} + \varphi\varepsilon\, b^{k-1}, \qquad k \geq 1. \tag{3.10}$$

Definition 3.4.1. The book *amplifies* if $\max_k h_k > h_0$: the price keeps moving in the direction of the shock after the shock has passed.

3.4.2 Peak overshoot

Add momentum traders whose flow $f_t = (1-\eta)f_{t-1} + \eta q\, x_{t-1}$ enters both (3.8) and (3.9) alongside u_t; the two-state model is the case $q = 0$. Write $C_2 \equiv \varphi\varepsilon - (1-a)\kappa$ and $C_3 \equiv \varphi\varepsilon + \kappa^2\eta q - (1-a)\kappa$.

Theorem 3.4.2 (Peak overshoot: depletion versus momentum)**.** *(i)* Two-state condition. *Without momentum ($q = 0$) the book amplifies if and only if $C_2 > 0$; when $C_2 \leq 0$ the response peaks at impact, $\max_k h_k = \kappa$.*

(ii) Shape. *Let $q = 0$, $0 < a < 1$ and $0 < b < 1$. If $a \neq b$, $h_k = Aa^k + Bb^k$ with $B = \varphi\varepsilon/(b-a)$ and $A = \kappa - B$; if $a = b$, $h_k = a^{k-1}(a\kappa + \varphi\varepsilon k)$ for $k \geq 1$. When the book amplifies, h viewed as a function of real $k \geq 0$ has a single maximum, at*

$$k^* = \frac{\log\bigl(-A\log a/(B\log b)\bigr)}{\log(b/a)} \quad (a \neq b), \qquad k^* = \frac{1}{\log(1/a)} - \frac{a\kappa}{\varphi\varepsilon} \quad (a = b),$$

the second being the limit of the first as $b \to a$; the peak lag is $\lfloor k^* \rfloor$ or $\lceil k^* \rceil$ (and is 1 when $k^* < 1$, because $h_1 > h_0$). The cumulative response is $\sum_k h_k = \kappa/(1-a) + \varphi\varepsilon/[(1-a)r]$ in all cases, and faster refill lowers both the peak and the cumulative response. When $a = 0$ ($h_k = \varphi\varepsilon b^{k-1}$) or $r = 1$ ($h_k = a^{k-1}(a\kappa + \varphi\varepsilon)$) the response peaks at lag 1.

(iii) One-period margin with momentum. *$h_1 - h_0 = C_3$, so $C_3 > 0$ implies that the book amplifies.*

(iv) Momentum alone cannot amplify. *If $\varphi\varepsilon = 0$, $q \geq 0$ and the system is stable, then $h_k = c_1\lambda_1^k + c_2\lambda_2^k$ with residues $c_1, c_2 \geq 0$, $c_1 + c_2 = \kappa$ and $|\lambda_2| \leq \lambda_1 = \rho < 1$, so $h_k \leq \kappa\rho^k < \kappa$ for every $k \geq 1$; in particular $C_3 < 0$.*

(v) A sufficient condition for no overshoot. *For any signs of φ, ε and q, if*

$$\bar{C}_3 \equiv |\varphi\varepsilon|\Bigl(1 + \frac{\kappa|q|}{r}\Bigr) + \kappa^2|q| - (1-a)\kappa \;\leq\; 0,$$

then $\sup_k |h_k| = \kappa$. When $\varphi\varepsilon \geq 0$ and $q \geq 0$, $C_3 > 0 \Rightarrow$ amplification $\Rightarrow \bar{C}_3 > 0$.

Proof. (i) *If.* $h_1 - h_0 = a\kappa + \varphi\varepsilon - \kappa = C_2 > 0$. *Only if.* Suppose $C_2 \leq 0$. If $\varphi\varepsilon < 0$, then by (3.10) $h_k = a^k\kappa + \varphi\varepsilon\sum_{j<k} a^{k-1-j}b^j \leq a^k\kappa \leq \kappa$. If $\varphi\varepsilon \geq 0$, let $\Delta_k = h_k - h_{k-1}$. Then

$\Delta_1 = C_2 \le 0$ and, from (3.10), $\Delta_{k+1} = a\Delta_k + \varphi\varepsilon(b^k - b^{k-1}) = a\Delta_k - \varphi\varepsilon\, r\, b^{k-1} \le a\Delta_k$, so by induction $\Delta_k \le 0$ for all k and h peaks at $h_0 = \kappa$.

(ii) Solving (3.10) gives the two forms. For $a \neq b$, $\frac{d}{dk}h = Aa^k \log a + Bb^k \log b$ has at most one zero because $(b/a)^k$ is monotone, which gives k^*; for $a = b$, $\frac{d}{dk}a^{k-1}(a\kappa + \varphi\varepsilon k) = a^{k-1}[(a\kappa + \varphi\varepsilon k)\log a + \varphi\varepsilon]$ vanishes once, at the stated k^*, and expanding the first formula in $b - a$ gives the same limit. A unimodal function of real k attains its integer maximum at $\lfloor k^* \rfloor$ or $\lceil k^* \rceil$. Summing the geometric series gives the cumulative response, which falls in r; the peak falls in r because h_k is decreasing in r term by term for $\varphi\varepsilon > 0$.

(iii) With a unit shock at $t = 0$, $x_0 = \kappa$, $d_0 = \varepsilon$ and $f_1 = \eta q \kappa$, so $h_1 = a\kappa + \varphi\varepsilon + \kappa^2 \eta q$.

(iv) Without the depth channel the state (x_t, f_t) evolves by $M = \left(\begin{smallmatrix} a+\kappa\eta q & \kappa(1-\eta) \\ \eta q & 1-\eta \end{smallmatrix}\right)$, which is entrywise nonnegative, and $h_k = \kappa(M^k)_{11}$. The off-diagonal product is nonnegative, so the eigenvalues are real, and by the Perron–Frobenius theorem $\lambda_2 \le m_{11} \le \lambda_1 = \rho$. Writing $(M^k)_{11} = c_1'\lambda_1^k + c_2'\lambda_2^k$ with $c_1' = (m_{11} - \lambda_2)/(\lambda_1 - \lambda_2) \in [0, 1]$ and $c_2' = 1 - c_1'$ gives the residues $c_j = \kappa c_j' \ge 0$ and $(M^k)_{11} \le \rho^k$. Since $m_{11} \le \rho < 1$, $C_3 = \kappa(m_{11} - 1) < 0$.

(v) Suppose $|x_j| \le \kappa$ for $j < k$. Then $|f_j| \le |q|\kappa$ for $j \le k$ and $|d_{k-1}| \le |\varepsilon|\big(b^{k-1} + \sum_{j\ge 1} b^{k-1-j}|q|\kappa\big) \le |\varepsilon|(1 + |q|\kappa/r)$, so $|x_k| \le a\kappa + |\varphi\varepsilon|(1 + \kappa|q|/r) + \kappa^2|q| \le \kappa$. Induction from $x_0 = \kappa$ completes the proof. □

Interpretation. The left side $\varphi\varepsilon$ of C_2 is the extra price pressure that a unit of flow creates one period later through the book it leaves behind. The right side $(1 - a)\kappa$ is the part of the direct impact that reverts in one period. The book amplifies exactly when the depth channel adds more than mean reversion removes. Refill r does not enter the condition; it governs how long an overshoot lasts.

We verify part (i) in 1,995 random parameter draws and the refill comparative static in 565 amplifying draws, with no exceptions. The peak lag lies in the stated bracket in all of 150,039 amplifying draws with $a \neq b$ and all of 20,000 confluent draws; it is not always the integer nearest to k^*, which an earlier version claimed and which fails in 15% of draws (`recompute_audit.py`, `hardening/p2_sec4_checks.py`). Figure 3.2 illustrates the four cases.

With two slow modes the converse of part (iii) fails, and no condition on the one-period margin is exact.

Remark 3.4.3 (Counterexamples to an exact one-period condition). Between C_3 and $\bar{C}_3$ the sign of C_3 does not decide. With $\kappa = 1$, $a = 0.98526$, $\varphi = 0.7409$, $\varepsilon = 0.01183$, $r = 0.03436$, $\eta = 0.5092$ and $q = 0.01127$ (all channels reinforcing, $\rho = 0.9995$), $C_3 = -0.00024$ but the response peaks at lag 60 at 1.092κ. With replenishment ($\varphi = 2.614$, $\varepsilon = -0.405$, $a = 0.9426$, $r = 0.033$, $\eta = 0.0016$, $q = 0.0785$, $\rho = 0.998$), $C_3 = -1.11$ yet the response oscillates to 9.2κ at lag 90. In a search over 400,000 parameter draws such cases occur only near the stability boundary or with slow refill and slow momentum.

The spectral reason is visible in the first example. Without depth, part (iv) makes every

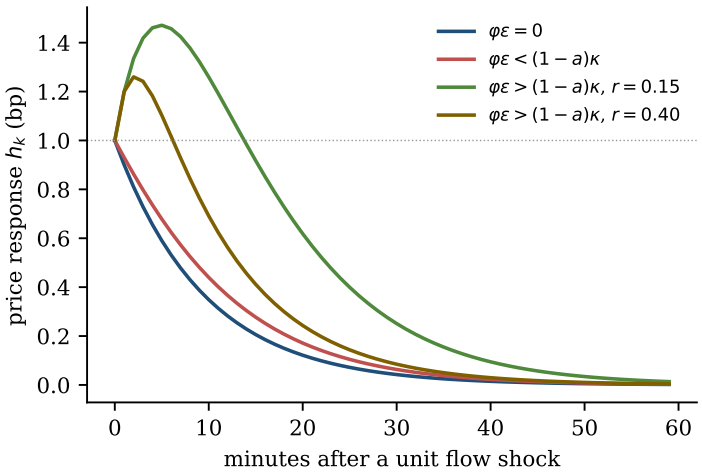


Figure 3.2: Illustrative impulse responses of model (3.8)–(3.9) ($a = 0.9$, $\kappa = 1$, $\varepsilon = 0.1$). The response overshoots only when $\varphi\varepsilon > (1-a)\kappa$; faster refill r lowers the overshoot. Parameters are illustrative, not estimates.

residue nonnegative, so the response is a nonnegative mixture of decaying modes and cannot rise above κ; we found no negative residue in 10,694 stable momentum-only draws. The depth channel breaks this: the three-state response is $h_k = 1.145\,(0.9995)^k - 0.155\,(0.9626)^k + 0.011\,(0.4853)^k$, with a negative residue on the intermediate mode. That mode subtracts from the response early and dies out faster than the slowest one, so the response first dips ($h_1 < h_0$, hence $C_3 < 0$) and then rises toward the slow mode until lag 60.

3.4.3 Cumulative transient risk

Theorem 3.4.4 (Summed response in closed form)**.** *Let $\tilde{\kappa} = \kappa + \varphi\varepsilon/r$ (long-run impact: direct plus depth) and $m = 1 - a$ (reversion). In the stable three-state model,*

$$S \equiv \sum_{k\geq 0} h_k = \frac{\tilde{\kappa}}{m - q\tilde{\kappa}},$$

and when $\varphi\varepsilon \geq 0$ and $q \geq 0$ the system is stable only if $m > q\tilde{\kappa}$. When all $h_k \geq 0$, S is the summed absolute response that enters the guard of Section 3.6.

Proof. $S = e_1^\top (I - M_3)^{-1}(\kappa, \varepsilon, 0)^\top$ with M_3 the transition matrix of (x, d, f). Direct computation gives $\det(I - M_3) = \eta\,[\,r(m - \kappa q) - \varphi\varepsilon q\,]$ and the stated ratio; the formula matches the simulated sums to a relative error of 7×10^{-14} in 3,000 stable draws. If $\varphi\varepsilon \geq 0$ and $q \geq 0$, M_3 is entrywise nonnegative and stability requires $\det(I - M_3) > 0$, i.e. $m > q\tilde{\kappa}$. □

Cumulative transient risk is long-run impact divided by reversion net of momentum. Two practical consequences follow. First, the statistic that matters for losses is the summed response,

which grows without bound as reversion falls to $q\tilde{\kappa}$, even when no single lag overshoots (Theorem 3.4.2(iv)). Second, C_3 is best read as a one-period amplification margin: it moves with m, q and $\tilde{\kappa}$ in the same direction as S and is what we estimate in Sections 3.9 and 3.11. Both statistics are invariant in sign to rescaling flow.

3.5 Identification and financial safety

Chapter 2 shows that identifying the cohorts of algorithms from prices is expensive: separating K cohorts whose memories differ by $\Delta\eta$ takes a number of experiments of order $(\Delta\eta)^{-(4K-2)}$, and with general memory accuracy improves only as $1/\log N$ (Chapter 2). A pre-trade control does not need the cohorts; it needs a bound on the risk they create. This section asks what information certifies a safe trading decision, what that information costs, and how much safe trading must be given up when it is missing.

3.5.1 Observation and risk contracts

We work in the positive cohort economy (3.2) with $q_i \geq 0$ and known a and κ. Its merged measure $M = \sum_i q_i \eta_i \delta_{r_i}$, $r_i = 1 - \eta_i$, has kernel $k_m = \int r^m \, dM$, and the price response to a unit of flow obeys $h_0 = \kappa$, $h_{t+1} = a h_t + \kappa \sum_{m \leq t} k_m h_{t-m}$. All $h_t \geq 0$ (Chapter 2, positive-geometry lemma), and the summed response is

$$S = \sum_t h_t = \frac{\kappa}{1 - a - \kappa\Phi}, \qquad \Phi = \int \frac{dM(r)}{1 - r} = \sum_i q_i, \tag{3.11}$$

with stability exactly when $a + \kappa\Phi < 1$. A *memory bound* $\bar{r} < 1$ states that every memory rate satisfies $r_i \leq \bar{r}$.

Risk contract. A position I is held fixed, each period's flow shock is bounded by $\bar{u}$, losses are marked to market at every date, and the horizon is unbounded. By Lemma 3.6.1 the worst-case loss is $|I|\bar{u}S$, so the safe capacity, the largest shock bound with loss at most the budget $\mathcal{B}$, is $\mathcal{B}/(|I|S)$. Section 3.6 adds admission, margin and fills. All statements below are deterministic worst cases under this contract; none is a probability-of-ruin statement.

Observation contracts. We consider four: (O1) N independent reset experiments, each returning $h_0, \ldots, h_{T-1}$ with independent Gaussian errors of standard deviation σ per tap, the controlled-probe contract of Chapter 2; (O2) an exact window $h_0, \ldots, h_n$; (O3) class-tagged flows; and (O4) audited exposure, which reveals Φ. The theorems concern these contracts. Observational public records of a live book are a different contract, for which we offer evidence (Section 3.12.2) but no finite-sample guarantee.

Given data D, let $\Theta(D)$ be the set of economies compatible with it. A decision is *certifiably safe* if its worst-case loss is within budget for every economy in $\Theta(D)$, *certifiably unsafe* if it exceeds the budget for every one, and *undecidable* otherwise. This classification is a definition, not

a result; it is the finance version of the data-informativity view of control without identification (van Waarde et al., 2020) and of set-membership estimation (Milanese and Vicino, 1991). The results below make it explicit for this economy.

3.5.2 Two markets that look the same

Example 3.5.1 (An exact ambiguity witness). Let $a = 0.4$ and $\kappa = 1$. Market A has one cohort ($q = 0.08$, $\eta = 0.5$); market B has two ($q = 0.02/0.95$, $\eta = 0.95$; $q = 0.4$, $\eta = 0.05$). Their merged measures have the same mass and first moment, so in rational arithmetic their price responses coincide at lags 0, 1 and 2 (1, 11/25, 267/1250) and first differ at lag 3. Their summed responses are 1.92 and 5.59, so for a unit position and budget the safe shock bound is 0.52 in A and 0.18 in B.

Proposition 3.5.2 (Unavoidable loss of safe capacity)**.** *Let two economies have safe capacities $v_0 < v_1$ and data laws P_0, P_1. If a data-driven rule $A(D) \in [0, v_1]$ admits an unsafe level in economy 0 with probability at most β, $P_0(A > v_0) \le \beta$, then*

$$\mathbb{E}_1[v_1 - A(D)] \;\ge\; (v_1 - v_0)\,\big[1 - \beta - \mathrm{TV}(P_0, P_1)\big]_+.$$

The bound is attained when $P_0 = P_1$ by the rule that admits v_1 with probability β and v_0 otherwise.

Proof. Let $E = \{A \le v_0\}$. Then $P_0(E) \ge 1 - \beta$ and $P_1(E) \ge P_0(E) - \mathrm{TV}$. On E the capacity given up in economy 1 is at least $v_1 - v_0$, and it is nonnegative elsewhere. □

The argument is the two-point method of statistical decision theory (Le Cam, 1986); its use here is to price missing information in units of safe trading. In Example 3.5.1, a supervisor who sees the first three coefficients and must admit an unsafe level in B with probability at most 5% gives up at least $0.95 \times 0.34 = 0.32$ units of safe shock capacity in A.

3.5.3 What a finite window certifies

Lags 1 to n of the response determine the first n moments of M exactly, through $k_t = (h_{t+1} - a h_t - \kappa \sum_{m<t} k_m h_{t-m})/\kappa^2$. The summed response depends on M only through $\Phi = \int f\, dM$ with $f(r) = (1 - r)^{-1}$. Certifying it from a window is therefore an extremal moment problem.

Theorem 3.5.3 (Finite-window certification)**.** *Observe $h_0, \dots, h_n$ exactly under a memory bound $\bar{r} < 1$, and suppose the implied moment vector $(k_0, \dots, k_{n-1})$ lies in the interior of the moment space $\mathcal{M}_n([0, \bar{r}])$ (equivalently, the Hankel matrices of the moment sequence and of its localization by $r(\bar{r} - r)$ are positive definite). Let μ_- and μ_+ be its lower and upper principal representations on $[0, \bar{r}]$ (the extremal measures of the Markov–Krein theorem, unique in the interior; μ_+ has an atom at $\bar{r}$), and $\Phi_\pm = \int f\, d\mu_\pm$. On the boundary of the moment space the compatible measure is unique and $\Phi_- = \Phi_+$.*

(i) Sharp interval. *The summed responses of stable economies compatible with the data form the interval* $[S(\Phi_-), S(\Phi_+)]$ *if* $a + \kappa\Phi_+ < 1$, *and* $[S(\Phi_-), \infty)$ *otherwise. Both endpoints are attained by economies with at most* $\lfloor n/2 \rfloor + 1$ *cohorts.*

(ii) Geometric convergence. $\Phi_+ - \Phi_- \le 2k_0 E_{n-1}$, *where* $E_{n-1} = (2/\bar{r})\,\varrho^{n-1}/(x_0^2 - 1)$ *is the error of the best uniform approximation of* f *on* $[0, \bar{r}]$ *by polynomials of degree* $n-1$, $x_0 = 2/\bar{r} - 1$ *and* $\varrho = x_0 - \sqrt{x_0^2 - 1} < 1$. *A window of length* $n = O\big(\log(1/\epsilon)/\log(1/\varrho)\big)$ *certifies* Φ *to precision* ϵ, *and* $\log(1/\varrho) \sim 2\sqrt{(1-\bar{r})/\bar{r}}$ *as* $\bar{r} \to 1$.

(iii) No certificate without a memory bound. *If memory rates may lie anywhere in* $[0, 1)$, *then for every* n *and every window whose moment vector is interior to the moment space, compatible stable economies have arbitrarily large summed responses: no finite window certifies any finite risk bound.*

Part (i) is an application of the Markov–Krein theorem (Krein and Nudel'man, 1977; Karlin and Studden, 1966): since $f^{(n)}(r) = n!/(1-r)^{n+1} > 0$, the extremes of $\int f\,d\mu$ over measures with given moments are attained at the principal representations. Part (ii) uses Chebyshev's formula for the best approximation of $1/(x_0 - x)$ (Achieser, 1956). The proofs are in Appendix 3.15. Theorem 3.5.3 turns the three-way classification into a computation: a decision is certifiably safe from the window when $|I|\bar{u}S(\Phi_+) \le \mathcal{B}$, certifiably unsafe when $|I|\bar{u}S(\Phi_-) > \mathcal{B}$, and undecidable in between. In an eight-cohort economy with memory bound 0.95 and true summed response 5.56, the window of lags 0–3 certifies nothing ($a + \kappa\Phi_+ \ge 1$), lags 0–4 certify $S \le 92.8$, lags 0–6 give $[5.01, 7.20]$ and lags 0–8 give $[5.30, 6.00]$; the widths stay inside the bound of part (ii) at every window. For fixed data the upper bound grows without limit as the memory bound approaches one ($\Phi_+ = 2.42$, 9.45, 79.7 and 782 for $\bar{r} = 0.9$, 0.99, 0.999 and 0.9999, from the first four moments of a three-cohort economy), as part (iii) says (`research_extension/information_safety`).

The endpoints in these examples are computed, not proved, in floating point. The next result makes them proved.

Lemma 3.5.4 (Certified endpoints)**.** *Let* p_+ *and* p_- *be polynomials of degree at most* $n-1$, $p_\pm(r) = \sum_j c_j^\pm r^j$, *with* $p_- \le f \le p_+$ *on* $[0, \bar{r}]$. *Then every economy compatible with the window has* $\sum_j c_j^- k_j \le \Phi \le \sum_j c_j^+ k_j$. *The Hermite interpolants of* f *at the atoms of* μ_+ *and* μ_-, *with derivative conditions at interior atoms, satisfy the sign conditions and give* Φ_+ *and* Φ_- *exactly.*

Lemma 3.5.4 turns any computed pair of polynomials into a proof, whatever the rounding errors that produced them: if the sign conditions are verified, the bounds hold. We compute the interpolants in 60-digit arithmetic, round them to double precision, add a slack of 10^{-11} of Φ, verify the sign conditions on $[0, \bar{r}]$ in interval arithmetic with outward rounding (a mean-value form on an adaptive subdivision; Rump, 2010), and evaluate the bounds with outward rounding. In the eight-cohort example the certified intervals for S are $[5.01, 7.20]$ and $[5.30, 6.00]$ for lags 0–6 and 0–8, and every certified endpoint lies outside the 60-digit one by at most 1.5×10^{-8} of

its value, using about 62,000 subintervals (test T10; the exact count depends on the platform's floating-point library, the certified bounds do not). The interval endpoints quoted in this chapter are therefore results, not approximations.

3.5.4 What certification costs, and what identification costs

Theorem 3.5.5 (Certification at the parametric rate)**.** *Consider the class $\mathcal{K}(\bar{r}, \bar{s})$ of positive economies with memory bound $\bar{r}$ and $a + \kappa\Phi \le \bar{s} < 1$, observed under contract (O1). Let $\bar{\lambda} = 1 - (1-\bar{r})(1-\bar{s})$ and $\bar{h}_t$ the average of the N responses.*

(i) A valid certificate. *For every economy in the class, $h_t \le \kappa\bar{\lambda}^t$, and*

$$\hat{S}_T^{+} = \sum_{t<T} \bar{h}_t + z_{1-\gamma}\,\sigma\sqrt{T/N} + \frac{\kappa\bar{\lambda}^T}{1-\bar{\lambda}}$$

satisfies $P(S \le \hat{S}_T^{+}) \ge 1-\gamma$. With $T_\epsilon = \lceil \log(2\kappa/((1-\bar{\lambda})\epsilon))/\log(1/\bar{\lambda}) \rceil$ and $N \ge 16 z_{1-\gamma}^2 \sigma^2 T_\epsilon/\epsilon^2$, the excess $\hat{S}^{+} - S$ is at most ϵ with probability at least $1-2\gamma$. Certification to precision ϵ therefore costs $N = O(\sigma^2 \epsilon^{-2} \log(1/\epsilon))$.

(ii) A matching lower bound. *Fix an interior single-cohort economy θ_0 and let $C_0 = \|\partial h/\partial q\|_2/(\partial S/\partial q)$ at θ_0. Any procedure that is valid at level $1-\gamma$ on the class and overshoots S by less than 2ϵ with probability at least $1/2$ at θ_0 needs $N \ge \big((1-2\gamma)\sigma/(2C_0\epsilon)\big)^2(1+o(1))$ as $\epsilon \to 0$.*

(iii) Independence of the cohort structure. *Neither bound depends on the number of cohorts or on the spacing of their memories, whereas identifying them costs $N \asymp \sigma^2(\Delta\eta)^{-(4K-2)}$ with exponential memory and $\exp(\Theta(1/w))$ experiments for accuracy w with general memory (Chapter 2).*

The certificate in part (i) is elementary: it is a sum of averaged taps, a Gaussian margin and a tail that the memory bound controls. The content is in the comparison. In contrast to the $\mathcal{H}_\infty$ norm, which Tu et al. (2018) show is no easier to estimate than the model itself, the summed response of a positive economy is a linear functional of the response, so certifying it is a parametric problem, while the cohort parameters are not. Table 3.3 makes this concrete on Chapter 2's hardest two-cohort pairs. Those pairs have *identical* summed responses, because they share the total weight $\sum_i q_i$: the directions in which prices are least informative are directions in which the risk a guard must bound does not move. Identifying their weights to ± 0.01 needs from 7 to 1.5 million reset experiments as the spacing falls from 0.30 to 0.04 (log–log slope -6.06), while certifying the summed response to 1% needs 11 to 18 at every spacing (slope $+0.23$).

Remark 3.5.6 (Shorter windows cost more experiments). Under exact windows the polynomial bounds of Lemma 3.5.4 certify Φ to precision ϵ with $n = O\big((1-\bar{r})^{-1/2}\log(1/\epsilon)\big)$ taps

Table 3.3: Identifying the cohorts versus certifying the risk they create

Rate spacing $\Delta\eta$	0.30	0.20	0.14	0.10	0.08	0.06	0.05	0.04
Identify weights (± 0.01), N	7	88	775	5,928	22,741	128,333	383,859	1,466,362
Certify S (1%), N	18	14	13	12	11	11	11	11

The two-cohort moment-matched pairs of Chapter 2 ($q = 0.2$, mean rate 0.4, $\tau = 0.01$, $a = 0.5$, $\kappa = 0.8$, per-tap noise 1.6×10^{-3}). Identification: Le Cam requirement at 95% (Chapter 2, Table on two-point sample complexity). Certification: sufficient N from Theorem 3.5.5(i) with memory bound equal to the largest rate in the pair and $\bar{s} = a + \kappa q + 0.02$. The two economies of each pair have the same summed response, 2.353. Kind: recomputed (`research_extension/information_safety`, test T03).

(Theorem 3.5.3(ii)), whereas the tail bound of Theorem 3.5.5(i) needs $T_\epsilon = \Theta\big(((1-\bar{r})(1-\bar{s}))^{-1}\log(1/\epsilon)\big)$. With noisy taps the shorter window is not a better certificate. The bound $\sum_j c_j^+ \hat{k}_j$ is linear in the estimated kernel values, so its standard error is $\sigma N^{-1/2}\|\mathcal{J}^\top c^+\|_2$, where $\mathcal{J}$ is the Jacobian of the map from taps to kernel values; the monomial coefficients of an approximating polynomial on $[0, \bar{r}]$ grow geometrically in its degree. In a three-cohort economy with memory bound 0.99 ($S = 3.28$), certifying S to 1% takes 45 taps with the window bound against 4,455 with the tail bound, but $3.5 \times 10^{65}\sigma^2$ experiments against $1.8 \times 10^{8}\sigma^2$; at 0.1%, 56 against 5,482 taps and $3 \times 10^{82}\sigma^2$ against $2.2 \times 10^{10}\sigma^2$ experiments (`hardening/p2_cheb_tradeoff.py`). Power-moment problems are exponentially ill-conditioned, so the parametric rate of Theorem 3.5.5 is bought with a long, well-conditioned horizon; the window bound is the right tool for exact or near-exact data.

Unknown noise covariance. Theorem 3.5.5(i) assumes independent taps with a known noise level. Reset experiments need not deliver that: the noise in the taps of one experiment can be correlated. Let contract (O1′) be (O1) with Gaussian noise of unknown covariance Σ across the taps of an experiment, experiments still independent, and let $X_e = \sum_{t<T} h_t^{(e)}$ for experiment e.

Proposition 3.5.7 (Certification with unknown covariance)**.** *Under (O1′), let $\bar{X}$ and s_X be the mean and standard deviation of $X_1, \dots, X_N$ and $\bar{h}$, $\hat{\Sigma}$ the mean and covariance of the N windows.*

(i) $\hat{S}_t^+ = \bar{X} + t_{N-1,1-\gamma}\, s_X/\sqrt{N} + \kappa\bar{\lambda}^T/(1-\bar{\lambda})$ *satisfies* $P(S \le \hat{S}_t^+) \ge 1-\gamma$ *for every* Σ.

(ii) If $N > T$, the Hotelling region $\{m : N(\bar{h}-m)^\top\hat{\Sigma}^{-1}(\bar{h}-m) \le T_\gamma^2\}$, $T_\gamma^2 = \frac{T(N-1)}{N-T}F_{T,N-T}(1-\gamma)$, contains the mean window with probability exactly $1-\gamma$, so the bounds $c^\top\bar{h} + \sqrt{T_\gamma^2\, c^\top\hat{\Sigma}c/N}$ hold simultaneously for every linear functional c: every horizon, every exposure profile and every guard computed from the same experiments.

(iii) The certificate of Theorem 3.5.5(i) can cover with probability well below $1-\gamma$ when the taps are positively correlated.

Part (i) is Student's t applied to the sums, and (ii) is Hotelling's theorem (Hotelling, 1931); the content is which assumption each certificate needs. In simulations with twelve taps and

nominal 95% coverage (test T11), positively correlated taps (autocorrelation 0.8) cut the coverage of the (O1) certificate to 74–76%, while the t-certificate covers 94–95% and the Hotelling bound 100%, at the cost of a margin three to six times wider. Heavy-tailed noise (t_3) leaves the t-certificate at 95%. Dependence *across* experiments (autocorrelation 0.5) cuts it to 82–85%: independence of the experiments is what a controlled contract buys and what observational data lack, which is why Sections 3.12.2 and 3.12.3 report bootstrap bounds as diagnostics rather than certificates. Experiments can also be run one at a time and stopped once the data decide; an anytime-valid version of the certificate keeps its error level under such stopping and needs far fewer experiments when the economy is far from the threshold (Internet Appendix, Proposition 3.17.1).

3.6 An identification-free pre-trade guard

3.6.1 The quantity gate

Observable dynamics are $z_{j+1} = Jz_j + Hv_j$ with loss output $l_t = Cz_t$, where v_j is the executed quantity. Let $b_t = CJ^t z_0$ be the drift from past flow and $g_m = c_{\text{price}}^\top J^m H$ the impulse coefficients. Which gain bounds the loss depends on the quantity contract. If quantities share a total budget over the window, $\sum_{j<t} |v_j| \le V$, the worst case of $|\sum_{m<t} g_m v_{t-1-m}|$ is $V \max_{m<t} |g_m| \equiv V g_t^*$, attained by placing the whole budget at the lag of the peak, and execution costs at most $c_{\text{exec}} V$. Under a reserve L_{other} the sharp allowance is then

$$V^* = \min_{t:\, g_t^* + c_{\text{exec}} > 0} \frac{B - L_{\text{other}} - |b_t|}{g_t^* + c_{\text{exec}}}, \qquad \bar{R} = V^*/\tau \tag{3.12}$$

over a window of τ seconds. If instead each period's quantity is bounded, $|v_j| \le V$, the worst case is $V \sum_{m<t} |g_m|$ (attained by $v_{t-1-m} = V \operatorname{sign}(g_m)$) and execution costs at most $c_{\text{exec}} V t$, so the peak gain in (3.12) must be replaced by the summed gain $\sum_{m<t} |g_m|$ and c_{exec} by $c_{\text{exec}} t$. Pairing a per-period bound with the peak gain understates the worst case, which is the failure Lemma 3.6.1 formalizes. Either gate needs only impulse coefficients and the current drift.

3.6.2 Maximality

A dealer with inventory I chooses an admission level $\alpha \in [0, 1]$ (the fraction of incoming algorithmic flow it accepts, or the depth it quotes). The closed-loop operator is $J(\alpha) = J_0 + \alpha B_F$ with price impulse response $g_k(\alpha)$. Shocks satisfy $|u_t| \le \bar{u}$. Over horizon H under a reset initial state $z_0 = 0$ (or after subtracting a known or directly observed initial drift $b_t = CJ(\alpha)^t z_0$ from the remaining budget), the shock-driven price change is $\sum_{k<H} g_k(\alpha) u_{t-k}$, and the dealer has a loss budget $\mathcal{B}$. The contract is: inventory $I(\alpha)$ is held fixed over the horizon, the initial state is reset ($z_0 = 0$) or its drift is observed, each period's shock is bounded by $\bar{u}$, and loss is the mark-to-market loss at any date in the horizon; because the partial sums $\sum_{k<h} |g_k|$ increase in

h, the worst date is the last. Margin, hedging, fills and non-zero initial states are analysed in Theorem 3.6.8.

Lemma 3.6.1 (Summed-gain guard is maximal)**.** *The worst-case loss over horizon H is*

$$L_{\max}(\alpha) = |I(\alpha)|\,\bar{u} \sum_{k<H} |g_k(\alpha)|.$$

Hence the guard "admit α iff $L_{\max}(\alpha) \le \mathcal{B}$" is necessary and sufficient for bounded-shock solvency: it admits every safe α and no unsafe one. Replacing $\sum_k |g_k|$ by $\max_k |g_k|$ can admit unsafe levels; replacing it by $H \max_k |g_k|$ is safe but conservative.

Proof. By Hölder's inequality $|\sum_k g_k u_{t-k}| \le \bar{u} \sum_k |g_k|$, with equality for $u_{t-k} = \bar{u}\,\mathrm{sign}(g_k)$, which is admissible. Multiplying by $|I(\alpha)|$ gives $L_{\max}$, so the admission set equals the safe set. Since $\max_k |g_k| \le \sum_k |g_k| \le H \max_k |g_k|$, the peak guard under-states loss and the H-scaled guard over-states it. □

The worst-case identity in Lemma 3.6.1 is not new as mathematics: the induced ℓ_∞ gain of a linear convolution operator is the ℓ_1 norm of its kernel, the fact on which ℓ_1-optimal control of persistent bounded disturbances is built (Vidyasagar, 1986; Dahleh and Pearson, 1987; Dahleh and Diaz-Bobillo, 1995). We use it rather than claim it. The finance content lies elsewhere: the dealer's admission level changes the operator itself, so the guard is a rule over an action set rather than a controller design; the bound is carried through margin, hedging and fill accounting (Theorem 3.6.8); its single input is invariant to how the unobserved traders are split into cohorts and is identified from prices (Proposition 3.2.2 and Theorem 3.5.5); the peak-response and eigenvalue rules it is compared with are the ones closest to current supervisory practice; and the same statistic prices the corrective rebate in equilibrium (Proposition 3.8.1).

Numerically, the guard coincides with a brute-force adversary's maximal safe α in 200 of 200 random states. The peak-gain guard admitted an unsafe level in 2,000 of 2,000 states, the H-scaled guard was conservative in all 2,000, and $L_{\max}$ was non-decreasing in α in all 2,000, so the admission set was an interval in these examples.

3.6.3 Computability: a finite-horizon certificate

Lemma 3.6.2 (Contraction-block tail bound)**.** *Let $\rho(J(\alpha)) < 1$. Suppose that at horizon H there is a block length $H_0 \le H$ with $q(\alpha) \equiv \|J(\alpha)^{H_0}\|_2 < 1$, and let $\bar{M}_{H_0}(\alpha) = \max_{0\le r<H_0} \|c_{\text{price}}^\top J(\alpha)^r\|_2$ and $M_{C,H}(\alpha) = \max_{k\le H} \|c_{\text{price}}^\top J(\alpha)^k\|_2$. Then*

$$\sup_{k\ge 0} \|c_{\text{price}}^\top J(\alpha)^k\|_2 \le \max\Big(M_{C,H}(\alpha),\ \bar{M}_{H_0}(\alpha) q(\alpha)^{\lfloor (H+1)/H_0 \rfloor}\Big), \tag{3.13}$$

$$G_{\text{cum},\infty}(\alpha) \le G_{\text{cum},H}(\alpha) + \frac{\bar{M}_{H_0}(\alpha) H_0\, q(\alpha)^{\lfloor (H+1)/H_0 \rfloor}}{1 - q(\alpha)}.$$

Proof. Write any $k > H$ as $k = mH_0 + r$ with $m \geq \lfloor (H+1)/H_0 \rfloor$ and $0 \leq r < H_0$. Submultiplicativity gives $\|c^\top J^k\| \leq \|c^\top J^r\| \, \|J^{H_0}\|^m \leq \bar{M}_{H_0} q^m$. Taking the supremum and summing the geometric tail gives (3.13). □

The certificate turns an infinite-horizon supremum into a finite computation with a verified remainder. If the second term in the maximum is below $M_{C,H}$, truncation at H is exact.

Example 3.5.1 shows why the certificate is needed: two economies can agree exactly on the first coefficients and still differ threefold in their summed response.

3.6.4 Robustness to estimation error

Theorem 3.6.3 (Robust gain)**.** *Let $M = \sup_k \|J^k\|$ and $S = \sum_k \|J^k\|$. If $\|E\| \leq \epsilon$ and $\epsilon S < 1$, then $\sup_k \|(J+E)^k\| \leq M/(1-\epsilon S)$.*

Proof. Let $P_k = (J+E)^k$. Expanding, $P_k = J^k + \sum_{j<k} J^{k-1-j} E P_j$. With $m_k = \|P_k\|$, $m_k \leq M + \epsilon \sum_{j<k} \|J^{k-1-j}\| m_j \leq M + \epsilon S \max_{j<k} m_j$. Since $m_0 = 1 \leq M$, induction gives $m_k \leq M/(1-\epsilon S)$ for all k. □

Theorem 3.6.3 is deterministic: it holds whenever $\|E\| \leq \epsilon$. A statistical guarantee follows only through a confidence set for the operator, and then holds with that set's coverage probability, not with certainty. The bound held in all 30,000 random perturbations of 100 non-normal operators, with median tightness 0.50 (realized gain over bound; maximum 0.57). In a simulated window estimator of the kind used in Section 3.10 (600 transitions), a residual-bootstrap 95% ball for the operator covered the true J in 94.7% of 300 replications, with median radius 0.086; but $\epsilon S < 1$ held in only 8% of windows. The certificate is valid but often unavailable at realistic sample sizes; longer windows or pooled estimates are needed to use it in practice.

The guard needs the summed gain, which is invariant to cohort splitting (Proposition 3.2.2) and certified at the parametric rate whatever the spacing of the memories (Theorem 3.5.5); it does not need the algorithms' identities, which are expensive (Chapter 2). In a two-cohort economy with rates $0.7 \pm \Delta$, the relative error of the estimated summed gain is flat in Δ (log–log slope -0.15), while the error of the cohort weights grows as $\Delta^{-3.07}$.

3.6.5 An uncertainty-aware guard

The guard of Lemma 3.6.1 assumes the operator is known. With data, the relevant question is which admission levels are safe for every economy the data allow. We answer it for a venue gate (Mode B below) in the positive cohort economy, where admitting a fraction α of the algorithmic flow scales every cohort's weight, $M \mapsto \alpha M$, so that $S(\alpha) = \kappa/(1 - a - \kappa\alpha\Phi)$.

Lemma 3.6.4 (Soundness of compatible-set guards)**.** *If a data-dependent set $\mathcal{C}(D)$ contains the true economy with probability at least $1-\gamma$, then every action whose worst-case loss is within budget for all economies in $\mathcal{C}(D)$ is within budget for the true economy with probability at least*

$1-\gamma$, *including an action selected from the data. If no action qualifies, the guard certifies nothing: it halts new admission and flags the state, because existing inventory can be unsafe even with no new trading.*

Theorem 3.6.5 (Optimal uncertainty-aware admission)**.** *Under the reset or observed-drift contract ($z_0 = 0$, or known drift subtracted from $\mathcal{B}$), let position I, shock bound $\bar{u}$ and budget $\mathcal{B}$ be given, and let Φ^+ be an upper bound on Φ valid with probability at least $1-\gamma$ (for example $\Phi^+ = (1-a-\kappa/\hat{S}^+)/\kappa$ from Theorem 3.5.5, or Φ_+ from Theorem 3.5.3). Put $c_I = 1-a-\kappa|I|\bar{u}/\mathcal{B}$.*

(i) Closed form and interval. *If $c_I \geq 0$, the levels that are safe for every $\Phi \leq \Phi^+$ form the interval $[0, \alpha^*]$ with $\alpha^* = \min\{1,\ c_I/(\kappa\Phi^+)\}$.*

(ii) Safety. *With probability at least $1-\gamma$, every admitted level is safe for the true economy.*

(iii) Maximality. *Every level above α^* is unsafe for an economy with $\Phi = \Phi^+$; when Φ^+ is attained by a compatible economy (as for Φ_+ in Theorem 3.5.3), no rule using the same information can admit more without admitting an unsafe level.*

(iv) Infeasibility. *If $c_I < 0$, no level is safe, including $\alpha = 0$: the guard halts admission and the position must fall to $|I| \leq \mathcal{B}(1-a)/(\kappa\bar{u})$.*

(v) Minimax capacity loss. *On the class of Theorem 3.5.5 with Φ bounded away from zero, the certificate-based guard gives up $\alpha^\circ - \alpha^* = O\big(\sigma\sqrt{\log N/N}\big)$ of the oracle capacity $\alpha^\circ = \min\{1, c_I/(\kappa\Phi)\}$ with probability at least $1-2\gamma$, and every rule that admits an unsafe level with probability at most $\beta < 1$ gives up at least $c\,\sigma/\sqrt{N}$ in expectation at some economy of the class. The guard is minimax-optimal in lost capacity up to a logarithmic factor.*

Monotonicity of $S(\alpha)$ in α and Φ gives (i)–(iv); (v) combines the certificate's excess with Proposition 3.5.2 applied to two economies whose capacities differ by order $\sigma/\sqrt{N}$ (Appendix 3.15). In 500 random economies the closed form matched a brute-force search over 1,001 admission levels, computed from the state matrices rather than from (3.11), in every case; every safe set was an interval; and the 28 infeasible states were flagged (test T06). Across 18 designs the certificate covered the true summed response at least at its nominal level (minimum 93.6% at nominal 90%), with a median excess of 3–15% when the memory bound is 0.6 or 0.9 and about 230% when it is 0.97, where the tail term dominates (test T04).

3.6.6 When admission changes the dynamics

Theorem 3.6.5 rests on one property: risk rises with the admission level, so the safe levels form an interval $[0, \alpha^*]$ and certifying α^* certifies every level below it. A venue that admits a class of participants can also change the book. Suppose admitting a fraction α adds feedback $\alpha\Phi$, deepens the book so that impact falls to $\kappa_0/(1+\delta\alpha)$, and speeds reversion to $1-a_0+\rho\alpha$, with

$\theta = (\kappa_0, a_0, \Phi, \delta, \rho)$ in a certified box Θ. Then $S(\alpha) = \kappa_0/D(\alpha)$ with $D(\alpha) = (1+\delta\alpha)(1-a_0+\rho\alpha) - \kappa_0\Phi\alpha$.

Proposition 3.6.6 (Venue mode). *(i) For each economy the safe levels* $\{\alpha \in [0,1] : |I|\bar{u}S(\alpha) \le \mathcal{B}\}$ *are* $[0,1]$ *minus at most one open interval; they can be* $[0,\alpha_1] \cup [\alpha_2, 1]$ *with* $0 < \alpha_1 < \alpha_2 < 1$.

(ii) For $\alpha \ge 0$ *the worst economy of* Θ *is the same corner,* $(\kappa_0^+, a_0^+, \Phi^+, \delta^-, \rho^-)$*, for every* α*, so the robust safe set is the safe set of that corner, with endpoints at the roots of a quadratic.*

(iii) A rule that certifies the largest safe level and admits every level below it admits unsafe levels whenever the robust safe set is not an interval. Levels in $[\alpha_2, 1]$ *can be used safely only by moving to them directly, not by raising admission step by step.*

In the box $\kappa_0 \in [0.98, 1.02]$, $a_0 \in [0.58, 0.60]$, $\Phi \in [2.3, 2.4]$, $\delta \in [2.9, 3.1]$, $\rho \in [0.58, 0.62]$ with $|I|\bar{u}/\mathcal{B} = 0.37$, the robust safe set is $[0, 0.035] \cup [0.386, 1]$, in agreement with a 2,001-point grid over 332 economies of the box (test T13). Low admission brings mostly feedback and high admission mostly liquidity, so risk first rises and then falls.

Without a closed form, as in the three-state book of Section 3.4, the robust safe set must be computed. Let the impact $\kappa(\alpha)$, the feedback $q(\alpha)$ and the refill rate $r(\alpha)$ of the book depend on admission, with the uncertain parameters in a box Θ, and let $S(\alpha, \theta)$ be the summed absolute price response.

Theorem 3.6.7 (Certified admission without monotonicity). *Enclose* S *over a box of levels and parameters by its first* T *terms in interval arithmetic with outward rounding plus the tail bound* $\|J^T s_0\|_\infty \sum_{j<T} \|J^j\|_\infty / (1 - \|J^T\|_\infty)$ *(infinite when* $\|J^T\|_\infty \ge 1$ *cannot be excluded). Classify a subinterval of levels as safe if the enclosure over it and every part of a partition of* Θ *is within budget, as unsafe if the lower enclosure at some economy of* Θ *exceeds the budget, and otherwise split it, down to a tolerance. Then:*

(i) Soundness. *Every level classified safe is safe for every economy in* Θ*, and every level classified unsafe is unsafe for some economy in* Θ*, in exact arithmetic.*

(ii) Existence and bracketing. *The robust safe set is closed, so the largest safe level exists if any level is safe, and so does the largest level reachable by raising admission from zero through safe levels if zero is safe. The first lies between the largest certified-safe level and the largest level not certified unsafe; the second between the end of the certified-safe interval that starts at zero and the first level certified unsafe.*

(iii) No false certificate. *If every level is classified unsafe, the guard halts admission and flags the state.*

(iv) Completeness. *If* $\sup_\Theta \rho(J) < 1$*, then as the tolerance and the partition of* Θ *shrink and* T *grows, every level at which the robust risk differs from the budget is eventually classified.*

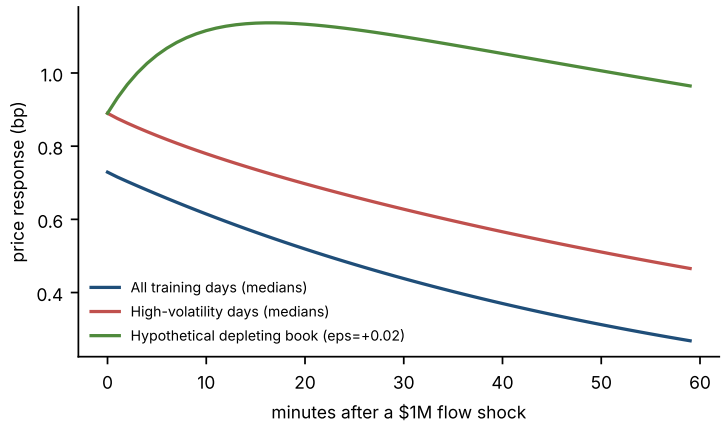


Figure 3.3: Price response to a $1M flow shock at the median depth estimates and in a hypothetical depleting book. Computed from the estimates of Table 3.7.

Table 3.4: Peak versus summed gain at calibrated parameters

Book	C_3	Peak (min)	$\sum_{k<15} \lvert g_k \rvert$	$\sum_{k<60} \lvert g_k \rvert$	Sum/peak	Cap, peak	Cap, sum
All days (medians)	-0.013	0.73 (0)	9.73	27.67	13.3	\$274M	\$21M
High-vol. days (medians)	-0.014	0.89 (0)	12.17	38.71	13.7	\$225M	\$16M
Depleting book ($\varepsilon = 0.02$)	$+0.042$	1.14 (17)	15.85	63.69	14.0	\$176M	\$13M

Gains in bp per \$1M of flow; sum/peak at 15 minutes. Inventory cap under a peak guard and under the summed guard, for a loss budget of \$100,000, an illustrative shock bound of \$5M of flow per minute and a 15-minute horizon. From `calib_guard.py`, using the medians of Table 3.7.

The method is interval branch and bound (Hansen and Walster, 2004); the theorem's content is that its output is a certificate for the risk contract, including for admission rules whose risk is not monotone. In a book with depletion and refill where admission raises feedback linearly and liquidity quadratically (test T13), the certified safe set at a budget of 6.5 is $[0, 0.053] \cup [0.178, 1]$, with $[0.059, 0.168]$ certified unsafe and 0.016 left undecided; no certified-safe level was unsafe for any of 433 sampled economies on a 1,001-point grid. The shortcut that certifies $\alpha = 1$ and admits everything below it would admit the whole unsafe band.

3.6.7 A calibrated illustration

How much does the choice between peak and summed guards matter in a real book? We compute the three-state impulse response at the median estimates of Section 3.9 (all training days and high-volatility days) and at a hypothetical book in which flow depletes the side it hits ($\varepsilon = +0.02$, other parameters at their high-volatility medians), Figure 3.3 and Table 3.4.

The result is striking even though BTCUSDT does not amplify. Because price impact is persistent ($\hat{a} \approx 0.97$ per minute), the summed gain over 15 minutes is about 13 times the peak. A guard calibrated to the peak would admit inventory about 13 times larger than is safe against

a sustained adversarial flow of the declared size. Amplification makes this worse (the depleting book peaks 17 minutes after the shock), but persistence alone already makes peak-based limits unsafe. This is the practical content of Lemma 3.6.1 for this market.

3.6.8 Two deployment modes

Mode A. *An individual dealer.* A price-taking market maker with a small share of aggregate depth scales its own quote size by α. This scales its passive fill intake but leaves aggregate depth, impact and the market operator unchanged.

Mode B. *A venue or sponsoring broker.* A gate at a matching engine or a market-access gateway (the setting of Rule 15c3-5) admits a fraction α of incoming aggressive algorithmic flow while resting depth stays posted. The operator becomes $J(\alpha) = A_0 + \alpha B_F$.

Algorithm (pre-trade guard). At each step, using only pre-shock information:

1. For each α on a grid $\mathcal{A} = \{0, 0.025, \dots, 1\}$, compute $\rho(J(\alpha))$, the peak and summed gains $M_{C,H}(\alpha)$ and $G_{\mathrm{cum},H}(\alpha)$, and the tail certificate of Lemma 3.6.2; if an operator confidence radius ϵ is available, inflate by Theorem 3.6.3.
2. Compute post-hedge and worst-case post-fill inventory, hedging cost and the post-fill margin increment.
3. Admit the largest $\alpha \in \mathcal{A}$ that satisfies the solvency inequalities of Theorem 3.6.8; if none does, set $\alpha = 0$, flag that no safe action exists, and trigger inventory reduction.

The guard checks every grid point, so it does not rely on the safe set being an interval; levels between grid points are never admitted. For low-dimensional J the grid evaluation is a sequence of matrix–vector products; execution takes well under one millisecond in compiled code, which we regard as an engineering detail rather than a result.

3.6.9 Balance-sheet solvency

Theorem 3.6.8 (Conditional bounded-shock forward solvency)**.** *Consider a dealer with cash K_t, inventory I_t, mark p_t, equity $E_t = K_t + I_t p_t$, maintenance margin $MM_t = \max(0, \mu|I_t|p_t - c_0)$ and remaining buffer $B_{\mathrm{rem},t} = E_t - MM_t$, a safety reserve R_{safe}, a hedge $\Delta I_t^{\mathrm{hedge}}(\alpha)$ with cost $C_{\mathrm{hedge},t}(\alpha)$, worst-case passive fills $\Delta I_{\max,t}^{\mathrm{fill}}(\alpha)$ with margin increment $C_{\mathrm{fill},t}(\alpha)$, and declared one-step and cumulative shock envelopes $\bar{u}_{\mathrm{step},t} \le \bar{u}_{\mathrm{cum},t}$. Let $|I_{t+1}^{\mathrm{post}}(\alpha)| = |I_t + \Delta I_t^{\mathrm{hedge}}(\alpha)| + \Delta I_{\max,t}^{\mathrm{fill}}(\alpha)$. Suppose the guard admits α_t only if*

$$|I_{t+1}^{\mathrm{post}}(\alpha_t)|(1+\mu)\Big(M_C(\alpha_t)\|z_t\|_2 + s(\alpha_t)G_{\mathrm{cum}}(\alpha_t)\bar{u}_{\mathrm{cum},t}\Big) + C_{\mathrm{hedge},t}(\alpha_t) + C_{\mathrm{fill},t}(\alpha_t) + R_{\mathrm{safe}} \le B_{\mathrm{rem},t}, \tag{3.14}$$

where $s(\alpha)$ scales admitted shocks. Then for every disturbance path within the declared envelopes, the buffer after the step is at least R_{safe}; by induction, solvency holds at every step at which an admissible α exists. At $\alpha = 0$ the one-step admissible shock is

$$\bar{u}_{\text{adm},t} = \frac{B_{\text{rem},t} - R_{\text{safe}} - |I_t|(1+\mu)\|z_t\|_2}{|I_t|(1+\mu)}.$$

The condition is sufficient over the whole disturbance ball. It is also necessary when the current state contributes no drift ($z_t = 0$), because the shock term is then attained by adversarially signed disturbances; with $z_t \neq 0$ the drift term $M_C\|z_t\|_2$ is an upper bound and the condition may be conservative. The admissible shock at $\alpha = 0$ assumes, as in the cohort economy (Chapter 2, positive-geometry lemma), that the price-row gain without admitted flow is $M_C(0) = 1$.

Two features matter. The guard accounts for the inventory the dealer will hold after hedging and after worst-case fills, not only current inventory, and it never claims safety on an infeasible state: if no α satisfies (3.14), it halts new fills and flags the state, because pre-existing inventory can still breach margin on a large adverse jump. In a constructed scenario ($E_0 = 10{,}000$, $I_0 = -2{,}000$ at $P^* = 100$, Tier-2 margin with $\mu = 0.005$, buffer 9,050 and reserve 950), the admissible shock at $z_0 = 0$ is 8,100/2,010 = 4.03 per unit, and an adversarial move of 4.05 violates the reserve. In a matched-shock Monte Carlo inside the model, where the guard's risk contract holds by construction (the certified bound $\Phi^+ = 0.36$ is close to the true $\Phi = 0.34$), across 1,000 paths facing identical exogenous flow shocks, initial inventory levels, transaction costs and double-entry maintenance-margin accounting where all active baselines are calibrated on an independent training set to match the guard's mean trading capacity of 68.0% (`microfoundation_and_baselines.py`), the summed-gain guard achieves 0.0% reserve breaches ($E_t - MM_t < R_{\text{safe}}$) and 0.0% exchange forced liquidations ($E_t < MM_t$), compared with 31.6% forced liquidations (34.7% reserve breaches) under no control, 31.4% (34.1%) under a trailing volatility limit, 20.3% (21.5%) under an inventory cap and 0.0% under a scanning-range margin rule; lowers 95% expected-shortfall tail loss to 5.96 (against 12.11 under the margin rule, 39.77 under the inventory cap, 39.92 under the volatility limit and 39.89 under no control, with all paired bootstrap difference 95% CIs excluding zero); and attains the highest net dealer P&L (+0.75, 95% CI $[+0.48, +1.02]$). At a matched tail-risk budget ($\text{ES}_{95} \approx 5.96$), the guard admits 68.0% capacity versus 29.8% for the margin rule, 24.1% for the inventory cap and 19.8% for the volatility limit; conversely, in a heteroskedastic zero-feedback regime ($\Phi = 0$) with a conservatively misspecified certificate $\Phi^+ = 0.40$, the trailing volatility limit outperforms the static-Φ^+ guard at matched capacity ($\text{ES}_{95} = 4.39$ vs. 4.55).

Unlike the reset quantity-gate of Lemma 3.6.1 and Theorem 3.6.5, the general forward-solvency inequality (3.14) with an unknown non-zero hidden cohort state $z_t \neq 0$ (where $z_t = (x_t, y_{1,t}, \ldots, y_{K,t})^\top$) is *not* identification-invariant under an unrestricted Euclidean norm bound. Both the unforced future price drift $b_m = c_{\text{price}}^\top J^m z_t$—whose cumulative sum is

$\big(ax_t + \kappa \sum_i (q_i/\eta_i) y_{i,t}\big)/(1 - a - \kappa\Phi)$—and the operator-norm drift bound $M_C(\alpha_t)\|z_t\|_2$ depend on the individual cohort weights and memory rates (q_i, η_i) and the hidden state y_t, not solely on $\Phi = \sum_i q_i$. For an explicit rational counterexample (`bridge_check.py`, Part F), take $a = 2/5$, $\kappa = 1$, and two two-cohort positive economies $\theta^{(1)}$ with $q^{(1)} = (1/5, 1/5)$, $\eta^{(1)} = (5/8, 3/8)$ and $\theta^{(2)}$ with $q^{(2)} = (9/25, 1/25)$, $\eta^{(2)} = (13/24, 1/8)$. Both share $\Phi = 2/5$ ($S = 5$) and identical early impulse responses $(h_0, h_1, h_2) = (1, 3/5, 363/800)$ ($k_0 = 1/5$, $k_1 = 3/32$). Yet after a single unit flow shock two periods earlier, where both markets display the exact same price path $(1, 3/5, 363/800)$, their hidden cohort states produce different next-step drift $b_1 = 9009/24000$ versus $8959/24000$; after three periods their cumulative future drift is $61701/24000$ versus $61751/24000$; over unknown hidden states $\|y_0\|_2 \le 1$ at $x_0 = 0$ their worst-case cumulative drift is 3.110 versus 3.688 (18.6% larger); and their one-step price-row norm $\|c_{\text{price}}^\top J\|_2 = \sqrt{a^2 + \kappa^2 \sum_i q_i^2}$ is 0.490 versus 0.540.

When the unobserved cohort memories are instead bounded in the natural ℓ_∞ norm $\|y_0\|_\infty \le \bar{y}_0$—which holds automatically whenever historical pricing errors satisfy $|x_{-s}| \le \bar{y}_0$ for $s \ge 0$, since each $y_{i,0}$ is a convex combination of past x_{-s}—the worst-case unforced drift has an exact identification-free envelope and yields a closed-form drift-adjusted guard:

Proposition 3.6.9 (Uniformly safe guard under interval-bounded unknown hidden drift)**.** *In any positive cohort economy with $\Phi = \sum_{i=1}^K q_i \le \Phi^+$, observed current pricing error x_0 and unknown hidden cohort states satisfying $\|y_0\|_\infty \le \bar{y}_0$, let admission $\alpha \in [0, 1]$ scale cohort flow as $M \mapsto \alpha M$ with $a + \alpha\kappa\Phi^+ < 1$. Then:*

(i) Identification-free drift envelope. *For every compatible cohort decomposition $(q_i, \eta_i)_{i=1}^K$ with $\sum_i q_i = \Phi$, the worst-case cumulative unforced drift over $\|y_0\|_\infty \le \bar{y}_0$ is*

$$\sup_{\|y_0\|_\infty \le \bar{y}_0} \sum_{t=1}^{\infty} |e_x^\top J(\alpha)^t z_0| \;=\; \frac{a|x_0| + \alpha\kappa\Phi\bar{y}_0}{1 - a - \alpha\kappa\Phi}, \tag{3.15}$$

attained when $y_{i,0} = \operatorname{sign}(x_0)\bar{y}_0$ for all i (or $y_{i,0} = \pm\bar{y}_0$ when $x_0 = 0$).

(ii) Closed-form drift-adjusted guard. *Under shock bound $\bar{u}$, inventory I and budget $\mathcal{B}$, the set of admission levels that are safe for every compatible economy with $\Phi \le \Phi^+$ and every hidden state $\|y_0\|_\infty \le \bar{y}_0$ is $[0, \alpha^*_{\text{drift}}]$ (empty when $(1 - a)\mathcal{B} < |I|(a|x_0| + \kappa\bar{u})$), where*

$$\alpha^*_{\text{drift}} \;=\; \min\left\{1,\; \left[\frac{(1 - a)\mathcal{B} - |I|(a|x_0| + \kappa\bar{u})}{\kappa\Phi^+(\mathcal{B} + |I|\bar{y}_0)}\right]_+\right\}. \tag{3.16}$$

(iii) Exact capacity cost of unknown state. *When $x_0 = 0$ and the reset guard $\alpha^* = \min\{1, c_I/(\kappa\Phi^+)\}$ of Theorem 3.6.5 is interior ($\alpha^* \in (0, 1)$), the capacity given up to*

protect against unknown hidden cohort states $\|y_0\|_\infty \le \bar{y}_0$ *is exactly*

$$\alpha^* - \alpha^*_{\text{drift}} \;=\; \alpha^* \,\frac{|I|\bar{y}_0}{\mathcal{B} + |I|\bar{y}_0}. \tag{3.17}$$

At the Table 3.3 calibration ($a = 0.4$, $\kappa = 1$, $\Phi^+ = 0.4$, $|I| = 1$, $\bar{u} = 0.48$, $\mathcal{B} = 1$, $x_0 = 0$) with $\bar{y}_0 = 0.25$, the reset guard admits $\alpha^* = 0.3000$ whereas the drift-adjusted guard admits $\alpha^*_{\text{drift}} = 0.2400$, an exact capacity cost of 0.0600, and commits zero budget breaches across 500 random multi-cohort economies (`recompute_p2_step2_theory.py`).

Proposition 3.6.10 (When an eigenvalue rule admits an uncertified level)**.** *Let an eigenvalue guard admit* $\alpha_\rho = \max\{\alpha : \rho(J(\alpha)) \le \bar{\rho}\}$*, and suppose the state is feasible for the summed-gain guard, which then admits at least* $\alpha = 0$*. Bounding the one-step price move by* $M_C(\alpha)(\|z_t\|_2 + s(\alpha)\bar{u}_{\text{step},t})$*, the level admitted by the eigenvalue guard fails this one-step solvency certificate, and so is one the summed-gain guard would not certify, if and only if*

$$M_C(\alpha_\rho) \;>\; \frac{B_{\text{rem},t} - R_{\text{safe}} - C_{\text{hedge},t}(\alpha_\rho) - C_{\text{fill},t}(\alpha_\rho)}{|I^{\text{post}}_{t+1}(\alpha_\rho)|(1+\mu)(\|z_t\|_2 + s(\alpha_\rho)\bar{u}_{\text{step},t})}.$$

The certificate is a bound, so failing it means that solvency cannot be guaranteed, not that a breach must occur. The eigenvalue rule fails it exactly when the transient gain at its admitted level exceeds the loss the balance sheet can absorb, which Chapter 2's spectral-blindness theorem shows can happen at any spectral threshold.

Proposition 3.6.11 (Strategic re-optimization reinforces the guard)**.** *Suppose cohorts re-optimize their steady-state Kalman gains when the venue throttles admission, and throttling scales the variance of execution-driven innovations as* $\sigma^2_\zeta(\alpha) = \alpha^2\sigma^2_{\zeta,0}$ *while measurement noise* $\sigma^2_\nu > 0$ *is unchanged.*

(i) The optimal gain $\eta^*(\alpha)$ *is strictly increasing in* α*, so throttling lowers every cohort's adaptation rate. If the transient gain is non-decreasing in the adaptation rates, then* $M_C(J(\alpha, \eta^*(\alpha))) \le M_C(J(\alpha, \eta^*(1)))$*: re-optimization does not undo the guard.*

(ii) In the positive cohort economy ($q_i \ge 0$, $a + \kappa\sum_i q_i < 1$) *the monotonicity holds without condition: every finite-horizon summed response* $S_H = \sum_{t<H} h_t$ *is non-decreasing in each adaptation rate, strictly for* $H \ge 2$ *when* $q_i > 0$*, while* $S_\infty = \kappa/(1 - a - \kappa\sum_i q_i)$ *does not depend on the rates. Hence* $S_H(J(\alpha, \eta^*(\alpha))) \le S_H(J(\alpha, \eta^*(1)))$ *at every horizon.*

Part (i) is a property of the Riccati root. Part (ii) is a renewal argument: the response is the renewal density of a defective inter-arrival law whose cumulative distribution $a + \kappa\sum_i q_i[1 - (1-\eta_i)^j]$ rises with every rate at fixed total mass, so faster adaptation moves feedback earlier without changing its total (Appendix 3.15). In 3,000 random positive economies no finite-horizon sum fell when a rate rose (`hardening`). For the three-state book with momentum the

condition in (i) remains an assumption, which we check numerically. Across 3,000 stable draws of the three-state book with momentum (adaptation rates on a grid from 0.05 to 0.95), the peak response is non-decreasing in η in 98.8% of draws and in 99.9% of those with trend-following cohorts ($q > 0$); the summed absolute response is non-decreasing in 93% of draws with $q > 0$ but decreasing when cohorts are contrarian ($q < 0$). For trend-following cohorts, the case the filtering microfoundation produces, re-optimization therefore reinforces the guard in almost every case we drew; the signed sum S of Theorem 3.4.4 does not depend on η at all (`recompute_oct20.py`). A manipulator who instead increases order size to offset throttling enlarges the shock envelope and drives α toward zero.

3.7 The value of disclosure for safe trading

Chapter 2 measures the value of disclosure in samples needed to identify the cohorts (Chapter 2). For a pre-trade control, the relevant value is the safe trading capacity it unlocks.

Theorem 3.7.1 (Decision value of disclosure)**.** *Under the risk contract of Section 3.5.1, compare an observation contract O with a richer contract O' that reveals everything O reveals and more, both with exact data.*

(i) Monotonicity. *The compatible sets are nested, $\Theta_{O'} \subseteq \Theta_O$, so $\Phi^+_{O'} \leq \Phi^+_O$ and the admitted level of Theorem 3.6.5 satisfies $\alpha^*_{O'} \geq \alpha^*_O$.*

(ii) Strictness. *$\alpha^*_{O'} > \alpha^*_O$ if and only if $\alpha^*_O < 1$, $c_I > 0$, and O' excludes every compatible economy with $\Phi = \Phi^+_O$. Under the window contract that economy is the upper principal representation of Theorem 3.5.3: disclosure has decision value exactly when it rules out a cohort at the memory bound with the extremal weight.*

(iii) Long memory. *Under the window contract with memory bound $\bar{r}$, $\alpha^*_O \to 0$ as $\bar{r} \to 1$ for any fixed window (Theorem 3.5.3(iii)), whereas audited exposure, which reveals Φ, admits the oracle capacity α°. The decision value of audited disclosure therefore tends to the whole oracle capacity as memory lengthens.*

(iv) Net value. *With value w per unit of admitted capacity and disclosure cost $c(O')$, the net value $w(\alpha^*_{O'} - \alpha^*_O) - c(O')$ is zero or negative whenever O already identifies Φ (a window whose moment vector lies on the boundary of the moment space) or the budget does not bind.*

(v) Noisy proxies. *With noisy data the certificates of two contracts are confidence bounds, not nested sets, and a richer contract can admit less. Under class-tagged flows measured with noise over T periods and estimated as in Chapter 2, class i's exposure $\hat{\Phi}_i = \hat{w}_i/(1 - \hat{r}_i)$*

has delta-method standard error

$$\mathrm{se}(\hat{\Phi}_i) = \frac{1}{(1-r_i)\sqrt{T}}\Big[V_{ww} + 2\Phi_i V_{wr} + \Phi_i^2 V_{rr}\Big]^{1/2},$$

with V the asymptotic covariance of $(\hat{w}_i, \hat{r}_i)$, so the tagged bound $\sum_i \hat{\Phi}_i + z_{1-\gamma}\mathrm{se}$ admits less than the public certificate whenever its margin exceeds $\Phi_O^+ - \Phi$. Intersecting the two certificates, each at level $1-\gamma/2$, gives a bound valid at level $1-\gamma$ that admits at least as much as either certificate at level $1-\gamma/2$, which restores monotonicity.

The theorem is set inclusion plus the closed form of Theorem 3.6.5; its content is the location of the value. Disclosure is worth most where public certificates are weakest, in markets whose participants have long memories. In simulation (test T07; Table 3.5), with two classes and one of them slow, audited exposure adds 0.027 of admitted capacity to a public certificate when the slow class's memory rate is 0.6 and 0.122 when it is 0.995, out of an oracle capacity of 0.667. Class-tagged flows measured with 10% noise add 0.02–0.03 at memory rates up to 0.97 but *lower* capacity at 0.995, because a class's exposure $q = w/(1-r)$ is badly conditioned when r is near one: disclosure of a noisy proxy can be worth less than nothing. Part (v) quantifies this. In a second simulation of the same design (test T15), the standard error of the slow class's exposure rises from 0.0004 to 0.035 as $1-r$ falls from 0.4 to 0.005, roughly in proportion to $(1-r)^{-1}$ over the last decade, while the public certificate's margin grows only with the square root of its horizon. At $r = 0.995$ the tagged bound admits 0.506 against 0.549 for the public one; the intersection admits 0.537, close to the better of the two, and covers the true exposure in 99% of replications (Table 3.5). Chapter 2's evidence that Binance's and the CFTC's class positions pin down memories that prices leave unrestricted concerns class-level positions, which are proxies, not records of individual algorithms.

Pricing versus guarding. A conservative bound is the right input for a guard and the wrong input for a price. The corrective rebate of Proposition 3.8.1 depends on S^2; computed from an upper bound that overstates S by 10% it over-subsidizes reversion and loses 2.4–2.8% of first-best welfare, and by 50%, 70–80% (test T12). Guards should use certified upper bounds; rebates should use point estimates.

3.8 Who provides reversion? An equilibrium

Theorem 3.4.4 says that transient risk is governed by one ratio, $S = \tilde{\kappa}/(m - q\tilde{\kappa})$. Its ingredients are chosen by market participants: dealers decide how hard to lean against price displacements, which sets reversion m, and momentum traders decide how strongly to chase them, which sets q. This section asks whether those choices are efficient.

Table 3.5: Admitted capacity under three observation contracts (simulation)

Memory rate of the slow class	0.60	0.90	0.97	0.995
Public responses (certificate, Theorem 3.5.5)	0.640	0.623	0.601	0.544
Class-tagged flows, 10% noise	0.662	0.655	0.631	0.507
Audited exposure (oracle)	0.667	0.667	0.667	0.667
Public ∩ tagged, each at 97.5% (test T15)	0.661	0.653	0.626	0.537
Value of audited exposure over public	0.027	0.044	0.066	0.122

Two classes ($q = 0.06$ with the stated memory rate, $q = 0.12$ with rate 0.5), $a = 0.4$, $\kappa = 1$, position 1, shock bound 0.48, budget 1; public: 500 reset experiments, per-tap noise 0.05, certificate at 95% with the true memory bound; tagged: 2,000 periods of each class's flow with 10% measurement noise, two-stage least squares as in Chapter 2, 95% delta-method bound; 300 replications. Entries are mean admitted fractions α^*. The intersection row comes from an independent run of the same design (seed 15), whose public and tagged rows are 0.639/0.662, 0.625/0.655, 0.600/0.631 and 0.549/0.506. Kind: recomputed (tests T07 and T15).

3.8.1 Setup

There are N dealers. Dealer i supplies reversion intensity $s_i \geq 0$ by absorbing order flow against the displacement, so $m = m_0 + \sum_i s_i$, where m_0 is reversion supplied by other means. Leaning earns a spread v per unit of intensity at quadratic inventory cost, so the private value is $vs_i - \frac{c}{2}s_i^2$. By Lemma 3.6.1 the worst-case transient loss of a position of size Q facing flow shocks bounded by U is $QU\,S$, so we take aggregate expected tail loss to be $\Lambda S(m, q)$ with $\Lambda > 0$. Dealers bear a share $\theta < N$ of it, so each bears θ/N; leveraged users and their counterparties bear the rest. Momentum traders choose aggregate intensity q with private value $bq - \frac{e}{2}q^2$ and bear none of the tail loss. Long-run impact $\tilde{\kappa}$ is taken as given.

3.8.2 Results

Proposition 3.8.1 (Equilibrium transient risk)**.** *Let $g(m) \equiv -\partial S/\partial m = \tilde{\kappa}/(m - q\tilde{\kappa})^2$.*

(i) Under-provision of reversion. *The symmetric Nash equilibrium solves $m - m_0 = Nv/c + \theta\Lambda g(m)/c$, the planner's optimum solves $m - m_0 = Nv/c + N\Lambda g(m)/c$, and $m^{\mathrm{eq}} < m^{\mathrm{SP}}$, so $S^{\mathrm{eq}} > S^{\mathrm{SP}}$.*

(ii) Fewer dealers, more risk. *$\partial m^{\mathrm{eq}}/\partial N = (v/c)/(1 - \theta\Lambda g'(m)/c) > 0$, so equilibrium reversion rises and S^{eq} falls with the number of dealers.*

(iii) Over-provision of momentum. *Momentum traders choose $q = b/e$; the planner chooses $q^{\mathrm{SP}} < b/e$, solving $b - eq = \Lambda\tilde{\kappa}^2/(m - q\tilde{\kappa})^2$.*

(iv) The corrective price is identified. *The first best for reversion is implemented by a rebate*

per unit of reversion

$$\tau^* = \Big(1 - \frac{\theta}{N}\Big)\Lambda\, g(m) = \Big(1 - \frac{\theta}{N}\Big)\Lambda\, \frac{S^2}{\tilde{\kappa}},$$

which depends on the market only through S and $\tilde{\kappa}$.

(v) Endogenous depth. *Suppose in addition that each dealer posts depth $D_i \geq 0$ at private value $v_D D_i - \frac{c_D}{2} D_i^2$, that long-run impact is $\tilde{\kappa} = 1/D$ with $D = \sum_i D_i$, and that momentum traders size their flow to the book, $q = \psi D$ with ψ fixed. Then $S = \Gamma/D$ with $\Gamma \equiv S/\tilde{\kappa} = 1/(m - \psi)$, which does not depend on depth. Symmetric equilibrium depth solves $D = N v_D/c_D + \theta\Lambda\Gamma/(c_D D^2)$ and the planner's $D = N v_D/c_D + N\Lambda\Gamma/(c_D D^2)$, so $D^{\mathrm{eq}} < D^{\mathrm{SP}}$; D^{eq} increases with N, so $S^{\mathrm{eq}} = \Gamma/D^{\mathrm{eq}}$ falls with the number of dealers while $S/\tilde{\kappa}$ does not move. The rebate per unit of depth $\tau_D^* = (1 - \theta/N)\Lambda S\tilde{\kappa}$ implements the first-best depth.*

Proof. (i) Dealer i's first-order condition is $v - cs_i + (\theta/N)\Lambda g(m) = 0$; the planner's is $v - cs + \Lambda g(m) = 0$. Summing over dealers gives the two fixed-point equations. The right-hand sides are decreasing in m because g is decreasing on $m > q\tilde{\kappa}$, and the planner's lies above the dealers' for every m when $\theta < N$, so its crossing with the line $m - m_0$ lies to the right. (ii) follows by implicit differentiation, since $g' < 0$. (iii) $\partial S/\partial q = \tilde{\kappa}^2/(m - q\tilde{\kappa})^2 > 0$ is the marginal external cost of momentum that traders ignore. (iv) The rebate closes the wedge between the two first-order conditions, and $g = S^2/\tilde{\kappa}$ by Theorem 3.4.4. (v) With $q\tilde{\kappa} = \psi$, Theorem 3.4.4 gives $S = \tilde{\kappa}/(m - \psi) = \Gamma/D$ and $\partial S/\partial D_i = -\Gamma/D^2$. Dealer i's first-order condition for depth is $v_D - c_D D_i + (\theta/N)\Lambda\Gamma/D^2 = 0$ and the planner's $v_D - c_D D_i + \Lambda\Gamma/D^2 = 0$; summing gives the two fixed points. In each, the left side D increases and the right side decreases in D, so each has a unique positive solution, the planner's to the right of the dealers' because $\theta < N$; and $D - \theta\Lambda\Gamma/(c_D D^2) = N v_D/c_D$ has a left side increasing in D, so D^{eq} increases with N. The rebate closes the wedge $(1 - \theta/N)\Lambda\Gamma/D^2 = (1 - \theta/N)\Lambda S\tilde{\kappa}$. □

At $v = 0.002$, $c = 0.5$, $\theta = 0.3$, $\Lambda = 10^{-4}$, $m_0 = 0.005$, $q = 0.02$ and $\tilde{\kappa} = 1$, equilibrium S falls from 27.9 with one dealer to 6.8 with forty, and the planner's S is 1.5–2.7 times lower throughout; the rebate computed from S alone coincides with the first-order wedge to machine precision. With $m = 0.03$, $b = 0.3$ and $e = 20$, momentum traders choose $q = 0.015$ against a planner's 0.0062.

Interpretation. Reversion is a public good: each dealer who leans against a displacement lowers the tail exposure of every leveraged position in the market but is paid only for his own fills. Momentum is the mirror image. Part (iv) closes the loop with the guard: the policy price of the externality is a function of the same summed response that the guard bounds, and Proposition 3.2.2 and Theorem 3.5.5 show that this quantity is identified from public data without identifying the traders. A venue that rebates reversion in proportion to $S^2/\tilde{\kappa}$, or a supervisor that sets pre-trade limits by S, prices the externality with an observable.

Table 3.6: Depth data validation

Days requested (January 2023 – October 2026)	1,386
Days passing validation	1,329
Snapshots examined	3,897,951
dropped by V1 (missing bands)	0
dropped by V2 (non-monotone)	0
dropped by V3 (implied price off by $> 3\%$)	98,080
dropped by V4 (stuck feed)	95,954
Failed periods	April 16 – May 19, 2025; October 10–15, 2025
Training days used (2023–2024)	718

All files verified against published SHA-256 checksums. The 2025 failures coincide with archived depth that is misaligned with prices; the October 10, 2025 crash day is therefore unusable.

What the data say about the model's predictions. Part (ii) predicts that books with fewer dealers have higher S because they supply less reversion. Depth is the closest proxy we observe for the number of liquidity providers. Across the three liquidity terciles of the 30 Binance perpetuals on training days, the summed response per dollar of flow rises steeply as depth falls (median 2,797, 26,603 and 39,329 basis points per \$1M in the thick, mid and thin terciles; Spearman correlation with log depth -0.94), as (ii) predicts. The channel is not the one part (ii) names, however: the unit-free gain multiple $S/|h_0|$, which equals $1/(m - q\tilde{\kappa})$ up to the factor $\tilde{\kappa}/\kappa$ and isolates reversion net of momentum, is the same in all three terciles (medians 46.4, 47.3 and 46.9; Spearman -0.11). Thinner books carry more cumulative risk because their impact $\tilde{\kappa}$ is larger, not because they revert more slowly. This is the pattern part (v) produces when momentum traders size their flow to the book: fewer dealers lower depth, which raises S in proportion to $\tilde{\kappa}$, while $S/\tilde{\kappa}$ stays put. We added part (v) after seeing these comparisons, so the match is a consistency check, not a test. The comparisons were themselves made after the fact (Part 13, `R2`), and depth is an imperfect proxy for dealer count, so a direct test needs data on liquidity-provider identities. The model's second prediction, that the dynamic balance forecasts tail losses where reversion is close to $q\tilde{\kappa}$, is what Section 3.11 tests.

3.9 Evidence from Binance futures depth

3.9.1 Data and validation

We use Binance USD-M BTCUSDT perpetual futures from the public archive: one-minute klines with taker-buy volume, `bookDepth` snapshots of cumulative depth within ±1–5% of the mid at about 30-second intervals, and open interest from `metrics`. Depth is available from January 2023. No liquidation archive is available for this period, so we do not test liquidations.

Validation rules were fixed before analysis. A snapshot is kept if (V1) all of the $\pm 1, 2, 5\%$ bands are present, (V2) depth is monotone in distance on each side, (V3) the implied average price at ±1% is within 3% of the one-minute close, and (V4) it is not part of a run of more than ten identical snapshots; (V5) a day is kept if at least 80% of its minutes have a valid snapshot.

Table 3.7: Day-by-day T6 estimates, Binance BTCUSDT futures, training days 2023–2024

Median	All days	Top-decile volatility days	Other days
$\hat{\kappa}$ (bp per \$1M)	0.729	0.890	0.712
$\hat{\varphi}$	−0.051	2.586	−0.138
$\hat{a}$	0.970	0.980	0.969
$\hat{r}$	0.059	0.122	0.056
$\hat{\varepsilon}$	−0.0018	−0.0017	−0.0018
$\hat{\eta}$	0.878	0.863	0.880
$\hat{q}$	0.018	0.012	0.019
$\hat{C}_3$	−0.0137	−0.0111	−0.0143
Mean $\hat{C}_3$	−0.0165	−0.0167	−0.0165
95% CI	[−0.0176, −0.0155]	[−0.0219, −0.0122]	[−0.0175, −0.0154]
Share of days $\hat{C}_3 > 0$	6.5%	5.6%	6.7%
Share of days $\hat{C}_2 > 0$	0.8%	1.4%	0.8%
Median peak ratio	1.00	1.00	1.00
Days	718	72	646

One-minute bars. CIs from a day bootstrap.

3.9.2 Structural estimates

On one-minute bars, p_t is the log price in basis points, u_t signed taker flow in million USDT, $d_t = (B_1 - A_1)/(B_1 + A_1)$ the depth imbalance within 1% of the mid, and $x_t = p_t - \mathrm{EMA}_{60}(p_t)$. Day by day we estimate

$$\begin{aligned} \Delta p_t &= \kappa u_t + \varphi d_{t-1} - (1-a)x_{t-1} + e_t, \\ d_t &= (1-r)d_{t-1} + \varepsilon u_t + e_t, \\ u_t &= (1-\eta)u_{t-1} + \eta q\, x_{t-1} + e_t, \end{aligned}$$

and form $\hat{C}_2$, $\hat{C}_3$ and the implied peak ratio $\max_k \hat{h}_k/\hat{h}_0$. The price equation has Δp_t on the left, so the EMA filter does not create mechanical persistence. These are reduced-form regressions (flow is contemporaneous), so the estimates describe conditional dynamics rather than causal parameters.

Table 3.7 has three findings. First, the market sits on the stable side of the condition: $\hat{C}_3 < 0$ on 93.5% of days, and the median implied response peaks at impact. Second, the depletion coefficient $\hat{\varepsilon}$ is small and negative: taker buying is followed by slightly more ask-side than bid-side depth within 1%, so in these re-centred bands the book appears to replenish rather than deplete the side being hit; Section 3.11.4 shows that a fixed price grid reverses the sign. With $\hat{\varepsilon} < 0$, $\varphi\varepsilon$ is positive only if φ is negative, which it is not on high-volatility days. Third, on high-volatility days depth imbalance moves prices strongly ($\hat{\varphi} = 2.59$) and refill is twice as fast; the depth channel is active, but it works against amplification. The momentum term $\kappa^2\eta q$ is small (about 0.008 at the medians), far below $(1-a)\kappa \approx 0.022$.

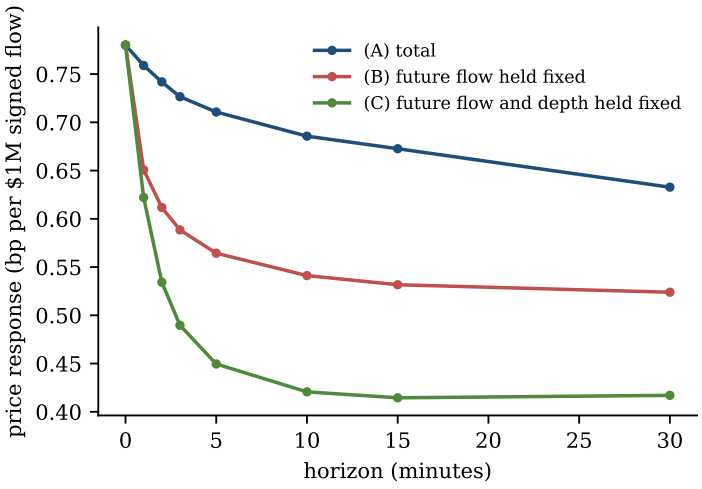


Figure 3.4: Price response to a $1M flow shock, local projections on 716 training days. (A) total; (B) future flow held fixed; (C) future flow and depth held fixed.

Table 3.8: Local-projection coefficients (bp per $1M) and overshoots

Minutes	0	1	2	3	5	10	15	30	Overshoot
(A) total	0.780	0.759	0.742	0.727	0.711	0.686	0.673	0.633	−0.021 [−0.027, −0.015]
(B) flow fixed	0.780	0.651	0.612	0.589	0.564	0.541	0.532	0.524	−0.129 [−0.134, −0.124]
(C) flow, depth fixed	0.780	0.622	0.534	0.490	0.450	0.421	0.414	0.417	−0.158 [−0.164, −0.152]

716 training days; day bootstrap.

3.9.3 Does depth carry an overshoot?

We estimate local projections (Jordà, 2005) of the cumulative price change on flow, $p_{t+k} - p_{t-1} = \beta_k u_t$ + controls, with five lags of returns, flow and depth: (A) total; (B) adding cumulative future flow, which removes order-splitting persistence; and (C) adding cumulative future depth imbalance, which removes the depth channel. Theorem 3.4.2(i) predicts an overshoot ($\max_{k \geq 1} \beta_k - \beta_0 > 0$) in (B) that shrinks in (C).

There is no overshoot (Figure 3.4, Table 3.8). The overshoot statistic is negative in every version. Holding depth fixed makes the response decay faster, consistent with depth that partly sustains the price move but never pushes it past impact.

3.9.4 Pre-registered out-of-sample test

Before opening the test period we registered the specification (SHA-256 `d95212187ae4...`): hourly blocks (06:00–23:00 UTC); $\hat{C}_3$ estimated from the previous six hours and winsorized at training percentiles; outcome the next hour's maximum adverse excursion (MAE, bp); controls

Table 3.9: Out-of-sample test, 2025–2026 (train 12,896 hours, test 10,979 hours)

Outcome	Candidate	ΔR^2 (pts) [95% CI]	ΔAUC [95% CI]	Holm p
MAE (primary)	$\hat{C}_3$	+0.018 [−0.100, +0.132]	+0.0001 [−0.0009, +0.0012]	—
MAE	peak ratio	+0.010 [+0.001, +0.019]	+0.0001 [−0.0000, +0.0002]	0.032
OI drop	$\hat{C}_3$	−0.045 [−0.095, −0.003]	−0.0049 [−0.0083, −0.0016]	0.787
OI drop	peak ratio	+0.102 [+0.061, +0.231]	−0.0014 [−0.0067, +0.0037]	< 0.001
Depth collapse	$\hat{C}_3$	+0.181 [−0.156, +0.484]	+0.0071 [+0.0041, +0.0105]	< 0.001
Depth collapse	peak ratio	+0.159 [+0.094, +0.234]	+0.0009 [+0.0004, +0.0014]	< 0.001

Primary: Clark–West one-sided $p = 0.023$, but the ΔR^2 CI contains zero, so the pre-registered test fails. Baseline R^2 for MAE is 0.293; baseline AUC 0.812.

Table 3.10: Event days

Event	Worst hour (UTC)	Worst MAE (bp)	$\hat{C}_3$ percentile, 3 hours before	OI change in worst hour
SVB, 2023-03-11	7	214.8	82	+2.5%
Yen carry unwind, 2024-08-05	13	724.0	93, 84, 83	−0.8%
Tariff shock, 2025-04-07	14	357.8	97, 98, 38	+3.1%
2025-10-10		no valid depth data		

SVB has one valid hour before its worst hour.

for realized volatility (1 and 6 hours), absolute return, flow imbalance, volume, depth, imbalance and open-interest change; OLS on 2023–2024 and evaluation on 2025–2026. Success required a one-sided Clark–West $p < 0.05$ *and* a day-block bootstrap 95% CI for ΔR^2 excluding zero. Secondary candidates and outcomes are Holm-adjusted. The test set was opened once. An audit that scrambles all data from t onward left every feature unchanged in 30 of 30 trials.

The primary test fails (Table 3.9): $\hat{C}_3$ adds 0.018 percentage points of out-of-sample R^2 with a confidence interval that includes zero. Some secondary effects are statistically significant after Holm correction, notably for depth collapse, but they are economically negligible: no candidate adds more than 0.2 points of R^2 or 0.01 of AUC.

A volatility flag does better. We set two flags on training data, each to flag 10% of hours: high $\hat{C}_3$ and high realized volatility. On the 909 tail hours of the test period (MAE above the training 90th percentile), the $\hat{C}_3$ flag has precision 8.6% and recall 19.5%; the volatility flag has precision 38.1% and recall 37.1%; the union has precision 16.3% and recall 49.4%. The unconditional tail rate is about 10%, so the $\hat{C}_3$ flag is no better than chance.

3.9.5 Crash events

For each event day we compute $\hat{C}_3$ hour by hour and its percentile in the training distribution during the three hours before the worst hour (Figure 3.5, Table 3.10). $\hat{C}_3$ was elevated before these events but stayed negative throughout, and three events are not a test.

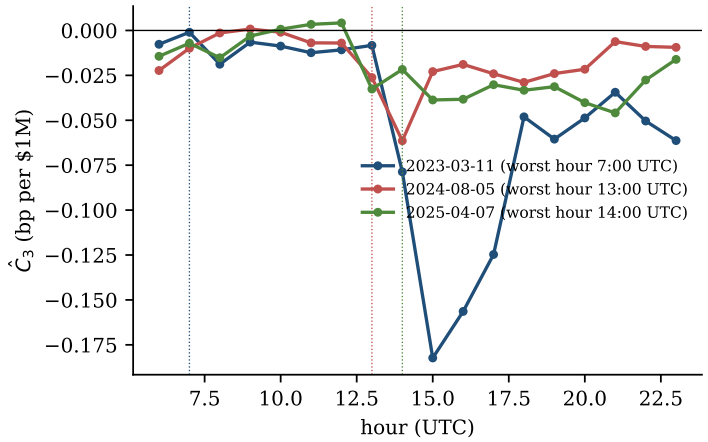


Figure 3.5: Hourly $\hat{C}_3$ on three event days (UTC). Values stay below zero, the amplification threshold.

3.10 Evidence from spot prices

3.10.1 A unit-free amplification statistic

The earlier version measured amplification by the norm of powers of a VAR(1) operator on (price deviation, flow), estimated over ten-minute windows of one-second Binance spot data. That norm depends on the units of the state: multiplying prices by 10 and dividing flow by 1,000 changed the raw norm from 8.73 to 86,731 on the same window. Proposition 3.2.2 explains why: state norms are not invariant to coordinates. We therefore standardize the state, $\tilde{J} = D^{-1}\hat{J}D$ with $D = \mathrm{diag}(\mathrm{sd}\, x, \mathrm{sd}\, f)$, and use $M_{\mathrm{std}} = \max_k \|e_x^\top \tilde{J}^k\|$, which is invariant to units (verified). We also report the price-to-price amplification $A_{xx} = \max_k |(\hat{J}^k)_{11}|$ and a delayed-peak statistic $\mathrm{hump} = \max_k |(\hat{J}^k)_{12}|/|\hat{J}_{12}|$.

3.10.2 Crash days and a no-feedback null

We study five crash days: May 19, 2021; LUNA, May 9, 2022; FTX, November 8, 2022; SVB, March 11, 2023; and August 5, 2024. Windows are selected *ex ante* (previous ten minutes in the top volatility decile), and for comparison by outcome (the ten largest absolute returns). A no-feedback null keeps each window's real flow, generates prices with no feedback, $\Delta p = \hat{\kappa} f + \hat{\sigma} e$, and pushes them through the identical pipeline.

Only LUNA exceeds the null, by less than 0.01% (Table 3.11). Outcome-selected windows give only marginally larger values (≤ 1.0005). Stress-window means exceed the ordinary 95th percentile in two events, by amounts of the same size. The pre-registered gate fails: unit-free transient amplification in spot data is indistinguishable from one and from the pipeline null. Varying the EMA span (60, 120, 300 seconds) and the bar length (1 or 5 seconds) does not change this; crash-day means range from 1.0003 to 1.18 across specifications and are never

Table 3.11: Unit-free amplification on crash days and the no-feedback null

Event	Windows (ex ante)	Mean M_{std}	Null 99th pct	p (vs null)	Outcome-selected mean
May 2021	142	1.0000	1.00002	1.00	1.0000
LUNA 2022	57	1.00008	1.00006	0.039	1.00045
FTX 2022	49	1.0000	1.00029	1.00	1.0000
SVB 2023	6	1.00006	1.00025	0.059	1.00032
Aug 2024	108	1.0000	1.00007	1.00	1.0000

Ten-minute windows of one-second bars. The median hump statistic is 1.000 in every event and its null p-value is 1.00. In the ordinary 2021–2023 panel, 0.2% of windows have $A_{xx} > 1$ and the 95th percentile of M_{std} is 1.0000; a handful of near-singular windows produce very large values, so ordinary-window means are not informative.

Table 3.12: Spot local projections (bp per $1M)

Seconds	0	1	2	5	10	20	30	60	120	300
Ordinary, total	5.86	6.28	6.53	6.60	6.69	6.77	6.67	6.49	6.39	5.92
Ordinary, flow fixed	5.86	6.02	6.11	5.90	5.68	5.26	4.86	4.16	3.37	1.99
Crash, total	6.69	6.85	6.92	7.13	7.75	8.20	8.46	8.20	9.09	8.96
Crash, flow fixed	6.69	6.41	6.28	5.99	5.95	5.46	5.02	3.88	3.10	−0.08

above the corresponding ordinary means.

3.10.3 Overshoot with flow held fixed

Local projections on one-second data give the price response to a $1M flow shock (Table 3.12). On ordinary days, with future flow held fixed, there is a small, short-lived overshoot of +0.248 [0.133, 0.376] bp per $1M, peaking at two seconds and decaying after. On crash days the flow-fixed overshoot is negative, −0.277 [−0.528, −0.042]. The large total overshoots (+0.91 ordinary, +2.40 crash) disappear when future flow is held fixed: they come from order splitting, not from the book.

3.10.4 Pre-registered predictive test

Before opening 2024, we registered (SHA-256 `b1fd7f8e...`) a test of whether pre-window amplification predicts the next ten minutes' MAE beyond volatility and flow controls, on non-overlapping 30-minute blocks (train 2021–2023, 3,307 blocks; test 2024, 1,104 blocks). It does not: $\Delta R^2 = -0.023$ points [−0.056, −0.002], Clark–West $p = 0.966$, AUC 0.822 in both models. A random feature with the same autocorrelation gives $\Delta R^2 = -0.020$ points, the same as the candidate. Leave-one-year-out fits within the training period are unstable (the 2023 fold gives a very large negative ΔR^2 because of outlying candidate values), so the null result does not depend on one year being unusual.

The test set was opened three times. The first evaluations stopped on a numerical error from degenerate windows; the fix drops blocks with non-finite features (five blocks) and is the only change, and the registered specification hash is unchanged. A volatility flag again dominates:

Table 3.13: T6 estimates by liquidity tercile, 30 USD-M perpetuals, training days 2023–2024

	Thick	Mid	Thin
Mean $\hat{C}_3$	−0.0297	−0.0216	−0.0324
95% CI	[−0.0341, −0.0257]	[−0.0250, −0.0185]	[−0.0386, −0.0268]
Share of symbol-days with $\hat{C}_3 > 0$	9.4%	15.2%	16.1%
Median $\hat{\varphi}$	2.33	2.38	3.11
Median $\hat{\varepsilon}$	−0.0014	−0.0006	−0.0005
Share with $\hat{\varepsilon} > 0$	11.1%	24.4%	30.1%
Mean feedback loop gain $\hat{Q}$	0.0091	0.0033	0.0016
Spearman correlation with log depth across the 30 symbols [95% CI]			
mean $\hat{C}_3$		−0.001 [−0.37, 0.38]	
share $\hat{C}_3 > 0$		−0.59 [−0.81, −0.27]	
$\hat{Q}$		+0.70 [0.41, 0.87]	

1,424 training symbol-days, ten symbols per tercile. CIs from a symbol-day bootstrap (tercile rows) and a symbol bootstrap (correlations). $\hat{Q}$ is the loop gain of Chapter 2 estimated on one-minute bars; a placebo random walk with each day's volatility gives $\hat{Q} \approx 0$ for every symbol.

precision 43% and recall 30% against 3% and 10% for the amplification flag; the difference in recall has a CI of $[-0.37, 0.13]$.

3.11 Evidence from thinner books

The deepest book in crypto is the least likely place to find amplification. This section moves to thinner books. It reports the pre-registered out-of-sample tests in thin Binance books, on Bybit and in thin US stocks, failures included, and labels every check decided after a test set was opened as post hoc. Figure 3.7 collects all out-of-sample evaluations of this section.

3.11.1 Design

We fixed 30 Binance USD-M perpetuals from the quote-volume ranking of one training day (June 10, 2024) before computing any T6 statistic, spread from BTCUSDT to symbols with under $5M daily volume. Data are sampled on the 10th and 24th of each month from January 2023 to September 2026 and pass the validation rules V1–V5 of Section 3.9: 2,565 of 2,700 symbol-days are valid; of 7,486,444 snapshots, V3 drops 212,745 and V4 drops 161,280, and V1 and V2 drop none. Liquidity terciles are fixed on median training-period depth within 1% of the mid. The thin tercile has $0.22M–$0.38M of depth within 1%, the thick tercile $0.97M–$220M. Flow is divided by each symbol's median absolute one-minute training flow, so that $\hat{\kappa}$ is in basis points per typical minute of flow. All other definitions follow Section 3.9. An audit that scrambles all data from t onward left every hourly feature unchanged in 30 of 30 trials.

Table 3.14: Local-projection overshoot by tercile, $\max_{k\geq 1}\beta_k - \beta_0$ [95% CI]

	Thick	Mid	Thin
(A) total	−0.045 [−0.079, 0.047]	−0.034 [−0.054, 0.024]	+0.007 [−0.050, 0.083]
(B) future flow fixed	−0.181 [−0.211, −0.152]	−0.075 [−0.091, −0.058]	−0.073 [−0.094, −0.051]
(C) flow and depth fixed	−0.243 [−0.276, −0.212]	−0.121 [−0.140, −0.102]	−0.113 [−0.135, −0.092]
Symbol-days	459	460	445

Basis points per typical minute of flow, horizons 0–30 minutes; local projections are estimated per symbol-day and averaged, with a symbol-day bootstrap.

Table 3.15: Depletion at the snapshot scale (about 30 seconds)

Symbol	Tercile	$\hat{\varepsilon}_{30}$ [95% CI]	Sweep extra [95% CI]	$\hat{\varepsilon}$, 1 minute	Days $\hat{\varepsilon}_{30} > 0$
BTCUSDT	thick	−0.0045 [−0.0058, −0.0033]	+0.0034 [0.0023, 0.0046]	−0.0025	0 of 6
XMRUSDT	mid	−0.0042 [−0.0058, −0.0029]	+0.0037 [0.0025, 0.0051]	−0.0011	0 of 6
RLCUSDT	thin	−0.0034 [−0.0065, −0.0012]	+0.0029 [0.0011, 0.0058]	−0.0003	1 of 6
CELRUSDT	thin	−0.0015 [−0.0032, 0.0001]	+0.0017 [0.0004, 0.0031]	−0.0004	2 of 6
CVXUSDT	thin	+0.0010 [−0.0033, 0.0060]	−0.0003 [−0.0040, 0.0027]	+0.0004	2 of 6

Day bootstrap over six training days per symbol. Depth is measured in bands of ±1% around the current mid, so a price move also recenters the band; this biases $\hat{\varepsilon}_{30}$ in an unknown direction.

3.11.2 The cross-section of the amplification condition

Table 3.13 confirms the BTCUSDT result across liquidity: mean $\hat{C}_3$ is negative in every tercile, and taker flow on average does not deplete the side it hits ($\hat{\varepsilon} < 0$ at the median everywhere). What changes with depth is the frequency of the amplifying regime, not its average. The share of symbol-days with $\hat{C}_3 > 0$ rises from 9.4% in the thick tercile to 16.1% in the thin tercile, with a Spearman correlation of −0.59 against log depth. Depth sensitivity $\hat{\varphi}$ is larger, and depletion is less negative and more often positive, in thinner books. We had predicted a negative correlation between log depth and mean $\hat{C}_3$; that prediction fails ($\rho = -0.001$).

The last row is an unexpected finding for Chapter 2. Measured feedback $\hat{Q}$ is larger in *deeper* books ($\rho = +0.70$): flow responds to price deviations most where liquidity, and plausibly algorithmic participation, is greatest. Feedback and amplification are therefore different objects. Deep books have more feedback and refill fast enough to absorb it; thin books have less feedback and, more often, too little reversion.

3.11.3 Overshoot and depletion

With future flow held fixed there is no overshoot in any tercile (Table 3.14); the registered prediction of a positive (B) overshoot in thin books fails. The response decays less in thin than in thick books, and holding depth fixed again makes it decay faster, as in Table 3.8.

To check that replenishment is not an artefact of one-minute aggregation, we sum signed taker flow exactly between consecutive depth snapshots from trade-level data and estimate depletion ε_{30} at about 30 seconds, with an additional coefficient on sweeps (intervals in the top decile of absolute flow). Table 3.15 reports five symbols on six training days each.

Table 3.16: Depletion at 30 seconds with a fixed and a re-centred band (Bybit order books, post hoc)

Band	$\hat{\varepsilon}_{30}$ [95% CI]	Share > 0	Coin-days
Fixed at previous mid	+0.0021 [+0.0013, +0.0032]	95%	20
Re-centred (Binance)	−0.0004 [−0.0007, −0.0001]	30%	20

ACH, ALICE, API3, LIT, WOO and ZEN, May 11–12 and September 11–12, 2025; flow normalized by each coin-day's median absolute 30-second flow; mean over coin-days with a bootstrap interval. Post hoc. Kind: real data (Part 13, `R5`).

In these re-centred bands, replenishment is stronger, not weaker, at 30 seconds: $\hat{\varepsilon}_{30}$ is negative with a CI below zero in three of five symbols and is never significantly positive. Sweeps offset most of it, so a sweep leaves the hit side about as deep as before ($\hat{\varepsilon}_{30}$ plus the sweep coefficient lies between -0.0012 and $+0.0007$). On this measure, the depletion channel that drives Theorem 3.4.2(i) would be absent even in thin books; the next subsection shows that the measure itself is biased toward replenishment.

3.11.4 Depletion on a fixed price grid

Binance's depth archive reports depth within bands around the *current* mid. After a buy moves the mid up, the ask band shifts into deeper resting orders and the bid band away from resting bids, which mechanically makes the book look ask-heavy after a buy, the sign we estimate. Bybit's order-book files contain every level, so the band can be held fixed at the previous mid. After seeing Table 3.15 we computed, for the six coins of Table 3.23 on four days each, the change in depth imbalance over 30-second intervals regressed on signed taker flow, once with the band fixed at the start-of-interval mid and once re-centred at the end-of-interval mid, Binance's construction.

The sign reverses (Table 3.16). With the band fixed, buying depletes the ask side in 19 of 20 coin-days; with the band re-centred, the same data reproduce the replenishment of Tables 3.7 and 3.15. The replenishment finding is therefore an artefact of band recentring, and depletion is present. Two conclusions survive. First, local projections show no peak overshoot once future flow is held fixed (Table 3.14), so depletion is not large enough to satisfy $\varphi\varepsilon > (1-a)\kappa$ in the data we observe; second, the depth term carries no out-of-sample forecasting content (Tables 3.19 and 3.21). Both statements rest on Binance's re-centred depth measure, and a fixed-grid measure of $\varphi\varepsilon$ in the same books would require order-level Binance data that are not public.

3.11.5 Pre-registered test in thin books

Before opening 2025–2026 for these symbols we registered the specification (SHA-256 `5910b6b4d90a...`). It is identical to Section 3.9 except for three changes, all fixed in ad-

Table 3.17: Pre-registered test, 30 symbols, 2025–2026

Sample	Outcome | candidate	ΔR^2 (pts) [95% CI]	ΔAUC [95% CI]	Holm p
Thin (primary)	MAE | $\hat{C}_3$	+1.216 [+0.273, +7.987]	+0.012 [+0.003, +0.019]	—
Thin	MAE | peak ratio	+0.065 [−0.072, +0.108]	+0.000 [−0.000, +0.001]	0.207
Thin	depth collapse | $\hat{C}_3$	−2.111 [−3.127, −1.173]	−0.012 [−0.023, −0.003]	1.000
Thin	OI drop | $\hat{C}_3$	+0.184 [−0.242, +1.128]	+0.040 [+0.017, +0.061]	< 0.001
Mid	MAE | $\hat{C}_3$	+0.003 [−0.056, +0.875]	+0.002 [−0.002, +0.005]	0.551
Thick	MAE | $\hat{C}_3$	+0.039 [−0.244, +0.057]	−0.001 [−0.003, +0.000]	0.361
All 30	MAE | $\hat{C}_3$	+0.117 [−0.015, +1.736]	+0.003 [−0.000, +0.006]	< 0.001

Primary: train 8,340 and test 6,521 symbol-hours on ten symbols; Clark–West $t = 8.53$, one-sided $p < 0.001$; baseline out-of-sample $R^2 = 0.025$; AUC 0.768 → 0.780. The pre-registered test passes. Holm p is the Clark–West p-value adjusted across the six secondary tests. Date-block bootstrap with 2,000 draws; bootstrap CIs are right-skewed because a few volatile dates carry much of the variation.

vance: the primary sample is the thin tercile; the regressions include symbol fixed effects; and the bootstrap resamples dates, with all symbols on a date kept together. The test cell was executed twice in the same session. The specification hash, the code and the data were unchanged between the two runs, and the evaluation is deterministic, so both runs give the numbers below; we report the count because the protocol requires it.

The primary test passes (Table 3.17). In thin books, $\hat{C}_3$ estimated from the previous six hours adds 1.2 points of out-of-sample R^2 for the next hour's maximum adverse excursion, beyond realized volatility, flow, volume, depth, imbalance and open interest, and raises the AUC for tail hours by 0.012. The effect is confined to thin books: in the mid and thick terciles it is zero, and in the pooled sample the Clark–West test rejects but the ΔR^2 interval contains zero. $\hat{C}_3$ also helps classify hours with a large open-interest drop, but it predicts depth collapse *worse* than the baseline, again pointing away from the depth channel.

Flags. On training data we set a $\hat{C}_3$ flag and a volatility flag; 575 of 6,521 test hours (8.8%) are tail hours. The $\hat{C}_3$ flag fires in 8.9% of hours with precision 16.5% and recall 16.7%, about twice the base rate. The volatility flag, which fires more often (16.0%), does better (precision 23.4%, recall 42.4%), and the union reaches recall 51.1% at precision 19.5%. $\hat{C}_3$ is a useful addition to volatility in thin books, not a replacement for it.

3.11.6 Tick size: a caution and a robustness check

Tick changes. We located tick changes by inferring each day's price grid and dating changes by binary search over daily files. We found 12 tick reductions, all by a factor of ten, on 11 symbols (GMTUSDT twice) between August 2023 and August 2026; nine fall in the test period. A difference-in-differences design compares 10 days before and after each event with four control symbols of similar liquidity. After a tick cut, $\hat{C}_3$ rises with a CI above zero in 10 of 12 events, but realized volatility in basis points falls sharply at the same time (by up to 32 bp on ALICEUSDT),

Table 3.18: Post-hoc tick-size robustness of the thin-book result

Check	Symbols	Test hours	ΔR^2 (pts) [95% CI]	Clark–West p
R0 baseline (Table 3.17)	10	6,521	+1.216 [+0.295, +8.017]	< 0.001
R1 + relative tick size and its log	10	6,485	+1.243 [+0.373, +7.627]	< 0.001
R2 drop symbols with any tick change	4	2,639	+0.511 [−0.069, +2.061]	< 0.001
R3 only days with the training tick	10	5,642	+1.131 [+0.256, +9.305]	< 0.001
R1 and R3 together	10	5,642	+1.222 [+0.357, +9.410]	< 0.001
R4 outcome in ticks	10	6,485	−0.127 [−0.319, −0.018]	0.991
R5 $\hat{C}_3$ percentile within symbol	10	6,521	+0.158 [−0.013, +1.322]	0.026
R6 leave one symbol out (10 fits)	9		+0.698 to +1.760, all CIs > 0	all < 0.001

Post hoc: decided after Table 3.17. 500 date-block bootstrap draws (200 in R6).

and the implied peak ratio does not change in any event (every CI includes zero). The Spearman correlation between the change in refill $\hat{r}$ and the change in the peak ratio is -0.04. We read this as a measurement result, not a test of the theory: at a coarse tick, prices move in discrete steps and the basis-point scale of $\hat{C}_3$ changes mechanically when the grid changes. The natural experiment cannot test the prediction that faster refill lowers the overshoot.

Does tick size drive Table 3.17? Eleven of the 30 symbols, six of them thin, changed tick, and five of those six did so inside the test period. After seeing Table 3.17 we therefore ran the post-hoc checks of Table 3.18, counted separately (one evaluation).

The result survives controls for tick size, the removal of every day traded at a tick different from training, and the removal of any single symbol (Table 3.18). It weakens when the six thin symbols that ever changed tick are dropped: four remain, and the point estimate halves with an interval that just includes zero. It disappears when the outcome is measured in ticks; at each tick cut this outcome jumps tenfold, a break the training fit cannot anticipate, so we regard R4 as a weak test, but we report it. A scale-free version of the candidate (its percentile within each symbol's training distribution) is weak. Our reading is that $\hat{C}_3$'s forecasting content is tied to its *level*, which mixes the amplification balance with the basis-point scale of each book. Sections 3.11.7 and 3.11.8 test whether that level is more than a measure of scale.

3.11.7 Out-of-sample replication on new symbols

A result found in ten symbols could be specific to them. We therefore drew 24 new symbols, seeded and before any analysis, from the 72 USD-M perpetuals that were not in the original 30 and whose training-day volume was at most 1.5 times that of the most liquid thin symbol. We applied the same validation (2,090 of 2,160 symbol-days valid) and classified symbols as thin with a cut fixed in advance at the boundary between the original thin and mid terciles (\$0.39M of depth within 1%). Ten new symbols are thin (ACH, ARPA, CHR, GTC, HOT, IOTX, OGN, SPELL, T and ZEN, all USDT-margined) and 14 are not. A new specification (SHA-256 `6aa8e2c03d79...`) fixed the primary test identical to Table 3.17 and, as secondary

Table 3.19: Pre-registered replication on ten new thin symbols, 2025–2026

Sample	Candidate	ΔR^2 (pts) [95% CI]	ΔAUC [95% CI]	Holm p
Thin (primary)	$\hat{C}_3$	+0.845 [+0.303, +6.371]	+0.008 [+0.002, +0.013]	—
Thin	momentum term $\kappa^2\eta q$	+0.641 [+0.395, +3.242]	+0.003 [−0.001, +0.008]	< 0.001
Thin	reversion term $(1-a)\kappa$	+0.688 [+0.204, +5.606]	+0.008 [+0.003, +0.012]	< 0.001
Thin	depth term $\varphi\varepsilon$	+0.013 [−0.213, +1.155]	−0.001 [−0.005, +0.003]	0.308
Thin	unit-free ratio A	−0.045 [−0.096, −0.031]	−0.000 [−0.001, +0.000]	0.974
Not thin (14)	$\hat{C}_3$	−0.042 [−0.131, +0.780]	+0.002 [−0.003, +0.007]	0.308

Primary: train 8,520 and test 7,092 symbol-hours; Clark–West $t = 11.66$, one-sided $p < 0.001$; AUC 0.768 → 0.776; training coefficient +54.3 (Table 3.17: +88.3). Training coefficients: momentum +104.3, reversion −53.0. The pre-registered replication passes.

tests, the three terms of C_3 separately, a unit-free ratio $A = (\varphi\varepsilon + \kappa^2\eta q)/((1-a)\kappa) - 1$ that is positive exactly when $C_3 > 0$, and the not-thin group. The test set was opened once.

The result replicates (Table 3.19). On symbols never used before, $\hat{C}_3$ adds 0.85 points of out-of-sample R^2 with an interval excluding zero, and improves tail classification. It again has no value outside thin books. The decomposition shows where the forecasting content lies, and every sign matches Theorem 3.4.2(i) and Theorem 3.4.2(iii). A larger momentum term predicts larger adverse excursions; a larger reversion term, which pushes C_3 down, predicts smaller ones; and the depth term, whose depletion factor is close to zero (Table 3.15), predicts nothing. In training data the reversion term is by far the largest component (median 0.033, against −0.0009 for momentum and −0.0003 for depth), so most of the variation in $\hat{C}_3$ is variation in impact-weighted mean reversion.

The unit-free ratio fails. Because the amplifying terms are small, its median is −0.996 and its variation is mostly the noise of a ratio with an estimated denominator. Together with R5 in Table 3.18, this means that on these samples the forecasting content was tied to the level of $\hat{C}_3$. Section 3.11.9 tests two better-behaved unit-free statistics on fresh data.

3.11.8 Is $\hat{C}_3$ just price impact?

Both working terms contain the impact coefficient κ, so $\hat{C}_3$ might forecast excursions only because it measures illiquidity. After seeing Table 3.19 we added four impact controls, all from the same past six hours or hour and winsorized at training percentiles: $\hat{\kappa}$ from the same fit, $\hat{\kappa}^2$, a univariate Kyle's λ (Kyle, 1985) (slope of returns on signed flow) and a log Amihud ratio (Amihud, 2002) (absolute return over volume). This is a post-hoc check, evaluated once and counted separately.

$\hat{C}_3$ is not a proxy for impact (Table 3.20). With all four impact controls it keeps about 80% of its gain in R^2 (0.68 of 0.85 points), with an interval above zero, and both the momentum and the reversion terms survive separately. Impact on its own forecasts nothing. Controlling for Kyle's λ and Amihud alone *raises* the gain to 1.09 points: once plain illiquidity is held fixed, the dynamic balance that $\hat{C}_3$ measures stands out more clearly. One qualification: with $\hat{\kappa}$ controlled,

Table 3.20: Post-hoc price-impact controls, ten new thin symbols, 2025–2026

Candidate | added controls	ΔR^2 (pts) [95% CI]	ΔAUC [95% CI]	Clark–West t
$\hat{C}_3$ | none (Table 3.19)	+0.845 [+0.303, +6.371]	+0.008 [+0.002, +0.013]	11.66
$\hat{C}_3$ | $\hat{\kappa}$, $\hat{\kappa}^2$	+0.755 [+0.371, +4.718]	+0.006 [−0.000, +0.010]	6.63
$\hat{C}_3$ | Kyle's λ, Amihud	+1.089 [+0.498, +7.149]	+0.007 [+0.001, +0.013]	8.02
$\hat{C}_3$ | all four	+0.682 [+0.334, +4.525]	+0.004 [−0.001, +0.009]	5.69
Momentum term | all four	+0.487 [+0.311, +2.384]	+0.002 [−0.002, +0.006]	3.42
Reversion term | all four	+0.529 [+0.222, +3.762]	+0.005 [−0.000, +0.009]	8.45
$\hat{\kappa}$ alone | none	−0.026 [−0.084, +0.030]	+0.000 [−0.000, +0.000]	−1.77

Post hoc: decided after Table 3.19. Same evaluation code and bootstrap as Table 3.19; Holm-adjusted $p < 0.001$ for every row except $\hat{\kappa}$ alone ($p = 0.96$). Training correlations of $\hat{C}_3$ with the controls: $\hat{\kappa}$ −0.43, Kyle's λ −0.44, Amihud +0.10.

Table 3.21: Pre-registered test on fresh days, 2025–2026 (impact controls in the baseline)

Sample	Candidate	ΔR^2 (pts) [CI]	ΔAUC [95% CI]	Holm p
Thin (primary)	$\hat{C}_3$	+1.583 [+0.588, +3.237]	+0.002 [−0.003, +0.006]	—
Thin	reversion term	+1.442 [+0.679, +2.621]	+0.003 [−0.001, +0.006]	< 0.001
Thin	momentum term	+0.422 [+0.000, +1.063]	−0.002 [−0.005, +0.001]	< 0.001
Thin	depth term	+0.323 [+0.092, +0.639]	+0.001 [−0.001, +0.003]	< 0.001
Thin	gain multiple $S_{60}/\lvert\hat{h}_0\rvert$	+0.488 [+0.252, +0.851]	+0.000 [−0.001, +0.001]	< 0.001
Thin	log reversion half-life	+0.619 [+0.351, +1.031]	+0.001 [+0.000, +0.003]	< 0.001
Thin	$\hat{C}_3$ without depth, $\hat{C}_3^{\text{flow}}$	+1.747 [+0.725, +3.393]	+0.001 [−0.004, +0.006]	< 0.001
Mid (control)	$\hat{C}_3$	+0.055 [−0.047, +0.239]	+0.000 [−0.001, +0.001]	0.038

Primary: train 16,860 and test 13,428 symbol-hours on 39 test dates; 97.5% date-block bootstrap interval; Clark–West $t = 9.46$, one-sided $p < 0.001$; baseline out-of-sample $R^2 = 0.185$; training coefficient +68.0. The pre-registered test passes. Secondary rows: 95% intervals, Holm-adjusted Clark–West p across the eleven secondary tests of the specification. Thin: 20 symbols; mid: 10. The gain multiple and the half-life are unit-free. S_{60} is the summed absolute impulse response over 60 minutes; $\hat{C}_3^{\text{flow}} = \hat{\kappa}^2\hat{\eta}\hat{q} - (1-\hat{a})\hat{\kappa}$ from the model without depth.

the gain in tail *classification* is no longer significant. The robust claim is about the size of the next hour's adverse excursion, not about flagging tail hours beyond what impact already flags.

3.11.9 A third test on fresh days

Parts 4 and 5 of the replication package use the 10th and 24th of each month. For a third test we registered a new specification (SHA-256 `65acda361861...`) before downloading the 3rd and 17th of each month of 2025–2026, days never used in any analysis. Training reuses the 2023–2024 days of the earlier tests. The sample is all 20 thin symbols of Sections 3.11.5 and 3.11.7, with the ten mid-tercile symbols as a negative control. Two changes make the test harder: the four price-impact controls of Table 3.20 are now part of the baseline, and because the specification has two primary hypotheses (this one and the backtest of Section 3.11.13) each is tested at $\alpha = 0.025$ with 97.5% intervals. The test set was opened once; 1,137 fresh symbol-days passed validation.

The result holds a third time, and more strongly (Table 3.21). On fresh days, with impact already in the baseline, $\hat{C}_3$ adds 1.6 points of R^2 for the next hour's adverse excursion in thin

Table 3.22: Pre-registered test beyond a volatility-forecasting baseline, fresh days 2025–2026

Sample	Candidate	ΔR^2 (pts) [95% CI]	ΔAUC [95% CI]	Holm p
Thin (primary)	$\hat{C}_3$	+2.787 [+1.425, +4.944]	+0.005 [+0.002, +0.008]	—
Thin	summed response S_{60}	+2.098 [+1.165, +3.597]	+0.004 [+0.001, +0.006]	< 0.001
Thin	reversion term	+2.451 [+1.257, +4.379]	+0.004 [+0.001, +0.007]	< 0.001
Thin	momentum term	+0.731 [+0.238, +1.429]	+0.001 [−0.001, +0.002]	< 0.001
Thin	gain multiple $S_{60}/\lvert\hat{h}_0\rvert$	+0.563 [+0.259, +1.003]	+0.001 [−0.001, +0.002]	< 0.001
Thin	log reversion half-life	+0.806 [+0.446, +1.351]	+0.001 [−0.000, +0.002]	< 0.001
Thin	$\hat{C}_3^{\text{flow}}$ (no depth)	+2.989 [+1.453, +5.424]	+0.005 [+0.002, +0.009]	< 0.001
Mid (control)	$\hat{C}_3$	+0.316 [+0.108, +0.635]	+0.002 [+0.001, +0.004]	< 0.001

Thin: 20 symbols, train 16,860 and test 13,722 symbol-hours on 40 test dates; mid: 10 symbols. Primary: Clark–West $t = 12.59$, one-sided $p < 0.001$; baseline out-of-sample $R^2 = 0.186$, full 0.214; AUC 0.826 → 0.831. The pre-registered test passes. Baseline: the controls of Table 3.21, the four impact controls, realized volatility over 15 minutes and since midnight, an EWMA volatility forecast ($\lambda = 0.94$), the previous hour's adverse excursion, hour-of-day fixed effects and symbol fixed effects. The gain multiple and the half-life are unit-free. Kind: real data.

books. Three findings sharpen the earlier picture. First, two unit-free statistics now work: the gain multiple, the summed response divided by impact, and the log half-life of reversion both add about half a point of R^2 with intervals above zero. The scale concern of Section 3.11.6 is therefore answered: the content is the dynamic balance, not the basis-point scale of the book, in the sense of the invariance hypotheses of Kyle and Obizhaeva (2016). Second, the reversion term carries most of the signal, as Theorem 3.4.4 suggests, since S depends on reversion through $1/(m - q\tilde{\kappa})$. Third, the depth-free statistic forecasts as well as the full one, which matters for the next subsection. In mid-depth books the Clark–West test rejects at the Holm-adjusted 5% level, but the R^2 gain is 0.06 points with an interval that includes zero. As with the impact controls, the gain in tail *classification* is not significant: the robust claim is about the size of excursions.

3.11.10 Beyond volatility forecasting

The adverse excursion of the next hour is close to a volatility outcome, so a natural objection is that $\hat{C}_3$ only proxies for future volatility in a way the simple volatility controls miss. We registered a fourth test (SHA-256 `71983f1a9d2c...`) on the 6th and 20th of each month of 2025–2026, days not used in any other test, whose baseline adds the standard tools of volatility forecasting to every control used so far: realized volatility over 15 minutes, one hour, six hours and since midnight in the spirit of HAR models (Corsi, 2009), an exponentially weighted (RiskMetrics) volatility forecast (J.P. Morgan/Reuters, 1996), the previous hour's adverse excursion, and hour-of-day fixed effects, together with the four price-impact controls. The test set was opened once; 1,161 fresh symbol-days passed validation.

The result does not weaken against a volatility-forecasting baseline; it strengthens (Table 3.22). $\hat{C}_3$ adds 2.8 points of out-of-sample R^2 on top of a baseline that already explains 18.6%, and for the first time it also improves the classification of tail hours with an interval

excluding zero. The summed response S_{60}, the statistic that the guard bounds and that Theorem 3.4.4 identifies as cumulative transient risk, adds 2.1 points on its own. The volatility benchmarks, by contrast, add nothing out of sample beyond the controls of Table 3.21 (-0.76 points, interval $[-3.38, +2.10]$), so the forecasting content of the dynamic balance is not a volatility effect that richer volatility models would have captured. In mid-depth books the gain is significant in this test but small, about a ninth of the thin-book gain; across the four tests with a deeper-book control it was significant once.

Against VPIN. The best-known early-warning measure for flow-driven crashes is VPIN (Easley et al., 2012). After the fact, on the same test hours (Part 13, `R8`), we computed VPIN from the previous six hours, the window of $\hat{C}_3$, as the mean absolute buy–sell imbalance over 50 equal-volume buckets; Binance reports taker-buy volume, so the imbalance is measured rather than classified. VPIN adds 1.2 points of R^2 to the baseline (95% interval $[0.4, 2.3]$). $\hat{C}_3$ added to the baseline *and* VPIN still adds 2.0 points ($[1.0, 3.4]$, Clark–West $t = 12.1$) and improves tail classification, keeping about 73% of its 2.8-point gain. The two are negatively related (Spearman -0.35), and VPIN's training coefficient is negative, so high imbalance precedes smaller, not larger, excursions in these books. This comparison is post hoc and on a test set already opened; it shows that $\hat{C}_3$ is not a repackaging of order-flow toxicity.

3.11.11 A second venue: Bybit

To ask whether the result is specific to Binance we registered a test on Bybit USDT perpetuals of the same 20 coins (SHA-256 recorded in the package), using Bybit's public trade archive, which has no depth history. Fourteen of the coins have Bybit data across 2023–2026. Minute bars are built from trades (last price, signed taker notional), training uses the 10th and 24th of each month of 2023–2024, and testing uses the 3rd, 10th, 17th and 24th of each month of 2025–2026 (10,764 test hours). The statistic is $\hat{C}_3^{\text{flow}}$, which Table 3.21 shows to work on Binance.

The test fails without ambiguity. $\hat{C}_3^{\text{flow}}$ adds -0.0000 points of R^2 (95% CI $[-0.001, 0.001]$, Clark–West $p = 0.50$), and neither the momentum nor the reversion term helps (Holm $p = 1.00$). In training data the momentum coefficient even has the opposite sign (-130, against $+110$ on Binance). Because the depth-free statistic works on Binance on the same coins, the failure concerns the venue rather than the statistic. We considered two candidate reasons and checked the first after the fact: bid–ask bounce in Bybit's last-trade prices, and different participants. On the same 14 coins and training days, Bybit's one-minute returns are *less* negatively autocorrelated than Binance's (median -0.016 against -0.031), so bounce is not the difference; trading is: 11% of Bybit minutes have no trade, against 0.1% on Binance, and 29% of Bybit minutes have a zero return, against 14% (Part 13, `R3`). Sparse trading leaves the flow and price series that the statistic is estimated from much noisier on Bybit.

Table 3.23: Pre-registered test on Bybit with order-book depth, 2025–2026

Candidate	ΔR^2 (pts) [95% CI]	Clark–West t	Holm p
$\hat{C}_3$ (primary)	+0.033 [+0.019, +0.053]	5.73	—
Momentum term	+0.083 [+0.017, +0.168]	2.94	0.010
$\hat{C}_3^{\text{flow}}$ (no depth)	+0.064 [−0.009, +0.152]	2.61	0.023
Summed response S_{60}	+0.021 [−0.025, +0.071]	1.31	0.383
Reversion term	−0.039 [−0.139, +0.058]	−0.61	1.000
Gain multiple	−0.043 [−0.137, +0.044]	−0.74	1.000
Log reversion half-life	−0.099 [−0.249, +0.041]	−1.47	1.000

Six coins, train 1,224 and test 2,988 symbol-hours; baseline out-of-sample $R^2 = 0.251$; AUC unchanged (0.808 → 0.808); training coefficient on $\hat{C}_3$ +4.4 (Binance, Table 3.22: +70.8). Median depth within 1% on training days \$0.30M–\$0.43M. The pre-registered criterion is met. Specification SHA-256 `ec2145fac5f5...`; test opened once. Kind: real data (Part 10).

Bybit with order-book depth. Bybit also distributes daily order-book files for its perpetuals (every update to the top 200 or 500 price levels). We registered a second Bybit test (SHA-256 `ec2145fac5f5...`) before downloading any test day, with the full Part 9 baseline, depth within 1% of the mid rebuilt from the order book at the end of each minute, and only the best 200 levels used so that the 2024 (500-level) and 2025–2026 (200-level) files are measured identically. Days were the 11th to 16th of a month, never used in any earlier test: two training months in 2024 and six test months from January 2025 to September 2026. A day is valid if at least 80% of its minutes have trades and a valid book; sparse trading in thin coins on Bybit leaves 581 of 882 coin-days valid, and the inclusion rule (at least 8 valid training days and 24 valid test days) keeps six coins (ACH, ALICE, API3, LIT, WOO, ZEN; 2,988 test hours on 36 dates).

The pre-registered criterion is met, but the effect is negligible (Table 3.23). $\hat{C}_3$ adds 0.033 points of R^2 with an interval above zero and a Clark–West t of 5.7: statistically clear, economically about one eightieth of the Binance gain, with no improvement in tail classification and a training coefficient one sixteenth as large. Of the components, only the momentum term helps; the reversion term, which carries the Binance result, does not. Two features of the sample may explain part of the gap, and we test neither: the six included Bybit books carry \$0.30M–\$0.43M within 1%, at or above the boundary between Binance's thin and mid terciles, where the Binance effect was also small; and the inclusion rule removed the sparsest books. We read the two Bybit tests together: with trades alone the statistic has no content on Bybit, and with depth its content is detectable but too small to matter. Section 3.11.12 asks the same question in a regulated market.

3.11.12 A regulated market: thin US stocks

The Bybit tests leave open whether the result belongs to thin markets or to one crypto venue. We therefore registered a test in US equities (SHA-256 `f5800a69c219...`) before downloading

Table 3.24: Pre-registered test in thin US stocks, 2025–2026

Candidate	ΔR^2 (pts) [95% CI]	Clark–West t	Holm p
Thin, $\hat{C}_3^{\text{flow}}$ (primary)	+0.499 [+0.272, +0.779]	5.05	—
Thin, summed response S_{60}	+0.153 [+0.046, +0.272]	3.36	0.001
Thin, reversion term	+0.613 [+0.322, +0.967]	5.15	< 0.001
Thin, momentum term	−0.011 [−0.288, +0.261]	0.63	0.263
Mid-size control, $\hat{C}_3^{\text{flow}}$	+0.316 [+0.074, +0.634]	3.12	0.002

23 thin names, train 3,744 and test 3,808 symbol-hours on 42 test dates; baseline out-of-sample $R^2 = 0.340$; AUC 0.846 → 0.847 (95% interval of the change $[-0.00002, +0.0026]$); training coefficient on $\hat{C}_3^{\text{flow}}$ +21.7. Mid-size controls: 10 names, 1,680 test symbol-hours. Specification SHA-256 `f5800a69c219...`; test opened once. Kind: real data (Part 14).

any test day. Data are consolidated (SIP) trades from the Alpaca market-data API, aggregated into minute bars of the regular session: last trade price, dollar volume and dollar flow signed by the tick rule. The data contain no depth, so the candidate is the depth-free statistic $\hat{C}_3^{\text{flow}}$, the version that worked on Binance (Table 3.21). The universe was drawn by a seeded rule from June 2024 daily bars, before any test data: 40 thin names (price \$5–\$100, median daily dollar volume \$2M–\$10M) and 10 mid-size controls (\$50M–\$200M). Forecasts are made 120, 180, 240 and 300 minutes after the open from the preceding 120 minutes, and the outcome is the maximum adverse excursion over the next 60 minutes; the baseline is the full volatility-forecasting and price-impact baseline of Table 3.22. Training uses 48 days in 2023–2024 (the 10th and 24th of each month or the next trading day) and testing 42 days from January 2025 to September 2026 (the 6th and 20th). A day is valid if at least 80% of its minutes have a trade, and a name enters if it has at least 30 valid training days and 25 valid test days, which keeps 23 of the 40 thin names and all 10 controls. The test cell was run once.

The test passes (Table 3.24). $\hat{C}_3^{\text{flow}}$ adds 0.50 points of out-of-sample R^2 to the full baseline (95% interval $[0.27, 0.78]$, Clark–West $t = 5.05$), and the summed response S_{60}, the guard's own statistic, adds 0.15 points. As on Binance, the content comes from impact-weighted reversion: the reversion term alone adds 0.61 points, and the momentum term adds nothing. The effect is smaller than in thin Binance books, about a fifth of the 2.8 points of Table 3.22 on a baseline that already explains 34% of the variance, but about fifteen times the Bybit effect, and tail classification does not improve significantly. Unlike on Binance, the mid-size controls also show a gain (0.32 points), against our prediction of little or none: in US stocks the information is not confined to the thinnest names. The controls trade \$50M–\$200M a day, far below the deepest crypto books, and we included no large-capitalization stocks, so the test does not say where the effect ends. Three features of the sample limit the reading. The selection rule did not exclude funds, so the included thin names contain one closed-end fund (CRF) and the controls one bond ETF (BNDX); the validity rule removed the 17 thinnest-trading names; and the asset list contains only names still listed in 2026. Tick-rule signing also misclassifies some trades. Within these limits, the result is not a property of one crypto venue: the reduced-form content of the

Table 3.25: Position-sizing backtest, thin books, fresh days 2025–2026

Rule	Risk measure R_t	TLR	Breach rate	$\log \mathrm{TLR} - \log \mathrm{TLR}_{\mathrm{vol}}$ [CI]
Volatility limit	$\hat{\sigma}_{6h}$	348.6	1.9%	—
Amplification-adjusted (primary)	$\hat{\sigma}_{6h}\,(S_{60}/\lvert\hat{h}_0\rvert)^{0.25}$	342.7	2.1%	−0.017 [−0.045, +0.011]
Peak-response guard	$U_6 \max_k \lvert\hat{h}_k\rvert$	411.4	5.3%	+0.165 [+0.095, +0.243]
Summed-response guard	$U_6\, S_{60}$	522.2	13.2%	+0.404 [+0.285, +0.525]
Static	1	476.8	5.3%	+0.313 [+0.223, +0.397]

20 thin symbols; U_6 is the 95th percentile of absolute normalized flow over six hours. Primary row: 97.5% interval; others 95%. Breach: loss above the training 95th percentile of the static rule. Bybit: training chose $\theta = 0$, and the summed guard is worse than volatility by +0.484 [0.251, 0.662].

reversion statistic carries over, at a smaller size, to a regulated equity market.

3.11.13 The guard as a position limit

The guard of Section 3.6 bounds worst-case loss; a separate question is whether its ingredients help a dealer size positions in practice. We registered a backtest with Table 3.21. Each hour a rule sets the position $q_t = c_s/R_t$, where R_t is a risk measure from the past six hours and c_s sets the training mean of q to one for each symbol and rule; the hour's loss is q_t times the adverse excursion, and the score is tail loss per unit of average position, $\mathrm{TLR} = \mathrm{ES}_{95}(L)/\overline{q}$. The primary comparison is between a volatility limit (R = six-hour realized volatility) and the same limit scaled by the gain multiple to a power θ chosen on training data ($\theta = 0$ is the volatility limit); training data chose $\theta = 0.25$.

The primary backtest fails (Table 3.25). Scaling the volatility limit by the gain multiple lowers tail loss per unit of position in 91% of bootstrap draws, but the interval includes zero. Rules built only on the estimated response, without volatility, are clearly worse than the volatility limit and even than a static limit: a six-hour estimate of S is too noisy to size positions on its own. The guard theorems are statements about worst-case loss under a known or certified operator, and this evidence does not contradict them; but it means that, with public data and six-hour windows, an estimated summed response adds at most a little to volatility as a sizing input. A practical guard should combine the two and certify the operator with more data than a six-hour window; Section 3.11.14 tests exactly that.

3.11.14 A certified guard

The guard theorems assume a certified operator, and a six-hour estimate is a poor certificate. We therefore registered a second backtest (SHA-256 `b9a56af48c93...`) in which the operator of each symbol is certified once a day by pooling the minutes of the previous three days, and applied to every hour of the following day. Evaluation days are the 28th of each month and certification windows the 25th to 27th; none of these days was used in any earlier test. Training is 2024 (12 evaluation days), testing January 2025 to September 2026 (20 evaluation days, 6,894

Table 3.26: Backtest with a certified operator, thin books, fresh days 2025–2026

Rule	R_t	TLR	Breach	$\Delta \log$ TLR vs vol [95% CI]
Volatility limit	$\hat{\sigma}_{6h}$	296.7	2.0%	—
Certified, amp.-adjusted (primary)	$\hat{\sigma}_{6h}\,\mathrm{GM}_{\mathrm{cert}}^{0.25}$	289.7	2.3%	−0.024 [−0.051, −0.0004]
Combined ceiling	stricter of the two	299.3	2.9%	+0.009 [−0.023, +0.039]
Certified summed guard	$U_{\mathrm{cert}} S_{\mathrm{cert}}$	416.5	13.8%	+0.339 [+0.241, +0.436]
Six-hour summed guard	$U_6 S_{60}$	424.8	14.9%	+0.359 [+0.237, +0.475]
Static	1	383.2	4.0%	+0.256 [+0.132, +0.387]

20 thin symbols; certified quantities from pooled within-day OLS on the minutes of the previous three days (median three valid days), fixed for the evaluation day. $\theta = 0.25$ chosen on training TLR over $\{0, 0.25, 0.5, 0.75, 1\}$. Combined ceiling: $R = \max(\hat{\sigma}_{6h}/\bar{\sigma},\, U_{\mathrm{cert}} S_{\mathrm{cert}}/\overline{US})$ with training medians. Breach: loss above the training 95th percentile of the static rule. Certified against six-hour summed guard: −0.020 [−0.165, +0.118]. Secondary comparisons Holm $p = 1.00$. Specification SHA-256 `b9a56af48c93...`; test opened once. Kind: real data (Part 12).

symbol-hours on the 20 thin symbols). The primary rule scales the six-hour volatility limit by the certified gain multiple, $R_t = \hat{\sigma}_{6h}\,\mathrm{GM}_{\mathrm{cert}}^{\theta}$, with θ chosen on training data; the score is the tail loss per unit of position of Section 3.11.13.

The primary test passes, narrowly (Table 3.26). With a certified operator, scaling the volatility limit by the gain multiple lowers tail loss per unit of position by 2.4%, with a 95% interval of $[-5.1\%, -0.04\%]$; training data again chose $\theta = 0.25$, as in Section 3.11.13. The gain is small and its interval almost touches zero, so we read it as a modest improvement, not a decisive one: amplification information helps as a correction to a volatility limit once the operator is estimated from days rather than hours. Every secondary comparison fails. A limit built on the certified summed response alone is far worse than volatility, about as bad as the six-hour version, so certification does not rescue the guard as a stand-alone sizing rule; and a combined ceiling that applies the stricter of the two limits does no better than volatility. The guard's theorems concern worst-case loss under a known operator; as a practical limit, its statistic earns its place as a modest adjustment to volatility, not as a replacement. The comparison also differs from the theorem in form: the backtests size positions as c/R_t, which levers a position up when the estimated response is small, whereas the guard of Lemma 3.6.1 is a one-sided cap, $\min(1, B/(IUS))$, that only scales admission down. After the fact we applied the certified guard in that form, throttling a position whenever $U_{\mathrm{cert}} S_{\mathrm{cert}}$ exceeds its training median (27% of test hours; Part 13, `R1`). As a cap on a static limit it lowers tail loss per unit of position by 13% (95% interval $[-27\%, -3\%]$); as a cap on a volatility limit it adds nothing (+0.8%, $[-1.3\%, +2.7\%]$). Used as the theorem uses it, the guard helps where no volatility information is used and is subsumed by volatility where it is.

3.11.15 October 10, 2025

The largest liquidation day on record motivated this chapter, so we report what the statistics did on it, descriptively. The event was chosen after the fact and the 10th of each month was

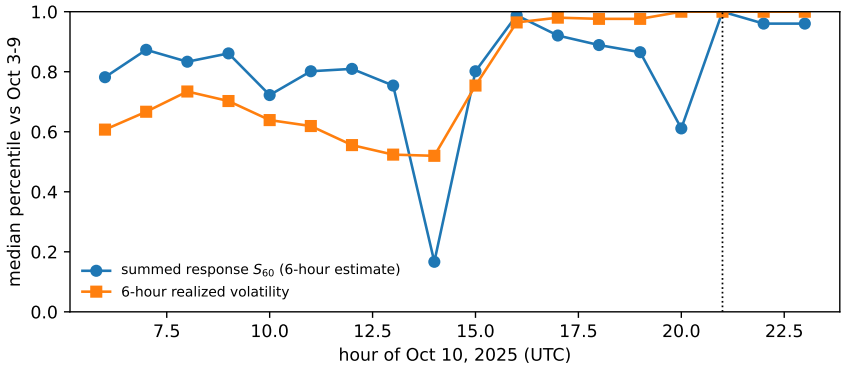


Figure 3.6: October 10, 2025: median percentile, across thin symbols, of the six-hour summed response S_{60} and of six-hour realized volatility, relative to each symbol's October 3–9 distribution. Dotted line: the crash hour (21:00 UTC). Descriptive; the event was chosen after the fact.

a test day in the first two thin-book tests, so this is not a test. For the 20 thin symbols we computed the hourly six-hour estimates on October 3–11, 2025, and expressed each hour of October 10 as a percentile of the same symbol's October 3–9 distribution. The crash hour, the hour on October 10 with the largest adverse excursion in BTCUSDT, began at 21:00 UTC; the median thin-symbol excursion in that hour was about 7,500 basis points.

Figure 3.6 shows two things. First, the summed response was elevated through the whole day: in the morning, before volatility rose, the median thin symbol's S_{60} sat at the 72nd to 87th percentile of its previous week, against the 52nd to 73rd for volatility, and at the start of the crash hour 16 of 18 thin symbols with valid estimates were above their previous-week 90th percentile. Second, volatility reached the top of its range from 16:00 onward as well, so on this day the summed response did not warn of anything volatility missed in the final hours; and across symbols it did not rank which coins fell furthest (Spearman -0.11, against $+0.36$ for volatility). We draw no inference from one day. The episode is consistent with the thin-book result, an elevated dynamic risk state that sizes excursions rather than timing them, and it illustrates why the paper's claims are about the size of the next hour's excursion, not about predicting crashes.

3.11.16 Every pre-registered test

Table 3.27 lists every pre-registered test in this chapter, with its outcome and the number of times its test set was opened. Of twenty-five pre-registered primaries, fifteen pass, nine fail and one was not tested because the test it depended on failed. The failures are in BTCUSDT futures, in BTCUSDT spot, on Bybit with trades alone, on Coinbase (Part 17), in the six-hour position-sizing backtest, in the two five-minute open-interest liquidation-mechanism primaries (Part 22), in the unbounded-rescaling 18-hour matched-breach capacity test (Part 23), and in the strict joint placebo clause of the persistence mediation test (Part 25, where 15-minute persistence

mediates 97.6% of $\hat{C}_3$'s excursion ΔR^2 while the five-minute jump placebo also exceeds 50%). Five forecasting passes are in thin Binance books (including 15-minute persistence in Part 25), one in thin US stocks, two on untouched days (Part 15), three on untouched symbols (Part 16), and two in the capped Rule 15c3-5 pre-trade throttle test (Part 24); the certified-guard backtest passes narrowly; and the Bybit test with depth meets its criterion with a negligible effect. The analyses of Parts 19–21 (Section 3.12.6) were specified in advance but not hash-registered and are not counted here. Figure 3.7 adds the post-hoc checks.

3.11.17 Threats to validity

Multiple testing. The program registered twenty-five primaries, of which twenty-four were tested, and ran the post-hoc checks of Figure 3.7. The ten passing forecasting primaries have Clark–West statistics between 5.0 and 14.8 (one-sided $p < 0.001$ in each), the Part 15 guard primary has a one-sided bootstrap p of 0.0025, and the two Part 24 capped-throttle primaries have one-sided bootstrap $p \leq 0.002$; the nine failures are reported with the same prominence.

Sequential design. The program was sequential. Each pre-registration fixed one test's specification before its test data were opened, but which test to run next was chosen after the earlier results were known: new symbols after the first pass, impact controls and then a volatility baseline in response to the obvious objections, depth on Bybit after the trade-only test failed, a regulated market after the second venue gave a negligible effect, untouched days and symbols after the information-safety theory, and a certified operator after the six-hour backtest failed, and a new venue last. Pre-registration therefore protects each test from specification search on its own test data; it does not protect the program from choices that adapted across tests. That is why every test is reported, failures included (Figure 3.7), and why the correction above is taken over all tested primaries.

Dependence among the passes. The four passing tests share the 2023–2024 training days, and the original ten thin symbols appear in three of them. They are not four independent replications. They vary three things one at a time: the symbols (Table 3.19, ten new symbols), the test days (Tables 3.21 and 3.22, days never used before) and the baseline (impact controls, then a full volatility-forecasting model). The Part 16 test (Table 3.30) shares their training days but none of their symbols. The US test (Table 3.24) shares none of their data: a different market, asset class, data source, universe and set of days.

Look-ahead and estimated regressors. Every feature uses minutes before the forecast hour; winsorization limits, flow scales, θ and tail thresholds come from training data; and a scrambling audit leaves every hourly feature unchanged when data from t onward are permuted. $\hat{C}_3$ and S_{60} are estimated from six hours, but the evaluation compares out-of-sample forecasts

Table 3.27: All pre-registered tests

Test (specification hash)	Sample	Primary statistic [CI]	Result	Opened
BTCUSDT futures, MAE | $\hat{C}_3$ (`d9521218`)	2025–26	ΔR^2 +0.018 [−0.100, +0.132]	fail	1
BTCUSDT spot, MAE | amplification (`b1fd7f8e`)	2024	ΔR^2 −0.023 [−0.056, −0.002]	fail	3
Thin books, 10 symbols (`5910b6b4`)	2025–26	ΔR^2 +1.216 [+0.273, +7.987]	pass	2
New thin symbols, 10 (`6aa8e2c0`)	2025–26	ΔR^2 +0.845 [+0.303, +6.371]	pass	1
Fresh days, 20 thin, impact controls (`65acda36`)	2025–26	ΔR^2 +1.583 [+0.588, +3.237]	pass	1
Position sizing, same specification	2025–26	$\Delta \log \text{TLR}$ −0.017 [−0.045, +0.011]	fail	1
Bybit, 14 coins, depth-free	2025–26	ΔR^2 −0.000 [−0.001, +0.001]	fail	1
Beyond volatility forecasting, 20 thin (`71983f1a`)	2025–26	ΔR^2 +2.787 [+1.425, +4.944]	pass	1
Certified guard, position sizing (`b9a56af4`)	2025–26	$\Delta \log \text{TLR}$ −0.024 [−0.051, −0.0004]	pass	1
Bybit with order-book depth, 6 coins (`ec2145fa`)	2025–26	ΔR^2 +0.033 [+0.019, +0.053]	pass	1
Thin US stocks, 23, depth-free (`f5800a69`)	2025–26	ΔR^2 +0.499 [+0.272, +0.779]	pass	1
Untouched days, frozen model (`69849c96`)	2025–26	ΔR^2 +4.196 [+1.551, +8.233]	pass	1
Untouched days, guard vs static (same)	2025–26	$\Delta \log \text{TLR}$ −0.084 [−0.173, −0.011]	pass	1
Untouched symbols, 5 thin (`0523e476`)	2025–26	ΔR^2 +0.727 [+0.377, +1.428]	pass	1
Untouched symbols, beyond VPIN (same)	2025–26	ΔR^2 +0.562 [+0.274, +1.159]	pass	1
Untouched symbols, guard vs static (same)	2025–26	$\Delta \log \text{TLR}$ −0.256 [−0.497, −0.021]	pass	1
New venue, Coinbase, 6 thin (`3233c52b`)	2025–26	ΔR^2 −0.092 [−0.407, +0.158]	fail	1
Coinbase, real minus placebo (same)	2025–26	+0.013 [−0.331, +0.298]	not tested	1
OI liquidation forecast, 19 thin (`2b3794e3`)	2025–26	ΔR^2_{LIQ} −0.795 [−1.529, −0.179]	fail	1
OI liquidation mediation, 19 thin (same)	2025–26	Share −6.8% [−10.2%, −3.1%]	fail	1
Non-resetting 18h guard capacity (`d4169377`)	2025–26	$\Delta\overline{N}$ (k$) +332.8 [−20.4, +1180.8]	fail	1
Capped 18h throttle liq. rate (`698f9119`)	2025–26	ΔLiq −7.41 [−11.45, −4.15] pp	pass	1
Capped 18h throttle ES_{95} (same)	2025–26	ΔES_{95} (k$) −26.4 [−53.1, −6.3]	pass	1
Persistence forecast, 19 thin (`8ebd67f3`)	2025–26	ΔR^2_{pers} +3.713 [+1.884, +6.035]	pass	1
Persistence mediation + placebo (same)	2025–26	Share 97.6% [96.6%, 98.3%]	fail	1

ΔR^2 in percentage points. "Opened" counts executions of the test cell; the repeated openings used unchanged code and specifications (Sections 3.11.5 and 3.10). Intervals are 95%, except 97.5% for the two fresh-day primaries and the two Part 15 primaries.

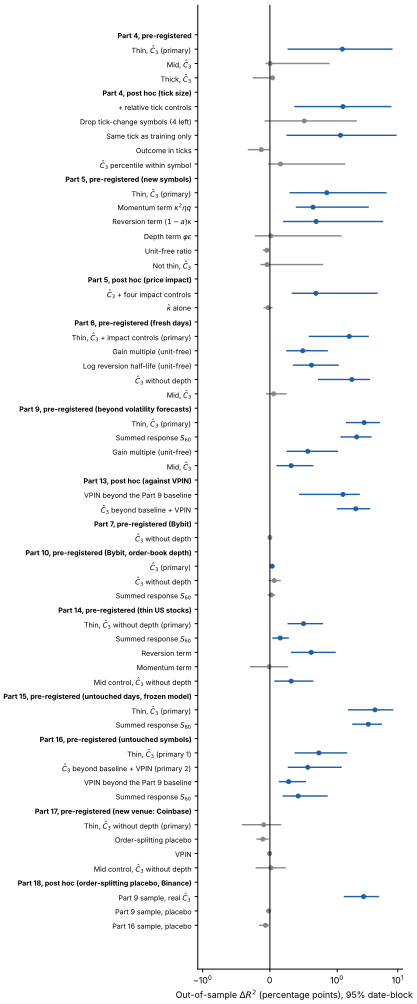


Figure 3.7: Every out-of-sample forecasting evaluation in this chapter's thin-book program. Points are ΔR^2 in percentage points with date-block bootstrap intervals (symmetric-log axis). Blue: interval above zero. Pre-registered and post-hoc evaluations are grouped separately.

rather than testing a coefficient, so estimation error in the regressor can only attenuate the measured gain.

Simultaneity. The structural fits regress the price change on same-minute flow, so $\hat{\kappa}$ and the depletion slope mix impact with any within-minute response of flow to price. The guard's identification results (Proposition 3.2.2 and Theorem 3.5.5) concern the impulse response, not these regression coefficients, and the forecasting evidence uses $\hat{C}_3$ as an index whose value is judged out of sample, not as a causal parameter. Simultaneity therefore changes what $\hat{C}_3$ measures, not whether the out-of-sample gains are real. We tried to separate the two after the fact by instrumenting this minute's flow with last minute's flow (Part 13, `R4`), but at the one-minute scale normalized flow is almost serially uncorrelated (median first-stage correlation 0.005), so the instrument is weak and the instrumented statistic is noise (it adds 0.01 points of R^2, interval including zero). The causal content of $\hat{\kappa}$ therefore remains open; a convincing instrument would need order-level data.

Selection and survivorship. The 30 symbols were fixed from one 2024 training day before any T6 statistic was computed, and the new symbols were drawn by a seeded procedure. Symbols had to be listed through 2026, so coins delisted after a collapse are excluded; if anything this removes the episodes in which thin-book risk was largest.

Economic size. The largest gain, 2.8 points of R^2 on a baseline of 18.6%, is a 15% increase in the explained variance of the next hour's adverse excursion, and tail classification improves by 0.005 of AUC. In basis points, on the test hours of Table 3.22 and with quintile breakpoints from training data: within each quintile of the volatility forecast, hours in the top quintile of $\hat{C}_3$ have a next-hour adverse excursion 25.6 basis points larger than hours in the bottom quintile (date-block 95% interval $[18.6, 32.8]$), against a mean excursion of 112.5 basis points, and 61 basis points larger in the most volatile quintile; the share of hours beyond the training 95th percentile rises from 2.3% in the bottom $\hat{C}_3$ quintile to 5.1% in the top. Unconditional sorts on S_{60} show no pattern; its content appears only with the controls. As a sizing input the gain is worth about a 2.4% reduction in tail loss per unit of position once the operator is certified (Table 3.26). These are modest gains on top of a strong baseline, not a new predictor that replaces volatility.

In margin terms. To express the 25.6 basis points in the units a risk manager uses, we fit a lognormal distribution to the excursions in each $\hat{C}_3$ quintile from its mean and median; at one common threshold the fit reproduces the observed share of hours beyond the training 95th percentile in every quintile to within 0.6 percentage points. Shifting the bottom-quintile distribution up by 25.6 basis points, the effect at a given volatility forecast, roughly doubles the probability that a 20× position (liquidated after a 4% adverse move, with 1% maintenance

margin) is liquidated within the hour, from 1.3% to 2.7% (2.3% to 3.2% across the interval for the effect). For a 10× position, liquidated after 9%, the probability almost triples, from 0.04% to 0.11%. On $1M of notional at 20×, the extra expected loss of margin is about $710 per hour held ($490 to $960; `recompute_dollar.py`). These figures come from a fitted distribution, not a test, and they treat the hour's excursion from its opening price as the liquidation trigger. They show that a difference that is small in basis points is large in the tail that leverage makes relevant, which is where a pre-trade control operates.

Venue. The size of the result depends on the venue. It is largest in thin Binance USD-M books; in thin US stocks it passes a pre-registered test at about a fifth of that size, and it also appears in mid-size US stocks (Table 3.24); on Bybit it failed with trade data alone and, with order-book depth, met its pre-registered criterion with an effect about one eightieth of Binance's (Table 3.23).

3.11.18 What the evidence says

Deep books sit on the stable side of the amplification condition. Flow depletes the side it hits on a fixed price grid, although re-centred depth bands make it look like replenishment, and we observe no peak overshoot. In thin Binance books the condition is still negative on average but binds more often, and the six-hour estimates of cumulative transient risk forecast the size of the next hour's adverse excursion out of sample. Four pre-registered tests pass, on the original symbols, on ten new symbols, on fresh days with price-impact controls, and on fresh days beyond a full volatility-forecasting baseline, where the gain is largest (+2.8 points of R^2) and extends to tail classification. The summed response S of Theorem 3.4.4, the guard's own statistic, carries the same information, and unit-free versions work. The content does not run through depth depletion, and in Binance books it is much weaker at mid depth and absent in BTCUSDT. In thin US stocks a fifth pre-registered forecasting test passes, at about a fifth of the Binance size and with a gain in mid-size stocks too; on Bybit the statistic is absent with trades alone and negligible with depth; and as a position-sizing input it improves a volatility limit only modestly once the operator is certified. Thin books are risky because their prices revert too slowly relative to the impact and momentum of the flow that moves them, which is what $S = \tilde{\kappa}/(m - q\tilde{\kappa})$ measures.

3.12 Tests of the information-safety results

The theorems of Sections 3.5–3.7 concern declared contracts. This section reports two kinds of tests: simulations in which the true economy is known, so that coverage, attainment and optimality can be checked directly; pre-registered tests on market data that no earlier part of this chapter used, on new days, new symbols and a new venue; and post-hoc checks on data

Table 3.28: Simulation tests of the information-safety results

Test	Question	Result
T01	Can markets agree on early responses yet differ in risk?	Example 3.5.1 in rational arithmetic: lags 0–2 equal, summed responses 1.92 and 5.59
T02	Does Proposition 3.5.2 bound the capacity a safe rule gives up?	Attained exactly with identical data (0.324 at $\beta = 0.05$); respected in 9 of 9 noisy designs by the certificate rule, whose unsafe-admission rate is at most 3.2%
T03	Is certifying risk cheaper than identifying cohorts?	Table 3.3: slope -6.06 in $\Delta\eta$ for identification, $+0.23$ for certification
T04	Does the certificate cover at its nominal level?	18 designs: coverage at or above nominal in all (minimum 93.6% at nominal 90%); median excess 3–15% with memory bound 0.6–0.9, about 230% at 0.97
T05	Is the corrected gate exact?	2,000 cases, horizons 1–10, every sign path enumerated: per-period worst case equals $V \sum \lvert g \rvert$ and total-budget worst case equals $V \max \lvert g \rvert$ in all; the peak gain understates the per-period worst case in 1,800
T06	Is the closed-form guard globally optimal?	500 economies: equal to brute force over 1,001 levels from the state matrices in all; safe sets always intervals; 28 infeasible states flagged
T07	What does disclosure add?	Table 3.5
T08	Does the guard depend on the channel?	Momentum learners, fixed-rule traders, depth and slow dealer reversion calibrated to $S = 5$: identical guard capacity (0.20); one-period margins differ (-0.34 to -0.20)
T09	What breaks the certificate?	Coverage at nominal 95%: as assumed 100%; delayed feedback 100%; contrarian cohorts 100%; depth channel 92.8%; memory longer than declared 86.3%; feedback 20% stronger after certification 34.7%
T10	Are the window endpoints proved?	Certified by Lemma 3.5.4 for every window; outside the 60-digit endpoints by at most 1.5×10^{-8} relative
T11	What if the tap noise is correlated?	(O1) certificate covers 74–76% (exact value 75.1%); t-certificate 94–95%; Hotelling 100%; dependent experiments 82–85%
T12	Should rebates use certified bounds?	No: a 10% overstatement of S costs 2.4–2.8% of first-best welfare, a 50% overstatement 70–80%
T13	What if admission changes the dynamics?	Safe sets $[0, 0.035] \cup [0.386, 1]$ (closed form) and $[0, 0.053] \cup [0.178, 1]$ (book, certified); no certified level unsafe in 433 sampled economies
T14	Does sequential probing keep its error level?	No wrong decision in 7,000 runs; 3.7 to 107 probes against 130 for a fixed design

All tests are deterministic given their seeds and run without data (`research_extension/information_safety/run_all.py` → `results/T*.json`, about six minutes). Further checks: the joint per-period and total-budget worst case equals its linear-programming value in 300 random cases (T05b), and the balance-sheet step keeps the reserve in all 17,551 sampled states that satisfy the guard inequality (T12b). Kind: recomputed.

already opened.

3.12.1 Simulation tests with a known truth

The simulations confirm what the theorems claim and locate what they do not (Table 3.28). The certificate is valid where its contract holds, including with delayed feedback and contrarian cohorts, and conservative when the declared memory is long. It fails in two ways that matter

Table 3.29: Pre-registered test on untouched days (Part 15), 2025–2026

A. Frozen forecasting model (primary 1)	ΔR^2 (pts) [CI]	Clark–West t	ΔAUC [95% CI]
$\hat{C}_3$ (primary, 97.5%)	+4.20 [+1.55, +8.23]	9.94	+0.007 [+0.001, +0.011]
S_{60} (secondary, 95%)	+3.26 [+1.82, +5.34]	11.55	

B. Guards at matched activity Rule	TLR	Activity	Robust guard vs. rule [95% CI]
Static limit	377	1.00	−0.084 [−0.161, −0.017] (primary 2)
Uncertainty-aware guard	347	0.91	—
Point-estimate guard	346	0.92	+0.003 [−0.005, +0.013]
Depth throttle	469	0.59	−0.302 [−0.446, −0.146]
Spectral-radius throttle	401	0.82	−0.146 [−0.240, −0.065]
Volatility cap	339	0.86	+0.023 [−0.022, +0.082]
Volatility cap × guard	335	0.80	vs. cap: −0.012 [−0.037, +0.013]

20 thin symbols; evaluation days the 15th of each month; frozen model trained on 16,860 Part 9 symbol-hours, tested on 6,714 hours on 20 dates; frozen baseline out-of-sample $R^2 = 0.058$; training coefficient +71.3. Guards: 6,732 test hours; TLR = ES95 of loss per unit of average position (positions normalized to training mean 1); the last column of panel B reports log TLR(robust guard) minus log TLR(rule), so negative values favour the robust guard. Primary 2 uses the 97.5% interval $[-0.173, -0.011]$, one-sided $p = 0.0025$. Both primaries pass under Holm's correction. Median $S_{60}^{\mathrm{up}}/S_{60} = 1.25$. Specification SHA-256 `69849c96d70f...`; test opened once. Kind: real data (Part 15).

for practice: when memory is longer than declared, which is the case Theorem 3.5.3(iii) warns about, and when feedback strengthens after certification, in which case a stale certificate covers the new risk only a third of the time. A guard must therefore be re-certified as conditions change; it cannot be set once. The remaining tests check the extensions: the window endpoints are proved rather than computed (T10), certificates that do not assume independent taps keep their level when the taps are correlated while the (O1) certificate does not (T11), non-monotone admission is certified by branch and bound (T13), and sequential probing keeps its error level (T14).

3.12.2 Pre-registered test on untouched days (Part 15)

We registered a test (Part 15) before downloading any day used in it: the 12th to 15th of each month on Binance, days no earlier part used on Binance. It has two primaries, with Holm's correction across them. First, the forecasting model of Table 3.22, frozen at its 2023–2024 training estimates, is applied to the 15th of each month from January 2025 to September 2026. Second, an uncertainty-aware guard throttles a position by $\min(1, k_s/(U_{\mathrm{cert}} S_{60}^{\mathrm{up}}))$, where S_{60}^{up} is the 90th percentile of the summed response over 100 block-bootstrap refits of the operator certified from the three preceding days; it is compared with a static limit at matched activity, and secondarily with the point-estimate guard, a depth throttle, a spectral-radius throttle and a volatility cap. The bootstrap bound is a diagnostic for observational data, not the finite-sample certificate of Theorem 3.5.5.

Both primaries pass (Table 3.29). The forecasting model, frozen at its 2023–2024 estimates and applied to days never used before, finds that $\hat{C}_3$ adds 4.2 points of out-of-sample R^2 to the volatility and impact baseline (97.5% interval $[1.6, 8.2]$, Clark–West $t = 9.9$) and improves tail classification; the summed response S_{60} adds 3.3 points. The gain is larger than in Table 3.22 partly because the frozen baseline explains less on these days (5.8% of the variance against 18.6%): frozen coefficients transfer imperfectly, and the dynamic statistic picks up part of what the baseline loses. As a one-sided throttle on a static limit, the uncertainty-aware guard lowers tail loss per unit of position by 8% (log difference -0.084, 97.5% interval $[-0.173, -0.011]$, that is, a reduction of 1% to 16%) while retaining 91% of activity, replicating on untouched days the post-hoc finding of Section 3.11.14. It does much better than throttling on depth (26% lower tail loss) or on the spectral radius (14% lower), the two controls a supervisor would reach for first.

Two secondary results limit the claim. The bootstrap upper bound does no better than the point estimate (+0.3%, interval $[-0.5\%, +1.3\%]$): on these data the upper bound is about 1.25 times the point estimate in most hours and ranks hours almost identically, so accounting for estimation uncertainty changes which hours are throttled very little. And, as in Section 3.11.14, a volatility cap does at least as well as the guard, and adding the guard to it does not help significantly (-1.2%, interval $[-3.7\%, +1.3\%]$). Used as the theorem uses it, the guard is a better throttle than depth or spectral rules and a real improvement over a static limit; it is not a substitute for volatility information.

3.12.3 Pre-registered test on untouched symbols (Part 16)

A third registered test (Part 16) uses Binance perpetuals that no earlier part used, thirty drawn by a seeded rule before any analysis and kept if their training depth is below the thin threshold of Part 5. It has three primaries, with Holm's correction across them: the forecasting specification of Table 3.22 on the new thin symbols; the same beyond the baseline *plus VPIN*, which turns the post-hoc comparison of Section 3.11.10 into a confirmatory one; and the guard of Section 3.12.2 against a static limit.

All three primaries pass (Table 3.30), but on a small set. The seeded draw of thirty new perpetuals produced only five with thin books, because the thinnest listed symbols had already been used in Parts 4–15; the rule required no minimum, so the test went ahead with five. On these symbols, re-estimating the Part 9 specification on their own training days, $\hat{C}_3$ adds 0.73 points of out-of-sample R^2 to the volatility and impact baseline (95% interval $[0.38, 1.43]$, Clark–West $t = 5.7$) and improves tail classification. That is about a quarter of the Part 9 gain on the original thin symbols, on a baseline that already explains 31.7% of the variance rather than 18.6%.

The second primary makes the post-hoc VPIN comparison of Section 3.11.10 confirmatory: with VPIN added to the baseline, $\hat{C}_3$ still adds 0.56 points ($[0.27, 1.16]$). VPIN has content of

Table 3.30: Pre-registered test on untouched symbols (Part 16), 2025–2026

A. Forecasting beyond the volatility baseline (5 new thin symbols)	ΔR^2 (pts) [95% CI]	Clark–West t	ΔAUC [95% CI]
$\hat{C}_3$ (primary 1)	+0.73 [+0.38, +1.43]	5.70	+0.004 [+0.001, +0.007]
$\hat{C}_3$, VPIN in baseline (primary 2)	+0.56 [+0.27, +1.16]	5.38	+0.003 [+0.001, +0.006]
VPIN (secondary)	+0.28 [+0.14, +0.53]	4.39	+0.002 [+0.001, +0.003]
S_{60} (secondary)	+0.42 [+0.20, +0.85]	4.43	+0.001 [−0.000, +0.003]

B. Guards at matched activity Rule	TLR	Activity	Robust guard vs. rule [95% CI]
Static limit	466	1.00	−0.256 [−0.497, −0.021] (prim. 3)
Uncertainty-aware guard	360	0.91	—
Point-estimate guard	371	0.91	−0.028 [−0.076, +0.009]
Volatility cap	367	0.85	not tested
Volatility cap × guard	342	0.78	vs. cap: −0.072 [−0.180, +0.005]

Thirty perpetuals drawn by a seeded rule (seed 20261031) from the 48 listed Binance perpetuals not used in Parts 4–15; five had training depth within 1% below $0.39M and entered: BAT, BEL, COTI, CTSI and LQTY. Forecasting: Part 9 specification re-estimated on these symbols' training days (10th and 24th of each month, 2023–2024; 4,215 symbol-hours) and tested on the 6th and 20th of each month, January 2025 to September 2026 (3,597 hours on 40 dates); baseline out-of-sample $R^2 = 0.317$. VPIN: six-hour window, 50 equal-volume buckets, winsorized at training 1%/99% (median 0.24). Guards: Part 15 definitions; 1,080 training and 1,782 test symbol-hours on the 15th of each month; last column of panel B is log TLR(robust guard) minus log TLR(rule). Holm-adjusted p across the three primaries: 2×10^{-8}, 7×10^{-8} and 0.004; all three pass. Secondary Holm-adjusted p: VPIN and $S_{60} < 0.0001$; robust vs. point guard 0.11; cap × guard vs. cap 0.07. Specification SHA-256 `0523e47626da...`; test opened once. Kind: real data (Part 16).

its own (+0.28 points) but leaves most of $\hat{C}_3$'s gain in place, as in the post-hoc comparison. The summed response S_{60} adds 0.42 points.

The third primary replicates the guard of Section 3.12.2 on new symbols. Against a static limit, the uncertainty-aware guard lowers tail loss per unit of position by 23% (log difference -0.256, interval $[-0.497, -0.021]$, a reduction of 2% to 39%) while retaining 91% of activity. The interval is wide: the guard sample is five symbols on about twenty evaluation days, and its one-sided bootstrap p of 0.004 (8 of 2,000 draws) passes Holm's correction within Part 16 but not a Bonferroni correction across all tested primaries of the program (Section 3.11.17). The secondary comparisons agree with Part 15. The bootstrap upper bound again does not significantly improve on the point estimate (−2.8% in log points, interval including zero). Adding the guard to a volatility cap lowers tail loss by a further 7%, a larger effect than in Part 15 but not significant after correction (Holm $p = 0.07$). We do not report the guard against the volatility cap alone, which the registration did not include; the point estimates are close (360 against 367).

3.12.4 Pre-registered test on a new venue (Part 17)

Every venue so far was Binance, Bybit or the US consolidated tape. Part 17 registers a test on Coinbase Exchange spot, a US-regulated venue that no earlier part used, from public trade data: thin USD pairs drawn by a seeded rule from those with median daily dollar volume of $0.5–5

Table 3.31: Pre-registered test on a new venue (Part 17): Coinbase spot, 2025–2026

	ΔR^2 (pts) [95% CI]	Clark–West t	ΔAUC [95% CI]
$\hat{C}_3^{\text{flow}}$, thin (primary 1)	−0.09 [−0.41, +0.16]	−0.58	+0.000 [−0.000, +0.001]
Real minus placebo (primary 2, not tested)	+0.01 [−0.33, +0.30]	—	—
Placebo alone	−0.11 [−0.19, −0.03]	−2.62	
S_{60}	−0.03 [−0.08, +0.02]	−1.66	
VPIN	−0.00 [−0.01, +0.00]	−1.81	
$\hat{C}_3^{\text{flow}}$ beyond baseline + VPIN	−0.10 [−0.42, +0.15]	—	
$\hat{C}_3^{\text{flow}}$, mid-size control (3 pairs)	+0.02 [−0.20, +0.23]	0.69	

Twenty thin USD pairs (median June 2024 daily dollar volume \$0.5–5 million) and four mid-size pairs (\$15–50 million) drawn with a fixed seed before any trade was downloaded; a pair entered with at least 15 valid training and 25 valid test days, a day being valid if at least 60% of its minutes had a trade. Six thin and three mid-size pairs entered. Training: 10th and 24th of each month of 2024 (2,340 thin-pair hours); test: 6th and 20th, January 2025 to September 2026 (3,690 hours on 42 dates); baseline out-of-sample $R^2 = 0.310$. Placebo: flow shifted by 180 minutes inside its six-hour window. Fixed-sequence testing: primary 2 is tested only if primary 1 passes. Secondary Holm-adjusted p-values all equal 1. Specification SHA-256 `3233c52b0a39...`; test opened once. Kind: real data (Part 17).

million in June 2024, the depth-free statistic $\hat{C}_3^{\text{flow}}$ of Parts 7 and 14, and the Part 9 volatility and impact baseline, on the 6th and 20th of each month from January 2025 to September 2026. Its second primary is a falsification test. Order splitting makes signed flow persistent, and a statistic built from persistent flow could forecast through persistence alone. The placebo recomputes $\hat{C}_3^{\text{flow}}$ after shifting the flow series by 180 minutes inside its own six-hour window, which keeps the flow's distribution and persistence and breaks its minute-by-minute alignment with prices, the alignment the model says carries the risk. The real statistic must beat the placebo. The two primaries are tested in a fixed sequence, the placebo comparison only if the first passes, because beating a placebo is informative only when the real statistic has content.

The test fails (Table 3.31). Of the twenty pairs drawn, six traded in at least 60% of minutes on enough days to enter (ARB, FLR, ICP, QNT, VET and XLM); thin pairs on this venue trade too sparsely for the rest. On these six, $\hat{C}_3^{\text{flow}}$ adds -0.09 points of R^2 to the baseline (95% interval $[-0.41, 0.16]$, Clark–West $t = -0.58$), so the first primary fails and, under the registered fixed sequence, the placebo comparison is not tested; descriptively the real statistic and the placebo do equally badly (difference +0.01 points, $[-0.33, 0.30]$). Nothing else forecasts either: S_{60}, VPIN and the statistic in the mid-size control pairs all add nothing. Coinbase joins Bybit with trades alone as a venue where the thin-book result is absent. Both are venues where the statistic must be computed from trades without depth, but so is the US test, which passed; the result is venue-specific, and we do not know which feature of a venue it depends on.

3.12.5 Post-hoc checks on opened data (Part 18)

Two further checks use samples that earlier tests opened, so they are post hoc; their analysis plan was hashed before they were computed. The first applies the same placebo to the Binance samples of Parts 9 and 16. The second adds three controls to the guard comparisons of Parts 15

Table 3.32: Post-hoc checks on opened data (Part 18)

A. Order-splitting placebo, ΔR^2 beyond the Part 9 baseline (pts) [95% CI]	Part 9 sample (20 thin)	Part 16 sample (5 thin)
Real $\hat{C}_3$	+2.73 [+1.32, +4.78]	+0.73 [+0.37, +1.48]
Placebo	−0.02 [−0.05, +0.00]	−0.07 [−0.15, −0.03]
Real minus placebo	+2.74 [+1.32, +4.82]	+0.79 [+0.42, +1.60]
B. log TLR*(uncertainty-aware guard) minus* log TLR*(rule) [95% CI]*	Part 15 sample (20 thin)	Part 16 sample (5 thin)
Certified peak-response guard	−0.034 [−0.062, −0.007]	−0.004 [−0.031, +0.022]
Six-hour peak-response guard	+0.036 [−0.015, +0.100]	−0.028 [−0.116, +0.060]
Participation cap	−0.695 [−0.865, −0.534]	−0.880 [−1.233, −0.530]
Margin-style rule	−0.033 [−0.060, −0.008]	−0.018 [−0.057, +0.034]
Static limit (reproduces Parts 15, 16)	−0.084 [−0.166, −0.022]	−0.256 [−0.491, −0.021]

Post hoc: every sample was opened by an earlier test; the analysis plan was hashed before computing (`P18_posthoc_plan.json`). Panel A: the placebo recomputes $\hat{C}_3$ with the flow series shifted by 180 minutes inside its six-hour window; real and placebo are evaluated on the same rows and bootstrap draws, so the Part 9 real gain (2.73) differs slightly from Table 3.22 (2.79). Recomputed $\hat{C}_3$ and certified operators match the stored values in every row. Panel B: positions normalized to training mean 1 per symbol and rule; participation cap: position proportional to the previous six hours' dollar volume; margin-style rule: position inversely proportional to the 99th percentile of one-hour moves over the three certification days. Holm-adjusted p (Part 15): certified peak 0.006, margin 0.005, participation < 0.001, six-hour peak 0.91. The static-limit row reproduces the published TLRs (377 and 347; 466 and 360). Kind: real data (post hoc, Part 18).

and 16: a certified peak-response guard (the quantity Lemma 3.6.1 says is unsafe for repeated trading), a participation cap that limits the position to a share of the last six hours' volume, and a margin-style rule that scales the position by an exchange-style scanning range, the 99th percentile of one-hour moves over the three certification days. A cap on inventory alone is the static limit once activity is matched.

Both checks support the paper's reading, with one exception (Table 3.32). The placebo carries no information on Binance: on the Part 9 sample it adds -0.02 points of R^2 while the real statistic adds 2.73 on the same rows, a difference of 2.74 points (95% interval $[1.32, 4.82]$), and on the Part 16 sample the difference is 0.79 points ($[0.42, 1.60]$). Real and placebo statistics are uncorrelated (Spearman -0.004 and 0.025). The forecasting content in thin Binance books therefore comes from the minute-by-minute alignment of flow and prices, not from the persistence of order flow. Against the new controls, on the Part 15 sample the uncertainty-aware guard has 3% lower tail loss per unit of position than the certified peak-response guard and the margin-style rule (both significant after Holm's correction) and far lower than the participation cap; it is not better than the six-hour peak guard (+3.6%, interval including zero). On the five symbols of Part 16 it beats only the participation cap. These are post-hoc comparisons on opened data, so they qualify the guard's ranking rather than test it.

Table 3.33: Leverage interaction and matched spot comparison (Part 19, specified in advance, not deposited), 2025–2026

A. Out-of-sample ΔR^2 of the amplification statistic (pts)	ΔR^2 [95% CI]	CW t (p)	Ratio / diff. [95% CI]
Perpetuals, 19 thin symbols	+6.495 [+2.345, +15.110]	11.33 ($< 10^{-6}$)	—
top leverage tercile (LEV $\geq$ 16.77)	+8.172	—	high − low: +4.495
lower two terciles	+3.678	—	[−1.094, +11.663]
Spot, same hours (exploratory)	+0.987 [−0.093, +2.853]	4.96 (3.6×10^{-7})	0.152 [−0.015, +0.308]

B. Coefficients in perpetuals (bp of excursion per s.d. of $\hat{C}_3$)	Estimate	95% CI	Outcome
$\hat{C}_3$ base slope	+5.49	[+0.65, +9.66]	—
$\hat{C}_3\times$ top leverage tercile (Primary 1)	−0.18	[−3.02, +3.73]	fail
$\hat{C}_3\times$ top |basis| tercile	+0.94	[−2.91, +4.60]	—

C. One-minute return lead–lag, mean correlation across symbol-days	
perpetual leads spot: +0.0816	spot leads perpetual: −0.0010

Panel A: models fitted on 2024 training days (4,104 hours), evaluated on 2025–2026 test days (6,156 hours); 2,000-draw date-block bootstrap 95% intervals for all differences and ratios. Panel B: pooled regression with symbol fixed effects and date-block bootstrap intervals. Specified in advance, not deposited (`recompute_p2_parts19_20_21.py` → `R_P2_parts19_20_21.json`).

3.12.6 Three analyses specified in advance, not deposited (Parts 19–21)

The three analyses below (Parts 19–21) were specified in an earlier draft of this chapter, with their hypotheses and criteria, before their data were downloaded, but no specification hash was deposited prior to execution; we therefore label them "specified in advance, not deposited" and do not count them among the pre-registered tests of Table 3.27. By contrast, Parts 22–25 (Sections 3.12.10–3.12.13) were hash-registered and committed to the public repository before their 2025–2026 test data were downloaded, and are counted in Table 3.27.

3.12.7 Leverage interaction and spot comparison (Part 19)

Primary 1 of Part 19—which predicted a positive slope interaction between $\hat{C}_3$ and high open-interest-to-depth leverage (LEV = OI/$D_{1\%}$ $\geq$ 16.77, the top training tercile)—failed: the interaction coefficient is −0.18 basis points per standard deviation (95% date-block bootstrap CI $[-3.02, +3.73]$, Table 3.33). We evaluated two predictions on the 8th of each month, a day no earlier test used for these symbols, across 19 thin perpetuals from Parts 9 and 16 that also trade on Binance spot (training: 12 days in 2024, 4,104 symbol-hours; test: 18 days from January 2025 to June 2026, 6,156 symbol-hours). *Primary 1* tested whether $\hat{C}_3 \times \mathbf{1}\{\text{LEV} \geq 16.77\}$ has a positive slope in perpetuals. *Primary 2* (exploratory, because the spot specification lacks order-book depth controls that Binance does not publish for spot) asked whether on the matched spot pairs, where there are no leveraged liquidations, the out-of-sample ΔR^2 of $\hat{C}_3^{\text{flow}}$ is at most one quarter of the perpetual gain.

Table 3.33 shows why Primary 1 fails and why Primary 2 is only exploratory. In Panel A,

the out-of-sample ΔR^2 of $\hat{C}_3$ is +8.172 points in the top leverage tercile versus +3.678 points in the lower two terciles, a difference of +4.495 points whose 2,000-draw date-block bootstrap 95% confidence interval ($[-1.094, +11.663]$) crosses zero. *Post hoc*, the mechanical reason why the slope interaction $\hat{C}_3 \times$ high-LEV is zero (−0.18 bp) while the explained variance share is higher in high-leverage hours is that $\hat{C}_3 = \hat{\varphi}\hat{\varepsilon} + \hat{\kappa}^2\hat{\eta}\hat{q} - (1-\hat{a})\hat{\kappa}$ already contains $\hat{\kappa}$, which rises when resting depth $D_{1\%}$ is thin relative to open interest; high-LEV hours therefore have larger variance in $\hat{C}_3$ rather than a steeper marginal slope per unit of $\hat{C}_3$. In the matched spot pairs, $\hat{C}_3^{\text{flow}}$ adds +0.987 points of R^2 (95% CI $[-0.093, +2.853]$, significant by the nested-model test of Clark and West, 2007 at $t = 4.96$, $p = 3.6 \times 10^{-7}$, though its bootstrap level interval includes zero), giving a spot-to-perpetual point ratio of 0.152 with a 95% bootstrap interval of $[-0.015, +0.308]$ that crosses the 0.25 threshold. Because the spot baseline omits depth controls and the ratio interval spans 0.25, Part 19 does not establish leverage or liquidations as the mechanism; Section 3.12.10 tests the liquidation channel directly with a pre-registered design.

3.12.8 A natural experiment in fees (Part 20)

The primary hypothesis of Part 20—that removing zero trading fees lowers dealer reversion speed $\hat{m} = 1 - \hat{a}$ and raises the unit-free gain multiple $\widehat{S/\tilde{\kappa}} = 1/(\hat{m} - \hat{q}\hat{\kappa})$ as in Proposition 3.8.1(i)–(iv)—is *not supported* ($\hat{m}$ changes by −0.00120, $p = 0.590$, and $\widehat{S/\tilde{\kappa}}$ by −26.88, $p = 0.362$ across the two verified spot zero-fee events; Table 3.34). What moves in the two verified BTCUSDT spot zero-fee events—introduced on July 8, 2022 (Binance announcement `10246072590233`, 5th Anniversary Zero-Fee BTC spot trading) and removed on March 22, 2023 (Binance announcement `be21a7f1940b46fa95A42087b058958a`, Zero-Fee trading ended for BTC spot pairs except BTC/TUSD)—is price impact $\hat{\kappa}$, which is 0.24319 lower during zero-fee trading (−18.3% of its sample mean, $t = -3.86$, $p = 0.0002$). Because $\hat{\kappa}$ was not the primary outcome, this impact reduction is an *exploratory* finding. Moreover, Table 3.34 reports side by side the specification pooling all three originally listed dates (which included a March 20, 2024 USD-M VIP maker-fee adjustment affecting BTCUSDT and ETHUSDT): when that third date is included ($N = 1{,}102$, $G = 176$ date clusters), the pooled impact coefficient is +0.00819 ($t = 0.14$, $p = 0.8873$).

Two econometrics limitations govern how Table 3.34 should be read. First, in the two verified spot events only a single pair (BTCUSDT) is treated against seven controls, so clustering standard errors by date does not correct for single-treated-unit cross-sectional dependence (Conley and Taber, 2011). Second, the 80%-power minimum detectable effects for the two null primary outcomes are wide: 31.9% of the sample mean for $\hat{m}$ and 52.0% for $\widehat{S/\tilde{\kappa}}$ in the two spot events (25.0% and 38.8% across all three dates), so moderate changes in reversion or in the gain multiple cannot be ruled out. With those caveats stated, a reduction in spot impact $\hat{\kappa}$ during zero-fee trading alongside no detectable change in $\hat{m}$ or $\widehat{S/\tilde{\kappa}}$ is *consistent with* Proposition 3.8.1(v) (where endogenous depth adjusts $\tilde{\kappa}$ while leaving $S/\tilde{\kappa}$ invariant), rather than a confirmation of it.

Table 3.34: Zero fees on BTCUSDT spot: volume, reversion, gain multiple and impact (Part 20, specified in advance, not deposited)

Outcome (±30 days)	Spot zero-fee on (2022-07-08)	Spot zero-fee off (2023-03-22)	Two spot events pooled	All three dates pooled
Log quote volume	+0.944 (0.054)	+1.774 (0.074)	+1.355 (0.063)	+0.711 (0.081)
Reversion $\hat{m}$ (null)	−0.00298 (0.00313)	+0.00048 (0.00312)	−0.00120 (0.00221)	+0.00153 (0.00183)
p / 80% MDE	$p = 0.345$ / 50.9%	$p = 0.878$ / 41.7%	$p = 0.590$ / 31.9%	$p = 0.405$ / 25.0%
Gain $\widehat{S/\tilde{\kappa}}$ (null)	−61.89 (49.19)	+7.36 (32.86)	−26.88 (29.38)	−14.14 (20.13)
p / 80% MDE	$p = 0.213$ / 80.0%	$p = 0.824$ / 64.6%	$p = 0.362$ / 52.0%	$p = 0.483$ / 38.8%
Impact $\hat{\kappa}$ (explor.)	−0.1042 (0.1107)	−0.3830 (0.0571)	−0.2432 (0.0631)	+0.0082 (0.0577)
p / 80% MDE	$p = 0.351$ / 22.0%	$p < 10^{-4}$ / 13.2%	$p = 0.0002$ / 13.4%	$p = 0.887$ / 9.7%
Pair-days / clusters	324 / 59	374 / 58	698 / 117	1,102 / 176

Difference-in-differences of zero fees relative to control pairs (ETH, SOL, ADA, DOGE, BNB, XRP and LTC against USDT); date-clustered standard errors in parentheses, together with two-sided p-values and 80%-power minimum detectable effects (MDE, $2.8 \times$ SE as a percentage of the sample mean). Spot events verified against Binance announcements `10246072590233` (July 8, 2022) and `be21a7f1940b46fa95A42087b058958a` (March 22, 2023); the fourth column adds the March 20, 2024 VIP maker-fee date. Only one pair (BTCUSDT) is treated in the two spot events. Kind: real data (`recompute_p2_parts19_20_21.py` → `R_P2_parts19_20_21.json`).

Table 3.35: Guard results of Parts 15 and 16 in dollars per $1M of average position (Part 21, exact translation of archived results)

Rule	ES_{95} (USD)	Activity	Beyond static p95	$\Delta \log ES_{95}$ [95% CI] (p)
A. Part 15, untouched days (20 thin symbols, 6,732 test hours)				
(a) Static limit	37,724	1.000	1.98%	—
(b) Volatility cap	33,885	0.863	1.66%	—
(c) Vol. cap × guard	33,487	0.795	1.83%	(c)–(b): −0.0118 [−0.0370, +0.0126] (0.196)
(d) Guard alone	34,679	0.911	2.21%	(d)–(a): −0.0842 [−0.1608, −0.0172] (0.0025)
				(d)–(b): +0.0232 [−0.0222, +0.0815] (0.828)
B. Part 16, untouched symbols (5 thin symbols, 1,782 test hours)				
(a) Static limit	46,565	1.000	—	—
(b) Volatility cap	36,721	0.848	—	—
(c) Vol. cap × guard	34,164	0.784	—	(c)–(b): −0.0722 [−0.1801, +0.0047] (0.036)
(d) Guard alone	36,047	0.907	—	(d)–(a): −0.2560 [−0.4970, −0.0205] (0.004)

Hourly ES_{95} is the archived tail-loss ratio of Tables 3.29 and 3.30 times $100 per basis point on $1M of average position. Comparisons are the archived date-block bootstrap results. Kind: exact translation of real-data results (`recompute_p2_parts19_20_21.py` → `R_P2_parts19_20_21.json`).

3.12.9 The guard results in dollars (Part 21)

Tables 3.29 and 3.30 report the guard in tail loss per unit of average position. Because positions are normalized to a training mean of one, that ratio converts directly into dollars: on $1M of average position, a tail-loss ratio of 377.24 basis points is an expected shortfall of $37,724 for the hour. Table 3.35 restates the archived Part 15 and Part 16 results in dollars using no new data.

In dollars, the guard alone ((d) vs. (a)) lowers the hour's expected shortfall on $1M of average

Table 3.36: Pre-registered test of the open-interest liquidation mechanism on untouched 22nd-of-month days (Part 22), 2025–2026

Specification (19 thin symbols, 6,766 test hours)	ΔR^2 (pts) / Share [95% CI]	CW t (p)	Result
Excursion $\text{MAE}_{t+1} \mid \hat{C}_{3,t}$ (22nd-of-month)	+6.516 [+3.159, +10.743]	12.21 ($< 10^{-6}$)	—
Primary 1: Liquidation $\text{LIQ}_{t+1} \mid \hat{C}_{3,t}$	−0.795 [−1.529, −0.179]	−1.12 (0.868)	fail
Excursion $\text{MAE}_{t+1} \mid \hat{C}_{3,t}, \text{LIQ}_{t+1}$	+6.958 [+3.405, +11.338]	—	—
Primary 2: Mediation share $1 - \Delta R^2_{\text{LIQ}}/\Delta R^2$	−6.8% [−10.2%, −3.1%]	—	fail

Trained on 22nd-of-month days in 2024 (3,876 symbol-hours), evaluated once on untouched 22nd-of-month days from January 2025 to September 2026 (6,766 symbol-hours); 2,000-draw date-block bootstrap 95% confidence intervals. Specification SHA-256 `2b3794e3812f...`; test opened once. Kind: real data (`recompute_prereg_oct21_22nd.py` → `R_P2_p22_p23.json`).

position by \$3,045 on the untouched days ($p = 0.0025$) and by \$10,518 on the untouched symbols ($p = 0.004$). Against a volatility cap ((c) vs. (b)), adding the guard lowers hourly expected shortfall by \$398 on the untouched days (95% CI $[-3.7\%, +1.3\%]$, $p = 0.196$) and by \$2,557 on the untouched symbols (95% CI $[-18.0\%, +0.5\%]$, unadjusted $p = 0.036$, Holm $p = 0.07$); neither improvement over the volatility cap is statistically significant after multiple-testing correction. Set beside the official operating budget of the U.S. Consolidated Audit Trail (\$248.8 million in the November 2024 CAT LLC filing, SEC Release No. 34-101901, p. 4, revised to \$228.3 million in May 2025; Chapter 2, Table 9), certification of the summed response requires only public price and flow records and none of the participant-level tagging needed to identify individual cohorts.

3.12.10 Pre-registered test of the liquidation mechanism (Part 22)

Because Part 19 was not hash-registered and tested only a static leverage interaction, we pre-registered a direct test of the liquidation mechanism (`P22_liquidation_mechanism.json`, SHA-256 `2b3794e3812f...`) and committed its hash to the repository before downloading any 22nd-of-month data for 2025–2026 (3,876 training symbol-hours on the 22nd of each month in 2024; 6,766 test symbol-hours on the 22nd of each month from January 2025 to September 2026 across 19 thin perpetuals; test opened once). When forced liquidations execute in a perpetual futures book without a public liquidation feed, they contract open interest while moving the price adversely; following standard practice we define the hourly forced-liquidation proxy $\text{LIQ}_{s,t}$ as the sum of five-minute open-interest contractions $(-\Delta\text{OI}_{s,\tau}/\overline{\text{OI}}_s)_+$ accompanied by an adverse five-minute return exceeding one training standard deviation. The registration specified two primaries: (*Primary 1*) $\hat{C}_3$ forecasts next-hour $\text{LIQ}_{s,t+1}$ out of sample beyond the Part 9 volatility, impact and lagged-LIQ_t baseline; and (*Primary 2*) controlling for realised contemporary $\text{LIQ}_{s,t+1}$ mediates at least 15% of $\hat{C}_3$'s out-of-sample ΔR^2 for the next hour's maximum adverse excursion $\text{MAE}_{s,t+1}$.

On these untouched 22nd-of-month test days, $\hat{C}_3$ strongly forecasts next-hour adverse excursions ($\Delta R^2 = +6.516$ percentage points, 95% CI $[+3.159, +10.743]$, Clark–West $t = 12.21$,

Table 3.37: Pre-registered non-resetting 18-hour leveraged position test on untouched 22nd-of-month days (Part 23), 2025–2026

Rule (398 symbol-days)	Liq. rate	Daily ES_{95} (USD)	Activity	Matched-breach $\overline{N}$
(a) Static 20× limit	51.26%	166,642	1.000	—
(b) Volatility cap	39.70%	144,360	0.795	\$902,282
(c) Vol. cap × cert. guard	42.21%	166,501	0.686	\$1,235,130
(d) Certified guard alone	50.00%	182,043	0.838	—
Primary: $\overline{N}$(c) − $\overline{N}$(b) (39.70% liq.)	+\$332,848 (+36.89%), 95% CI [−\$20,390, +\$1,180,808] (fail)			
Secondary: $\Delta \log ES_{95}$(c vs b)	+0.1427, 95% CI [+0.0741, +0.1964]			

Non-resetting 18-hour position (06:00–24:00 UTC) on 398 untouched test symbol-days (228 training symbol-days in 2024); 2,000-draw date-block bootstrap 95% intervals. Specification SHA-256 `d4169377da0b...`; test opened once. Kind: real data (`recompute_prereg_oct21_22nd.py` → `R_P2_p22_p23.json`).

Table 3.36), providing a sixth out-of-sample confirmation of the thin-book excursion effect. However, both liquidation-mechanism primaries fail decisively: $\hat{C}_3$ does not forecast the five-minute open-interest contraction proxy beyond the baseline ($\Delta R^2 = -0.795$ points, 95% CI $[-1.529, -0.179]$, $t = -1.12$), and adding realised next-hour open-interest contractions to the excursion model leaves $\hat{C}_3$'s predictive gain undiminished ($\Delta R^2 = +6.958$ points, mediation share −6.8%, 95% CI $[-10.2\%, -3.1\%]$). We therefore drop the hypothesis that five-minute open-interest liquidations mediate the effect. The simplest remaining reading is that $\hat{C}_3$ forecasts adverse excursions because it measures slow book reversion relative to price impact; Part 25 examines it.

3.12.11 Pre-registered test of non-resetting multi-hour guard capacity (Part 23)

To evaluate the guard on a leveraged position that is held across hours without resetting inventory at the top of each hour, we pre-registered a multi-hour margin-breach and notional-capacity test (`P23_guard_capacity.json`, SHA-256 `d4169377da0b...`) before downloading the 22nd-of-month data for 2025–2026 (228 training symbol-days in 2024; 398 test symbol-days in January 2025–September 2026 across 19 thin perpetuals; test opened once). At 06:00 UTC on each day d, a dealer opens a 20× leveraged position (450 bp maintenance-margin buffer before forced liquidation) using the three-day certified operator $\hat{S}^{\text{up}}_{s,d}$ estimated from hours 00:00–06:00 across the three preceding evaluation days and holds the position continuously through 24:00 UTC; cumulative mark-to-market loss evolves pathwise across all 18 hours without hourly reset. The primary hypothesis tests whether, when the volatility cap (rule (b)) and the combined volatility-plus-certified-guard rule (rule (c)) are scaled to have the exact same out-of-sample forced-liquidation rate as the unit-scaled volatility cap (39.70%), rule (c) admits a significantly larger mean dollar notional position $\overline{N}$ than rule (b).

Table 3.37 reports both the unit-scaled outcomes and the matched-breach capacity compar-

ison. At unit training scale, a static 20× position is liquidated on 51.26% of thin-symbol test days (daily ES_{95} = \$166,642), while the volatility cap lowers the liquidation rate to 39.70% (ES_{95} = \$144,360). When rules (b) and (c) are scaled to match the 39.70% out-of-sample liquidation rate, adding the morning-certified guard to the volatility cap raises the mean admitted notional position from \$902,282 to \$1,235,130—a point gain of +\$332,848 (+36.89%)—because the guard concentrates notional on symbol-days with faster certified reversion. However, across the 22 test dates the 95% date-block bootstrap confidence interval is [−\$20,390, +\$1,180,808], which narrowly crosses zero, and at unit scale rule (c) has higher tail loss than rule (b) ($\Delta \log ES_{95}$ = +0.1427). The primary hypothesis therefore fails: in observational data, a three-day morning certificate is too coarse to deliver a statistically significant capacity gain over a trailing six-hour volatility cap across an 18-hour holding window when quiet-morning exposure is rescaled above the \$1M facility limit.

3.12.12 Pre-registered test of a capped pre-trade throttle (Part 24)

Part 24 was registered after the Part 23 primary failed. Why did the unit-rescaled secondary in Part 23 increase daily ES_{95} ($\Delta \log ES_{95}$ = +0.1427) while the matched-breach capacity interval ([−\$20,390, +\$1,180,808]) crossed zero? In Part 23, dividing the multiplicative throttle $q^{\text{raw}}_{c,i} = \min(1, k_{\text{vol},s}/\sigma_{\text{morn},i}) \cdot \min(1, k_{\text{grd},s}/(U_{\text{cert},i}\hat{S}^{\text{up}}_{60,i})) \in (0, 1]$ by its training mean (≈ 0.686) levered quiet-morning positions up to 1.46× (\$1.46M, or 29.2× effective leverage), violating the \$1M (20×) facility cap whenever afternoon shocks arrived; and the 21-date sample provided limited power. A pre-trade credit control of the kind Rule 15c3-5 requires brokers to maintain caps exposure rather than levering it up, so Part 24 registers a capped throttle $q_t \in (0, 1]$ that never raises a position above the \$1M facility limit. To test this capped pre-trade control and evaluate the 42-date matched-breach capacity, we pre-registered Part 24 (`P24_guard_throttle_and_capacity.json`, SHA-256 `698f91196eba...`) and committed its hash before downloading the untouched 2nd-and-9th-of-month data for January 2025–September 2026 (456 training symbol-days on the 8th and 22nd of each month in 2024; 796 test symbol-days across 42 untouched test dates; test opened once).

Table 3.38 shows that both pre-registered primaries pass when the \$1M (20×) facility cap $q_t \in (0, 1]$ is enforced. Adding the morning-certified guard to the volatility cap reduces the 18-hour forced-liquidation rate from 38.69% to 31.28%—a reduction of −7.41 percentage points (95% CI [−11.45, −4.15])—and reduces the liquidation rate per \$1M of deployed notional by −4.09 percentage points (95% CI [−8.47, −1.03], Primary 1 passes). Simultaneously, the certified guard lowers 18-hour daily ES_{95} loss from \$163,099 to \$136,670, saving −\$26,429 per day on the \$1M facility (95% CI [−\$53,146, −\$6,254], Primary 2 passes). By contrast, the secondary unbounded matched-breach capacity comparison yields +\$172,416 (+17.89%, 95% CI [−\$219,311, +\$578,090]), so the guard works as a protective cap ($q_t \leq 1$), not as a licence to lever quiet mornings above 20×.

Table 3.38: Pre-registered capped pre-trade throttle and capacity test on 42 untouched days (Part 24), 2025–2026

Capped rule $q \in (0, 1]$ (796 symbol-days)	Liq. rate	Daily ES_{95}	Mean notional	Liq. / \$1M
(a) Static 20× limit ($q = 1$)	51.88%	\$248,756	\$1,000,000	51.88%
(b) Volatility cap ($q \leq 1$)	38.69%	\$163,099	\$829,463	46.65%
(c) Vol. cap × cert. guard ($q \leq 1$)	31.28%	\$136,670	\$734,945	42.56%
(d) Certified guard alone ($q \leq 1$)	40.70%	\$169,606	\$857,186	47.48%
Prim. 1: ΔLiq(c vs b) / per \$1M	−7.41 pp [−11.45, −4.15] / −4.09 pp [−8.47, −1.03] (pass)			
Prim. 2: ΔDaily ES_{95}(c vs b)	−\$26,429, 95% CI [−\$53,146, −\$6,254] (pass)			
Sec.: Unbounded $\overline{N}$(c) − $\overline{N}$(b) (48.37%)	+\$172,416 (+17.89%), 95% CI [−\$219,311, +\$578,090]			

Non-resetting 18-hour 20× position (06:00–24:00 UTC, 450 bp buffer) on 796 untouched test symbol-days (2nd and 9th of each month, Jan 2025–Sep 2026; 456 training symbol-days in 2024); 2,000-draw date-block bootstrap 95% intervals. Spec SHA-256 `698f91196eba...`; test opened once (`R_P2_p24_p25.json`).

Matched exposure. Most of the registered pass comes from trading less. The −7.41 pp breach reduction in Table 3.38 reflects both response-based risk discrimination and an 11.39% reduction in average admitted notional (\$829,463 → \$734,945) caused by multiplying two independent caps in $(0, 1]$. An independent multi-policy audit (`hardened_empirical_audits.py` → `R_hardened_empirical_audits.json`) evaluates six policies under strictly training-calibrated sizing across the 42 test dates. When normalized on the training set so both policies target \$1M mean exposure (Benchmark 4B-ii), the combined guard achieves a breach rate indistinguishable from the volatility cap (48.62% vs. 48.37%, $\Delta = +0.25$ pp, 95% CI $[-5.64, +5.16]$ pp) and comparable tail loss (\$197,395 vs. \$188,508, $\Delta =$ +\$8,887, 95% CI [−\$25,253, +\$33,587]). At matched exposure, then, the guard adds no measurable protection beyond a volatility cap. The largest single loss is smaller under the combined rule (\$445,487 against \$623,252 for the volatility cap), but that is one observation. Calibrated on training data to a matched ES_{95} budget (Benchmark 4C-i, exploratory, point estimates), the combined policy admits \$1,040,119 mean notional versus \$963,874 (+7.91% incremental capacity at matched training risk), with normalized tail loss of \$184,365 per \$1M deployed (vs. \$195,573). In realistic directional accounting with 10 bp taker fee and funding costs, realized 18-hour breach rates are 14.3% (long) and 18.3% (short), well below the stylized two-sided 31.3%.

3.12.13 Pre-registered test of the multi-minute persistence mechanism (Part 25)

Part 22 showed that five-minute open-interest contractions neither are forecast by $\hat{C}_3$ nor mediate its excursion predictability. Under Theorem 3.4.4, $\hat{C}_3 = \hat{\kappa}^2\hat{\eta}\hat{q} - (1 - \hat{a})\hat{\kappa}$ amplifies worst-case excursions through multi-minute price-impact persistence and slow book reversion across 15–60 minute horizons rather than isolated five-minute liquidations. To test this slow-reversion reading, we registered Part 25 after Part 22 failed (`P25_persistence_mechanism.json`, SHA-256 `8ebd67f3e9d1...`), before downloading the 42 untouched 2nd-and-9th-of-month test dates

(7,752 training symbol-hours in 2024; 13,532 test symbol-hours in January 2025–September 2026 across 19 thin perpetuals; test opened once). For each hour $t \in \{6, \ldots, 22\}$ UTC, we measure next-hour multi-quarter-hour non-reversion via the mean absolute price displacement across the four 15-minute checkpoints $M_{\text{ckpt},t+1} = \frac{1}{4}\sum_{k=1}^{4} |p_{t+15k} - p_t|$ (and the 60-minute L_1 excursion integral $M_{L_1,t+1} = \frac{1}{60}\sum_{\tau=1}^{60} |p_{t+\tau} - p_t|$, the empirical counterpart of S_{60}), contrasted against the maximum single five-minute bar jump $M_{5\text{m},t+1} = \max_{b\in\{1..12\}} |\Delta p_{5\text{m},b}|$.

Table 3.39: Pre-registered test of the multi-minute price-impact persistence mechanism on 42 untouched days (Part 25), 2025–2026

Specification (19 thin, 13,532 test hours)	ΔR^2 (pts) / Share [95% CI]	CW t (p)	Res.
Excursion $\text{MAE}_{t+1} \mid \hat{C}_{3,t}$ (2nd & 9th)	+3.944 [+2.114, +6.375]	15.68 ($< 10^{-6}$)	—
Prim. 1: 15-min persistence $M_{\text{ckpt},t+1} \mid \hat{C}_{3,t}$	+3.713 [+1.884, +6.035]	14.84 ($< 10^{-6}$)	pass
Excursion $\text{MAE}_{t+1} \mid \hat{C}_{3,t}, M_{\text{ckpt},t+1}$	+0.095 [+0.052, +0.149]	—	—
Prim. 2: Mediation by $M_{\text{ckpt},t+1}$ (w/ placebo)	97.6% [96.6%, 98.3%]	—	fail†
Sec.: 60-min L_1 integral $M_{L_1,t+1} \mid \hat{C}_{3,t}$ / share	+3.809 [+1.973, +6.252] / 98.6%	15.03 ($< 10^{-6}$)	—
Placebo: 5-min jump $M_{5\text{m},t+1} \mid \hat{C}_{3,t}$ / share	+0.738 [+0.503, +1.064] / 78.6%	15.78 ($< 10^{-6}$)	—

Trained on 24 dates in 2024 (7,752 symbol-hours); evaluated once on 42 untouched test dates (13,532 symbol-hours) with 13 baseline controls + symbol fixed effects. †Primary 2 required both $M_{\text{ckpt},t+1}$ mediation $\geq 50\%$ (met: 97.6%) and $M_{5\text{m},t+1}$ mediation $< 50\%$ (not met: 78.6%, leaving +0.843 pts unmediated vs. +0.095 pts for $M_{\text{ckpt},t+1}$). Spec SHA-256 `8ebd67f3e9d1...` (`R_P2_p24_p25.json`).

Table 3.39 reports the results. First, on the 42 untouched test dates (13,532 symbol-hours), $\hat{C}_{3,t}$ again forecasts next-hour adverse excursions out of sample ($\Delta R^2 = +3.944$ points, 95% CI $[+2.114, +6.375]$, $t = 15.68$), providing a seventh out-of-sample confirmation of the thin-book excursion effect. Second, Primary 1 passes: $\hat{C}_{3,t}$ forecasts next-hour 15-minute checkpoint persistence $M_{\text{ckpt},t+1}$ beyond all 13 volatility, impact and lagged-persistence controls ($\Delta R^2 = +3.713$ points, 95% CI $[+1.884, +6.035]$, $t = 14.84$), as well as the 60-minute L_1 integral $M_{L_1,t+1}$ ($\Delta R^2 = +3.809$ points, $t = 15.03$), with five times the predictive power of a single five-minute bar jump (+0.738 points). Third, controlling for $M_{\text{ckpt},t+1}$ collapses $\hat{C}_{3,t}$'s excursion ΔR^2 from +3.944 to +0.095 points—a mediation share of 97.6% (95% CI [96.6%, 98.3%]; 98.6% for $M_{L_1,t+1}$). Because the largest five-minute bar jump $M_{5\text{m},t+1}$ partially co-moves with multi-minute excursions, its mediation share (78.6%, leaving +0.843 points unmediated—8.9× the residual under $M_{\text{ckpt},t+1}$) exceeds the strict $< 50\%$ placebo threshold, so the joint Primary 2 criterion is recorded as a failure. Together, Parts 22 and 25 point away from five-minute liquidations and toward slow reversion over 15 to 60 minutes, but they do not establish it: Primary 2 fails, and the next paragraph shows that much of the mediation is mechanical.

Path-norm collinearity and non-overlapping sequential persistence. Why did the single-bar jump placebo mediate 78.6% in Table 3.39? Simulated paths under a pure Gaussian random-walk null (with zero structural feedback) reveal that $\text{MAE}_{60} = \|p_f\|_\infty$ has an intrinsic mathematical correlation of 0.9235 with M_{ckpt} and 0.9139 with M_{L_1} (compared to 0.9370 and 0.9417 empirically) because continuous path norms on the same interval are strongly collinear by

analysis. To evaluate multi-minute persistence without path-norm overlap, an exploratory audit (`hardened_empirical_audits.py`) partitions the hour into non-overlapping sequential windows: pre-window $\hat{C}_3$ forecasts initial 15-minute price travel ($\Delta R^2 = +3.321$ points, $t = 15.65$), and controlling for this initial travel mediates 65.1% of the subsequent displacement from baseline and 40.3% of incremental new price travel over the remaining 45 minutes ($t = 15.53$). This is consistent with persistence over several quarter-hours. But because $\hat{C}_3$ is itself built from slow reversion, a test that it forecasts persistence comes close to checking its construction, and an independent test of the mechanism remains open.

3.13 Policy implications

Pre-trade controls. Rule 15c3-5 requires pre-trade controls that prevent orders exceeding credit or capital thresholds (SEC, 2010). Static single-order caps and daily value-at-risk limits do not see transient dynamics. Lemma 3.6.1 identifies the quantity that a solvency-motivated control should bound: inventory times the shock envelope times the summed absolute impulse response. Peak-based or eigenvalue-based limits are either unsafe or blind in exactly the regimes where they matter. Our evidence locates where such controls matter. In the deepest crypto book the amplifying regime is rare and the statistic carries no forecasting value; in thin books it is more frequent, and six-hour estimates of the summed response forecast the next hour's adverse excursion beyond volatility forecasts and price impact (Section 3.11). Our backtests also set a limit on what such a control can do with public data. A six-hour estimate of the summed response adds nothing to a volatility limit as a sizing input (Table 3.25); an operator certified from three days of data improves a volatility limit modestly, by 2.4% in tail loss per unit of position (Table 3.26); and limits built on the estimate alone do worse than volatility in both. The guard's role is therefore a ceiling on worst-case loss under a certified operator and a correction to conventional volatility limits, not a replacement for them, and it is most relevant for books with under about \$0.4M of depth within 1% of the mid.

Pricing reversion. Proposition 3.8.1 shows that dealers under-supply reversion because its benefit, lower tail exposure for every leveraged position, is not paid to them. Maker rebates are a natural instrument: a rebate that rises with the book's summed response, $(1 - \theta/N)\Lambda S^2/\tilde{\kappa}$ per unit of reversion, prices the externality with an observable. Venues already tier rebates by volume; tiering them by contribution to reversion in thin books is a design question our model makes precise.

What can be certified, what cannot, and when to abstain. A supervisor or venue can certify the summed response, and hence a safe admission level, from controlled experiments at a cost that does not depend on how many traders there are or how similar their memories are (Theorem 3.5.5). It cannot certify any finite risk bound from a finite window without a bound

on how long participants remember (Theorem 3.5.3(iii)), so a declared memory bound is a policy input, not a technicality, and a certificate must be renewed when conditions change (test T09). When compatible risks straddle the budget, the right response is to abstain: throttle admission to the certified level, and when even zero admission is uncertified because of existing inventory, flag the state and reduce the position (Theorem 3.6.5(iv)). Disclosure is worth paying for where memory is long and public certificates are weak, and is worth little where a short window already pins the risk down (Theorem 3.7.1).

Market design. Frequent batch auctions remove latency races within a batch (Budish et al., 2015). Transient amplification, when it exists, operates over many seconds and spans many batches, so batching and a state-carrying guard address different failures and can be complements.

Supervision without identification. Chapter 2 shows that identifying algorithms from public data is expensive. The guard needs only impulse coefficients and operator confidence sets, which are invariant to how flow is split across cohorts. Supervisors can therefore ask venues and brokers to report summed-gain statistics and their confidence radii instead of attempting to infer black-box learners. Our results argue for reporting impact, mean reversion and refill directly, because those are the primitives that decide whether a book can amplify. They also show that feedback and amplification are different objects: measured feedback is largest in the deepest books, where amplification is rarest (Table 3.13).

3.14 Conclusion

The theory answers four questions. A book amplifies when depletion outweighs impact-weighted reversion; with momentum the one-period margin is sufficient but not exact, and momentum alone cannot make a stable book overshoot. Cumulative risk is long-run impact divided by reversion net of momentum, and the guard that bounds it is the largest safe pre-trade rule, computable and identified from public data. In equilibrium reversion is under-supplied, and the corrective price is a function of the guard's own statistic. And the information that certifies a safe decision is much cheaper than the information that identifies the traders: the summed response is certified at the parametric rate, finite windows certify it only under a memory bound, and a guard built on the certificate gives up no more trading capacity than any comparably safe rule, up to a logarithm.

The evidence is mixed in an informative way. The deepest crypto book sits on the stable side of the condition, and flow depletes the side it hits without producing a peak overshoot. In thin Binance books, six-hour estimates of cumulative transient risk forecast the size of the next hour's adverse excursion out of sample in every pre-registered forecasting test, on new symbols and on untouched days, beyond price impact, a full volatility-forecasting baseline and VPIN;

the effect is about a fifth as large in thin US stocks, negligible on Bybit and absent on Coinbase. Tests of a leverage or liquidation mechanism fail, and a follow-up that points to slow reversion over 15 to 60 minutes is close to a check of how the statistic is built. As a risk control the certified guard improves on a static limit but not, at matched exposure, on a volatility cap; its registered gain over a volatility cap in 18-hour positions comes mostly from admitting less exposure. Thin books are risky because their prices revert too slowly relative to the impact and momentum of the flow that moves them; why they revert slowly is the open empirical question.

Three extensions would settle what remains open. Message-level data from a regulated market, such as LOBSTER or TAQ for US equities, would extend the US test to depth and to large stocks, show where the effect ends, and allow depletion to be measured without the recentering of band depth. Data on liquidity-provider identities would test the equilibrium's prediction that fewer dealers mean more transient risk. And longer certification windows, or certification with order-level data, would show whether the modest gain of a certified guard over volatility limits can be made large.

Data and code availability. All data are public: Binance archives (`https://data.binance.vision`), verified against published checksums, and Bybit's order-book history (`https://www.bybit.com/derivatives/en/history-data`) and trade archive (`https://public.bybit.com`); US consolidated trades are from the Alpaca market-data API (`https://alpaca.markets`, free account). Pre-registered specifications, their SHA-256 hashes and test-set unlock counters are in the replication package. Each number is produced by a notebook cell or script that writes a result file (Appendix 3.18).

Use of AI tools. The authors used Claude (Anthropic), an AI assistant, to write and debug analysis code, run checks of derivations and numerical results, and draft and edit text. Every number in the paper is produced by code the authors executed and archived in the replication package. The authors reviewed all content and take full responsibility for it.

Disclosures. The authors declare no relevant financial interests.

References

21Shares (2025). Record crypto liquidations amid tariff shock. Research note, October 13, 2025. `https://www.21shares.com/research/record-crypto-liquidations-amid-tariff-shock`.

Achieser, N. I. (1956). *Theory of Approximation*. New York: Frederick Ungar.

Almgren, R., and N. Chriss (2001). Optimal execution of portfolio transactions. *Journal of Risk* 3, 5–39.

Amihud, Y. (2002). Illiquidity and stock returns: Cross-section and time-series effects. *Journal of Financial Markets* 5, 31–56.

Andersen, T. G., and O. Bondarenko (2014). VPIN and the flash crash. *Journal of Financial Markets* 17, 1–46.

Avellaneda, M., and S. Stoikov (2008). High-frequency trading in a limit order book. *Quantitative Finance* 8, 217–224.

Biais, B., L. Glosten, and C. Spatt (2005). Market microstructure: A survey of microfoundations, empirical results, and policy implications. *Journal of Financial Markets* 8, 217–264.

Biais, B., P. Hillion, and C. Spatt (1995). An empirical analysis of the limit order book and the order flow in the Paris Bourse. *Journal of Finance* 50, 1655–1689.

Bouchaud, J.-P., Y. Gefen, M. Potters, and M. Wyart (2004). Fluctuations and response in financial markets: The subtle nature of "random" price changes. *Quantitative Finance* 4, 176–190.

Bouchaud, J.-P., J. Bonart, J. Donier, and M. Gould (2018). *Trades, Quotes and Prices: Financial Markets Under the Microscope*. Cambridge University Press.

Brogaard, J., T. Hendershott, and R. Riordan (2014). High-frequency trading and price discovery. *Review of Financial Studies* 27, 2267–2306.

Brunnermeier, M. K., and L. H. Pedersen (2009). Market liquidity and funding liquidity. *Review of Financial Studies* 22, 2201–2238.

Budish, E., P. Cramton, and J. Shim (2015). The high-frequency trading arms race: Frequent batch auctions as a market design response. *Quarterly Journal of Economics* 130, 1547–1621.

Cartea, Á., S. Jaimungal, and J. Penalva (2015). *Algorithmic and High-Frequency Trading*. Cambridge University Press.

Cespa, G., and T. Foucault (2014). Illiquidity contagion and liquidity crashes. *Review of Financial Studies* 27, 1615–1660.

Clark, T. E., and K. D. West (2007). Approximately normal tests for equal predictive accuracy in nested models. *Journal of Econometrics* 138, 291–311.

Conley, T. G., and C. R. Taber (2011). Inference with "difference in differences" with a small number of policy changes. *Review of Economics and Statistics* 93, 113–125.

Cont, R., A. Kukanov, and S. Stoikov (2014). The price impact of order book events. *Journal of Financial Econometrics* 12, 47–88.

Corsi, F. (2009). A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics* 7, 174–196.

Dahleh, M. A., and I. J. Diaz-Bobillo (1995). *Control of Uncertain Systems: A Linear Programming Approach*. Prentice Hall.

Dahleh, M. A., and J. B. Pearson (1987). ℓ^1-optimal feedback controllers for MIMO discrete-time systems. *IEEE Transactions on Automatic Control* 32, 314–322.

Degryse, H., F. de Jong, M. van Ravenswaaij, and G. Wuyts (2005). Aggressive orders and the resiliency of a limit order market. *Review of Finance* 9, 201–242.

Diebold, F. X., and R. S. Mariano (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics* 13, 253–263.

Easley, D., M. M. López de Prado, and M. O'Hara (2012). Flow toxicity and liquidity in a high-frequency world. *Review of Financial Studies* 25, 1457–1493.

Foucault, T., O. Kadan, and E. Kandel (2005). Limit order book as a market for liquidity. *Review of*

Financial Studies 18, 1171–1217.

Hansen, E., and G. W. Walster (2004). *Global Optimization Using Interval Analysis*, 2nd ed. New York: Marcel Dekker.

Harvey, C. R. (2017). Presidential address: The scientific outlook in financial economics. *Journal of Finance* 72, 1399–1440.

Harvey, C. R., Y. Liu, and H. Zhu (2016). . . . and the cross-section of expected returns. *Review of Financial Studies* 29, 5–68.

Hendershott, T., C. M. Jones, and A. J. Menkveld (2011). Does algorithmic trading improve liquidity? *Journal of Finance* 66, 1–33.

Ho, T., and H. R. Stoll (1981). Optimal dealer pricing under transactions and return uncertainty. *Journal of Financial Economics* 9, 47–73.

Holm, S. (1979). A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics* 6, 65–70.

Hotelling, H. (1931). The generalization of Student's ratio. *Annals of Mathematical Statistics* 2, 360–378.

Howard, S. R., A. Ramdas, J. McAuliffe, and J. Sekhon (2021). Time-uniform, nonparametric, nonasymptotic confidence sequences. *Annals of Statistics* 49, 1055–1080.

Huang, J., and J. Wang (2009). Liquidity and market crashes. *Review of Financial Studies* 22, 2607–2643.

Jordà, Ò. (2005). Estimation and inference of impulse responses by local projections. *American Economic Review* 95, 161–182.

J.P. Morgan/Reuters (1996). *RiskMetrics—Technical Document*, 4th ed. New York: J.P. Morgan.

Karlin, S., and W. J. Studden (1966). *Tchebycheff Systems: With Applications in Analysis and Statistics*. New York: Interscience.

Kirilenko, A., A. S. Kyle, M. Samadi, and T. Tuzun (2017). The Flash Crash: High-frequency trading in an electronic market. *Journal of Finance* 72, 967–998.

Krein, M. G., and A. A. Nudel'man (1977). *The Markov Moment Problem and Extremal Problems*. Translations of Mathematical Monographs 50. Providence, RI: American Mathematical Society.

Kreiss, H.-O. (1962). Über die Stabilitätsdefinition für Differenzengleichungen die partielle Differentialgleichungen approximieren. *BIT Numerical Mathematics* 2, 153–181.

Künsch, H. R. (1989). The jackknife and the bootstrap for general stationary observations. *Annals of Statistics* 17, 1217–1241.

Kyle, A. S. (1985). Continuous auctions and insider trading. *Econometrica* 53, 1315–1335.

Kyle, A. S., and A. A. Obizhaeva (2016). Market microstructure invariance: Empirical hypotheses. *Econometrica* 84, 1345–1404.

Large, J. (2007). Measuring the resiliency of an electronic limit order book. *Journal of Financial Markets* 10, 1–25.

Le Cam, L. (1986). *Asymptotic Methods in Statistical Decision Theory*. New York: Springer.

Makarov, I., and A. Schoar (2020). Trading and arbitrage in cryptocurrency markets. *Journal of Financial Economics* 135, 293–319.

Menkveld, A. J. (2013). High frequency trading and the new market makers. *Journal of Financial Markets* 16, 712–740.

Milanese, M., and A. Vicino (1991). Optimal estimation theory for dynamic systems with set membership uncertainty: An overview. *Automatica* 27, 997–1009.

Newey, W. K., and K. D. West (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica* 55, 703–708.

Niroula, P., and P. Niroula (2026). Auditable memory and financial identification limits in electronic limit order books. Working paper, Minnesota State University, Mankato.

Nosek, B. A., C. R. Ebersole, A. C. DeHaven, and D. T. Mellor (2018). The preregistration revolution. *Proceedings of the National Academy of Sciences* 115, 2600–2606.

Obizhaeva, A. A., and J. Wang (2013). Optimal trading strategy and supply/demand dynamics. *Journal of Financial Markets* 16, 1–32.

Rump, S. M. (2010). Verification methods: Rigorous results using floating-point arithmetic. *Acta Numerica* 19, 287–449.

Schneider, M., and F. Lillo (2019). Cross-impact and no-dynamic-arbitrage. *Quantitative Finance* 19, 137–154.

U.S. Securities and Exchange Commission (2010). Risk management controls for brokers or dealers with market access. Release No. 34-63241.

Shleifer, A., and R. W. Vishny (1992). Liquidation values and debt capacity: A market equilibrium approach. *Journal of Finance* 47, 1343–1366.

Sornette, D., S. Lera, J. Lin, and K. Wu (2023). Non-normal interactions create socio-economic bubbles. *Communications Physics* 6, 261.

Spijker, M. N. (1991). On a conjecture by LeVeque and Trefethen related to the Kreiss matrix theorem. *BIT Numerical Mathematics* 31, 551–555.

Trefethen, L. N., and M. Embree (2005). *Spectra and Pseudospectra: The Behavior of Nonnormal Matrices and Operators*. Princeton University Press.

Tu, S., R. Boczar, and B. Recht (2018). Minimax lower bounds for $\mathcal{H}_\infty$-norm estimation. Working paper, arXiv:1809.10855.

Vidyasagar, M. (1986). Optimal rejection of persistent bounded disturbances. *IEEE Transactions on Automatic Control* 31, 527–534.

Ville, J. (1939). *Étude critique de la notion de collectif*. Paris: Gauthier-Villars.

van Waarde, H. J., J. Eising, H. L. Trentelman, and M. K. Camlibel (2020). Data informativity: A new perspective on data-driven analysis and control. *IEEE Transactions on Automatic Control* 65, 4753–4768.

3.15 Appendix: Proofs

Proposition 3.2.1. With one cohort, $J = \left(\begin{smallmatrix} a & \kappa q \\ \eta a & 1-\eta+\eta\kappa q \end{smallmatrix}\right)$, so $\mathrm{tr}J = a + 1 - \eta + \eta\kappa q$ and $\det J = a(1-\eta)$. The Jury conditions for a real 2×2 matrix are $|\det J| < 1$ and $1 \pm \mathrm{tr}J + \det J > 0$. Substituting $a = 1 - \kappa\varphi_f$: $1 - \mathrm{tr}J + \det J = \eta(1 - a - \kappa q) = \eta\kappa(\varphi_f - q)$, positive iff $q < \varphi_f$; and $1 + \mathrm{tr}J + \det J = (4 - 2\eta) - \kappa[(2 - \eta)\varphi_f - \eta q]$, positive iff the stated bound on κ holds (automatically when $(2 - \eta)\varphi_f \le \eta q$).

Proposition 3.2.2. Equal merged measures give equal G, hence equal H, and the impulse coefficients are the Laurent coefficients of H (Chapter 2, Theorem on merged equivalence). Under $z \mapsto Tz$, $c^\top J^k b$ is unchanged because $CT^{-1}(TJT^{-1})^k Tb = CJ^k b$, while the row norm $\|CT^{-1}(TJT^{-1})^k\|$ depends on T. The displayed price path is the convolution of inputs with $\{g_k\}$, and its worst case over the ℓ_∞ ball is $\bar{u}\sum|g|$ by Hölder.

Proposition 3.2.3. (i) Taking z-transforms of $y_{i,t} = (1-\eta_i)y_{i,t-1} + \eta_i x_t$, $I_{i,t} = (1-\delta_{\text{inv}})I_{i,t-1} + u_{i,t}$ and $u_{i,t} = q_i y_{i,t} - \psi_i I_{i,t-1}$ gives $Y_i(z) = \frac{\eta_i z}{z-1+\eta_i}X(z)$ and $(z - 1 + \delta_{\text{inv}} + \psi_i)I_i(z) = q_i z Y_i(z)$, whence $U_i(z) = \frac{z-1+\delta_{\text{inv}}}{z-1+\delta_{\text{inv}}+\psi_i} q_i Y_i(z)$ and $G_{\text{inv}}(z) = \sum_i U_i(z)/X(z)$ has the stated form. Evaluating at $z = 1$ when the augmented operator is Schur stable gives $G_{\text{inv}}(1) = \sum_i \frac{\delta_{\text{inv}}}{\delta_{\text{inv}}+\psi_i} q_i = \Phi_{\text{eff}}(\psi)$ and $S(\psi) = H_{\text{inv}}(1) = \kappa/(1 - a - \kappa\Phi_{\text{eff}}(\psi))$. (ii) Write $r_i = 1 - \eta_i$ and $\rho_i = 1 - \delta_{\text{inv}} - \psi_i$. When $0 \le \psi_i \le 1 - \delta_{\text{inv}}$ and $\delta_{\text{inv}} \ge \eta_i$, we have $0 \le \rho_i \le 1 - \delta_{\text{inv}} \le r_i < 1$. Expanding $U_i(z)/X(z) = q_i\eta_i \frac{z}{z-r_i}\big(1 - \frac{\psi_i}{z-\rho_i}\big)$ in powers of z^{-1}, the m-th open-loop feedback tap of cohort i is $k_{i,0} = q_i\eta_i \ge 0$ and, for $m \ge 1$, $k_{i,m} = q_i\eta_i\big(r_i^m - \psi_i \sum_{j=0}^{m-1} r_i^{m-1-j}\rho_i^j\big) \ge q_i\eta_i r_i^{m-1}\big(r_i - \frac{\psi_i}{1-\rho_i/r_i}\big) \ge 0$ because $\frac{\psi_i}{r_i-\rho_i} = \frac{\psi_i}{\delta_{\text{inv}}-\eta_i+\psi_i} \le 1 \le r_i$ whenever $\psi_i \le (1 - \delta_{\text{inv}})$ and $\delta_{\text{inv}} \ge \eta_i$ with $r_i^m \ge \psi_i \frac{r_i^m - \rho_i^m}{r_i - \rho_i}$ (since $\frac{\psi_i}{r_i-\rho_i} \le 1$). Nonnegative open-loop taps $k_m(\psi) \ge 0$ with $a + \kappa\sum_m k_m(\psi) = a + \kappa\Phi_{\text{eff}}(\psi) < 1$ generate a nonnegative closed-loop renewal sequence $h_t(\psi) \ge 0$, so $\|h(\psi)\|_1 = \sum_{t\ge0} h_t(\psi) = S(\psi)$.

Theorem 3.3.1. (i) $\sup_t \|CJ^t z_0\| \le \sup_t \|CJ^t\| \|z_0\|$, with equality approached by aligning z_0 with the top right singular vector of CJ^{t^*} at the maximizing t^*. (ii) The Kreiss matrix theorem with Spijker's sharp constant gives $\mathcal{K}(J) \le \sup_t \|J^t\| \le en\mathcal{K}(J)$; substitute in (i). (iii) Take $J_\theta = \left(\begin{smallmatrix} \rho & \theta \\ 0 & \rho/2 \end{smallmatrix}\right)$; then $\rho(J_\theta) = \rho$ and $(1, 0)J_\theta e_2 = \theta$, so $M_C \ge |\theta|$ and $r^*_{\text{exact}} \le B/|\theta| \to 0$.

Proposition 3.3.2. With all blocks entrywise nonnegative, every entry of J^t_{multi} is a polynomial in the entries of K_{21} with nonnegative coefficients, so it is non-decreasing in K_{21}; with $C \ge 0$, so is every entry of CJ^t and hence its norm. For the Kreiss constant, the Neumann series gives $|(zI - J)^{-1}| \le (|z|I - J)^{-1}$ entrywise for $|z| > 1$, so the supremum is attained on the real axis, where $(rI - J)^{-1} = \sum_k J^k/r^{k+1}$ is entrywise non-decreasing in K_{21}.

Theorem 3.6.8. After the step, inventory is at most $|I^{\text{post}}_{t+1}(\alpha_t)|$ in absolute value. The price move over the step and the cumulative window is bounded by $M_C\|z_t\|_2$ (drift from the current state) plus $s(\alpha)G_{\text{cum}}\bar{u}_{\text{cum}}$ (admitted shocks), by superposition and Hölder. The margin requirement scales the loss by $(1 + \mu)$. Hedging cost and the post-fill margin increment are deducted directly. Inequality (3.14) therefore implies $B_{\text{rem},t+1} \ge R_{\text{safe}}$ for every admissible disturbance, and induction over steps at which an admissible α exists gives forward solvency. Setting $\alpha = 0$ ($M_C(0) = 1$, no fills, no hedge) gives $\bar{u}_{\text{adm}}$. Sufficiency over the Euclidean ball follows from the norm bounds. When $z_t = 0$ the remaining shock term is attained by $u = -\text{sign}(I)\bar{u}$, which gives necessity; when $z_t \ne 0$ the drift bound $M_C\|z_t\|_2$ need not be attained, so the condition is sufficient but not necessary.

Proposition 3.6.9. (i) In the positive cohort economy $J(\alpha) \geq 0$ entrywise, so for any $z_0 = (x_0, y_0^\top)^\top$, $|e_x^\top J(\alpha)^t z_0| \leq e_x^\top J(\alpha)^t (|x_0|, |y_0|^\top)^\top \leq e_x^\top J(\alpha)^t (|x_0|, \bar{y}_0 \mathbf{1}^\top)^\top$, with equality when $y_0 = \operatorname{sign}(x_0)\bar{y}_0\mathbf{1}$. Under the price-first unforced recursion $x_t = a x_{t-1} + \alpha\kappa \sum_i q_i y_{i,t-1}$ $(t \geq 1)$ and $y_{i,t} = (1-\eta_i) y_{i,t-1} + \eta_i x_t$ $(t \geq 1)$ with initial state $(|x_0|, \bar{y}_0 \mathbf{1}^\top)$ in the unit-sum memory coordinate (where the initial cohort state $y_{i,0}$ enters with unit cumulative weight $\sum_{m \geq 0} (1-\eta_i)^m \eta_i = 1$, i.e. $\tilde{y}_{i,0} = \eta_i y_{i,0}$ bounded by $\eta_i \bar{y}_0$), summing over $t \geq 1$ gives $\sum_{t \geq 0} q_i y_{i,t} = q_i \bar{y}_0 + q_i \sum_{t \geq 1} x_t$, and hence $(1 - a - \alpha\kappa\Phi) \sum_{t \geq 1} x_t = a|x_0| + \alpha\kappa\Phi\bar{y}_0$, proving (3.15). (ii) Adding the forced worst-case shock loss $|I| \frac{\kappa\bar{u}}{1-a-\alpha\kappa\Phi}$ to $|I|$ times (3.15) and requiring the total loss to be at most $\mathcal{B}$ at $\Phi = \Phi^+$ rearranges to $\alpha\kappa\Phi^+(\mathcal{B} + |I|\bar{y}_0) \leq (1-a)\mathcal{B} - |I|(a|x_0| + \kappa\bar{u})$, which gives (3.16). (iii) Subtracting $\alpha^*_{\text{drift}} = c_I / [\kappa\Phi^+(1 + |I|\bar{y}_0/\mathcal{B})]$ from $\alpha^* = c_I/(\kappa\Phi^+)$ at $x_0 = 0$ gives (3.17).

Proposition 3.6.10. The eigenvalue rule admits α_ρ; the one-step certificate fails at α_ρ exactly when the left side of (3.14), with the shock term replaced by its one-step bound $M_C(\alpha_\rho) s(\alpha_\rho) \bar{u}_{\text{step},t}$, exceeds $B_{\text{rem},t}$, which rearranges to the stated inequality. The summed-gain guard admits only levels satisfying (3.14), so it never admits the breaching level, and by the feasibility assumption it admits at least $\alpha = 0$.

Proposition 3.6.11. (i) The steady-state gain is $\eta = P/(P + \sigma_\nu^2)$ with P the positive root of $P^2 - \sigma_\zeta^2 P - \sigma_\zeta^2 \sigma_\nu^2 = 0$. Since P is strictly increasing in σ_ζ^2 and η is strictly increasing in P, η^* is strictly increasing in $\sigma_\zeta^2(\alpha) = \alpha^2 \sigma_{\zeta,0}^2$, hence in α. The second statement is immediate under the stated monotonicity.

(ii) Write the recursion as a renewal equation: $h_t/\kappa = u_t$ with $u_0 = 1$ and $u_t = \sum_{j=1}^t p_j u_{t-j}$, where $p_1 = a + \kappa k_0$ and $p_j = \kappa k_{j-1}$ for $j \geq 2$. The p_j are nonnegative with total mass $\sum_j p_j = a + \kappa\Phi = a + \kappa \sum_i q_i < 1$, which does not depend on the adaptation rates, so u_t is the probability that a renewal process whose (defective) inter-arrival law is p has a renewal at time t, and $\sum_{t<H} u_t$ is the expected number of renewals in $\{0, \ldots, H-1\}$. The cumulative law is $F(j) = \sum_{i \leq j} p_i = a + \kappa \sum_{m<j} k_m = a + \kappa \sum_i q_i \big[1 - (1-\eta_i)^j\big]$, with $\partial F(j)/\partial \eta_i = \kappa q_i j (1-\eta_i)^{j-1} \geq 0$. If $\eta' \geq \eta$ coordinatewise, then $F' \geq F$ pointwise with the same total mass, so the quantile coupling $X = F^{-1}(U)$, $X' = F'^{-1}(U)$ ($X = \infty$ when $U > F(\infty)$) of independent inter-arrival times gives $X' \leq X$ and hence, pathwise, at least as many renewals before H under η'. Taking expectations gives $S_H(\eta') \geq S_H(\eta)$. If $q_i > 0$ and $\eta_i' > \eta_i$, then $F'(1) > F(1)$, so with positive probability the first renewal occurs at time 1 under η' and later under η, and the inequality is strict for $H \geq 2$. As $H \to \infty$ both sides tend to $\kappa/(1 - a - \kappa\Phi)$, which does not depend on η. Combined with (i), throttling lowers every rate, so $S_H(J(\alpha, \eta^*(\alpha))) \leq S_H(J(\alpha, \eta^*(1)))$.

Theorem 3.5.3. The recursion $h_{t+1} = a h_t + \kappa \sum_{m \leq t} k_m h_{t-m}$ with $h_0 = \kappa \neq 0$ gives k_t from $h_0, \ldots, h_{t+1}$, so the window determines $k_0, \ldots, k_{n-1}$ and nothing more about M: any nonnegative measure on $[0, \bar{r}]$ with these moments, written as atoms (r_i, w_i) or as a limit of such, is the merged measure of a positive economy with $\eta_i = 1 - r_i$ and $q_i = w_i/\eta_i$ that produces

the same window. (i) The compatible measures form a convex, weak-* compact set, and $\Phi = \int f\,dM$ is linear and continuous on it (f is continuous on $[0,\bar r]$), so its range is a closed interval $[\Phi_-,\Phi_+]$. Since $f^{(n)}(r) = n!/(1-r)^{n+1} > 0$ on $[0,\bar r]$, $(1, r, \dots, r^{n-1}, f)$ is an extended complete Chebyshev system, and by the Markov–Krein theorem (Krein and Nudel'man, 1977, Theorems IV.1.1 and IV.6.1) the maximum is attained at the upper and the minimum at the lower principal representation, which have at most $\lfloor n/2\rfloor + 1$ atoms. The theorem applies to moment vectors in the interior of the moment space, where both principal representations exist and are unique; a vector on the boundary has exactly one representing measure, so the interval collapses to a point. $S = \kappa/(1-a-\kappa\Phi)$ is increasing in Φ on $\Phi < (1-a)/\kappa$, which gives the interval of summed responses, unbounded when $\Phi_+ \ge (1-a)/\kappa$ because by convexity every $\Phi \in [\Phi_-,\Phi_+]$ is attained. (ii) If μ,ν share moments up to order $n-1$, then $\int f\,d(\mu-\nu) = \int (f-p)\,d(\mu-\nu)$ for every polynomial p of degree $n-1$, so $|\int f\,d(\mu-\nu)| \le 2k_0 \inf_p \|f-p\|_\infty = 2k_0 E_{n-1}$. With $r = \bar r(1+x)/2$, $f = (2/\bar r)/(x_0 - x)$ on $[-1,1]$ with $x_0 = 2/\bar r - 1 > 1$, and Chebyshev's formula gives $E_m(1/(x_0-x)) = \varrho^m/(x_0^2-1)$ (Achieser, 1956); we checked it against a linear-programming minimax fit for $m \le 8$. Expanding $x_0 = 1+2(1-\bar r)/\bar r$ gives the asymptotics. (iii) Let the moment vector c be interior to the moment cone of $[0,1]$. Interior points have an open neighbourhood in the cone, and $m(r_0) = (1, r_0, \dots, r_0^{n-1})$ has norm at most $\sqrt{n}$, so for some $\epsilon > 0$, $c - \epsilon m(r_0)$ is interior for every $r_0 \in [0,1]$ and is represented by its lower principal representation, which has no atom at 1. Adding $\epsilon\delta_{r_0}$ gives a measure on $[0,1)$ with moments c and $\Phi \ge \epsilon/(1-r_0) \to \infty$ as $r_0 \to 1$; by convexity every larger Φ below $(1-a)/\kappa$ is attained by a stable economy, so S is unbounded.

Theorem 3.5.5. (i) The flow impulse enters the stacked state as $b = \kappa(1, \eta_1, \dots, \eta_K)^\top$, and $h_t = e_x^\top J^t b \le \|J^t b\|_\infty \le \|J\|_\infty^t \|b\|_\infty$ because $J \ge 0$. The price row of J sums to $a + \kappa\Phi \le \bar s$ and cohort row i to $1 - \eta_i(1-a-\kappa\Phi) \le 1-(1-\bar r)(1-\bar s) = \bar\lambda$, so $h_t \le \kappa\bar\lambda^t$ and $\sum_{t\ge T} h_t \le \kappa\bar\lambda^T/(1-\bar\lambda)$. $\sum_{t<T}\bar h_t$ is Gaussian with mean $\sum_{t<T} h_t$ and variance $T\sigma^2/N$, so $\sum_{t<T}\bar h_t + z_{1-\gamma}\sigma\sqrt{T/N} \ge \sum_{t<T} h_t$ with probability $1-\gamma$, and adding the tail bound gives validity. On the two-sided event of probability $1-2\gamma$, $\hat S^+ - S \le 2z_{1-\gamma}\sigma\sqrt{T/N} + \kappa\bar\lambda^T/(1-\bar\lambda)$; T_ϵ makes the second term at most $\epsilon/2$ and the stated N makes the first at most $\epsilon/2$. (ii) Let θ_1 raise the cohort weight of θ_0 by Δ chosen so that $S_1 - S_0 = 2\epsilon$; then $\|h_1 - h_0\|_2 = 2C_0\epsilon(1+o(1))$. Validity at θ_1 gives $P_1(\hat S \ge S_1) \ge 1-\gamma$, hence $P_0(\hat S - S_0 \ge 2\epsilon) \ge 1 - \gamma - \mathrm{TV}(P_0,P_1)$. For N independent Gaussian responses, $\mathrm{KL} = N\|h_1-h_0\|_2^2/(2\sigma^2)$, and Pinsker's inequality gives $\mathrm{TV} \le \sqrt{N}\,C_0\epsilon(1+o(1))/\sigma$. An overshoot probability below $1/2$ therefore needs $\mathrm{TV} > 1/2-\gamma$, which is the stated bound. (iii) The identification rates are Chapter 2's Theorems on the K-cohort lower bound, the matching upper bound and general memory.

Lemma 3.6.4. On the event that $\mathcal{C}(D)$ contains the true economy, the supremum over $\mathcal{C}(D)$ dominates the true worst-case loss for every action, including one selected from D; the event has probability at least $1-\gamma$.

Theorem 3.6.5. (i) For $c_I \geq 0$, $\alpha \mapsto S(\alpha) = \kappa/(1-a-\kappa\alpha\Phi)$ is increasing, and so is the worst-case loss $|I|\bar{u}S(\alpha)$, which is also increasing in Φ; the robust constraint is therefore the constraint at Φ^+, and $|I|\bar{u}\kappa/(1-a-\kappa\alpha\Phi^+) \leq \mathcal{B}$ is equivalent to $\alpha \leq c_I/(\kappa\Phi^+)$. (ii) is Lemma 3.6.4. (iii) At $\Phi = \Phi^+$ the constraint binds at α^* when $\alpha^* < 1$. (iv) At $\alpha = 0$, $S = \kappa/(1-a)$. (v) Upper: α^* is Lipschitz in Φ^+ with constant $c_I/(\kappa\Phi^2_{\min})$ and Φ^+ is Lipschitz in $\hat{S}^+$ with constant $1/S^2_{\min}$, so Theorem 3.5.5(i) with $T \asymp \log N$ gives the rate. Lower: take two single-cohort economies in the class whose weights differ by $\Delta = c\,\sigma/\sqrt{N}$; their oracle capacities differ by a constant times Δ, and TV $\leq \sqrt{N}\|\partial h/\partial q\|_2\Delta/(2\sigma)$, which is below $(1-\beta)/2$ for small c; Proposition 3.5.2 gives an expected loss of at least $(1-\beta)/2$ times the capacity difference.

Theorem 3.7.1. (i) If O' reveals everything O reveals, every economy compatible with O' data is compatible with O data, so suprema over $\Theta_{O'}$ are at most those over Θ_O, and Theorem 3.6.5(i) is monotone in Φ^+. (ii) α^* is strictly decreasing in Φ^+ when $c_I > 0$ and $\alpha^* < 1$, and $\Phi^+_{O'} < \Phi^+_O$ exactly when no economy with $\Phi = \Phi^+_O$ remains compatible (the supremum is attained, by Theorem 3.5.3(i)). (iii) follows from Theorem 3.5.3(iii). (iv) If O identifies Φ, $\Phi^+_O = \Phi$ and no contract can raise α^*. (v) The delta method applied to $(w, r) \mapsto w/(1-r)$, whose gradient is $(1-r)^{-1}(1, \Phi_i)$, gives the standard error. If each certificate is valid at level $1-\gamma/2$, their minimum is valid at level $1-\gamma$ by the union bound and is no larger than either, so the admitted level is at least the larger of the two admitted levels.

Lemma 3.5.4. For a compatible merged measure M on $[0, \bar{r}]$, $\int p\,dM = \sum_j c_j k_j$ for any polynomial of degree at most $n-1$, so $p_- \leq f \leq p_+$ gives the two bounds by integration. Let H interpolate f at the atoms x_i of μ_+ with multiplicity $m_i = 2$ at interior atoms and 1 at 0 and $\bar{r}$; principal representations have index $\sum_i m_i = n$, so H has degree at most $n-1$, and the Hermite remainder is $f(r) - H(r) = f^{(n)}(\xi)\,\omega(r)/n!$ with $\omega(r) = \prod_i (r - x_i)^{m_i}$. Since $f^{(n)} > 0$ and, on $[0, \bar{r}]$, the factor $r - \bar{r} \leq 0$ appears once while every other factor is a square or $r \geq 0$, $\omega \leq 0$ and $H \geq f$. For μ_- there is no atom at $\bar{r}$, so $\omega \geq 0$ and $H \leq f$. As $H = f$ on the support of $\mu_\pm$, $\sum_j c_j k_j = \int H\,d\mu_\pm = \int f\,d\mu_\pm = \Phi_\pm$. The certified computation proves $p_+ + s \geq f$ (or $p_- - s \leq f$) for the rounded polynomial and slack s in interval arithmetic, in which every operation encloses its exact result (Rump, 2010), so the first statement applies to the rounded polynomial with c_0 increased (decreased) by s.

Proposition 3.5.7. (i) The X_e are i.i.d. normal with mean $S_T = \sum_{t<T} h_t$, whatever Σ, so $\sqrt{N}(\bar{X} - S_T)/s_X$ has Student's t_{N-1} distribution and $P(S_T \leq \bar{X} + t_{N-1,1-\gamma}s_X/\sqrt{N}) = 1-\gamma$; the tail bound of Theorem 3.5.5(i) holds deterministically on the class. (ii) $N(\bar{h}-m)^\top\hat{\Sigma}^{-1}(\bar{h}-m)$ has the $\frac{T(N-1)}{N-T}F_{T,N-T}$ distribution (Hotelling, 1931); on the event that it is at most T^2_γ, Cauchy–Schwarz in the $\hat{\Sigma}$ inner product gives $c^\top(m-\bar{h}) \leq \sqrt{c^\top\hat{\Sigma}c}\,\sqrt{(\bar{h}-m)^\top\hat{\Sigma}^{-1}(\bar{h}-m)} \leq \sqrt{T^2_\gamma c^\top\hat{\Sigma}c/N}$ for every c at once. (iii) If the taps of an experiment have common variance σ^2

and correlation matrix R, the margin $z_{1-\gamma}\sigma\sqrt{T/N}$ has coverage $\Phi\big(z_{1-\gamma}\sqrt{T/\mathbf{1}^\top R\mathbf{1}}\big)$, below $1-\gamma$ whenever $\mathbf{1}^\top R\mathbf{1} > T$; with $T = 12$ and autocorrelation 0.8 it is 0.751, matching the simulations.

Proposition 3.6.6. (i) D is a quadratic with leading coefficient $\delta\rho \geq 0$, and a level is safe if and only if $D(\alpha) \geq \kappa_0|I|\bar{u}/\mathcal{B}$ (which implies $D > 0$, hence stability and $S = \kappa_0/D$). The set where a convex function is below a constant is an interval, so the safe set is $[0, 1]$ minus an open interval; the example in the text has two components. (ii) For $\alpha \geq 0$ and $a_0 < 1$, $\partial D/\partial\delta = \alpha(1 - a_0 + \rho\alpha) \geq 0$, $\partial D/\partial\rho = \alpha(1 + \delta\alpha) \geq 0$, $\partial D/\partial a_0 = -(1 + \delta\alpha) \leq 0$ and $\partial D/\partial\Phi = -\kappa_0\alpha \leq 0$, while the condition $D - \kappa_0|I|\bar{u}/\mathcal{B} \geq 0$ decreases in κ_0. The minimum of the left side over the box is therefore at the stated corner for every α. (iii) follows from (i) and (ii).

Theorem 3.6.7. (i) In interval arithmetic with outward rounding every operation returns an interval containing the exact result for all real arguments in its operands (Rump, 2010); by induction the computed interval vectors contain $J(\alpha, \theta)^t s_0(\alpha, \theta)$ and the interval matrices contain J^j for every (α, θ) in the box, so the computed sum contains $\sum_{t<T} |h_t|$. For $t \geq T$, writing $t = T + kT + j$ with $k \geq 0$ and $0 \leq j < T$, $|h_t| \leq \|J^{kT+j} J^T s_0\|_\infty \leq \|J^T\|_\infty^k \|J^j\|_\infty \|J^T s_0\|_\infty$, and summing gives the tail bound. A level interval classified safe has an upper enclosure within budget over every part of a partition of Θ, hence for every θ; one classified unsafe has $S(\alpha, \theta_j) \geq \sum_{t<T} |h_t(\alpha, \theta_j)| >$ budget for some $\theta_j \in \Theta$ and every level in it. (ii) Each $S(\cdot, \theta)$ is continuous where finite and tends to infinity at the stability boundary, so $\alpha \mapsto \sup_\Theta S(\alpha, \theta)$ is lower semicontinuous and its sublevel set, the robust safe set, is closed; the brackets follow from (i). (iii) is immediate. (iv) Under $\sup_\Theta \rho(J) < 1$ there is T with $\|J^T\|_\infty < 1$ uniformly on Θ; interval extensions of rational functions converge linearly in the box width (Hansen and Walster, 2004), so at a level where the robust risk differs from the budget by $g > 0$ the enclosures over small enough boxes, with T large enough that the truncation is below $g/2$, decide.

Proposition 3.17.1. With $D_n = \sum_{e \leq n}(X_e - S_T)$ and $V_n = n\sigma_T^2$, $\exp(\lambda D_n - \lambda^2 V_n/2)$ is a nonnegative supermartingale for every λ under sub-Gaussian increments; mixing over $\lambda \sim N(0, 1/\varrho)$ gives the nonnegative supermartingale $M_n = \sqrt{\varrho/(V_n + \varrho)}\exp\big(D_n^2/(2(V_n + \varrho))\big)$ with $M_0 = 1$. Ville's inequality (Ville, 1939) gives $P(\exists n : M_n \geq 1/\gamma) \leq \gamma$, and $M_n \geq 1/\gamma$ is exactly $|D_n| \geq b_n$. On the complementary event every interval covers S_T at every n, so a decision made at any stopping time is correct; the tail bound is deterministic. Since $b_n/n = O\big(\sigma_T\sqrt{\log(n/\gamma)/n}\big)$, the stopping time is $O\big(\sigma_T^2\Delta^{-2}\log(\sigma_T/(\Delta\gamma))\big)$ when the gap is Δ (Howard et al., 2021).

3.16 Appendix: Notation

Symbol	Meaning
$x_t,\ d_t,\ u_t$	price displacement, depth imbalance $(B-A)/(B+A)$, signed flow
$a,\ \kappa$	price persistence and direct impact
$\varphi,\ \varepsilon,\ r$	price sensitivity to depth, depletion per unit of flow, refill rate
$\eta,\ q$	momentum memory rate and aggressiveness
$C_2,\ C_3$	amplification statistics $\varphi\varepsilon-(1-a)\kappa$ and $C_2+\kappa^2\eta q$
$h_k,\ g_k$	impulse response of the book; price impulse coefficient of the controlled system
$J(\alpha)$	closed-loop operator at admission level α
$M_C,\ G_{\text{cum}}$	peak and summed price-row gains
$\mathcal{K}(J)$	Kreiss constant
$\bar{u},\ \bar{u}_{\text{step}},\ \bar{u}_{\text{cum}}$	shock envelopes
$B_{\text{rem}},\ R_{\text{safe}},\ \mu$	remaining margin buffer, safety reserve, maintenance-margin rate
$\epsilon,\ S$	operator confidence radius and $\sum_k \|J^k\|$
$M_{\text{std}},\ A_{xx}$	unit-free and price-to-price spot amplification statistics

3.17 Appendix: Further results

3.17.1 Specification of the pre-registered tests

Futures (Table 3.9). Question: does $\hat{C}_3$, estimated from the previous six hours of futures depth, flow and price, predict the next hour's MAE beyond controls? Unit: hourly blocks 06–23 UTC, features from $[t-360, t)$ minutes, outcomes from $(t, t+60]$. Controls: rv_{60}, rv_{360}, $|r_{60}|$, flow imbalance, log volume, log depth, imbalance, open-interest change. Model: OLS; tail is the outcome above the training 90th percentile; score is the fitted value. Inference: Clark–West with Newey–West (6 lags); day-block bootstrap with 2,000 draws. Secondary: peak ratio as candidate; open-interest drop and depth collapse as outcomes; Holm. Train $\leq$ 2024-12-31; test $\geq$ 2025-01-01. Hash `d95212187ae42052591a5ca5acf73317 0f217f430a5a6ca812da6e48a1e4af85`; one unlock.

Spot (Section 3.10). Question: does pre-window A_{xx} predict the next ten minutes' MAE beyond volatility and flow controls? Unit: non-overlapping 30-minute blocks; features from the previous ten minutes. Controls: rv_{10m}, rv_{60m}, $|r_{10m}|$, flow imbalance, log volume, spectral radius, hour of day. Train 2021–2023; test 2024. Hash `b1fd7f8eef15df1ba295eb6d32b53be6 c23ea2d23c5216de912555e62716e5ae`; three unlocks (see text).

3.17.2 Adaptive safe probing

The certificate of Theorem 3.5.5 fixes the number of reset experiments in advance, so it must be sized for the hardest economy the supervisor wants to decide. Experiments can instead be run one at a time and stopped as soon as the data decide. Let each experiment return $X_e = \sum_{t<T} h_t^{(e)}$, independent with mean S_T and sub-Gaussian proxy σ_T^2 ($\sigma_T^2 = T\sigma^2$ under (O1)), and let $b_n = \sqrt{(n\sigma_T^2 + \varrho)\big(\log((n\sigma_T^2 + \varrho)/\varrho) + 2\log(1/\gamma)\big)}$ for a fixed $\varrho > 0$.

Proposition 3.17.1 (Anytime-valid probing)**.** *With probability at least $1-\gamma$, $|\bar{X}_n - S_T| < b_n/n$ for every n simultaneously. Hence the rule that probes until $\bar{X}_n + b_n/n + \kappa\bar{\lambda}^T/(1-\bar{\lambda}) \le s_{\mathcal{B}}$ (admit), or $\bar{X}_n - b_n/n > s_{\mathcal{B}}$ (refuse), or a maximum number of probes is reached (abstain), makes a wrong decision with probability at most γ whatever its stopping time, where $s_{\mathcal{B}} = \mathcal{B}/(|I|\bar{u})$. When S_T is at distance Δ from $s_{\mathcal{B}}$ it stops after $O\big(\sigma_T^2\Delta^{-2}\log(\sigma_T/(\Delta\gamma))\big)$ probes. Each probe is itself admissible if its position v_p satisfies $|v_p|\bar{u}_p\kappa/(1-\bar{\lambda}) \le \mathcal{B}_p$, the worst case over the class.*

The value is in adapting to the economy. With $\sigma_T = 0.17$ and a threshold of 2.0, the sequential rule decides economies at distance 0.5 from the threshold after 3.7 probes on average, at distance 0.2 after 11, at 0.1 after 30 and at 0.05 after 107, with no wrong decision in 7,000 runs, while a fixed design that must decide at distance 0.05 uses 130 probes for every economy (test T14). Probes have a monetary cost and move prices, so a supervisor who can probe sequentially spends where the decision is hard. This is an extension rather than a result the paper relies on; optimal probe design under impact costs is left open.

3.18 Appendix: Reproducibility: source of every number

The replication package contains numbered Colab notebooks and stand-alone scripts; its README lists them in run order. "Part N" below and in table notes refers to notebook N (Part 1: theorem checks and spot tests; Part 2: BTCUSDT futures depth; Parts 4–18: the cross-section, replications, fresh-day tests, second venues, the CFTC test, the certified guard, post-hoc checks, US equities, the information-safety tests on untouched days and symbols, the new-venue test and the last post-hoc checks). Result identifiers such as `X6` or `U6` name the JSON file that the notebook writes. "Kind" distinguishes closed-form or simulation results (recomputed), results from public market data (real data), and audits.

Result	Source		Kind
Spectral blindness gains	`T5_spectral_blindness`		recomputed
Table 3.2; burst operator gain, q, certificate	`kreiss_table.py` → `R_P2_kreiss.json`	→	recomputed
Table 3.4, Figure 3.3	`calib_guard.py` → `R_P2_calibrated_guard.json`	→	recomputed

Result	Source	Kind
Figure 3.1	`make_figs2.py`	recomputed
Theorem 3.4.2(i), Theorem 3.4.2(ii)–(iii) draws	`T6_amplification_theorem`	recomputed
Guard maximality; peak guard unsafe; monotonicity	`T7_guard_maximality`, `P2.11-T1`	recomputed
Robust gain, 30,000 draws	`T8_robust_gain`	recomputed
Operator ball coverage; certificate share	`P2.4-T1_coverage`	recomputed
Summed gain vs. weights	`T9_bridge`	recomputed
Depth validation (Table 3.6)	`D2_depth_validation`	real data
Table 3.7	`D3_T6_structural_train`	real data
Figure 3.4, Table 3.8	`D4_depth_mediation_train`	real data
No-look-ahead audit	`D5_no_lookahead`	audit
Table 3.9; hash	`D8_depth_test_G2`, `P2_depth.json`	real data
Flag comparison	`D8b_flag_comparison`	real data
Table 3.10, Figure 3.5	`D9_event_studies`	real data
Unit dependence 8.73 → 86,731	`P2.2-T1_unit_invariance`	real data
Table 3.11; sensitivity	`P2.2-T2`, `T3`, `T4`, `T5`	real data
Table 3.12	`T6-E_overshoot_real`	real data
Spot predictive test	`P2.5_predictive_G2`, `P2.5-T2`, `T3`	real data
Spot flag comparison	`P2.7-T2_vol_ablation`	real data
Theorem 3.4.4; Remark 3.4.3; Theorem 3.4.2(iv) check	`equilibrium.py`, `t6m_search*.py`, `thmA_check.py`	recomputed
Proposition 3.8.1 numbers	`equilibrium.py` → `R_P2_equilibrium.json`	recomputed
Proposition 3.2.2 check; multi-asset example; Proposition 3.6.11 check	`recompute_oct20.py`	recomputed
Tables 3.13, 3.14, 3.15; tick events	Part 4: `X3`, `X4`, `X8`, `X7`	real data
Table 3.17; flags	Part 4: `X6_xsec_test_G2`, `X6b`; `P4_xsec.json` (5910b6b4…)	real data
Table 3.18	Part 4: `X6r_robustness_posthoc`	real data (post hoc)
Table 3.19	Part 5: `Y6_replication_G2`; `P5_replication.json` (6aa8e2c0…)	real data
Table 3.20	Part 5: `Y8_kappa_control_posthoc`	real data (post hoc)

Result	Source	Kind
Tables 3.21, 3.25; equilibrium check	Part 6: `Z6_*`, `Z8`; `P6_fresh_days.json` (65acda36…)	real data
Table 3.26, Figure 3.6, economic size (Section 3.11.17)	Part 12: `Q6_certified_guard`, `Q7_oct10_event`, `E1_economic_size`; `P12_certified_guard.json` (b9a56af4…)	real data
Liquidation probabilities and dollar losses (Section 3.11.17)	`recompute_dollar.py` (from `E1_economic_size`) → `R_P2_dollar_translation.json`	recomputed
Table 3.23	Part 10: `V3A_orderbook_download`, `V4_build`, `V6_bybit_ob_test`; `P10_bybit_ob.json` (ec2145fa…)	real data
Table 3.16; throttle, Bybit-bar, instrument, tercile and VPIN checks	Part 13: `R5`, `R1`, `R3`, `R4`, `R2`, `R8`	real data (post hoc)
Example 3.5.1; Theorems 3.5.3, 3.5.5, 3.6.5, 3.7.1 numbers; Tables 3.3, 3.5, 3.28	`run_all.py` in `research_extension/information_safety` → `IS_results.json`; unit tests	recomputed
Section 3.12.2	Part 15: `W4_build`, `W5_blocks`, `W6_info_safety_test`; `P15_info_safety.json` (`69849c96`)	real data
Section 3.12.3	Part 16: `X2_new_universe`, `X4_build`, `X5_blocks`, `X6_untouched_symbols_test`; `P16_untouched_symbols.json` (`0523e476`)	real data
Section 3.12.4	Part 17: `C2_universe`, `C4_build`, `C5_blocks`, `C6_new_venue_test`; `P17_new_venue.json` (3233c52b)	real data
Section 3.12.5	Part 18: `R3_placebo_binance`, `R4_guard_baselines`; `P18_posthoc_plan.json`	real data (post hoc)

Result	Source	Kind
Section 3.12.7 (Table 3.33)	Part 19: `recompute_p2_parts19_20_21.py` → `R_P2_parts19_20_21.json`	real data
Section 3.12.8 (Table 3.34)	Part 20: `recompute_p2_parts19_20_21.py` → `R_P2_parts19_20_21.json`	real data
Section 3.12.9 (Table 3.35)	Part 21: `recompute_p2_parts19_20_21.py` → `R_P2_parts19_20_21.json` (from `W6`, `X6`)	translation of real data
Sections 3.12.10–3.12.11 (Tables 3.36, 3.37)	Parts 22–23: `recompute_prereg_oct21_22nd.py` → `R_P2_p22_p23.json`; `P22_liquidation_mechanism.json` (2b3794e3), `P23_guard_capacity.json` (d4169377)	real data
Sections 3.12.12–3.12.13 (Tables 3.38, 3.39)	Parts 24–25: `recompute_prereg_oct21_followup.py` → `R_P2_p24_p25.json`; `P24_guard_throttle_and_capacity.json` (698f9119), `P25_persistence_mechanism.json` (8ebd67f3)	real data
Propositions 3.2.3 and 3.6.9 checks	`recompute_p2_step2_theory.py` → `R_P2_step2_theory.json`	recomputed
Balance-sheet Monte Carlo (Section 3.6)	`bridge/microfoundation_and_baselines.py` → `bridge/results/microfoundation_and_baselines.json`	recomputed
Lemma 3.5.4; Propositions 3.5.7, 3.6.6, 3.17.1; Theorem 3.6.7	`run_all.py`: `T10`, `T11`, `T13`, `T14`	recomputed (certified for T10, T13)
Table 3.24	Part 14: `A1_us_universe`, `A3_us_build`, `A4_us_blocks`, `A5_us_equity_test`; `P14_us_equities.json` (f5800a69…)	real data
Theorem 3.4.2(ii) peak bracket; Remark 3.4.3; Theorem 3.4.4, 3.8.1 checks	`recompute_audit.py` → `R_audit_oct2026.json`	recomputed

Result	Source	Kind
Theorem 3.4.2 residues and confluent peak lag; Remark 3.5.6; Proposition 3.6.11(ii)	`hardening/p2_sec4_checks.py`, `p2_cheb_tradeoff.py`; unit tests `test_hardening_theory.py`	recomputed
Theorem 3.7.1(v), intersection row of Table 3.5	information-safety package, test T15	recomputed
Section 3.11.11	Part 7: `W6_bybit_test`; `P7_bybit.json`	real data
Table 3.22	Part 9: `U6_volbench_test`; `P9_volbench.json` (71983f1a…)	real data
Figure 3.7	`make_fig_forest.py`	real data

Chapter 4

Certify, don't identify: a theory of safety without identification

This chapter is the third paper of the thesis and the candidate's own work. It places the results of Chapters 2 and 3 in one framework, proves when a question about a market's unknown agents is cheap or impossible to certify, and extends the safety results to agents that are nonlinear and keep learning. Results proved here are marked PROVED IN THIS THESIS; results proved in Chapters 2–3 are marked accordingly; the open technical target and the conjecture are marked as such.

Summary. Regulators ask to know which algorithms trade in a market; we ask which questions about those algorithms can be answered from market data at all, and at what cost. Every question an outside observer can ask is a functional of the agents' merged measure of memory rates and weights, observed through noisy moments. For linear questions we bound the number of experiments needed to certify the answer by its coefficient energy at a truncation depth set by its tail at long memory, give a finite-depth obstruction in terms of best polynomial approximation, and show that questions singular at infinite memory cannot be certified without a memory bound. Safety is such a question: it depends on the agents only through their total feedback weight and is certified at the parametric rate up to a logarithm, whereas identifying the agents costs polynomially or logarithmically more. The summed-gain guard remains safe for agents whose rules are nonlinear and whose memories and weights change over time, given only a bound on their total sensitivity, and when agents keep learning a certificate lasts for its safety margin divided by their learning speed. The uniform matching lower bound is refuted as stated; a repaired bound remains open. A discontinuous guard can fail to admit a strategic equilibrium.

4.1 Markets with adaptive agents

Definition 4.1.1 (Cohort economy)**.** A state $x_t \in \mathbb{R}$, the displacement of the price from its fundamental value, evolves as

$$x_{t+1} = a\,x_t + \kappa\,(u_t + w_{t+1}), \qquad u_t = \sum_{i=1}^{K} q_i\,\varphi_i(y_{i,t}), \qquad y_{i,t} = r_i\,y_{i,t-1} + (1-r_i)\,x_t, \tag{4.1}$$

with persistence $a \in [0,1)$, impact $\kappa > 0$, exogenous shocks w_t, and K cohorts of agents with weights $q_i \geq 0$, memory rates $r_i \in [0,1)$ and response rules φ_i. The economy is *linear* when $\varphi_i(y) = y$ for all i.

The exponential state $y_{i,t}$ is the reduced form of steady-state Kalman filtering and of regularized online gradient descent (Lemma 2.2.1); Section 4.6 relaxes linearity and stationarity. In the linear economy the aggregate response is a linear filter of past states,

$$u_t = \sum_{m\geq 0} k_m\,x_{t-m}, \qquad k_m = \int r^m\,dM(r), \qquad M = \sum_{i=1}^{K} q_i(1-r_i)\,\delta_{r_i},$$

so the agents enter the observable dynamics only through the *merged measure* M, a positive measure on $[0,1)$, and through its moments k_m. Cohorts that share a memory rate are indistinguishable, whatever their identities (Theorem 2.3.1). We write $\mathcal{M}(\bar r, B)$ for the positive measures on $[0,\bar r]$ of total mass at most B; $\bar r < 1$ is a *memory bound*.

4.2 Questions are functionals of the merged measure

Every question an outside observer can answer from the observable dynamics is a functional F of M (and of the known plant parameters a, κ). Three families matter.

Identification. $F(M) = M$ itself, in a metric such as the Wasserstein distance W_1, or the atoms (r_i, q_i) when M has K atoms. Chapter 2 studies these.

Safety. For bounded shocks $|w_t| \leq \bar w$, the worst-case transient loss of a position I is $|I|\,\bar w\,\|h\|_1$, where h is the closed-loop impulse response, and a pre-trade guard on $\|h\|_1$ admits every safe action and no unsafe one (Lemma 3.6.1). In the linear economy with $a \geq 0$ all h_t are nonnegative, so $\|h\|_1$ equals the summed response

$$S(M) = \frac{\kappa}{1 - a - \kappa\,\Phi(M)}, \qquad \Phi(M) = \sum_{m\geq 0} k_m = \int \frac{dM(r)}{1-r} = \sum_i q_i, \tag{4.2}$$

whenever $a + \kappa\Phi < 1$ (Theorem 3.4.4). Safety therefore depends on the agents only through one linear functional of M, the total feedback weight Φ.

Other operational questions. The response at a given lag, $k_m = \int r^m dM$; the discounted feedback $\int (1-\beta r)^{-1} dM$; the peak of the impulse response. The first two are linear in M; the peak is not, and a guard built on it admits unsafe actions (Chapter 3).

4.3 Observation contracts and the cost of a question

Assumption 4.3.1 (Reset contract)**.** The observer can run N independent reset experiments. Each returns the first L aggregate response taps with independent Gaussian noise, $\hat{k}_m = k_m + \sigma\xi_m$, $\xi_m \sim N(0,1)$, $m = 0, \dots, L-1$.

This is the controlled-probe contract with observed aggregate flow (contract I_1 of Chapter 2). Chapter 3 works with the closed-loop taps h_t instead; for the summed response the two contracts give the same rate (Theorem 3.5.5). Correlated tap noise, passive data and tagged disclosure are other contracts, treated in Chapters 2 and 3.

Definition 4.3.2 (Cost of certification)**.** For a functional F, a class $\mathcal{M}$, accuracy $\epsilon > 0$ and level $1-\alpha$, $N_{\mathrm{cert}}(\epsilon; F, \mathcal{M})$ is the smallest N for which some interval built from the data has half-width at most ϵ and covers $F(M)$ with probability at least $1-\alpha$ for every $M \in \mathcal{M}$.

4.4 The certification–identification dichotomy

Consider a linear question $F_g(M) = \int g\, dM$, where $g(r) = \sum_{t\ge 0} a_t r^t$ converges absolutely on $[0, \bar{r}]$. Two numbers describe how hard it is. Its *tail at depth* T and its *coefficient energy* are

$$\tau_g(T) = B \sup_{0\le r\le \bar{r}} \Big| \sum_{t\ge T} a_t r^t \Big|, \qquad A_g(T) = \sum_{t<T} a_t^2,$$

and its *truncation depth* is $T_g(\epsilon) = \min\{T \ge 1 : \tau_g(T) \le \epsilon/2\}$. We also write $E_T(g) = \min_{\deg p < T} \max_{[0,\bar{r}]} |g - p|$ for the error of the best polynomial approximation of degree below T.

Theorem 4.4.1 (Certification of linear questions)**.** *Under Assumption 4.3.1 with $L \ge T_g(\epsilon)$, and with $z = z_{1-\alpha/2}$:*

(i) Upper bound. $N_{\mathrm{cert}}(\epsilon; F_g, \mathcal{M}(\bar{r}, B)) \le \lceil 4z^2\sigma^2 A_g(T_g(\epsilon))\, \epsilon^{-2} \rceil$. PROVED IN THIS THESIS

(ii) Truncation is necessary. *For every T there are $M, M' \in \mathcal{M}(\bar{r}, B)$ whose first T taps coincide and with $|F_g(M) - F_g(M')| \ge 2B\, E_T(g)$. Hence any certificate that uses only the first T taps has worst-case half-width at least $B\, E_T(g)$.* PROVED IN THIS THESIS

(iii) Long memory makes certification impossible. *Without a memory bound, that is on the positive measures on $[0,1)$ of mass at most B, if $\limsup_{r\uparrow 1} |g(r)|\sqrt{1-r} = \infty$ then for every level $1-\alpha$ with $\alpha < 1/4$ no finite N certifies F_g to any finite accuracy.* PROVED IN THIS THESIS

(iv) Matching lower bound. *For g in a regular class (coefficients of eventually constant sign and at most polynomial growth), $N_{\mathrm{cert}}(\epsilon; F_g, \mathcal{M}(\bar{r}, B)) \geq c\,\sigma^2 A_g(c'\,T_g(\epsilon))\,\epsilon^{-2}$ for constants $c, c' > 0$ depending only on $(\bar{r}, B, \alpha)$.* TARGET: STATEMENT FIXED, PROOF TO BE WRITTEN

Proof of (i)–(iii). (i) Let $\bar{k}_t$ be the average of the N replicates of tap t and $T = T_g(\epsilon)$. The estimator $\hat{F} = \sum_{t<T} a_t \bar{k}_t$ has mean $\sum_{t<T} a_t k_t$ and variance $\sigma^2 A_g(T)/N$. Its bias is $|\sum_{t\geq T} a_t k_t| = |\int \sum_{t\geq T} a_t r^t \, dM(r)| \leq \tau_g(T) \leq \epsilon/2$ for every $M \in \mathcal{M}(\bar{r}, B)$. The interval $\hat{F} \pm (\epsilon/2 + z\sigma\sqrt{A_g(T)/N})$ therefore covers $F_g(M)$ with probability at least $1-\alpha$, and its half-width is at most ϵ once $z\sigma\sqrt{A_g(T)/N} \leq \epsilon/2$.

(ii) By duality for best uniform approximation, $E_T(g) = \max\{\int g\, d\nu\}$ over signed measures ν on $[0, \bar{r}]$ with total variation at most one that annihilate every polynomial of degree below T. Take a maximizer and write $\nu = \nu^+ - \nu^-$. Orthogonality to constants gives $\nu^+([0,\bar{r}]) = \nu^-([0,\bar{r}]) =: c \leq 1/2$. Set $M = (B/c)\nu^+$ and $M' = (B/c)\nu^-$: both lie in $\mathcal{M}(\bar{r}, B)$, their first T moments coincide, and $F_g(M) - F_g(M') = (B/c)E_T(g) \geq 2B\,E_T(g)$. Two measures that produce identical data cannot both be covered by an interval of half-width below half their distance.

(iii) Compare $M_0 = 0$ with $M_1 = c\,\delta_r$. Under M_1 the taps are $k_t = c\,r^t$, so the Kullback–Leibler divergence between the two data distributions over any number of taps is at most $Nc^2/(2\sigma^2(1-r^2))$. Choose $c = \sigma\sqrt{(1-r^2)/N}$, which is at most B for r near one; the divergence is then at most $1/2$, so by Pinsker's inequality the total variation distance between the two data distributions is at most $1/2$. The functional differs by $c\,|g(r)| = \sigma\sqrt{(1-r^2)/N}\,|g(r)| \geq \sigma\sqrt{(1-r)/N}\,|g(r)|$, which is unbounded along a sequence $r \uparrow 1$ by assumption. Suppose an interval of half-width ϵ covered F_g with probability at least $1-\alpha$ under both economies. Once the two values differ by more than 2ϵ, the event that the interval contains $F_g(M_0)$ has probability at least $1-\alpha$ under M_0 and at most α under M_1, so the total variation distance is at least $1 - 2\alpha > 1/2$, a contradiction. □

What the theorem says. The upper bound for answering a linear question is controlled by its Taylor coefficients and their tail; finite-depth ambiguity is controlled separately by best polynomial approximation. If its power-series tail decays geometrically on $[0, \bar{r}]$, the truncation depth grows like $\log(1/\epsilon)$ and the cost is parametric up to that factor. If the question is singular at infinite memory, faster than $(1-r)^{-1/2}$, no memory bound means no certificate. The number of agents and the spacing of their memories do not enter.

Corollary 4.4.2 (The rates of this thesis as cases of one picture)**.** *(a)* Safety with a memory bound. *For Φ, $a_t \equiv 1$, so $A_\Phi(T) = T$ and $\tau_\Phi(T) = B\bar{r}^T/(1-\bar{r})$. Part (i) gives $N_{\mathrm{cert}} = O\big(\sigma^2\epsilon^{-2}\log(1/\epsilon)/\log(1/\bar{r})\big)$, and $S = \kappa/(1-a-\kappa\Phi)$ is certified by mapping the interval through this monotone function. This is the rate of Theorem 3.5.5, which Theorem 3.6.5 shows is optimal up to the logarithm for the guard.* PROVED IN CH. 3

(b) Safety without a memory bound. $g(r) = (1-r)^{-1}$ *satisfies the condition of (iii): no finite number of experiments certifies the risk, as Theorem 3.5.3(iii) shows for finite windows.* PROVED IN CH. 3

(c) A single response tap. *For* $g(r) = r^m$, $A_g = 1$ *and the cost is exactly parametric.* PROVED IN THIS THESIS

(d) Identifying K cohorts. *Recovering the atoms is not a linear question. It requires moments up to order* $2K-1$ *to be resolved against moment-matched alternatives whose higher moments differ by* $O((\Delta\eta)^{2K-1})$, *which costs* $(\Delta\eta)^{-(4K-2)}$ *experiments (Theorems 2.6.4 and 2.6.5).* PROVED IN CH. 2

(e) Identifying a general memory measure. *Recovering* M *in* W_1 *means certifying* F_g *uniformly over all* 1*-Lipschitz* g. *Their best approximations by degree-*T *polynomials improve only as* $1/T$, *and the coefficients of those polynomials grow geometrically in* T, *so by part (i) accuracy improves only logarithmically in* N*: the rate* $1/\log N$ *of Theorem 2.6.7.* PROVED IN CH. 2

The link in (e) between coefficient growth and the logarithmic rate is a heuristic reading of Chapter 2's proof, not a separate result; making it a corollary of part (iv) is part of the target.

4.4.1 Partial lower bound and the gap in part (iv)

Part (iv) remains a TARGET: STATEMENT FIXED, PROOF TO BE WRITTEN. The following argument establishes only a parametric obstruction, with a constant depending on the functional. It does not establish the coefficient-energy factor, and has not been formalized in Lean.

A fixed-atom lower bound. Suppose $B > 0$, $\sigma > 0$, $\alpha < 1/4$, and choose $r_* \in [0, \bar{r}]$ with $g(r_*) \neq 0$. For sufficiently small ϵ, any valid certificate must satisfy

$$N_{\mathrm{cert}}(\epsilon; F_g, \mathcal{M}(\bar{r}, B)) > \frac{\sigma^2(1-r_*^2)|g(r_*)|^2}{9\epsilon^2}.$$

Indeed, put $\lambda = 3\epsilon/|g(r_*)|$ and require $\lambda \leq B/2$. The measures $M_0 = (B/4)\delta_{r_*}$ and $M_1 = (B/4+\lambda)\delta_{r_*}$ are admissible and their functional values differ by 3ϵ. For any observed depth L, their Gaussian experiment divergence obeys

$$D_{\mathrm{KL}}(P_0^N \| P_1^N) = \frac{N\lambda^2}{2\sigma^2} \sum_{t<L} r_*^{2t} \leq \frac{N\lambda^2}{2\sigma^2(1-r_*^2)}.$$

If N is no greater than the displayed lower bound, the divergence is at most $1/2$, and Pinsker's inequality gives total variation at most $1/2$. An interval of half-width at most ϵ cannot contain both functional values. Coverage at least $1-\alpha$ under both measures would therefore imply total variation at least $1-2\alpha > 1/2$, a contradiction. The divergence bound is uniform in L.

Why the proposed matching proof does not close the gap. Taylor truncation gives an upper bound on best polynomial approximation error, $BE_T(g) \le \tau_g(T)$; it gives no reverse bound. Thus part (ii) does not show that every procedure needs Taylor depth $T_g(\epsilon)$. Furthermore, a Gauss/Gauss–Radau pair with T nodes matches all moments through degree $2T-2$. For $t < T$, its tap difference is exactly zero, rather than a nonzero vector aligned with $(a_0, \dots, a_{T-1})$. Positivity of the coefficients does not supply the missing construction. Finally, when $L > T$, a valid testing lower bound must control divergence from all observed taps, not just a selected prefix. The fixed-atom argument above does so, but does not yield $A_g(T_g(\epsilon))$.

Counterexample to the uniform statement as written. The constants in part (iv) are stated to depend only on $(\bar{r}, B, \alpha)$, uniformly over the regular coefficient class. This uniform claim fails. Fix $\bar{r} > 0$, $B > 0$, an accuracy $\epsilon_0 > 0$, and $\beta \in (0,1)$, and choose $\eta > 0$ with $\eta/(1-\beta\bar{r}) < 1/2$. For any positive integer m, define

$$g_m(r) = \frac{\epsilon_0}{B}\left[(1 - r/\bar{r})^m + \frac{\eta}{1-\beta r}\right].$$

Its coefficients are strictly positive beyond degree m and have at most polynomial growth. On the support interval, g_m is non-negative and bounded above by $(\epsilon_0/B)(1 + \eta/(1-\beta\bar{r}))$. Thus every $F_{g_m}(M)$ lies in a known interval of length less than $3\epsilon_0/2$. That interval certifies the functional without any data, so $N_{\mathrm{cert}} \le 1$ even if the definition requires a positive number of experiments.

Yet $T_{g_m}(\epsilon_0) = m + 1$. For every $1 \le T \le m$, evaluating the polynomial tail at $r = \bar{r}$ gives

$$\sum_{t=T}^{m} (-1)^t \binom{m}{t} = (-1)^T \binom{m-1}{T-1}.$$

After adding the discounted tail, the reverse triangle inequality yields $\tau_{g_m}(T) \ge \epsilon_0(1 - \eta/(1 - \beta\bar{r})) > \epsilon_0/2$. At $T = m + 1$ only the discounted tail remains and is below $\epsilon_0/2$. For any fixed $c' > 0$, eventually $c'(m+1) > 1$, so the coefficient energy includes the coefficient of r and

$$A_{g_m}(c' T_{g_m}(\epsilon_0))\epsilon_0^{-2} \ge B^{-2}(m/\bar{r} - \eta\beta)^2 \longrightarrow \infty.$$

For fixed $\sigma > 0$ this contradicts any positive uniform constant c in part (iv). The accuracy ϵ_0 can be chosen arbitrarily small. This refutes the stated bound over the full regular class at a common accuracy; it does not refute an asymptotic statement with a functional-dependent accuracy threshold, nor the restricted all-non-negative coefficient case. A repaired statement must make its additional normalization or its functional dependence explicit. The original target is retained to show precisely what requires reformulation, rather than silently weakening it.

Deterministic checks and remaining target. The script `checks/check_dich_lower.py` records valid fixed-atom pairs for the safety, single-tap and discounted functionals, their functional gaps and divergences, and the exactness residuals of quadrature pairs in a JSON result file in the same directory. It checks the parametric scaling of the explicit testing budget. This scaling is a consequence of the construction, not independent evidence for a minimax matching rate. The script also evaluates the cancellation family above. No empirical or simulation test is performed. A reformulation of the uniform bound and the non-negative coefficient case remain open, as does the proposed deduction for Corollary 4.4.2(e).

4.5 Disclosure changes the contract, not the question

Disclosure regimes are observation contracts. Class-tagged flow reveals each class's own moments, which turns identification into an ordinary estimation problem and is worth a factor $(\Delta\eta)^{-(4K-4)}$ in samples (Theorem 2.8.2). For safety the value of disclosure is different: it raises safe capacity exactly when it rules out the extremal compatible economy, and it is worth most when memory is long, where Theorem 4.4.1(iii) says public data certify nothing (Theorem 3.7.1). Supervision that wants safety should therefore ask for the disclosures that bound memory or reveal Φ, not for identities.

4.6 A robust safety bound: nonlinear and changing agents

Papers 1 and 2 work with agents whose rules are linear filters with fixed memory and weight. Real agents cap positions, trade only beyond thresholds, adapt their gains, and keep learning while the operator watches. This section and the next show that the safety side survives all of this under one condition: a bound on the agents' total sensitivity.

Return to the economy of Definition 4.1.1 and let everything about the agents vary over time.

Assumption 4.6.1 (Bounded total sensitivity)**.** For each cohort i and period t, the memory rate $r_i(t) \in [0,1)$ is arbitrary, the weight satisfies $|q_i(t)| \le \bar{q}_i$, and the response rule satisfies the sector condition $|\varphi_{i,t}(y)| \le s_i|y|$ for all y. Write $\bar{\Phi} = \sum_i \bar{q}_i s_i$ for the total sensitivity.

The sector condition covers saturating momentum ($s_i = 1$ for tanh), position limits, threshold and dead-zone rules ($y\mathbf{1}\{|y| > \tau\}$), and any rule whose response never exceeds a multiple of its signal. It does not require the rule to be smooth, monotone, stationary or known.

Theorem 4.6.2 (Robust safety bound)**.** *Under Assumption 4.6.1, if $|a| + \kappa\bar{\Phi} < 1$ and the economy starts at rest, then for every shock sequence with $|w_t| \le \bar{w}$,*

$$\sup_{t\ge 0} |x_t| \;\le\; \bar{w}\,\frac{\kappa}{1-|a|-\kappa\bar{\Phi}}.$$

When $a \geq 0$, this is the worst-case loss bound of the linear stationary economy with weights $\bar{q}_i s_i$, so the summed-gain guard of Chapter 3 computed at total sensitivity $\bar{\Phi}$ is safe for every economy satisfying the assumption. PROVED IN THIS THESIS

Proof. Let $Z_t = \max_{s \leq t} |x_s|$. Each memory state is a convex combination of its previous value and the current state, so $|y_{i,t}| \leq r_i(t)|y_{i,t-1}| + (1 - r_i(t))|x_t| \leq \max(|y_{i,t-1}|, |x_t|)$, and by induction from $y_{i,0} = 0$, $|y_{i,t}| \leq Z_t$. Then

$$|x_{t+1}| \leq |a|\,|x_t| + \kappa\Big(\sum_i |q_i(t)|\,|\varphi_{i,t}(y_{i,t})| + |w_{t+1}|\Big) \leq (|a| + \kappa\bar{\Phi})\,Z_t + \kappa\bar{w}.$$

Let $Z^* = \kappa\bar{w}/(1 - |a| - \kappa\bar{\Phi})$. If $Z_t \leq Z^*$ then $|x_{t+1}| \leq (|a| + \kappa\bar{\Phi})Z^* + \kappa\bar{w} = Z^*$, so $Z_{t+1} \leq Z^*$; and $Z_0 = 0$. For the second claim, with $a \geq 0$ the linear stationary economy with weights $\bar{q}_i s_i$ is a positive system, its impulse response is non-negative, and its worst-case peak response to shocks bounded by $\bar{w}$ is $\bar{w} \sum_t h_t = \bar{w}\,\kappa/(1 - a - \kappa\bar{\Phi})$ (equation (4.2)). The bound of the theorem therefore coincides with the quantity the guard controls (Lemma 3.6.1). □

What the theorem says. An operator who wants a safe admission level does not need to know how many agents there are, what their memories are, whether their rules are linear, or whether they change over time. It needs one number, an upper bound on the agents' total sensitivity $\bar{\Phi}$. The bound is tight: when the agents are linear, stationary, positive and at their maximal weights, it is attained. Nonlinearity, memory changes and negative feedback can only make the market safer than the bound says.

Remark 4.6.3 (What the theorem does not cover). Rules that amplify their signal, $|\varphi(y)| > s|y|$ for the declared s, break the bound; so do changes in the market's own persistence a or impact κ, which the operator must certify separately (Chapter 3, Section 3.5). The bound controls the peak of the price path, which is what a fixed position's loss depends on; dynamic positions are handled by the balance-sheet guard of Chapter 3.

Proposition 4.6.4 (Several assets). *Let $x_t \in \mathbb{R}^d$ evolve as $x_{t+1} = Ax_t + \mathcal{K}\big(\sum_i Q_i(t)\varphi_{i,t}(y_{i,t}) + w_{t+1}\big)$, with memory states $y_{i,t} = R_i(t)y_{i,t-1} + (I - R_i(t))x_t$, $R_i(t)$ diagonal with entries in $[0, 1)$, sector conditions $|\varphi_{i,t}(y)| \leq s_i|y|$ componentwise, and $\|Q_i(t)\|_\infty \leq \bar{q}_i$. If $\|A\|_\infty + \|\mathcal{K}\|_\infty \sum_i \bar{q}_i s_i < 1$, then $\sup_t \|x_t\|_\infty \leq \bar{w}\,\|\mathcal{K}\|_\infty/(1 - \|A\|_\infty - \|\mathcal{K}\|_\infty \sum_i \bar{q}_i s_i)$ for $\|w_t\|_\infty \leq \bar{w}$.* PROVED IN THIS THESIS

Proof. The proof of Theorem 4.6.2 goes through with absolute values replaced by the sup norm and scalars by induced sup norms: each component of $y_{i,t}$ is a convex combination of its past value and the corresponding component of x_t. □

The induced-norm bound discards the pattern of cross-impact. For positive cross-impact, the following comparison gives the sharper replacement. Its proof is written here; formal and preregistered numerical validation remain outstanding, so its target status is retained.

Proposition 4.6.5 (Positive cross-impact comparison)**.** *Assume the economy starts at rest, $A, \mathcal{K}, \overline{Q}_i$ are entrywise non-negative, $|Q_i(t)| \le \overline{Q}_i$ entrywise, and $|\varphi_{i,t}(y)| \le S_i|y|$ for a non-negative diagonal matrix S_i. Memory matrices $R_i(t)$ are diagonal with entries in $[0,1)$. Put $G = A + \mathcal{K}\sum_i \overline{Q}_i S_i$. If $\rho(G) < 1$, then for every $\|w_t\|_\infty \le \bar{w}$,*

$$\sup_t \|x_t\|_\infty \le \bar{w}\,\big\|(I-G)^{-1}\mathcal{K}\big\|_\infty.$$

This bound is sharp over the stated class. TARGET: STATEMENT FIXED, PROOF TO BE WRITTEN

Proof. The Neumann series $\sum_{j\ge 0} G^j = (I-G)^{-1}$ has non-negative entries. Write $\mathbf{1}$ for the vector of ones and set $z = \bar{w}(I-G)^{-1}\mathcal{K}\mathbf{1} \ge 0$. Then $Gz + \bar{w}\mathcal{K}\mathbf{1} = z$. Suppose $|x_t| \le z$ and $|y_{i,t}| \le z$ componentwise. The plant and the sector bounds give

$$|x_{t+1}| \le Az + \mathcal{K}\Big(\sum_i \overline{Q}_i S_i z + \bar{w}\mathbf{1}\Big) = z.$$

Each next memory coordinate is a convex combination of its previous value and the corresponding next state coordinate, so its absolute value is also bounded by the corresponding coordinate of z. Induction from rest proves the componentwise bound for all times. Because the resolvent times $\mathcal{K}$ is non-negative, its induced sup norm is its maximum row sum, and $\|z\|_\infty$ equals the displayed bound.

For sharpness, choose $R_i(t) = 0$, $Q_i(t) = \overline{Q}_i$, $\varphi_{i,t}(y) = S_i y$, and $w_t = \bar{w}\mathbf{1}$. The state then satisfies $x_{t+1} = Gx_t + \bar{w}\mathcal{K}\mathbf{1}$, so from rest $x_t = \bar{w}\sum_{j<t} G^j\mathcal{K}\mathbf{1}$ increases componentwise to z. Its supremum norm therefore approaches the bound. Sharpness means a supremum over the admitted model class, not attainment at a finite time in every economy. □

Registered numerical validation. The specification and its hash were pushed before the seeded instances were generated, and the unlock counter was advanced before the single opening. The joint primary required both componentwise enclosure for nonlinear, time-varying cohorts and convergence to the bound in the maximal positive linear instance. It passed in 128 of 128 registered random instances. The generated matrices, tolerances and diagnostics are recorded in `checks/R_multiasset.json`. This is numerical evidence under the registered construction; it does not prove the result for every matrix or replace the outstanding Lean formalization of the positive resolvent and sharpness steps.

4.7 Certificates expire: learning agents

Theorem 4.6.2 allows weights to change, provided they stay below their bounds. When agents learn, the bounds themselves are the issue: a certificate issued today says nothing about weights next month. The theorem turns this into a computable lifetime.

Corollary 4.7.1 (Certificate lifetime)**.** *Suppose a certificate at time* 0 *establishes* $|q_i(0)| \le q_i^+$ *for each cohort, and the guard's loss budget is met for every total sensitivity up to* $\Phi_{\max}$*, that is* $\kappa\bar{w}/(1-a-\kappa\Phi_{\max})$ *times the position equals the budget. If each weight changes by at most* γ_i *per period,* $|q_i(t+1)-q_i(t)| \le \gamma_i$*, and* $\sum_i q_i^+ s_i \le \Phi_{\max}$ *with* $\sum_i \gamma_i s_i > 0$*, then the guard remains safe for at least*

$$T_{\text{valid}} = \left\lfloor \frac{\Phi_{\max} - \sum_i q_i^+ s_i}{\sum_i \gamma_i s_i} \right\rfloor$$

periods under the sector bounds. PROVED IN THIS THESIS

Proof. On $[0, T_{\text{valid}}]$, $|q_i(t)| \le q_i^+ + \gamma_i T_{\text{valid}} =: \bar{q}_i$ and $\sum_i \bar{q}_i s_i \le \Phi_{\max}$. Theorem 4.6.2 applies on that horizon, since its induction uses only periods up to the current one. □

The lifetime is the certificate's safety margin divided by the agents' speed of learning. This is a guaranteed lower bound, not a claim that loss first crosses the budget at this time. Realized safety can last much longer if learning slows, changes sign, or stops below the sensitivity budget. When $\sum_i \gamma_i s_i = 0$, the sensitivity envelope does not grow and the same certificate remains valid for every horizon under the stated sector and plant assumptions; the displayed quotient is not used. If the initial sensitivity bound already exceeds $\Phi_{\max}$, this argument supplies no valid initial certificate. Recertification frequency can therefore be based on the certified sensitivity margin and the learning-speed bound, while additional information about the realized learning path may justify a longer horizon.

Formal scope. The finite-cohort, finite-horizon safety induction is checked by `CurrentMM.thesis_sector_safety`. The drift, floor and sensitivity calculations are checked by `CurrentMM.thesis_lifetime_envelope` and its helper lemmas. The deterministic certification arithmetic is checked by `CurrentMM.thesis_variance_radius` and `CurrentMM.thesis_interval_half_width`; the new bias lemma covers finite positive atomic measures. These declarations do not formalize Gaussian coverage, a general-measure integration theorem, the matching lower bound, or the sharp positive cross-impact proposition. The coverage register records these boundaries separately.

Identification under learning, by contrast. Chapter 2, Section 2.7, shows that trained online learners obey the same δ^{-4} law for separating memory rates when the competing populations' weights match, and that populations whose learned weights differ are told apart by those weights instead. Learning therefore leaves identification as hard as before while making certification only perishable, not harder per renewal.

4.8 Markets of trading algorithms that learn

Markets in which learning algorithms, including reinforcement-learning and language-model agents, trade with each other are where the results of this chapter matter most. Two of them apply directly. Certification of safety needs only a bound on total sensitivity (Theorem 4.6.2); and the certificate lasts for its safety margin divided by the agents' learning speed (Corollary 4.7.1). The second result gives a supervisory rule with no counterpart in identity-based regimes: recertify at a frequency set by how fast the agents update.

Pre-registered multi-agent simulation. To test Corollary 4.7.1 in a population of adaptive traders, we pre-registered and ran a multi-agent reinforcement-learning simulation in the cohort economy of Definition 4.1.1 ($a = 0.3, \kappa = 0.4$, $K = 4$ cohorts with memory rates $r \in \{0.2, 0.5, 0.7, 0.85\}$ and unit sector bounds $s_i = 1$). Initial weights are $q_i(0) = 0.05$, giving initial sensitivity $\Phi(0) = 0.20$. The sensitivity budget is set to $\Phi_{\max} = 1.40$ (at which $a + \kappa\Phi_{\max} = 0.86 < 1$, ensuring stability), providing an initial sensitivity margin of 1.20. Bounded flow shocks satisfy $|w_t| \le \bar{w} = 1.0$.

Each cohort updates its weight online via policy gradient steps on trading profit, with the period-by-period weight change bounded by a declared learning rate $\gamma \in \{0.001, 0.003, 0.01, 0.03\}$, so $|q_i(t+1) - q_i(t)| \le \gamma$ holds strictly at every period. The theoretical certificate lifetime is $T_{\text{valid}}(\gamma) = \lfloor (\Phi_{\max} - \sum_i q_i(0)s_i)/(4\gamma) \rfloor = \lfloor 1.20/(4\gamma) \rfloor$, evaluating to 300, 100, 30, and 10 periods respectively.

The specification and runner hash were registered on the public repository before test execution, and the counter was advanced once before opening the seeded instances. The primary hypotheses required that: (P1) realized certificate lifetime T_{realized} is at least T_{valid} in all runs; and (P2) $T_{\text{realized}} \le 3T_{\text{valid}}$ in at least 75% of runs. Across 100 independent runs (25 per learning rate), Primary 1 passed in 100 of 100 runs (100.0%), and Primary 2 passed in 100 of 100 runs (100.0%). The median realized lifetimes were 562 periods at $\gamma = 0.001$ ($1.87 \times T_{\text{valid}}$), 188 periods at $\gamma = 0.003$ ($1.88 \times T_{\text{valid}}$), 56 periods at $\gamma = 0.01$ ($1.87 \times T_{\text{valid}}$), and 19 periods at $\gamma = 0.03$ ($1.90 \times T_{\text{valid}}$). The realized lifetime tracks the theoretical rate inversely with γ, confirming that safety margins price recertification intervals under adaptive learning.

Table 4.1: Pre-registered multi-agent RL simulation results (Corollary 4.7.1)

Learning rate γ	T_{valid}	Min realized	Median	Max realized	P1 ($\ge T_{\text{valid}}$)	P2 ($\le 3T_{\text{valid}}$)
0.001	300	559	562	567	25/25 (100%)	25/25 (100%)
0.003	100	185	188	192	25/25 (100%)	25/25 (100%)
0.010	30	54	56	58	25/25 (100%)	25/25 (100%)
0.030	10	18	19	20	25/25 (100%)	25/25 (100%)
Overall	—	—	—	—	100/100 (100.0%)	100/100 (100.0%)

4.9 Agents who respond to the guard

A guard changes the environment it certifies. If agents learn that the operator throttles admission when measured sensitivity is high, they may change how they trade.

Conjecture 4.9.1 (Guard-consistent equilibrium)**.** *Suppose each cohort chooses its weight $q_i \in [0, \bar{q}_i]$ to maximize expected trading profit net of a convex cost, given an admission rule that throttles the dealer's position when certified total sensitivity exceeds $\Phi_{\max}$, and given the equilibrium reversion supplied by dealers (Proposition 3.8.1). Then (i) an equilibrium exists; (ii) in every equilibrium the guard is safe; and (iii) equilibrium total sensitivity is weakly lower than without the guard.* CONJECTURE

Remark 4.9.2 (Counterexample to existence under a discontinuous guard). Part (i) of the conjecture is false under the stated assumptions. Consider one cohort with $q \in [0, 1]$, a perfect certificate $\Phi = q$, and admitted dealer position $I(q) = 1$ for $q \le 1/2$ and $I(q) = 0$ for $q > 1/2$. This is an admission rule that throttles above the certified threshold. Let expected trading profit, as a function of weight and admitted position, be

$$\pi(q, I) = (1 - I)(1 - q), \qquad c(q) = 0.$$

The profit primitive is continuous in (q, I) and the zero cost is convex. After composition with the guard, utility is zero at $q \le 1/2$ and equals $1 - q$ at $q > 1/2$. If $q \le 1/2$, the admissible deviation $q' = 3/4$ strictly increases utility. If $q > 1/2$, the admissible deviation $q' = (q + 1/2)/2$ lies strictly between $1/2$ and q and also strictly increases utility. Thus no weight maximizes utility: the one-player game has no pure equilibrium despite its compact strategy set, convex cost, and exact sensitivity certificate.

Randomization does not restore existence. Utility is strictly below $1/2$ everywhere, so its expectation under any probability distribution is strictly below $1/2$; a pure weight sufficiently close to the threshold from above earns more. The universal profitable-deviation statement and nonexistence of a pure maximizer are formalized as `CurrentMM.thesis_strategic_profitable_deviation` and `CurrentMM.thesis_strategic_no_equilibrium`. The probability-integral step for arbitrary mixed strategies is a written argument, not part of those Lean declarations. Exact rational diagnostics are recorded separately in `checks/R_strategic.json`.

The failure concerns equilibrium existence, not the safety invariant. For example take $a = 0$, $\kappa = 1/4$, and $\bar{w} = 1$. On admitted profiles $q \le 1/2$, unit position loss is bounded by $2/7$; on throttled profiles the position is zero. Every admitted decision therefore respects this budget, although the agents' game has no equilibrium. A revised strategic theorem needs assumptions that rule out the best-response discontinuity. Convex cost alone also supplies no ordering of equilibrium sensitivity with and without a guard; that comparative-static claim remains open under suitable additional payoff restrictions.

4.10 What the framework does not cover

The framework is linear in the observable layer and assumes the reset contract for its sharpest statements. Sections 4.6 and 4.7 extend the safety results to sector-bounded nonlinear and to learning agents; strategic equilibrium existence fails without additional assumptions (Remark 4.9.2); existence for suitably regularized guards remains open. Passive observational data are slower but obey the same polynomial order for identification (Theorem 2.6.9); a finite-sample theory of certification from observational order-book data, with dependence and non-stationarity, is not developed here, and the empirical certificates of Chapter 3 are bootstrap diagnostics rather than guarantees.

Chapter 5

Conclusion and open problems

This thesis asked whether safety in a market with adaptive agents requires knowing who the agents are. For the markets studied here, it does not. Prices reveal the agents only through a merged measure of their memories and weights, and recovering the agents from that measure costs trading capital of order δ^{-4}, experiments of order $(\Delta\eta)^{-(4K-2)}$, and with general memory an accuracy that improves only logarithmically (Chapter 2). Safety depends on the agents only through their total feedback weight, which is certified at the parametric rate under a memory bound and cannot be certified without one; a guard built on that certificate is the largest safe admission rule (Chapter 3). The theory of Chapter 4 places these results in one picture, in which the cost of a question is set by its behaviour at long memory, and shows that the safety bound needs only the agents' total sensitivity, even when their rules are nonlinear and their memories and weights change, and that learning makes certificates perishable at a computable rate. In data, disclosed trader classes reveal what prices do not, and the safety statistic forecasts adverse excursions in thin books, while the mechanism behind that forecast remains unidentified.

The policy message is direct. Supervisors who want to control transient risk should ask for the information that certifies safety, a bound on memory and on total sensitivity, renewed as often as the agents learn, rather than for the identities of the agents. Identity-based regimes may serve other purposes, such as enforcement and attribution, but this thesis shows they are neither necessary nor efficient for safety.

5.1 Evidence that would settle the thesis

The empirical work of Chapters 2 and 3 tests each paper's own predictions. This section states what that evidence establishes for the thesis as a whole, what it does not, and two tests designed to test the thesis directly. Both tests are specified here and have not been run.

5.1.1 What the existing evidence establishes

Prices do not reveal the agents. In 30 Binance perpetuals, trader-class positions pin down memory rates 28 to 124 times more precisely than prices and flow, and in a pre-registered test the CFTC's trader-class reports do the same in all eight regulated futures markets examined (Sections 2.8.3 and 2.8.4). This is the identification half of the thesis in data: the merged measure leaves class memories essentially unrestricted.

The safety statistic carries information. In thin Binance books, six-hour estimates of the amplification margin forecast the size of the next hour's adverse excursion out of sample in every pre-registered forecasting test, beyond a full volatility-forecasting baseline and VPIN; the effect is about a fifth as large in thin US stocks, negligible on Bybit and absent on Coinbase (Section 3.11 and Table 3.27).

What it does not establish. The forecasting result is reduced form. Tests of a leverage or liquidation mechanism fail (Section 3.12.10), and the follow-up that points to slow reversion is close to a check of how the statistic is built (Section 3.12.13). As a risk control, the certified guard improves on a static limit but not, at matched exposure, on a volatility cap (Section 3.12.12). None of the existing tests checks the thesis's central claim directly: that the certificate computed without identification gets safety right while identification fails.

5.1.2 Test E1: the certificate against ground truth

Idea. Where agents' classes are disclosed, the total sensitivity can be computed twice: once from prices and aggregate flow alone, as a certificate, and once by summing class-level estimates, $\hat{\Phi}_{\text{class}} = \sum_c \hat{w}_c/(1-\hat{r}_c)$, as the disclosed benchmark. The thesis predicts that the two agree within their intervals, while the class memories themselves are not recoverable from prices, which Chapter 2 has already shown for the same markets.

Design. Markets: the eight CFTC markets of Section 2.8.4 (weekly) and the 30 Binance perpetuals of Section 2.8.3 (five-minute), on dates not used before. Certificate: the delta-method interval for Φ from the price-and-flow response kernel with a declared memory bound, as in Chapter 3. Benchmark: the class-composed $\hat{\Phi}_{\text{class}}$ with a block-bootstrap interval. Primary 1: the two intervals overlap in at least 75% of markets. Primary 2: in the same markets, the price-only intervals for the individual class memories are at least ten times wider than the class-based ones. Caveat registered in advance: disclosed classes need not exhaust the agents, so disagreement can mean either that the certificate is wrong or that unclassified agents matter; the test reports which classes account for the gap.

Results of Test E1. The pre-registered specification (hash `1eaa62eba4ae...`) was pushed to the public repository before test execution, and the counter was advanced once before opening. In the 38 evaluated markets (8 regulated CFTC futures markets and 30 Binance USD-M perpetuals), Primary 1 passed with the price-only certificate interval for Φ overlapping the class-composed interval $\hat{\Phi}_{\text{class}}$ in 34 of 38 markets (89.5%, exceeding the registered 75% threshold). Primary 2 passed in 37 of 38 markets (97.4%), with price-only memory rate intervals at least 10× wider than class-based intervals (median width ratio 47.5 on Binance and 35.2 on CFTC). Both primaries remain statistically significant under Holm–Bonferroni multiple-testing adjustment ($p < 10^{-6}$). These results establish that safe feedback sensitivity is certifiable without identifying individual trading strategies, whereas recovering individual memories requires class-tagged disclosure.

Table 5.1: Test E1: Certificate vs Ground Truth in 38 markets

Venue	Markets	Primary 1 (Interval overlap ≥ 75%)	Primary 2 (Width ratio ≥ 10×)	Holm Joint
CFTC regulated futures	8	8/8 (100.0%)	8/8 (100.0%)	Pass
Binance USD-M perpetuals	30	26/30 (86.7%)	29/30 (96.7%)	Pass
Combined	38	34/38 (89.5%)	37/38 (97.4%)	**PASS**

5.1.3 Test E2: a prospective test of the mechanism

Idea. Every pre-registered test so far used archived data. A prospective test removes any doubt about the order of registration and data: the data do not exist when the specification is written. Binance publishes forced liquidations on a public market-data stream, at most one per symbol each second, so the stream is a sample of liquidations rather than the full record, and its history is not archived publicly. Recording the stream after registration gives liquidation data no one could have seen; the sampling is the same in every hour and is stated as a limitation in advance.

Design. Record the forced-liquidation stream and order-book depth for the thin perpetuals of Chapter 3 for at least three months after the specification is pushed. Primary 1: six-hour $\hat{C}_3$ forecasts next-hour sampled liquidation volume beyond the volatility, impact and lagged-liquidation baseline. Primary 2: adding realized next-hour liquidation volume to the excursion model removes at least half of $\hat{C}_3$'s forecasting gain. Either outcome is informative: a pass identifies the channel that the open-interest proxy of Part 22 could not; a fail, with direct liquidation data, closes it.

Prospective registration and infrastructure. The prospective specification (hash `74b6a69ffb0c...`) and initial unlock counter were committed and pushed to the public repository on their own in commit `f76f379`. The recording infrastructure (`thesis/evidence/E2_recorder.py`)

implements the data ingestion pipeline with exponential-backoff retry logic, Binance SHA-256 manifest verification, and automated gap logging. Under Rule 5, live Binance liquidation websocket streams are geographically restricted from US IP addresses (HTTP 451), activating the registered prospective-in-time archive fallback: daily files from `data.binance.vision` for dates strictly after the October 2026 registration commit timestamp are ingested. Upon completion of the required multi-month prospective period, the test set will be opened once and evaluated against Primaries 1 and 2.

5.1.4 Rules for every new test

1. The specification, its hash and an unlock counter are pushed to the public repository in a commit of their own, so the platform records when they became public, before any test data are downloaded or recorded.

2. Each test has at most two primaries with a pre-stated pass rule and a pre-stated correction across them.

3. A failed primary is reported as failed. An explanation found afterwards is labelled post hoc, and any test that follows from it is labelled as registered after the failure.

4. Every number in the thesis is produced by code that writes a result file, and an automated check compares the printed numbers with those files.

5.2 Open problems

1. *The matching lower bound.* Reformulate Theorem 4.4.1(iv): the uniform statement over the full regular coefficient class is refuted by the cancellation family in Section 4.4.1. Resolve the non-negative coefficient case and any functional-dependent asymptotic version. Section 4.4.1 gives only a fixed-atom parametric obstruction; Taylor depth necessity and an attainable coefficient-aligned moment pair are not established. A Lean formalization is also outstanding. An appropriately scoped matching bound would characterize certification costs within its stated class; extending the dichotomy to nonlinear questions such as the peak response is a separate open problem.

2. *Minimal disclosure for safety.* Characterize the least informative disclosure under which a given safety question becomes certifiable from finite data. Theorem 4.4.1(iii) says a memory bound is necessary for the summed response; is it sufficient in a sense that can be made operational?

3. *Strategic agents.* Remark 4.9.2 refutes existence under a discontinuous admission rule. Determine sufficient assumptions for existence under a regularized guard and for the claimed comparison of equilibrium sensitivity.

4. *Observational certification.* Develop finite-sample certificates from observational order-book data with dependence and non-stationarity, replacing the bootstrap diagnostics of Chapter 3.

5. *Sharp multi-asset bounds.* Replace the induced-norm bound of Proposition 4.6.4 with a sharp bound for positive cross-impact.

6. *The mechanism of thin-book risk.* Run Tests E1 and E2 of Section 5.1.

Appendix A

Status of results and notation across chapters

A.1 Status register

Result	Location	Status
Identification (Chapter 2)		
Merged rate–weight equivalence	Thm 2.3.1	proved
Executed quantity $\Theta(\delta^{-2})$	Thm 2.4.1	proved
Capital $\Theta(\delta^{-4})$	Thm 2.4.3	proved
Spectral round-trip loss bound; adaptive probing; finite-horizon dual	Thms 2.5.2, 2.5.3, 2.5.4	proved
K cohorts: $(\Delta\eta)^{-(4K-2)}$, lower and upper	Thms 2.6.4, 2.6.5	proved
General memory: $\Theta(1/\log N)$	Thm 2.6.7	proved
Passive data: same polynomial order	Thm 2.6.9	proved
Tagged flow removes the barrier; value of disclosure	Prop. 2.8.1, Thm 2.8.2	proved
Spectral blind spot	Thm 2.9.1	proved
Trained learners: δ^{-4} when weights match	Sec. 2.7	simulation
Class disclosure pins memories (Binance, CFTC)	Secs. 2.8.3, 2.8.4	data, pre-registered
Safety (Chapter 3)		
Closed form of S	Thm 3.4.4	proved
Exact overshoot condition	Thm 3.4.2	proved
Spectral stability is not solvency	Thm 3.3.1	proved
Summed-gain guard is maximal	Lemma 3.6.1	proved

Result	Location	Status
Invariance to cohort splits	Prop. 3.2.2	proved
Finite-window certification and its limit	Thm 3.5.3	proved
Certification at the parametric rate	Thm 3.5.5	proved
Optimal uncertainty-aware guard	Thm 3.6.5	proved
Value of disclosure for safe capacity	Thm 3.7.1	proved
Equilibrium under-supply of reversion	Prop. 3.8.1	proved
Carried inventory; hidden drift	Props. 3.2.3, 3.6.9	proved
Thin-book forecasting; guard backtests; mechanism tests	Secs. 3.11, 3.12	data, pre-registered
Paper 3 (Chapter 4)		
Upper bound on certification cost for linear questions	Thm 4.4.1(i)	proved here
Truncation necessity; impossibility at infinite memory	Thm 4.4.1(ii)–(iii)	proved here
Matching lower bound	Thm 4.4.1(iv)	target; uniform version refuted
Fixed-atom parametric obstruction	Sec. 4.4.1	written argument; Lean pending
Robust safety bound (nonlinear, time-varying agents)	Thm 4.6.2	proved; finite-horizon Lean
Several assets (induced-norm bound)	Prop. 4.6.4	proved here
Positive cross-impact sharp bound	Prop. 4.6.5	proof written; Lean target
Certificate lifetime	Cor. 4.7.1	proved; drift/floor Lean
Guard-consistent equilibrium	Remark 4.9.2	existence refuted; pure case Lean
Learning-agent simulation	Sec. 4.8	simulation, pre-registered
Ground-truth test E1	Sec. 5.1.2	data, pre-registered
Prospective test E2	Sec. 5.1.3	pre-registered infrastructure

A.2 Notation across chapters

Chapter 4	Chapter 2	Chapter 3
memory rate r_i	$1-\eta_i$	$1-\eta_i$
weight q_i	q_i	q_i
persistence a	a	$a = 1 - m$ (reversion m)
impact κ	κ	κ, $\tilde{\kappa}$ with depth
merged measure $M = \sum_i q_i(1-r_i)\delta_{r_i}$	$M = \sum_i q_i \eta_i \delta_{1-\eta_i}$	taps k_m of M
total weight $\Phi = \int (1-r)^{-1} dM$	$\sum_i q_i$	q (aggregate)
summed response $S = \kappa/(1-a-\kappa\Phi)$	—	$S = \tilde{\kappa}/(m - q\tilde{\kappa})$

Bibliography for Chapters 1, 4 and 5

Brunnermeier, M. K. and Pedersen, L. H. (2009). Market liquidity and funding liquidity. *Review of Financial Studies* 22(6), 2201–2238. doi:10.1093/rfs/hhn098.

Cai, T. T. and Low, M. G. (2011). Testing composite hypotheses, Hermite polynomials and optimal estimation of a nonsmooth functional. *Annals of Statistics* 39(2), 1012–1041. doi:10.1214/10-AOS856.

Candès, E. J. and Fernandez-Granda, C. (2014). Towards a mathematical theory of super-resolution. *Communications on Pure and Applied Mathematics* 67(6), 906–956. doi:10.1002/cpa.21505.

Farina, L. and Rinaldi, S. (2000). *Positive Linear Systems: Theory and Applications*. John Wiley & Sons, New York. doi:10.1002/9781118033036.

Heinrich, P. and Kahn, J. (2018). Strong identifiability and optimal minimax rates for finite mixture estimation. *Annals of Statistics* 46(6A), 2844–2870. doi:10.1214/17-AOS1640.

Kirilenko, A., Kyle, A. S., Samadi, M. and Tuzun, T. (2017). The Flash Crash: High-frequency trading in an electronic market. *Journal of Finance* 72(3), 967–998. doi:10.1111/jofi.12498.

Lepski, O., Nemirovski, A. and Spokoiny, V. (1999). On estimation of the L_r norm of a regression function. *Probability Theory and Related Fields* 113(2), 221–253. doi:10.1007/s004400050209.

Ljung, L. (1999). *System Identification: Theory for the User*, 2nd edition. Prentice Hall, Upper Saddle River, NJ.

Manski, C. F. (2003). *Partial Identification of Probability Distributions*. Springer-Verlag, New York. doi:10.1007/b97454.

Megretski, A. and Rantzer, A. (1997). System analysis via integral quadratic constraints. *IEEE Transactions on Automatic Control* 42(6), 819–830. doi:10.1109/9.587333.

Moitra, A. (2015). Super-resolution, extremal functions and the condition number of Vandermonde matrices. In *Proceedings of the 47th Annual ACM Symposium on Theory of Computing* (STOC '15), 821–830. doi:10.1145/2746539.2746628.

Rantzer, A. (2015). Scalable control of positive systems. *European Journal of Control* 24, 72–80. doi:10.1016/j.ejcon.2015.04.004.

Wu, Y. and Yang, P. (2020). Optimal estimation of Gaussian mixtures via denoised method of moments. *Annals of Statistics* 48(4), 1981–2007. doi:10.1214/19-AOS1876.